

PHYSICS FALCON NOTES FOR CLASS SECOND YEAR

ABOUT PHYSICS

“Physics is not just a subject to study; it is a universe to understand.”

“Understand the concept, master the formula, and practice until the problem becomes easy.”

“Every equation tells a story; every concept reveals a law of nature.”

“Don’t memorize Physics—understand it, visualize it, and apply it.”

“Strong concepts today, confident results tomorrow.”

“Physics teaches us that behind every phenomenon, there is a reason waiting to be discovered.”

“Learn with curiosity, solve with logic, and achieve with consistency.”

ABOUT NOTES

“These notes are not just pages of Physics; they are a journey from confusion to clarity.”

ABOUT AUTHOR SIR IMRAN KHAN

M.PHIL IN PYSICS FROM PESHWAR UNIVERSITY

SPECIAL THANKS TO MY COLLEAGUES

I express my sincere gratitude’s to the managing director of the victorious college sir Tahir Khan M.phil Biology, Sir Farooq khan Mphil Mathematics and Sir Irfan khan M.Phil Chemistry to able me for the success of this notes.

Chapter #11 "Electrostatics"

Atom

An atom is the basic unit of a chemical element, consisting of a nucleus surrounded by electrons. The nucleus contains protons and usually neutrons, while the electrons orbit around the nucleus in Energy levels.

Neutral Atom:

A neutral atom is an atom that has an equal number of protons and electrons, resulting a net zero charge.

Charge on an electron and proton:

Charge is a natural property on an electron and proton, due to which a repulsive force is produced between two electrons or between two protons, when they are close to each other. OR

Charge is a natural property on an electron and proton, due to which an attractive force is produced between electron and proton. Such a charge property is quantitative property, and its unit is coulomb "C" and its representation symbol is "q". Further Millikan physicist calculated the charge on an electron experimentally, and its value is equal to $1.602 \times 10^{-19} \text{ C}$.

Further he also calculated the charge on a proton and its value is equal to the same electron charge value is $1.602 \times 10^{-19} \text{ C}$.

Further Millikan took negative sign with electron charge value, and took positive sign with proton charge value conventionally for the atom neutrality.

Now we can write as

$$q_e = -1.602 \times 10^{-19} \text{ C} \quad \text{and} \quad q_p = +1.602 \times 10^{-19} \text{ C}$$

Further in magnitude form we can write as

$$|q_e| = |q_p| = 1.602 \times 10^{-19} \text{ C} = e$$

Where $e = 1.602 \times 10^{-19} \text{ C}$ is called elementary charge of electron and proton.

Charge on a Materials

When a material loses the electrons, then positive charge will come on that material. Mathematically such a charge we can write as

$$q = ne \quad \text{----- (1)} \quad \text{Where } n = 1, 2, 3, 4, \dots \text{ and } e = 1.602 \times 10^{-19} \text{ C}.$$

Equation (1) represents if a material lost a single electron then the charge on it will be equal to $1e$. (i.e $q = 1e$). Similarly if a material lost "2" electrons then the charge on it will be equal to $2e$. (i.e $q = 2e$) and up to so on.

Now from equation (1) we can say that charge "q" is quantized its mean charge on a material is an integral multiple of elementary charge "e". Integral multiple means (1, 2, 3, 4, ...).

Now if a material gained the electrons, then negative charge will come on that material. Mathematically such a charge we can write as $q = -ne$ ----- (2)

Equation (2) represents if a material gained an electron then the charge on it will be equal to $-1e$. (i.e $q = -1e$). Similarly if a material gained "2" electrons then the charge on it will be equal to $-2e$. (i.e $q = -2e$) and up to so on.

Now from equation (2) we can also say that charge “q” is quantized its mean charge on a material is an integral multiple of elementary charge “e”.

Types of charges

Source charge

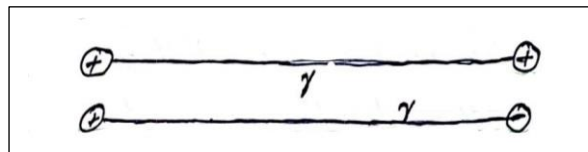
A charge which has greater magnitude of charge is called Source charge.

Test Charge

A charge which has negligible magnitude of charge is called Test charge.

Point Charge: A source charge which we study it from a greater distance is called point charge.

point charges: Those source charges which have greater distance in between and become look like as a points are called point charges.



Coulomb's Law

The quantitate measurement of the force (attractive or repulsive) between two electric point charges was first made by coulomb. He carried out series of experiment to measure the force between the electric point charges by using an apparatus known as torsion balance. Then coulomb expressed his experimental data in the form of a statement which is known as coulomb's law.

Statement: The magnitude of force between two electric point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. Mathematically we can writ as

$$F \propto q_1 q_2 \text{ ----- (1)}$$

$$F \propto 1/r^2 \text{ ----- (2)}$$

Combine (1) and (2) we get as

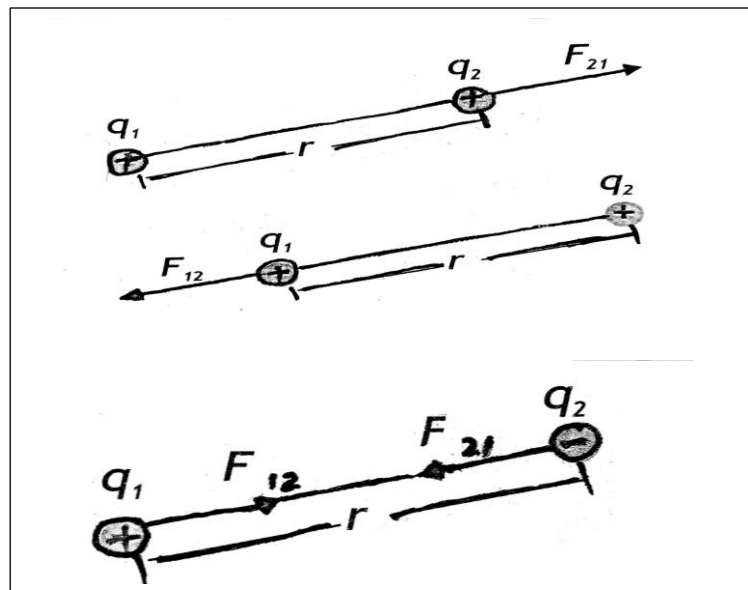
$$F \propto q_1 q_2 / r^2$$

$$F = K q_1 q_2 / r^2 \text{ ----- (3)}$$

Equation (3) represents Mathematical form of coulomb's law. In this equation “F” represents the attractive or repulsive force between the two electric point charges.

Explanation of constant “K”

In equation (3) “k” is a constant of proportionality and its value depends on



the surrounding medium around the point charges and it also balances the units to each side in equation. Then coulomb's calculated the value of "k" on experiment base in case of vacuum or free space around the point charges and it can be written as

$$K = 1/4\pi\epsilon_0 \text{ ----- (4) (on experiment base).}$$

Then coulomb's also calculated the value of " ϵ_0 " on experiment base in case of vacuum or free space around the point charges and it can be written as

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2 \text{ -----(5)}$$

In equation (5) " ϵ_0 " is called permittivity of free space. Now equation (4) we can write as

$$K = 1/4 \times 3.14 \times 8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2$$

$$K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Permittivity: The property of the medium such that to affect or resist the magnitude of force between two point charges is called permittivity.

Permittivity of free space of vacuum we will represent by " ϵ_0 ". And permittivity of any medium we will represent by " ϵ ".

Vector form of coulomb's law:

Vector form of coulomb's law means to explain the direction of forces between two point charges. For the direction of forces we will use unit vectors as shown in figure.

Now in figure $\hat{r}_{12}$ pointing from the charge " q_1 " towards the charge " q_2 " represents the direction of $\vec{F}_{12}$ force. $\vec{F}_{12}$ force mean force on " q_2 " charge due to " q_1 " charge.

Now in vector form $\vec{F}_{12}$ force we can write as

$$\vec{F}_{12} = k q_1 q_2 / r^2 \hat{r}_{12} \text{ -----(6)}$$

Similarly in figure $\hat{r}_{21}$ pointing from the charge " q_2 " towards the charge " q_1 " represents the direction of $\vec{F}_{21}$ force. $\vec{F}_{21}$ force mean force on " q_1 " charge due to " q_2 " charge.

Now in vector form $\vec{F}_{21}$ force we can write as

$$\vec{F}_{21} = k q_1 q_2 / r^2 \hat{r}_{21} \text{ -----(7)}$$

Now from figures we can write as

$$\hat{r}_{12} = -\hat{r}_{21} \text{ ----- (8)}$$

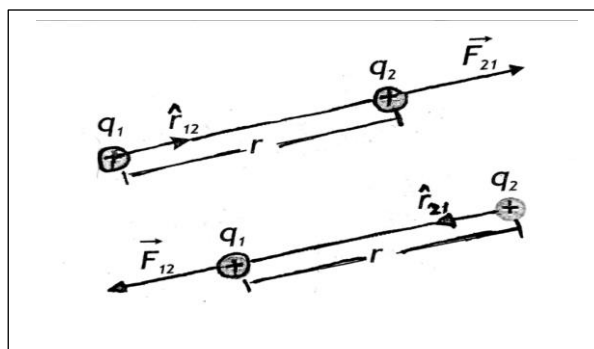
Now equation (8) we can write as by putting equation (6) and (7). i.e

$$\begin{aligned} \vec{F}_{21} r^2 / k q_1 q_2 &= -\vec{F}_{12} r^2 / k q_1 q_2 \\ \Rightarrow \vec{F}_{21} &= -\vec{F}_{12} \text{ ----- (8)} \end{aligned}$$

Now from equation (8) we can say that these two forces will be equal in magnitudes but will have the opposite directions.

Coulomb's law in material media

When the point charges are in free space or vacuum then the expression for the force we can write as



$$F_{\text{vacuum}} = q_1 q_2 / 4\pi\epsilon_0 r^2 \text{ -----(9)}$$

Now when the point charges are in material medium like water, Ammonia or Benzene etc then the expression for the force we can write as

$$F_{\text{med}} = q_1 q_2 / 4\pi\epsilon r^2 \text{ -----(10)}$$

Now we define relative permittivity such as “The ratio of the permittivity of any medium to the permittivity of free space is called relative permittivity”. i.e

$$\epsilon_r = \epsilon / \epsilon_0 \text{ -----(11)}$$

Relative permittivity just represents the comparison of any medium permittivity with respect to permittivity of free space of vacuum.

Now from equation (11) we can write as

$$\epsilon = \epsilon_r \epsilon_0 \text{ -----(12)}$$

Now put equation (12) in equation (10). We get as

$$F_{\text{med}} = q_1 q_2 / 4\pi \epsilon_r \epsilon_0 r^2$$

$$F_{\text{med}} = 1/\epsilon_r (q_1 q_2 / 4\pi \epsilon_0 r^2) \text{ -----(13)}$$

Now put equation (9) in equation (13) we get as

$$F_{\text{med}} = 1/\epsilon_r (F_{\text{vacuum}})$$

$$F_{\text{med}} = F_{\text{vacuum}} / \epsilon_r \text{ -----(14)}$$

Now from equation (14) we can say that F_{med} will be less than $1/\epsilon_r$ time than F_{vacuum} .

i.e $F_{\text{med}} > F_{\text{vacuum}}$.

Prove that Relative permittivity of vacuum of free space is 1

As we know that

$$\epsilon_r = \epsilon / \epsilon_0 \text{ -----(1)}$$

For vacuum we put $\epsilon = \epsilon_0$ in equation (1) we get as

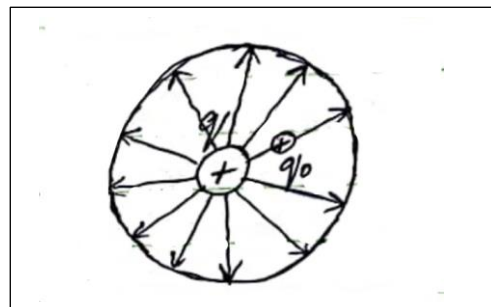
$$\epsilon_r = \epsilon_0 / \epsilon_0$$

$$\epsilon_r = 1 \text{ (proved)}$$

Electric field and its intensity

Electric field: The region around a source charge such that in which a test charge to an experience an electric force is called Electric field region and the field which existing in that region is said to be Electric field.

Explanation: The concept of Electric field was first proposed by Michael Faraday, in the 19th century. He stated that the electric field around a source charge is like a sphere with in which a test charge experiences an electric force. The direction of electric field we will decide from the motion of a test charge. In a figure a test charge “q” moves outside due to the repulsive force, so the direction of Electric field will be also in out word direction as shown in figure. Electric field is a vector quantity because due to which a test charge an experiences Electric force.



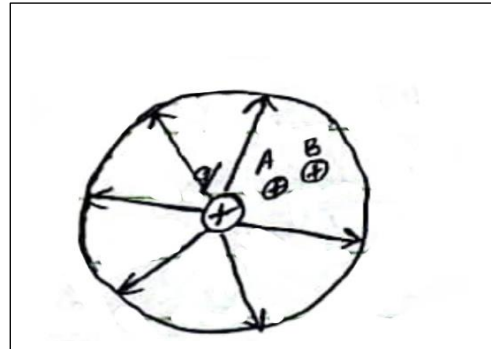
Electric field intensity or Electric field strength:

At any point in the Electric field region the Electric field intensity depends on the magnitude of an Electric force on a test charge. At a given point in the region, if the magnitude of an Electric force is greater on a test charge, then the electric field intensity or strength will be greater at that point-and vice versa.

In a figure the electric field intensity will be greater at “A” point and will be less at “B” point. Electric field intensity is also a vector quantity because due to which a test charge experiences an electric force. Now electric field intensity mathematically

We can write as

$$\vec{E} = \vec{F}/q_0$$



Relation of Electric field intensity with respect to distance

As we know that

$$\vec{E} = \vec{F}/q_0 \text{ ----- (1)}$$

In equation (1) F is an electric force on a test charge “q₀”. Now from coulomb’s law we can write as

$$\vec{F} = kq q_0 / r^2 \hat{r} \text{ ----- (2)}$$

In equation (2) “r” represents a distance from the source charge up to a given point in the electric field region with the existence of a test charge on that point. “ $\hat{r}$ ” unit vector represents the direction of an electric force.

Now put eq (2) in eq (1). we get as

$$\begin{aligned} \vec{E} &= k q q_0 / q_0 r^2 \hat{r} \\ \vec{E} &= k q / r^2 \hat{r} \text{ ----- (3)} \end{aligned}$$

From eq (3) we can say that the electric field strength or intensity is directly proportional to the magnitude of the source charge and is inversely proportional to the square of the distance.

So we can say that when a test charge moves away from a source charge along points, then the electric field strength will be decreases due to the increase in distance.

Unit of Electric field intensity: As we know that $E = F/q_0$. So the unit of an electric field intensity is N/C. The other unit of an electric field intensity is V/m.

Prove that N/C = V/m: As we know that

$$\text{Voltage} = \text{Work done} / \text{charge}$$

$$\text{Voltage} = Fd/q$$

$$\text{Volt} = \text{Nm}/\text{C}$$

$$V = \text{Nm}/\text{C}$$

$$\text{N/C} = \text{V/m. (Proved)}$$

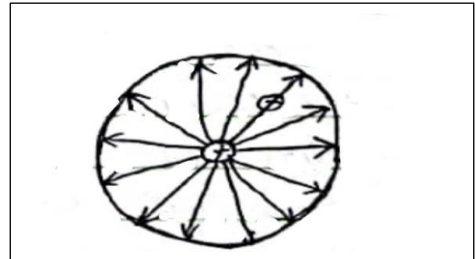
Representation of Electric field by lines

The pictorial representation of Electric field lines is called Representation of Electric field by lines. We discuss cases for the Representation of Electric field by lines.

Case 01: Positive source charge:

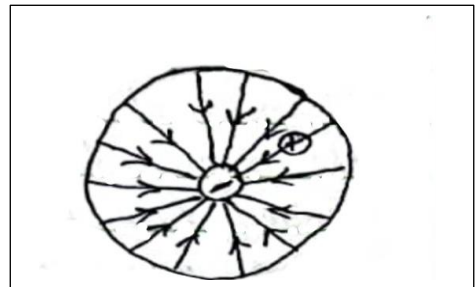
In case of positive source charge the representation of Electric field lines we will decide from the motion of a test charge.

In a given figure a test charge moves in outward direction, so electric field lines will be in outward direction from the positive Source charge.



Case 02: Negative Source Charge:

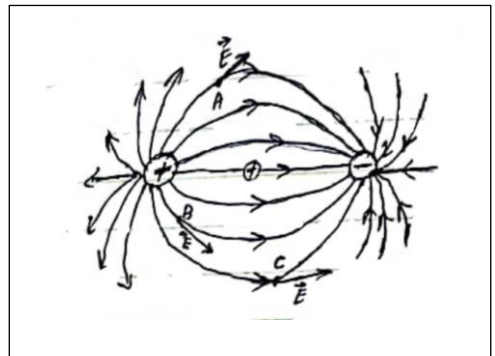
In case of negative source charge the representation of Electric field lines we will decide from the motion of a test charge. In a given figure a test charge moves in inward direction, so electric field lines will be in inward direction to the negative Source charge.



Case 03: Two opposite Source Charges:

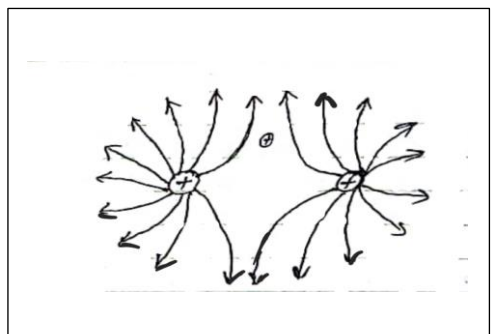
In case of two opposite source charges the representation of electric field lines we will decide from the motion of a test charge.

In a given figure a test charge moves from +ve source charge towards the -ve source charge. So the direction of Electric field lines will start from +ve source charge and will terminate on the -ve source charge. In this case the direction of Electric field E will be tangential to the lines as shown on points A, B and C.



Case 04: Two same +ve source charges:

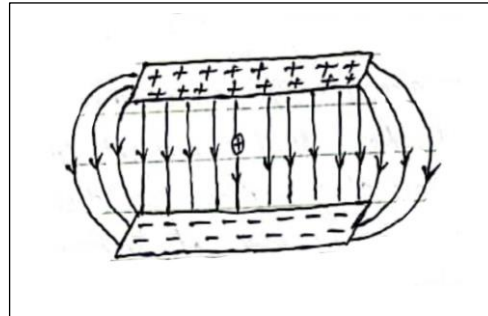
In case of two same +ve source charges the representation of electric field lines we will decide from the motion of a test charge. In a given figure a test charge moves out side from these two +ve source charges. So the direction of Electric field lines will be in outward direction and will be parallel to each other. In this case the direction of Electric field will be tangential to the lines.



Case 05: Two oppositely infinite charged parallel plates:

In case of two oppositely infinite charged parallel plates the representation of Electric field lines we will decide from the motion of a test charge.

In a gives figure a test charge moves from +ve charged plate towards a -ve charged plate. So the direction of Electric field lines from +ve charged plate and will be terminate on the -ve charged Plate. Now the electric field lines in between the plates give us the uniform Electric field.



Now uniform electric field is define as the field in which the electric field lines having

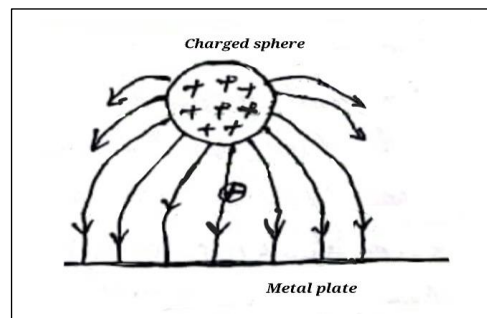
the same direction and having the same intensity is called uniform Electric field. Now the Electric field lines at the ends of the plates give us the non-uniform or fringing Electric field.

Fringing Electric field is define as the field in which the Electric field lines not having the same direction and also not having the same intensity is called fringing Electric field.

Case 06: Charged sphere:

In case of charged sphere the representation of Electric field lines we will decide from the motion of a test charge.

In the figure a charged sphere closed to the metal plate are shown. Now due to an electrostatic induction the charged sphere creates the -ve charge on the surface of the metal plate. Then a +ve charged sphere will repel the test charge and -ve charge on the surface of the metal will attract the test charge. So the test charge will move towards the metal plate Surface. Now we can say that the electric field lines will start from the charged sphere and will terminate perpendicularly on the surface of the metal plate.

**Application of Electrostatics****Photocopier**

A device which gives us the copies of documents is called photocopier.

Xerography: The copying process of photocopier is called xerography.

Construction: It has a light source, a glass stage and Aluminum drum with coated plastic layer of selenium material. Selenium is a material which behave is an insulator in dark and become conductor when expose to light. It is also called photoconductor.

Working principle: when we switch on a photocopier for the copy of some

Documents, then the rod get +ve charges. Then a side fan pushes the air atom between the rod and aluminum drum and then the air atoms collides with a rod and during collision the atoms will lost the electron and will get a +ve charge atom. Then the +ve charge of the rod will repel the +ve charge atom towards a selenium material and then such a +ve charge atom will stick on a selenium material because it will behave as an insulator when there is no light.

By this way a selenium material will get a +ve charge atom at all surface when the aluminum drum is rolling. Then after that a light start falling on the glass stage from the light source. Then the light will pass from the stage and from the white part of the documents and will not pass from the black part of the "A".

Then the selenium material will lost the +ve charge atoms at those places where the light falls on it. Because due to falling light selenium material will become conductor and then will give the electrons to the +ve charge atom.

Then the selenium material will have the +ve charge atoms at those places where the light does not fall on it. Because due to no light selenium material behave as an insulator and will maintain the +ve charge atom.

So the +ve charge atom will have the same shape as "A" on a Selenium material because of no falling light on these areas. Then after that the toner powder goes towards the selenium material and will stick on charged areas on the selenium in the shape of "A" and will slip down from the uncharged areas. Then after that a sheet of white paper passes over the drum and then the toner powders are attached with it.

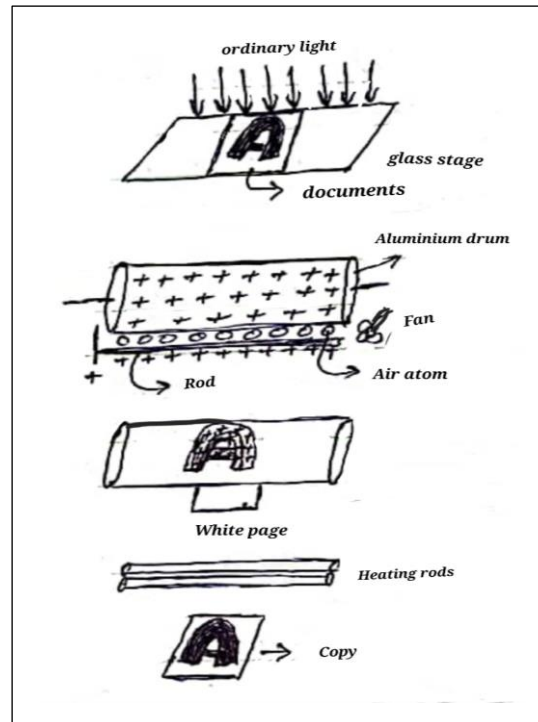
Then after that the paper passes from the heating rods and melt a toner powders in the shape of "A" on the document. By this way then we get a permanent copy of the document.

Laser printer

Laser printer has the same construction and same working principle as a photocopier.

But the difference in photocopier and laser printer is that photocopier works on an ordinary light and Laser printer works on Laser light.

Laser printer gives us high quality copies of the documents as compared with photo copies. Because laser light is monochromatic and have the same intensity.



Inkjet printer

A device which gives us a print copy with a help of liquid ink is called Inkjet printer.

Construction: It has the following parts.

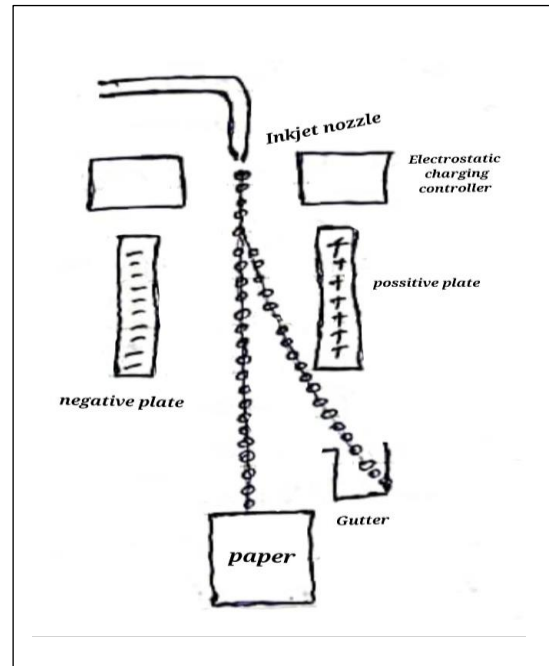
- ① Ink Supply ② pump
- ③ Print head (which reads or scans the documents in the computer)
- ④ Charging plates ⑤ Deflecting plates ⑥ Gutter

Working principle: An inkjet printer is another type of printer that uses electric charges in its operation.

In an inkjet printer tiny drops of ink are forced out from a nozzle, then during flight, the drops pass through two electrical components. One is called Electrostatic charging control and the second one is called deflecting parallel charged plates.

Now when the print head moves over the regions of the paper (Document) such that which have not colors, then the electrostatic charging control is turn on and give the electrons to the drops. Then the deflecting plates divert the charged drops into a gutter and thus prevent them from reaching the paper.

Now when the print head moves over the regions of the paper (Document) such that which have color letters such as (a ,b ,c), then the electrostatic charging control is turn off and does not give the electrons to the drops. Then the deflecting plates does not divert the charged drops into a gutter ,but the ink drops will move straight and will strike a paper, and by this way at last then we get the print copy of the Document.



Electric Flux

The measure of number of Electric field lines that pass through a certain area is called Electric flux.

Explanation: In a figure a number of electric field lines are passing through a certain area, so its measure from that area is called electric flux. Now electric flux (Φ) depends on the following factors.

- ① Surface area ② electric field intensity
- ③ On the angle between area vector A and Electric field intensity E.

Mathematically we can write as

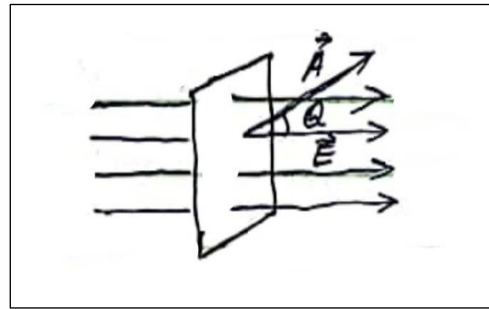
$$\Phi \propto A \text{ -----(1)}$$

$$\Phi \propto E \text{ -----(2)}$$

$$\Phi \propto \cos\theta \text{ -----(3)}$$

From these relations we can say that if area is greater then more number of electric field lines will be passing and electric flux will be greater and vice versa.

Similarly we can say that if Electric field intensity is greater then more number of electric field lines will be passing through area and electric flux will be greater and vice versa. Also we can say that if $\cos\theta$ value is greater then electric flux will be greater and vice versa.



Now combine (1), (2) and (3) we will get as

$$\Phi \propto EA \cos\theta$$

$$\Rightarrow \Phi = K EA \cos\theta$$

$$\Rightarrow \Phi = EA \cos\theta \quad \text{put } K=1$$

$$\Rightarrow \Phi = \vec{E} \cdot \vec{A} \text{ ----- (4)}$$

In equation (4) from the dot product of $\vec{E}$ and $\vec{A}$ we can say that the electric flux is a scalar quantity.

Dependence of Electric flux at angle " θ " :

1) Maximum flux: When area vector is in the direction of Electric field, then

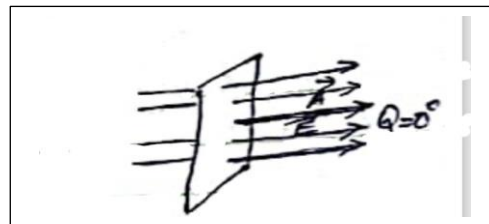
$\theta = 0^\circ$. Now we know that

$$\Phi = EA \cos\theta$$

$$\Rightarrow \Phi = EA \cos 0^\circ$$

$$\Rightarrow \Phi_{\text{max}} = EA$$

So for $\theta = 0^\circ$ the Electric flux will be maximum and it will be equal to EA.



(2) Zero flux: When area vector is perpendicular to the Electric field, then

$\theta = 90^\circ$. Now we know that

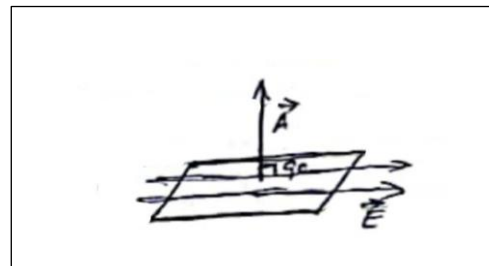
$$\Phi = EA \cos\theta$$

$$\Rightarrow \Phi = EA \cos 90^\circ$$

$$\Rightarrow \Phi = EA (0)$$

$$\Rightarrow \Phi = 0$$

So for $\theta = 90^\circ$ the electric flux will be equal to zero.



(3) Flux at any angle: When $90^\circ < \theta < 0^\circ$ between area vector and Electric field. Then in this case we can say that the Electric flux will depend on the value of " θ ".

Unit of Electric Flux: As we know that.

$$\Phi = EA \cos\theta.$$

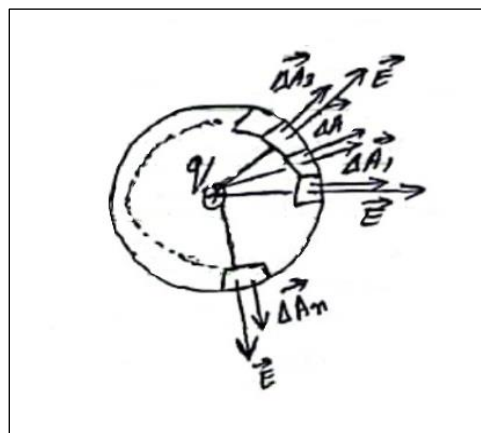
$$\Rightarrow \Phi = \text{N/C m}^2$$

$$\Rightarrow \Phi = \text{Nm}^2\text{C}^{-1}$$

Electric flux through a closed

A closed surface in the form of sphere and the charge at its center is shown in figures. Now we know that the whole Surface area of the Sphere is a scalar quantity.

Now for the electric flux we will convert the whole surface area of the sphere to an areas which are vector quantities. So for this we will divide the whole surface area of the sphere to very small flat areas as shown in figure 1. Now each flat areas ($\vec{\Delta A_1}$, $\vec{\Delta A_2}$, $\vec{\Delta A_3}$, ... $\vec{\Delta A_n}$) is a vector quantity and for each flat area the electric field intensity $E = kq/r^2$ is constant because “k”, “q” and “r” has the same values. Now for total electric flux from this closed surface we can write



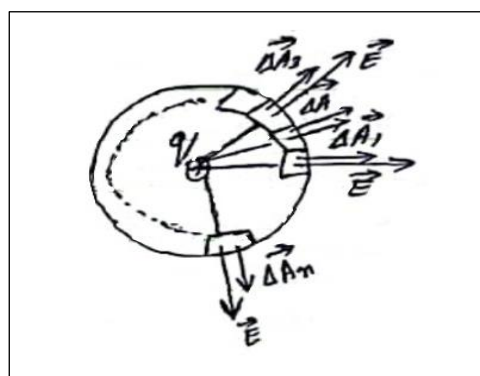
Now for total electric flux from this closed surface we can write

$$\begin{aligned}\Phi_T &= \Phi_1 + \Phi_2 + \Phi_3 + \dots + \Phi_n \\ \Rightarrow \Phi_T &= \vec{E} \cdot \vec{\Delta A_1} + \vec{E} \cdot \vec{\Delta A_2} + \vec{E} \cdot \vec{\Delta A_3} + \dots + \vec{E} \cdot \vec{\Delta A_n} \\ \Rightarrow \Phi_T &= E\Delta A_1 \cos 0^\circ + E\Delta A_2 \cos 0^\circ + E\Delta A_3 \cos 0^\circ + \dots + E\Delta A_n \cos 0^\circ \\ \Rightarrow \Phi_T &= E\Delta A_1 + E\Delta A_2 + E\Delta A_3 + \dots + E\Delta A_n \\ \Rightarrow \Phi_T &= E(\Delta A_1 + \Delta A_2 + \Delta A_3 + \dots + \Delta A_n) \\ \Rightarrow \Phi_T &= E \sum_{i=1,2,3 \dots n} \Delta A\end{aligned}$$

Gauss's Law

Statement: The total electric flux through any closed surface is equal to the $1/\epsilon_0$ times the total charge “q” enclosed by the surface.

Explanation: A closed surface in the form of sphere and the charge at its center is shown in figure 1. Now we know that the whole surface area of the sphere is a scalar quantity. Now for the electric flux we will convert the whole surface area of the sphere to an areas which are vector quantities. So for this we will divide the whole surface area of the sphere to a very very small flat areas as shown in figure 1.



Now each flat areas (ΔA_1 , ΔA_2 , ΔA_3 , ... ΔA_n) is a vector quantity and for each flat area the electric field intensity $E = k q/r^2$ is constant because “k”, “q” and “r” has the same values.

Now for total electric flux from this closed surface we can write as

$$\begin{aligned}\Phi_T &= \Phi_1 + \Phi_2 + \Phi_3 + \dots + \Phi_n \\ \Rightarrow \Phi_T &= \vec{E} \cdot \vec{\Delta A_1} + \vec{E} \cdot \vec{\Delta A_2} + \vec{E} \cdot \vec{\Delta A_3} + \dots + \vec{E} \cdot \vec{\Delta A_n} \\ \Rightarrow \Phi_T &= E\Delta A_1 \cos 0^\circ + E\Delta A_2 \cos 0^\circ + E\Delta A_3 \cos 0^\circ + \dots + E\Delta A_n \cos 0^\circ \\ \Rightarrow \Phi_T &= E\Delta A_1 + E\Delta A_2 + E\Delta A_3 + \dots + E\Delta A_n\end{aligned}$$

$$\Rightarrow \Phi_T = E(\Delta A_1 + \Delta A_2 + \Delta A_3 + \dots + \Delta A_n)$$

$$\Rightarrow \Phi_T = E \sum_{i=1,2,3 \dots n} \Delta A \text{ ----- (1)}$$

Now we know that

$$E = k q/r^2$$

$$\Rightarrow E = q/4\pi\epsilon_0 r^2 \text{ -----(2)}$$

We also know that whole surface area of the sphere = $\Sigma \Delta A = 4\pi r^2$ ----- (3)

Now put eq (2) and (3) in eq (1) we get as

$$\Phi_T = q/4\pi\epsilon_0 r^2 \times 4\pi r^2$$

$$\Rightarrow \Phi_T = 1/\epsilon_0 \times q$$

$$\Rightarrow \Phi_T = q/\epsilon_0 \text{ ----- (4) proved}$$

From equation (4) we can say that

- ① Total Electric flux does not depend upon the shape or geometry of closed surface.
- ② Total Electric flux only depends upon the total charge enclosed by that surface.

Gauss's law for irregular closed surface due to nth charges

In order to calculate the total electric flux, let us consider a closed surface of irregular shape enclosing point charges ($q_1, q_2, q_3, \dots, q_n$) as shown in figure.

In this case we can write as

$$\Phi_T = (\Phi_1) \text{ due to } q_1 + (\Phi_2) \text{ due to } q_2 + (\Phi_3) \text{ due to } q_3 + \dots + (\Phi_n) \text{ due to } q_n.$$

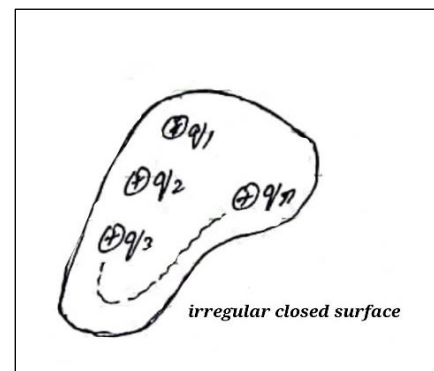
$$\Rightarrow \Phi_T = q_1/\epsilon_0 + q_2/\epsilon_0 + q_3/\epsilon_0 + \dots + q_n/\epsilon_0$$

$$\Rightarrow \Phi_T = (1/\epsilon_0) (q_1 + q_2 + q_3 + \dots + q_n)$$

$$\Rightarrow \Phi_T = 1/\epsilon_0 \times Q \text{ Where } Q = q_1 + q_2 + q_3 + \dots + q_n$$

$$\Rightarrow \Phi_T = Q/\epsilon_0$$

In eq (1) $Q = q_1 + q_2 + q_3 + \dots + q_n$ represents the total charge enclosed by that surface. Thus Gauss's law shows that the total electric flux through any closed surface is $1/\epsilon_0$ times the total charge enclosed in it.



Applications of Gauss's Law

(1) Location of excess charges on a conductor

The excess charges on the surface of a conductor are shown in figure. Now location of excess charges on conductor means to explain the effect of these charges inside the conductor as well as out side on the surface of the conductor.

Case 01: The location or effect of excess charges inside a conductor

Now to find the location or effect of excess charges inside a conductor for this we use Gauss's law. Now before the using of Gauss's law we will first draw a closed surface called Gaussian Surface as shown in figure. Now in Gaussian

closed surface the net electric field is zero because all field lines cancel each other in opposite direction. Now from Gauss's law we know that

$$\Phi_T = Q / \epsilon_0 \text{ ----- (1)}$$

We also know that total flux from closed surface we can write as

$$\Phi_T = E \Sigma \Delta A \text{ ----- (2)}$$

Now put eq (2) in eq (1). We get as

$$E \Sigma \Delta A = Q / \epsilon_0$$

$$\Rightarrow Q = \epsilon_0 E \Sigma \Delta A \text{ ----- (3)}$$

Now inside the conductor or Gaussian closed surface the net electric field is zero. i.e.

$$E=0 \text{ ----- (4)}$$

Now put eq (2) in eq (3). We get as

$$Q = \epsilon_0 \times 0 \times \Sigma \Delta A$$

$$\Rightarrow Q = 0 \text{ ----- (5)}$$

Now from eq (5) we can say that the location or effect of excess charges inside the conductor will be equal to zero.

Case (b): The location or effect of excess charges on the surface of a conductor:

Now in this case a Gaussian closed surface is shown closer to the surface of a conductor at out side.

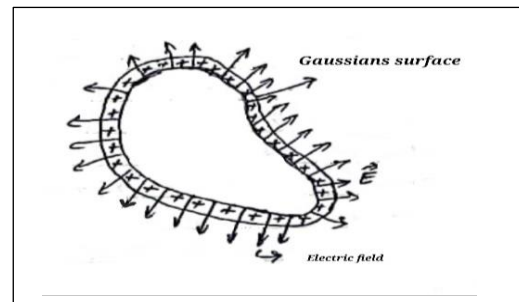
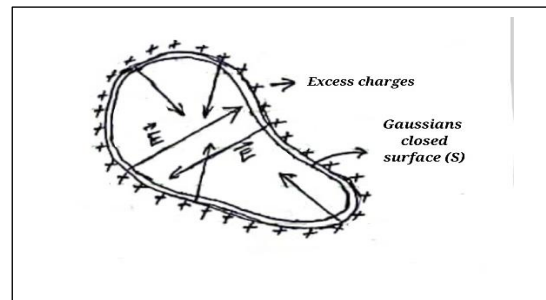
Also in this case the electric field out side the conductor on its surface is not equal to zero. i.e $E_{out} \neq 0$ -----(6)

Now we can write as

$$Q_{out} = \epsilon_0 E_{out} \Sigma \Delta A \text{ -----(7)}$$

In eq (7) if $E_{out} \neq 0$ then $Q_{out} \neq 0$ ----- (8)

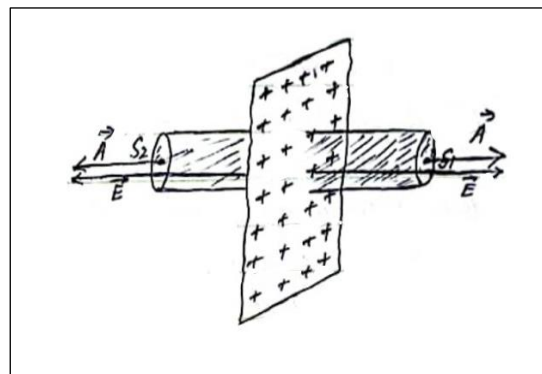
So from eq (8) we can say that the location or effect of excess charges will be only on the conductor outside.



(2) Electric field intensity due to infinite sheet of charge

The infinite (large) Sheet of charge is shown in figure. Now we want to find the Electric field intensity at a point "S₁" or at point "S₂" both points having equal distance from the sheet. For this we will use Gauss's law.

Before the using of Gauss's law first we will draw a closed surface in the form of cylindrical shape to the either side to enclose these points. Now we know that



from Gauss's law

$$\Phi_T = Q/\epsilon_0 \text{ ----- (1)}$$

Now in figure the electric field lines are passing through two circular faces only. Therefore there is two electric fluxes. So we can write as

$$\begin{aligned} \Phi_T &= \Phi_1 + \Phi_2 \\ \Rightarrow \Phi_T &= \vec{E} \cdot \vec{A} + \vec{E} \cdot \vec{A} \\ \Rightarrow \Phi_T &= EA \cos 0^\circ + EA \cos 0^\circ \\ \Rightarrow \Phi_T &= EA + EA \\ \Rightarrow \Phi_T &= 2EA \text{ ----- (2)} \end{aligned}$$

Now put eq (2) in eq (1) we get as

$$2EA = Q/\epsilon_0 \text{ ----- (3)}$$

Surface charge density: Total charge "Q" divided by total Area of the sheet is called Surface charge density (δ). i.e

$$\begin{aligned} \delta &= Q/A \\ \Rightarrow Q &= A\delta \text{ ----- (4)} \end{aligned}$$

Now put eq (4) in (3). We get as

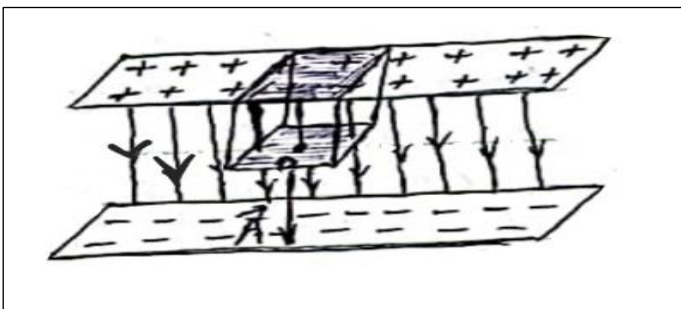
$$\begin{aligned} 2EA &= A\delta/\epsilon_0 \\ \Rightarrow E &= \delta/2\epsilon_0 \end{aligned}$$

This is the required Electric field intensity at "S₁" and "S₂" in magnitude form. Now in vector form we can write as

$$\vec{E} = \delta/2\epsilon_0 \hat{r} \text{ (Here } \hat{r} \text{ unit vector represents the direction of } \vec{E}.)$$

(3) Electric field intensity between two oppositely charged parallel plates:

Two oppositely charged parallel plates are shown in figure. These plates are assumed to be of infinite length in order to avoid fringing Electric field at ends. Under these conditions, the Electric field intensity will be uniform and normal to the plates. Now we want to find the Electric field intensity at point "P"



between the plates. For this we will use Gauss's law.

Before the using of Gauss's law first we will draw a closed Surface in the form of rectangular shape to enclose the point "p".

Now we know that from Gauss's law

$$\Phi_T = Q/\epsilon_0 \text{ ----- (1)}$$

Now in figure the Electric field lines are passing through one down face only. Therefore there is one Electric flux only.

So we can write as

$$\begin{aligned} \Phi_T &= \Phi_1 \\ \Rightarrow \Phi_T &= E.A \end{aligned}$$

$$\Rightarrow \Phi_T = EA \cos 0^\circ$$

$$\Rightarrow \Phi_T = EA \text{ ----- (2)}$$

Now put eq (2) in eq (1). we get as

$$EA = Q/\epsilon_0 \text{ ----- (3)}$$

$$\text{Now surface charge density} = \delta = Q/A \Rightarrow Q = A\delta \text{ ----- (4)}$$

Now put eq (4) in eq (3). we get as

$$EA = A\delta/\epsilon_0$$

$$\Rightarrow E = \delta/\epsilon_0$$

This is the required Electric field intensity at point "P" in magnitude form.

Now in vector form we can write as

$$\vec{E} = \frac{\delta}{\epsilon_0} \vec{r}$$

Here " $\vec{r}$ " unit vector represents the direction of Electric field intensity.

Electric Potential

The work done on a test charge " q_0 " against the electric field from infinity position is called Electric potential (V). i.e

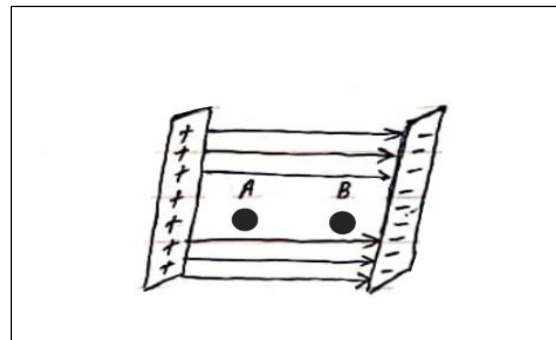
$$V = W/q_0$$

OR The energy gained by a unit positive test charge against the electric field is called Electric potential (V). i.e

$$V = U/q_0$$

Infinity position: The position at the end of Electric field is called infinity position.

Explanation: For the explanation of Electric potential we have two oppositely charged parallel plates as shown in figure. In figure the electric field lines start from +ve plate and end on the -ve plate. The negative plate we consider as infinity position because the electric field ends on this plate. Now if we do work on a test charge " q_0 " from negative plate (infinity position) up to point "B" against the Electric field then we will get an electric potential " V_B " on that point, which we can write as



$$V_B = W_B/q_0 \text{ ----- (1)}$$

This work will change to energy and it is said to be Electric potential energy with charge " q_0 ". So we can write as

$$W_B = U_B \text{ ----- (2)}$$

Now put eq (2) in eq (1) we get as

$$V_B = U_B / q_0 \text{ ----- (3)}$$

Now if we do work on a test charge q_0 from negative plate (infinity position) up to point "A" against the Electric field then we will get an electric potential " V_A " on that point, which we can write as

$$V_A = W_A / q_0 \text{ or } V_A = U_A / q_0 \text{ ----- (4)}$$

Now the electric potential at “A” point will be greater than the electric potential at “B” point. Because “ W_A ” or “ U_A ” will be greater than “ W_B ” or “ U_B ” from the negative plate (infinity position).

Electric Potential difference: The work done on a test charge inside the electric field between two points against the Electric field is called the Electric potential difference (ΔV). In electric potential difference work on a test charge will not start from infinity position but work will start from any position inside the electric field. Now we can write as

$$\begin{aligned}\Delta V &= W_{BA}/q_0 \\ \Rightarrow \Delta V &= (W_A - W_B)/q_0 \\ \Rightarrow \Delta V &= W_A/q_0 - W_B/q_0 \\ \Rightarrow \Delta V &= V_A - V_B \text{ ----- (5)}\end{aligned}$$

Now from equation (5) we can also say that the difference in electric potentials between the two points inside the electric field is also called Electric potential difference. Now work done on a test charge between the two points “B” and “A” will also change the Electric potential energy of the test charge. So we can write as

$$\begin{aligned}W_{BA} &= \Delta U \text{ -----(7)} \\ \Rightarrow W_{BA} &= U_A - U_B \text{ -----(8)}\end{aligned}$$

Now put eq (7) in eq (8) We get as

$$\begin{aligned}\Delta V &= \Delta U/q_0 \\ \Rightarrow \Delta V &= U_A - U_B/q_0 \text{ -----(9)}\end{aligned}$$

Now from eq (9) we can also say that the change in electric potential energies by a test charge between the two points inside the electric field is also called the potential difference.

Conversion of electric potential difference to electric potential:

As we know that the electric potential difference is

$$\Delta V = W_{BA}/q_0 \text{ -----(10)}$$

Now if we take “B” point as an infinity point or infinity position on negative plate then Electric potential difference will convert to Electric potential. And such an Electric Potential is said to be V_A . Now eq (10) for Electric

Potential we can write as

$$V_A = W_{\infty A}/q_0 \text{ or } V_A = W_A/q_0$$

Unit of Electric potential:

The unit of Electric potential is volt. if work is 1J on 1C test charge then Electric potential will be equal to 1V. i.e $1V = 1J/1C$.

Unit of potential difference: Unit of electric potential difference is also volt.

Voltage: The second name of potential difference is called Voltage.

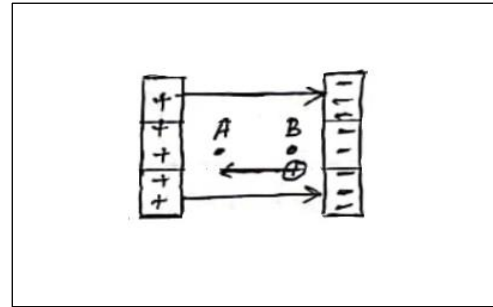
Electric field and potential Gradient

The rate of change of Electric potential with respect to displacement is called Electric potential gradient. i.e

$$\text{Electric potential Gradient} = - \Delta V/\Delta r$$

Relation between Electric field and Potential Gradient

Two oppositely charged plates are shown in figure. In figure the Electric field line starts from +ve plate and ends on the negative plate. Now let "A" and "B" two points which are very close to each other so that the electric field intensity is almost constant on these points.



Now the work done on a test charge " q_0 " from "B" point to "A" point against the Electric field we can write as

$$\Delta W = F_{\text{app}} \Delta r \text{ ----- (1) } \text{ "}\Delta r\text{" is the displacement.}$$

Now on a test charge " q_0 " there is two forces one is called " F_{app} " force due to which a charge covers a displacement " Δr " and the second one is an electric force " F_E " due to an Electric field.

Now for slow motion of a test charge we can write as

$$F_{\text{app}} = - F_E \text{ -----(2)}$$

Now put eq (2) in eq (1) we get as

$$\Delta W = - F_E \Delta r \text{ -----(3)}$$

Now we know that $E = F_E / q_0 \Rightarrow F_E = q_0 E$ ----- (4)

Now put eq (4) in eq (3) we get as

$$\Delta W = - q_0 E \Delta r \text{ ----- (5)}$$

Now we also know that the electric potential difference we can write as

$$\Delta V = \Delta W / q_0 \Rightarrow \Delta W = q_0 \Delta V \text{ ----- (6)}$$

Now compare eq (5) and (6) we get as

$$- q_0 E \Delta r = q_0 \Delta V$$

$$\Rightarrow E = -\Delta V / \Delta r \text{ ----- (7)}$$

Eq (7) represents the required relation between electric field intensity and potential gradient. In equation the negative sign indicates that the direction of the electric field is opposite to the direction in which the electric potential is increasing w.r.t displacement.

From eq (7) we can also say that if electric potential is increasing w.r.t displacement then electric field intensity will be also increasing in opposite direction and vice versa.

Equipotential Surfaces

Those surfaces around the charge which have the same electric potential at each point are called equipotential surfaces.

Equipotential lines: A lines which have the same electric potential at each point are called equipotential lines.

Explanation: Equipotential surfaces and lines are shown in figure. These surfaces and lines are imaginary in nature. Now we know that

$$V = W/q_0 \text{ ----- (1)}$$

So from equation (1) we can say that work done on a test charge from infinity position will be the same up to all points on equipotential surface or an equipotential line.

And when work is same on the test charge then electric potentials will be also same on all points on equipotential surface or an equipotential line. Now the work done between any two points on an equipotential surface or line will be equal to zero. As we know that

$$\Delta V = W/q_0$$

$$\Rightarrow W = q_0 \Delta V$$

$$\Rightarrow W = q_0 (V_2 - V_1) \text{ ----- (2)}$$

Now equipotential surface or line have the same Electric potential at all points.

So we can write as $V_1 = V_2 = V$ ----- (3)

Now put equation (3) in equation (2) we get as

$$W = q_0(V - V)$$

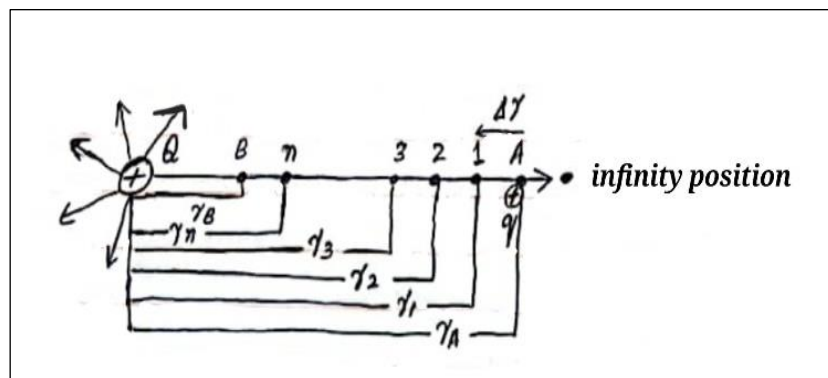
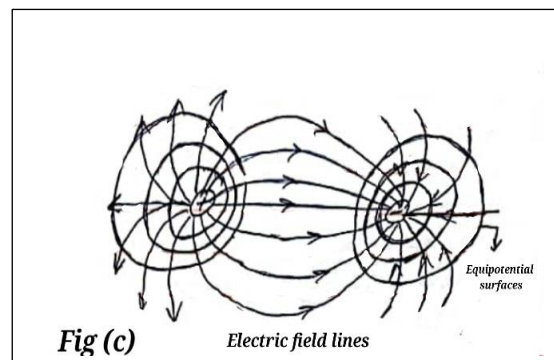
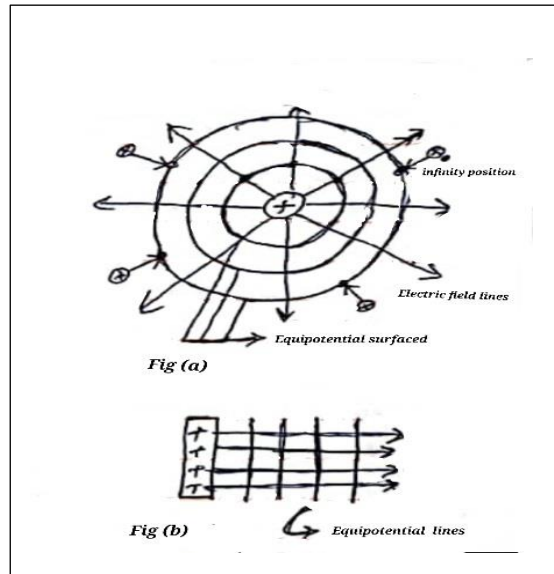
$$\Rightarrow W = 0$$

Now in case of two opposite charges the electric field and equipotential surfaces are shown in figure. If there are two or more charges present, then the net electric potential at any point is the sum of the electric potentials due to each charge. Electric potential is a scalar quantity. So electric potential can be either positive or negative depending upon the sign of the charge.

Electric potential is a scalar quantity. So electric potential can be either positive or negative depending upon the sign of the charge.

Electric potential Energy and potential due to a point charge:

A point charge is shown in a figure. Also points (A, 1, 2, 3, ... n, B) with a distance ($r_A, r_1, r_2, r_3, \dots, r_n, r_B$) from the point charge are shown in a figure inside the Electric field. Now let these points are very close to each



other, so that the electric field intensity is almost constant on these points. Now first we find the work done on a test charge “q” from “A” point to “1” point against the Electric field. So we can write as

$$\Delta W = F_{app} \cdot \Delta r \text{ ----- (1)}$$

Now on a test charge there will be two forces: one is called “F_{app}” force due to which a charge will cover a displacement “Δr” and the second one will be electric force “F_e” due to an Electric field.

Now for non accelerated slow motion we can write as $F_{app} = - F_e \text{ ----- (2)}$

Now put equation 2) in equation (1) we get as

$$\Delta W_{r_A \rightarrow r_1} = -F_e \cdot \Delta r \text{ ----- (3)}$$

We also know that $E = F/q \Rightarrow F_e = qE \text{ ----- (4)}$

Now put eq(3) in eq(2) We get as

$$\Delta W_{r_A \rightarrow r_1} = -qE \cdot \Delta r$$

$$\Rightarrow \Delta W_{r_A \rightarrow r_1} = -qE \cdot \Delta r$$

$$\Rightarrow \Delta W_{r_A \rightarrow r_1} = -qE \Delta r \cos\theta \text{ ----- (4)}$$

Now put $\theta = 180^\circ$ because “E” and “Δr” are in opposite direction. So put $\theta = 180^\circ$ in eq(4) we get as

$$\Delta W_{r_A \rightarrow r_1} = -qE \Delta r \cos 180^\circ$$

$$\Rightarrow \Delta W_{r_A \rightarrow r_1} = qE \Delta r \text{ ----- (5)}$$

Now we know that $E = Q/4\pi\epsilon_0 r^2 \text{ ----- (6)}$

Now put eq(6) in eq(5) we get as

$$\Delta W_{r_A \rightarrow r_1} = Qq/4\pi\epsilon_0 r^2 \times \Delta r \text{ ----- (7)}$$

Now in figure the distance of a test charge is changing from “r_A” distance point to “r₁” distance point. So we will take average distance in eq(7). i.e.

$$\Delta W_{r_A \rightarrow r_1} = Q/4\pi\epsilon_0 r_{avg}^2 \Delta r \text{ ----- (8)}$$

Now we find r_{avg}^2 . So from figure we can write as

$$r_{avg} = (r_A + r_1)/2 \text{ ----- (9). Now in fig}$$

$$\Delta r = r_A - r_1$$

$$\Rightarrow r_A = \Delta r + r_1 \text{ ----- (10)}$$

Now put eq (10) in eq (9) we get as

$$r_{avg} = (\Delta r + r_1 + r_1)/2 \Rightarrow r_{avg} = (\Delta r + 2r_1)/2 \text{ ----- (11)}$$

Now taking square to both side we get as

$$r_{avg}^2 = (\Delta r + 2r_1)^2/4 \Rightarrow r_{avg}^2 = \Delta r^2 + 4r_1^2 + 4r_1\Delta r/4$$

Now ignore Δr^2 because its value is very small due to a closer points.

$$\Rightarrow r_{avg}^2 = 4r_1^2 + 4r_1(r_A - r_1)/4$$

$$\Rightarrow r_{avg}^2 = 4r_1^2 + 4r_1 r_A - 4r_1^2/4 \Rightarrow r_{avg}^2 = r_1 r_A \text{ ----- (12)}$$

Now put eq(12) in eq(8) we get as

$$\Delta W_{r_A \rightarrow r_1} = Qq/4\pi\epsilon_0 (r_1 r_A) \times (r_A - r_1)$$

$$\Rightarrow \Delta W_{r_A \rightarrow r_1} = Qq/4\pi\epsilon_0 (r_A / r_1 r_A - r_1 / r_1 r_A)$$

$$\Rightarrow \Delta W_{r_A \rightarrow r_1} = Qq/4\pi\epsilon_0 (1/r_1 - 1/r_A)$$

Now other works between the points on a test charge we find from similarity.

$$\Delta W_{r_1 \rightarrow r_2} = Qq/4\pi\epsilon_0 (1/r_2 - 1/r_1) \text{ And}$$

$$\Delta W_{r_2 \rightarrow r_3} = Qq/4\pi\epsilon_0 (1/r_3 - 1/r_2)$$

$$\Delta W_{r_n \rightarrow r_B} = Qq/4\pi\epsilon_0 (1/r_n - 1/r_B)$$

Now we can write as

$$W_{r_A \rightarrow r_B} = \Delta W_{r_A \rightarrow r_1} + \Delta W_{r_1 \rightarrow r_2} + \Delta W_{r_2 \rightarrow r_3} + \dots + \Delta W_{r_n \rightarrow r_B}$$

$$W_{r_A \rightarrow r_B} = Qq/4\pi\epsilon_0 (1/r_1 - 1/r_A + 1/r_2 - 1/r_1 + 1/r_3 - 1/r_2 + \dots + 1/r_B - 1/r_n)$$

$$W_{r_A \rightarrow r_B} = Qq/4\pi\epsilon_0 (1/r_B - 1/r_A)$$

Now if we take "A" point as infinity. Then r_A distance we will take infinity. i.e

$$W_{\infty \rightarrow r_B} = Qq/4\pi\epsilon_0 (1/r_B - 1/\infty)$$

$$W_{\infty \rightarrow r_B} = Qq/4\pi\epsilon_0 (1/r_B - 0) \text{ ----- (13)}$$

Now eq(13) in general form we can write as

$$W = Qq/4\pi\epsilon_0 r \text{ ----- (14)}$$

Now this work on a test charge from infinity position against the electric field will convert to electric potential Energy "U". So eq (14) we can write as

$$U = Qq/4\pi\epsilon_0 r \text{ ----- (15)}$$

Electric potential: As we know that

$$V = U/q$$

$$\Rightarrow V = Qq/4\pi\epsilon_0 r q$$

$$\Rightarrow V = Q/4\pi\epsilon_0 r$$

$$\Rightarrow V = KQ/r$$

The Electron volt (eV)

The electron volt is a unit of energy or work which is defined as "The amount of energy gained or lost by an electron when it is displaced across the two points having the Potential difference of one volt.

Explanations: In figure a two points are shown in figure which having the potential difference of 1 volt. As we know that $V = kQ/r$ ---(1). In eq (1) "r" will be a distance from charged object. So from eq (1) we can say that electric potential at "A" point will be greater than from an electric potential at "B" point. Because at 'A' point the distance from an object is less and greater up to "B" point. Now if these two points having the potential difference of 1 volt, then the energy gained or lost by an electron during motion between these two points is called electron volt (eV).

If the electron moves from "B" points toward "A" points it will gain the electric potential energy, and will lost the electric energy if it moves from "A" points towards "B" point.

Now we know that

$$\Delta V = \Delta U/q$$

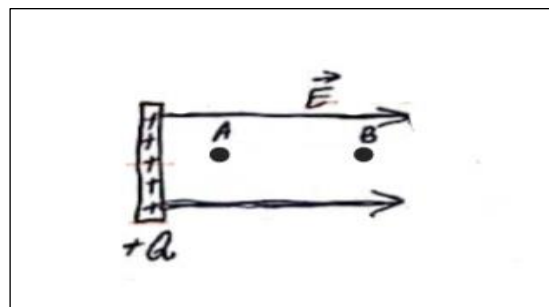
$$\Rightarrow \Delta U = q\Delta V \text{ ----- (1)}$$

Now put $q = \text{electron} = e$ and

$\Delta V = 1 \text{ volt}$ in eq(1). we get as

$$\Delta U = e \times 1V$$

$$\Rightarrow \Delta U = 1eV$$



or $\Delta K.E. = 1eV$

Now we calculate 1eV energy in unit of joule.

$$1eV = 1 \times 1.6 \times 10^{-19} \text{ CV}$$

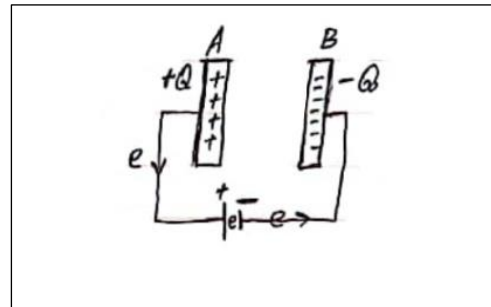
$$\Rightarrow 1eV = 1.6 \times 10^{-19} \text{ J.} \quad (\text{CV} = \text{Joule})$$

Capacitor

A device which is used for storing the electric charges is called Capacitor.

Construction: It consist of two parallel metal plates, separated by a small distance. The medium between the two plates is air or sheet of some insulating materials called dielectric materials.

Charging of a capacitor: In figure the capacitor plates are connected with the terminals of a battery. Then the +ve terminal of a battery will attract the electrons from "A" plate and create on it a +ve charge (+Q). Then due to an electrostatic induction of "A" plate to "B" plate and due to the work done of a battery the electrons will go to the "B" plate and create on it a -ve charge (-Q). The magnitude of (+Q) will be equal to the magnitude of (-Q) on both plates. The charging process will be continuous until the Potential difference or voltage across the plates will become equal to the P.D or voltage of a battery. After removal of a battery the charge effect will be maintain on the plates, and the plates will not move towards each other due to attraction because they are fixed in the construction.



Capacitance of a capacitor and its unit

The capacity of a capacitor to store the charges is called Capacitance.

Explanation: When a charge is stored on one of the plate of a capacitor then it causes the potential difference or voltage between the plates. Now if potential difference "V" between the plates is increasing, then the charge on each plate will be also increasing and vice versa. So we can write as

$$Q \propto V$$

$$\Rightarrow Q = CV$$

Where "C" is constant and it is called capacitance of a Capacitor.

Now we can write as

$$C = Q/V \text{ ----- (1)}$$

From eq (1) the capacitance is thus defined as the ratio of magnitude of charge on either plate to the potential difference produced between the plates.

The SI unit of a capacitance is called farad.

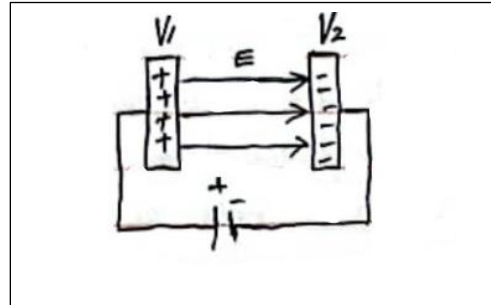
One farad is define as if the stored charge is 1C on either plate and having the potential difference of 1V, then capacitance will be equal to one farad.

Smaller unit of a capacitance are $1pF = 10^{-12} \text{ F}$ and $1nF = 10^{-6} \text{ F}$.



Capacitance of a parallel plate capacitor

In figure the capacitor plates are connected with the terminals of a battery. Then due to a battery and electrostatic induction the charging process will be continuous until the potential difference or voltage across the plates will become equal to the potential difference or voltage of a battery. Now let " V_1 " is the electric potential of positive plate and ' V_2 ' is the electric potential of a negative plate. Now for the electric field between the plates we can write as



$$E = - \Delta V / \Delta r \text{ ----- (1)}$$

Now $\Delta V = V_2 - V_1$ and let $\Delta r = d$ (separation b/w the plates)

$$E = - (V_2 - V_1) / d$$

$$\Rightarrow E = V_1 - V_2 / d$$

$$\Rightarrow E = V / d \text{ ----- (1)}$$

Now by using Gauss's law the electric field intensity between the plates we can write

$$E = \delta / \epsilon_0 \text{ ----- (2) where "}\delta\text{" represents surface charge density.}$$

Now we also know that

$$\delta = Q / A \text{ ----- (3)}$$

Now put eq (3) in eq (2). We get as

$$E = Q / \epsilon_0 A \text{ ----- (4)}$$

Now compare eq (2) and eq (4). We get as

$$V / d = Q / \epsilon_0 A$$

$$\Rightarrow Q = \epsilon_0 A V \text{ ----- (5)}$$

Now we know that for the capacitance of a capacitor in case of vacuum b/w the plates.

$$C_{vac} = Q / V$$

Now put eq (5) in eq (6) we get as

$$C_{vac} = \epsilon_0 A V / d$$

$$\Rightarrow C_{vac} = \epsilon_0 A / d \text{ ----- (6)}$$

Capacitance in case of dielectric medium:

As we know that in case of dielectric medium between the plates we will take " ϵ ", permittivity not " ϵ_0 " permittivity. So in case of medium eq (7) we can write

$$C_{med} = \epsilon A / d \text{ ----- (7)}$$

Now we also know that $\epsilon_r = \epsilon / \epsilon_0 \Rightarrow \epsilon = \epsilon_0 \epsilon_r \text{ ----- (8)}$

Now put eq (7) in eq (8) we get as

$$C_{med} = \epsilon_0 \epsilon_r A / d \text{ ----- (10)}$$

Now from eq (10) we can say that Capacitance in case of medium depends upon the area of the plates, the separation b/w the plates and the relative permittivity of the medium. Now eq (10) we can also write as

$$C_{\text{med}} = \epsilon_r C_{\text{vac}} \text{ ----- (11)}$$

From eq (11) we can say that “ C_{med} ” will be greater than “ C_{vac} ” time of ϵ_r . Now eq (11) we can also write as

$$\epsilon_r = C_{\text{med}} / C_{\text{vac}} \text{ ----- (12)}$$

Now from eq (12) the relative permittivity or dielectric constant or specific inductive capacity is define as “The ratio of the capacitance in case of medium to the capacitance of a vacuum is called relative permittivity”.

Combination of capacitors

We know that the capacitors can be connected either in series or in parallel. We want to find out an equivalent capacitance in each case that has the same capacitance as that of the combination of capacitors.

Series combination of capacitors:

When capacitors are connected plate to plate i.e the right plate of one capacitor is connected to the left plate of the next capacitor and so on as shown in figure then it is called series combination of capacitors.

OR

The combination of capacitors in such a circuit in which current does not divide along different paths is called series combination of capacitors.

Characteristics of a Series Combination of Capacitors:

1) Each capacitor of capacitance C_1 , C_2 and C_3 will store an equal amount of charge as shown in the figure.

So we can write as:

$$Q_1 = Q_2 = Q_3 = Q$$

2) Due to different capacitances, the voltages (P.D.) will be different across each capacitor, i.e.,

$$V_1 = Q_1 / C_1 \Rightarrow V_1 = Q / C_1$$

$$V_2 = Q_2 / C_2 \Rightarrow V_2 = Q / C_2$$

$$V_3 = Q_3 / C_3 \Rightarrow V_3 = Q / C_3$$

3) The sum of all voltages across each capacitor will be equal to the battery voltage, i.e.,

$$V = V_1 + V_2 + V_3$$

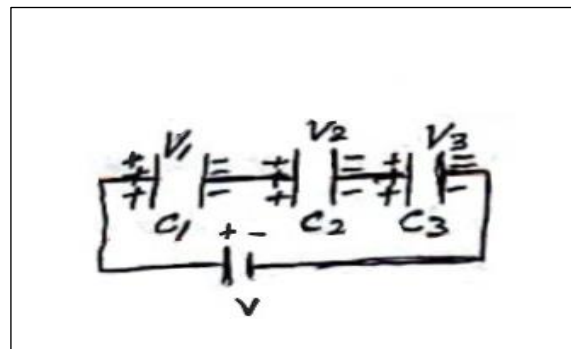
$$\Rightarrow V = Q/C_1 + Q/C_2 + Q/C_3$$

$$\Rightarrow V = Q (1/C_1 + 1/C_2 + 1/C_3)$$

$$\Rightarrow V/Q = 1/C_1 + 1/C_2 + 1/C_3$$

$$\Rightarrow 1/C_{\text{eq}} = 1/C_1 + 1/C_2 + 1/C_3 \text{ — (1)}$$

From equation (1), we can say that the effect of three capacitors of capacitances C_1 , C_2 and C_3 connected in series is equal to a single capacitor of equivalent capacitance C_{eq} . Also from equation (1), the value of C_{eq} is always less than the individual capacitances C_1 , C_2 and C_3 .



Parallel Combination of Capacitors

The combination of capacitors in such a circuit in which current divides along different paths is called parallel combination of capacitors.

Characteristics of Parallel Combination:

1) Each capacitor of capacitance C_1 , C_2 and C_3 will have equal voltage across it because all capacitors are connected to the same potential difference between points A and B.

So we can write:

$$V_1 = V_2 = V_3 = V$$

2) Due to different capacitances, the charges will be different on each capacitor:

$$Q_1 = C_1 V$$

$$Q_2 = C_2 V$$

$$Q_3 = C_3 V$$

3) The total charge on the capacitors is equal to the sum of charges stored on each capacitor:

$$Q = Q_1 + Q_2 + Q_3$$

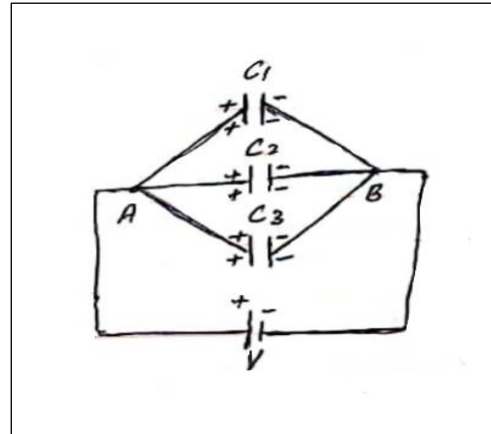
$$Q = C_1 V + C_2 V + C_3 V$$

$$Q = V (C_1 + C_2 + C_3)$$

$$Q/V = C_1 + C_2 + C_3$$

$$C_{eq} = C_1 + C_2 + C_3 \quad \text{--- (1)}$$

From eq (1) we can say that the effect of a single capacitor of equivalent capacitance in a parallel circuit must be equal to the three capacitors of capacitance C_1 , C_2 and C_3 . Also from eq (1) we can say that the value of " C_{eq} " will be always greater than the individual capacitance C_1 , C_2 and C_3 .

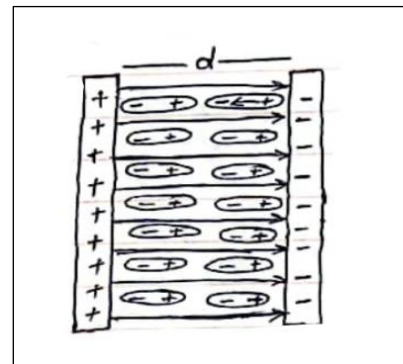


Electric Polarization

The displacement of electrons to one side and of the nucleus to the other side in the atom or molecule due to an external electric field is called Electric polarization.

Explanation: When a dielectric substance (insulator) is placed between two parallel plates with an applied electric field.

Then electric field disturb the negatively charged particles (electrons) towards a positive plate and the positive charged particles (Nucleus) towards the negative plate as shown in figure. Such a displacement of electrons to one side and of the nucleus to the other side in the atoms or molecules due to an external electric field is called electric polarization. Now the electric polarization process in a dielectric material will be continuous until



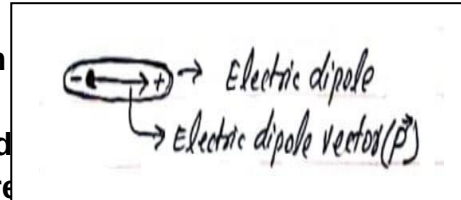
the sum of all electric fields produced in the atoms of molecules when becomes equal to the external electric field.

Electric dipole: The opposite charged System Separated by small distance is called Electric dipole.

Electric dipole moment or Electric dipole vector:

Electric dipole moment is a vector in electric dipole and such a vector represents the strength of electric polarization in the atoms or molecules of a dielectric material.

The magnitude of such a vector we will find from product of magnitude of charge “Q” with a separation distance . i.e $P = |Qd|$. If the magnitude “P” is greater, then it will represent a greater strength of electric polarization and vice versa.



Polar Molecules: Those molecules which have permanent electric polarization and also have permanent electric dipole moment after the removal of an external electric field is called polar molecules. Examples: H_2O molecules, NaCl molecule.

Non Polar Molecules: Those molecules which have not permanent electric polarization and also have not permanent electric dipole moment after the removal of an external electric field is called Non polar molecules. Example CH_4 molecule.

Polar dielectric Materials: These insulator materials which have polar molecules are called polar dielectric materials.

Non polar dielectric Materials: These insulator materials which have non polar molecules are called non polar dielectric materials.

Energy stored in a capacitor

Initially when capacitor plates have no charge +Q” and “-Q” Then voltage or P.D between the plates will be equal to zero. Now when a capacitor is fully charged, then at final stage the voltage or P.D between the plates must be equal to the battery voltage or P.D.

Now we know that $\Delta V = \Delta U$ or $V = U/Q$

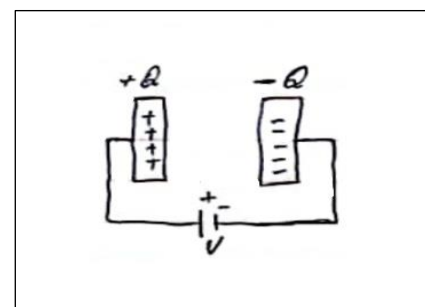
Now in eq(1) we will take average voltage, because it changes between the plates. So we can write as

$$\begin{aligned} V_{avg} &= U/Q \\ \Rightarrow V_i + V_f/2 &= U/Q \\ \Rightarrow 0 + V/2 &= U/Q \\ \Rightarrow V/2 &= U/Q \\ \Rightarrow U &= QV/2 \text{ ----- (1)} \end{aligned}$$

In eq (1) “U” represent the stored electrical energy in a capacitor. Now we also know that

$$Q = CV \text{ ----- (3)}$$

Now put eq (3) in eq (1). We get as



$$U = CVV / 2 \Rightarrow U = CV^2/2 \text{ ----- (4)}$$

$$\text{Now we also know that } Q = CV \Rightarrow V = Q/C \text{ ----- (5)}$$

Now put eq (5) in eq (4). We get as

$$U = CQ^2 / 2C^2 \Rightarrow U = Q^2/2C \text{ ----- (6)}$$

$$\text{Now we also know that } E = V/d \Rightarrow V = Ed \text{ ----- (7)}$$

$$\text{We also know that } C_{\text{med}} = A\epsilon_0\epsilon_r/d \text{ ----- (8)}$$

Now put eq (7) and (8) in eq (4). We get as

$$\begin{aligned} U_{\text{med}} &= \frac{1}{2} \times A\epsilon_0\epsilon_r (Ed)^2/d \\ \Rightarrow U_{\text{med}} &= \frac{1}{2} \epsilon_0\epsilon_r E^2 Ad \\ \Rightarrow U_{\text{med}} &= \frac{1}{2} \epsilon_0\epsilon_r E^2 V \text{ ----- (9) "V" represents volume of the plate.} \end{aligned}$$

For Vacuum between the plates eq (9) We can write as

$$U_{\text{vac}} = \frac{1}{2} \epsilon_0 E^2 V \text{ ----- (10)}$$

Energy density: Energy stored in the capacitor divided by the volume of plates is called Energy density. i.e

$$u = U/V \text{ ----- (11)}$$

Now put eq (9) in eq (11), we get as

$$U_{\text{med}} = \frac{1}{2} \epsilon_0\epsilon_r E^2 V / V \Rightarrow U_{\text{med}} = \frac{1}{2} \epsilon_0\epsilon_r E^2 \text{ ----- (12)}$$

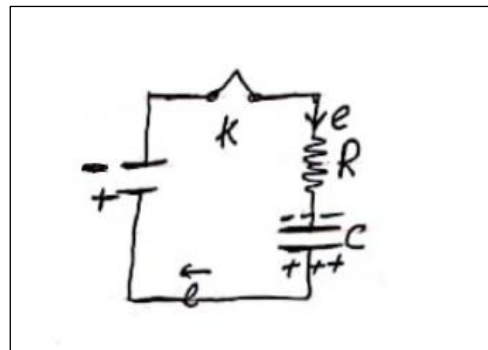
For vacuum between the plates, Eq (12) we can write as

$$u_{\text{vac}} = \frac{1}{2} \epsilon_0 E^2 \text{ ----- (13)}$$

Charging and discharging of a Capacitor:

Charging of a Capacitor: The storing of charge on the plates of the capacitor is called charging of a capacitor.

Now consider a circuit having a capacitor and resistor which are joined in Series with a battery of voltage "V" through a key "k" as shown in figure. Now when the key "k" is closed, the capacitor will start charging through resistor "R" and it will take some time for the fully charging process and such a time is called time constant for charging.



Time constant for charging: The time in which the capacitor is fully charged is called Time constant for charging. OR

The time in which a capacitor to store the charges up to 63.2% of the maximum value of charge is called time constant. Now Time constant depends on two constants resistance "R" and capacitance "C". In a circuit if "R" and "C" are greater then time constant will be greater and vice versa. Now we can write as

$$t \propto R \text{ ----- (1)}$$

$$t \propto C \text{ ----- (2)}$$

Now combine (1) and (2) and we as

$$t = KRC \quad \text{Put } k=1 \text{ then } t = RC \text{ ----- (3)}$$

So from (3) we can say that time constant is equal to RC.

Charging equation: The equation $q = q_0(1 - e^{-t/RC})$ is called charging equation. In such an equation “q” represents the instant charge and “ q_0 ” represents the maximum charge ($q_0 = CV$) and “t” is the instant time and RC is time constant and $e = 2.7182$ is called exponential constant.

Calculation for instant charge “q” when key is open or $t=0$

As we know that $q = q_0(1 - e^{-t/RC})$

$$\Rightarrow q = q_0(1 - 1/e^{t/RC}). \text{ Now put } t=0$$

$$\Rightarrow q = q_0(1 - 1/e^{0/RC})$$

$$\Rightarrow q = q_0(1 - 1/e^0)$$

$$\Rightarrow q = q_0(1 - 1)$$

$$\Rightarrow q = q_0(0)$$

$$\Rightarrow q = 0$$

$$\Rightarrow q = 0 \text{ ----- (1)}$$

Now from (1) we can say that instant charge $q = 0$ When Key is open or time $t=0$.

Calculation for instant charge when $t = RC$ (Time constant or fully charging time)

As we know that $q = q_0(1 - e^{-t/RC})$

$$\Rightarrow q = q_0(1 - 1/e^{t/RC}). \text{ Now put } t = RC$$

$$\Rightarrow q = q_0(1 - 1/e^{RC/RC})$$

$$\Rightarrow q = q_0(1 - 1/e)$$

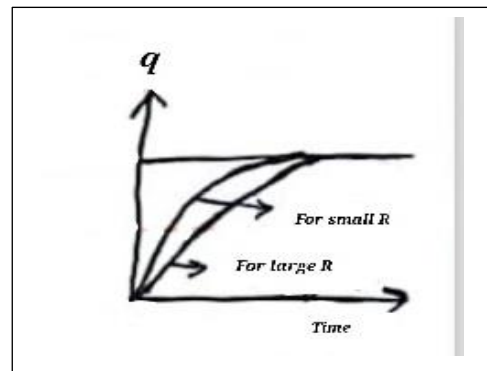
$$\Rightarrow q = q_0(1 - 1/2.7182)$$

$$\Rightarrow q = q_0 \times 0.632$$

$$\Rightarrow q = q_0 \times 0.632 \times 100/100$$

$$\Rightarrow q = q_0 \times 63.2\%$$

$$\Rightarrow q/q_0 = 63.2\% \text{ ----- (2)}$$



Now from eq (2) we can say that in time constant $t = RC$ the capacitor will store the instant charges “q” up to 63.2% of the maximum value of charge.

Note: Capacitor will not store the instant charge up to 100%. Because the negative plate will repel the further electrons instead of deposit on it.

Note: Maximum charge is to be considered 100%.

Maximum charge “ q_0 ” we can find as $q_0 = CV$. Where “C” represents capacitance of a capacitor and “V” represents the voltage of a battery.

Discharging of a Capacitor

The liberation of charges from the plates of a charged capacitor is called discharging of a capacitor.

Now consider a circuit having a charged capacitor and Resistor which are joined in series with each other through a key “k” as shown in fig.

Now when the key “k” is closed, then capacitor will start discharging itself through Resistor “R” and it will take some time for the fully discharging process and such a time is called time constant for discharging.

Time constant for discharging: The time in which a capacitor is fully discharged is called Time constant for discharging.

OR The time in which a capacitor is to lost 63.2% charges and left 36.8% charges of its maximum value of charge is called time constant for discharging.

Now time constant depends on two constants resistance "R" and capacitance

"C". In a circuit if "R" and "C" are greater then time constant will be greater and vice versa. Now we can write as

$$t \propto R \text{ ----- (1)}$$

$$t \propto C \text{ -----(2)}$$

Now combine (1) and (2) and we as

$$t = KRC \quad \text{Put } k=1 \text{ then } t = RC \text{ ----- (3)}$$

So from eq(3) we can say that the time constant is equal to RC.

Discharging equation: The equation $q = q_0 e^{-t/RC}$ is called discharging equation. In such an equation "q" represents the instant charge and "q₀" represents the maximum charge ($q_0 = CV$) and "t" is the instant time and RC is time constant and e = 2.7182 is called exponential constant.

Calculation for instant charge "q" when key is open or t=0

As we know that $q = q_0 e^{-t/RC}$

$$\Rightarrow q = q_0 \times 1/e^{t/RC} \text{ . Now put } t=0$$

$$\Rightarrow q = q_0 \times 1/e^{0/RC}$$

$$\Rightarrow q = q_0 \times 1/e^0$$

$$\Rightarrow q = q_0 \times 1$$

$$\Rightarrow q = q_0 \text{ ----- (1)}$$

Now from (1) we can say that instant charge $q = q_0$ When Key is open or time $t = 0$.

Calculation for instant charge when $t = RC$ (Time constant or fully discharging time):

As we know that $q = q_0 e^{-t/RC}$

$$\Rightarrow q = q_0 \times 1/e^{t/RC} \text{ . Now put } t=0$$

$$\Rightarrow q = q_0 \times 1/e^{RC/RC}$$

$$\Rightarrow q = q_0 \times 1/e$$

$$\Rightarrow q = q_0 \times 1/2.7182$$

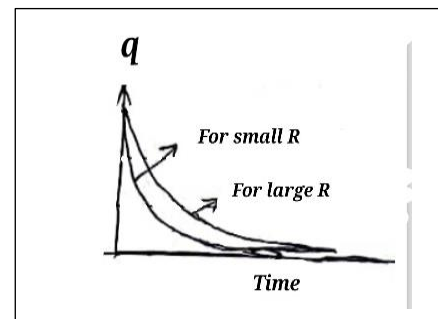
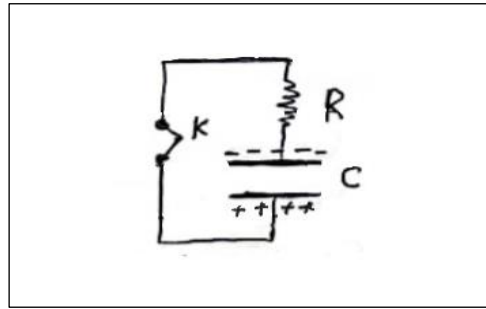
$$\Rightarrow q = q_0 \times 0.368$$

$$\Rightarrow q = q_0 \times 0.368 \times 100/100$$

$$\Rightarrow q = q_0 \times 36.8\%$$

$$\Rightarrow q/q_0 = 36.8\% \text{ ----- (2)}$$

Now from eq (2) we can say that in time constant $t = RC$ the capacitor will lost 63.2% charges and will left the instant charges "q" up to 36.8% of its maximum



value of charge. For the instant charge “q” the percentage 36.8% is called missing percentage. Such a percentage charges will not be present on the plates in real.

Applications: The charging and discharging of a capacitor has many applications. Capacitor discharge ignition (CDI) is a type of automobile electronic ignition system which is widely used in motorcycles, lawn mowers, chain saws, small engines, turbine powdered aircraft, and some cars. Other applications include electronic flashguns in cameras and wind shields of cars.

Examples and Assignments

Example 11.1 : Three charges $q_1 = -5 \mu\text{C}$, $q_2 = +10 \mu\text{C}$ and $q_3 = -12 \mu\text{C}$ are placed in a line as shown in the figure. Calculate the net electrostatic force on charge q_2 due to the other two charges.

Given: $q_1 = -5 \mu\text{C} = -5 \times 10^{-6} \text{ C}$

$q_2 = +10 \mu\text{C} = +10 \times 10^{-6} \text{ C}$

$q_3 = -12 \mu\text{C} = -12 \times 10^{-6} \text{ C}$

Required: $F_{\text{net}} = ?$ on charge q_2 .

Solution: First we find force on charge q_2 due to q_1 .

$$F_{21} = K q_1 q_2 / r^2$$

$$\Rightarrow F_{21} = (9 \times 10^9 \text{ Nm}^2/\text{C}^2)(-5 \times 10^{-6} \text{ C})(10 \times 10^{-6} \text{ C}) / (0.06 \text{ m})^2$$

$$\Rightarrow F_{21} = -125 \text{ N}$$

Now we find force on charge q_2 due to q_3 .

$$F_{23} = K q_2 q_3 / r^2$$

$$\Rightarrow F_{23} = (9 \times 10^9 \text{ Nm}^2/\text{C}^2)(10 \times 10^{-6} \text{ C})(-12 \times 10^{-6} \text{ C}) / (0.04 \text{ m})^2$$

$$\Rightarrow F_{23} = -675 \text{ N}$$

Now we find F_{net} . For this we can write as

$$F_{\text{net}} = F_{23} - F_{21} \text{ (Negative sign with } F_{21} \text{ is due to opposite direction w.r.t } F_{23})$$

$$\Rightarrow F_{\text{net}} = -675 + 125$$

$$\Rightarrow F_{\text{net}} = -550 \text{ N (Ans)}$$

The direction of the net force on charge “ q_2 ” is towards charge “ q_3 ”. The negative sign shows the force is attractive.

Assignment 11.2

Three charges $q_1 = 5 \mu\text{C}$, $q_2 = 10 \text{ nC}$ and q_3 are placed as shown in the figure. What is the electrostatic charge on q_3 if the net force on q_2 is zero?

Given: $q_1 = 5 \mu\text{C} = 5 \times 10^{-6} \text{ C}$, $q_2 = 10 \text{ nC} = 10 \times 10^{-9} \text{ C}$, $F_{\text{net}} = 0$

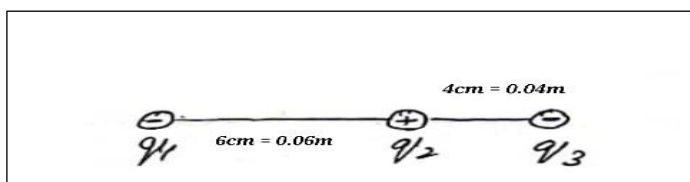
Required: $q_3 = ?$

Solution: It is given that the net force on q_2 is zero.

$$F_{\text{net}} = 0$$

$$\Rightarrow F_{23} - F_{21} = 0 \text{ (Negative sign with } F_{21} \text{ is due to opposite direction w.r.t } F_{23})$$

$$\Rightarrow F_{21} = F_{23}$$



$$K q_1 q_2 / r_{12}^2 = K q_2 q_3 / r_{23}^2$$

Cancel K and q_2 to both sides.

$$q_1 / r_{12}^2 = q_3 / r_{23}^2$$

$$\Rightarrow q_3 = q_1 (r_{23} / r_{12})^2$$

Substitute values:

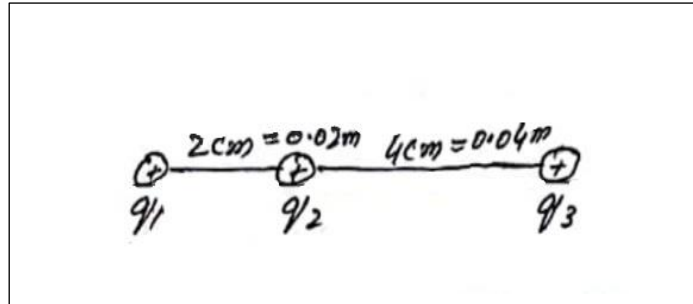
$$q_3 = (5 \times 10^{-6}) (0.04 / 0.02)^2$$

$$\Rightarrow q_3 = (5 \times 10^{-6}) (2)^2$$

$$\Rightarrow q_3 = (5 \times 10^{-6}) (4)$$

$$\Rightarrow q_3 = 20 \times 10^{-6} \text{ C}$$

$$\Rightarrow q_3 = 20 \mu\text{C} \text{ (Ans)}$$



Example 11.2: Find electric field at a distance of 30cm from a $3\mu\text{C}$ point charge.

Given: $r = 30 \text{ cm} = 0.3\text{m}$, $q_1 = 3\mu\text{C} = 3 \times 10^{-6}$

Required: Electric field intensity = $E = ??$

Sol: As we know that $E = K q_1 / r^2$

$$\Rightarrow E = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 3 \times 10^{-6} / (0.3\text{m})^2$$

$$\Rightarrow E = 3 \times 10^5 \text{ N/C}$$

Example 11.3: A metallic Sphere of diameter 40cm carries a charge of $600 \mu\text{C}$ at its center. Find the electric field Intensity (a) at a distance of 60cm from the center of the sphere (b) at the surface of sphere.

Given: $d = 40\text{cm}$ $r = 20\text{cm} = 0.2\text{m}$, $r_1 = 60\text{cm} = 0.6\text{m}$, $Q = 600\mu\text{C} = 600 \times 10^{-6}\text{C}$

Required: a) $E_1 = ??$ b) $E = ??$

Sol (a) As we know that

$$E_1 = k q / r^2$$

$$\Rightarrow E_1 = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 600 \times 10^{-6} \text{C} / (0.6\text{m})^2$$

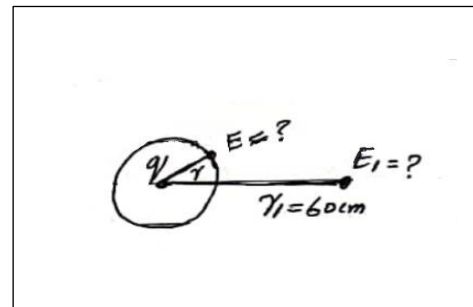
$$\Rightarrow E_1 = 1.5 \times 10^7 \text{ N/C} \text{ Ans}$$

Sol (b) Now we can write as

$$E = k q / r^2$$

$$\Rightarrow E = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 600 \times 10^{-6} \text{C} / (0.2\text{m})^2$$

$$\Rightarrow E = 13.5 \times 10^7 \text{ N/C. Ans}$$



Example: 11.4: You have a 12V motorcycle battery that can move 5000C of charge and a 12V car battery that can move 60,000C of charge. How much energy does each deliver?

Sol: As we know that

$$\Delta V = \Delta U / q$$

$$\Delta U = q / \Delta V \text{ ----- (1)}$$

In eq (1) ΔU represents the energy and ΔV represents the voltage. Now we can write as

$$\Delta U_{\text{m.cycle}} = q / \Delta V \Rightarrow \Delta U_{\text{m.cycle}} = 5000\text{C} \times 12\text{V} = 6 \times 10^4 \text{ J}$$

$$\Delta U_{\text{car}} = q / \Delta V \Rightarrow \Delta U_{\text{car}} = 60,000\text{C} \times 12\text{V} = 7.20 \times 10^5\text{J} \text{ (Ans)}$$

Assignment 11.2: When a 12V car battery powers a single 30 watt head light, how many electrons pass through it each second.

Given: $\Delta V = 12\text{V}$ Power = $P = 30\text{W}$

Req: No of electrons = $n = ??$

Sol As we know that $P = E / t$ ----- ①

Now take $E = \Delta U = q / \Delta V$

$$P = q / \Delta V t$$

$$\Rightarrow P = ne \Delta V / t \quad (q = ne)$$

$$\Rightarrow n = Pt / e \Delta V$$

$$\Rightarrow n = 30\text{W} \times 1\text{s} / 1.6 \times 10^{-19}\text{C} \times 12\text{V}$$

$$\Rightarrow n = 1.56 \times 10^{19} \text{ electrons Ans}$$

Example 11.5: A point charge of $3\mu\text{C}$ is placed at point "0" between "M" and "N" 3cm apart. Point "M" is 2cm from the charge and "N" is 1cm from the charge. What is the potential difference $V_M - V_N$?

Given: $q = 3\mu\text{C} = 3 \times 10^{-6}\text{C}$

$$r_1 = 0.02\text{m}, r_2 = 0.01\text{m}$$

Req: Potential difference = $V_M - V_N = ?$

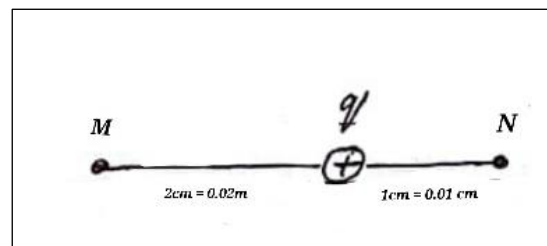
Solution: Now we can write as

$$V_M - V_N = kq / r_1 - kq / r_2$$

$$\Rightarrow V_M - V_N = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 3 \times 10^{-6} / (0.02\text{m})^2 - 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 3 \times 10^{-6} / (0.01\text{m})^2$$

$$\Rightarrow V_M - V_N = 13.5 \times 10^5 \text{ V} - 27 \times 10^5 \text{ V}$$

$$\Rightarrow V_M - V_N = -13.5 \times 10^5 \text{ V Ans}$$



Assignment 11.3: Two charges $5 \times 10^{-8}\text{C}$ and $-3 \times 10^{-8}\text{C}$ are located 16cm apart. At what points on the line joining the two charges is the electric potential zero? Take the potential at infinity to be zero.

Solution: At point "P" we consider the electric potential is zero. i.e

$$V_{\text{net}} = 0$$

$\Rightarrow V_1 + V_2 = 0$ (Electric potential is a scalar quantity, it will be added always on a point)

$$\Rightarrow Kq_1 / (16 - x) + kq_2 / x = 0$$

$$\Rightarrow Kq_1 / (16 - x) = -kq_2 / x$$

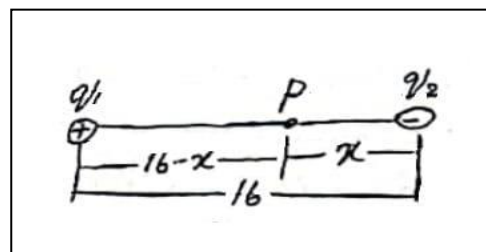
$$\Rightarrow q_1 / (16 - x) = -q_2 / x$$

$$\Rightarrow q_1 x = -(16 - x) q_2$$

$$\Rightarrow 5 \times 10^{-8}\text{C} x = -(16 - x) -3 \times 10^{-8}\text{C}$$

$$\Rightarrow 5x = 48 - 3x \Rightarrow 5x + 3x = 48 \Rightarrow 8x = 48$$

$$\Rightarrow x = 48/8 \Rightarrow x = 6\text{cm}$$



So the point where net electric potential V_{net} is zero lies at a distance of 6cm from charge " q_2 " and at a distance of $(16-6) = 10\text{cm}$ from charge q_1 .

Example 11.7: Figure shows a different combination of capacitors.

If the total charge is $600\mu\text{C}$, determine the values of V_1 and C_2 .

Given: $Q = 600\mu\text{C} = 600 \times 10^{-6} \text{ C}$, $C_1 = 8\mu\text{F} = 8 \times 10^{-6} \text{ F}$, $V_2 = 20\text{V}$

Required: $C_2 = ?$, $V_1 = ?$

Solution: First, find V_1 :

$$V_1 = Q/C_1$$

$$V_1 = (600 \times 10^{-6}) / (8 \times 10^{-6})$$

$$V_1 = 75\text{V}$$

Now total potential difference:

$$V = V_1 + V_2$$

$$V = 75\text{V} + 20\text{V}$$

$$V = 95\text{V}$$

Charge on capacitor C_3 :

$$Q_3 = C_3 V_2$$

$$Q_3 = (8 \times 10^{-6} \text{ F})(20\text{V})$$

$$Q_3 = 160 \times 10^{-6} \text{ C}$$

Now charge on capacitor C_2 :

$$Q_2 = Q - Q_3$$

$$Q_2 = 600 \times 10^{-6} \text{ C} - 160 \times 10^{-6} \text{ C}$$

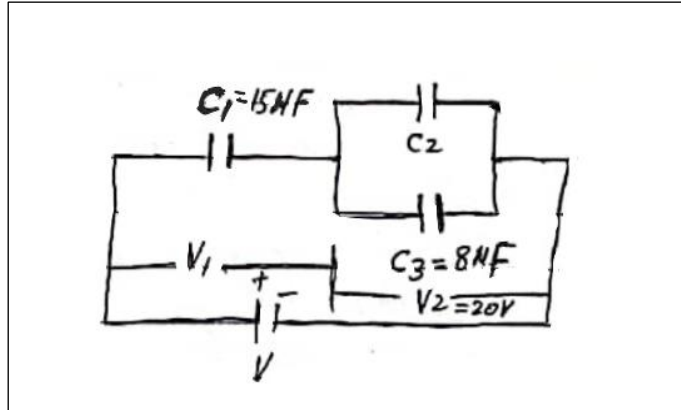
$$Q_2 = 440 \times 10^{-6} \text{ C} = 440\mu\text{C}$$

Now capacitance of capacitor C_2 :

$$C_2 = Q_2 / V_2$$

$$C_2 = (440 \times 10^{-6} \text{ C}) / 20\text{V}$$

$$C_2 = 22 \times 10^{-6} \text{ F} = 22\mu\text{F}$$



Example 11.8: A $6\mu\text{F}$ capacitor is charged to a P.D of 200V and then connected in parallel with an uncharged $3\mu\text{F}$ capacitor. Calculate the P.D across the parallel combination.

Given: $C_1 = 6\mu\text{F} = 6 \times 10^{-6} \text{ F}$, $C_2 = 3\mu\text{F} = 3 \times 10^{-6} \text{ F}$, $V = 200\text{V}$

Required: Common potential difference $V = ?$

Solution: Charge on capacitor C_1 :

$$Q = CV$$

$$Q = (6 \times 10^{-6} \text{ F})(200\text{V})$$

$$Q = 0.0012 \text{ C}$$

Equivalent capacitance of parallel combination:

$$C_{\text{eq}} = C_1 + C_2$$

$$C_{\text{eq}} = 6\mu\text{F} + 3\mu\text{F} = 9\mu\text{F}$$

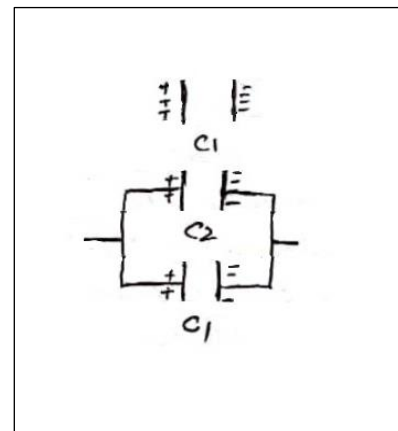
Since charge is conserved, the common P.D is:

$$V = Q / C_{\text{eq}}$$

$$V = 0.0012\text{C} / (9 \times 10^{-6} \text{ F})$$

$$V = 133.3\text{V}$$

Answer: 133.3V Ans



Exercise MCQS

- ① A charge “Q” is divided into two parts q_1 and $(Q - q_1)$ and separated by distance R. The force of repulsion between them will be maximum when

A) $q = Q/4$ B) $q = Q/2$ ✓ C) $q = Q$ D) None of these

Sol: As we know that $F = K q_1 q_2 / r^2$. Now taking $q_1 = q$ and $q_2 = Q - q$ and $r = R$.

i.e $F = K q (Q - q) / r^2$ ----- (1)

Now if we put $q = Q$ then the force will be zero. So option C is wrong. Now in values $Q/4$ and $Q/2$ the value $Q/2$ is greater. So by putting $q = Q/2$ we will get greater force.

- ② Three objects are brought close to each other, two at a time. When objects A and B are brought together, they attract. When objects B and C are brought together, they repel. From this, we conclude that

A) Objects A and C possess charges of the same sign
 B) Objects A and C possess charges of opposite sign ✓
 C) All three objects possess charges of the same sign
 D) One of the objects is neutral

- ③ Object A has a charge of $+2\mu\text{C}$ and object B has a charge of $+6\mu\text{C}$. Which statement is true about the electric forces on the objects?

A) $F_{AB} = -3F_{BA}$ B) $F_{AB} = -F_{BA}$ ✓ C) $3F_{AB} = -F_{BA}$ D) $F_{AB} = 3F_{BA}$

Sol : Forces will be equal in magnitude but opposite in direction as we explained in coloumb's law.

- ④ Some charge is being given to a conductor. Then its potential

A) is maximum at surface
 B) is maximum at Centre
 C) is remains same throughout the conductor ✓
 D) Is maximum somewhere between surface and center

- ⑤ The unit of electric field intensity is

A) Volt/meter B) newton/coulomb
 B) C) joule/coulomb.meter D) All of them ✓

- ⑥ A proton is about 1840 times heavier than an electron. When it is accelerated by a potential difference of 1 KV, its kinetic energy will be

A) 1840 KeV B) $1/1840$ KeV C) 1 KeV ✓ D) 920 KeV

Sol: As we know that

$$\Delta V = \Delta U / q_0$$

$$\Rightarrow \Delta U = q_0 \Delta V. \text{ Now put } q_0 = e \text{ and } \Delta V = 1\text{KV then we get as}$$

$$\Delta U = e \times 1\text{KV}$$

$$\Rightarrow \Delta U = 1\text{KeV}$$

- ⑦ The electric field at a distance of 20 cm from a $2\mu\text{C}$ charge is

A) $3 \times 10^5 \text{ N/C}$ B) $3 \times 10^3 \text{ N/C}$ C) $9 \times 10^3 \text{ N/C}$ D) $9 \times 10^5 \text{ N/C}$

Sol : As we know that $E = K q / r^2$. Now by putting $r = 0.2\text{m}$ and $q = 2 \times 10^{-6}\text{C}$ and $K = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$ then we will get $4.5 \times 10^5 \text{ N/C}$. So all options are wrong.

- ⑧ An electric charge q is placed at the centre of a cube of side a . The electric flux on one face will be

A) $q/6\epsilon_0$ ✓ B) $q/\epsilon_0 a^2$ C) $q/4\epsilon_0$ D) q/ϵ_0

Sol : As we know that total flux through cube from Gauss's law we can write as q/ϵ_0 . Now we also know that the cube has 6 faces. So for one face the electric flux will be equal to $q/6\epsilon_0$.

- ⑨ The electric flux through the surface of a sphere due to a charge q placed at its center depends upon

A) Radius of sphere B) Quantity of charge outside sphere
C) Surface area of sphere D) Quantity of charge inside sphere ✓

- ⑩ A $1\mu\text{F}$ capacitor is subjected to 4000 V potential difference. The energy stored in the capacitor is

A) 8 J ✓ B) 16 J C) 4×10^{-3} J D) 2×10^{-3} J

Sol: As we know that $U = CV^2/2$. Using this equation we will get 8J.

- ⑪ Two parallel metal plates are 8 m apart. Electric field between them is 4 N/C.

If an ion of charge $+2e$ is released at rest from one plate, its kinetic energy when it reaches the other plate is

A) 4 eV B) 64 eV ✓ C) 32 eV D) 16 eV

Sol : As we know that

$$\Delta V = \Delta U / q_0$$

$\Rightarrow \Delta U = q_0 \Delta V$. Now put $q_0 = 2e$ and $\Delta V = Ed$ then we get as

$$\Delta U = 2e \times Ed$$

$$\Rightarrow \Delta U = 2e \times 4\text{V/m} \times 8\text{m} \quad (\text{V/m} = \text{N/C})$$

$$\Rightarrow \Delta U = 64 \text{ eV}$$

- ⑫ The electric potential difference between two points A and B is ΔV . The work done W by the field in moving a charge q from A to B is

A) $W = -q\Delta V$ ✓ B) $W = q\Delta V$ C) $W = -\Delta V/q$ D) $W = \Delta V/q$

Sol : Work done by the field on charge is negative and our work done on the charge is positive.

Short Question

Question 01: The electric potential is constant through a given region of space. Is the electric field zero or non zero in this region?

Ans: The electric field may or may not be zero if the electric potential is constant through a given region of space.

Case 01: As we know that the relation between electric field and potential gradient is given by $E = -\Delta V / \Delta r$ ----- (1)

Now if the electric potential is constant through a given region of space. Then we can write as

$$V_1 = V_2 \Rightarrow \Delta V = V_2 - V_1 \Rightarrow \Delta V = 0 \text{ ----- (2)}$$

Now put eq (2) in eq (1). We get as

$$E = -0/\Delta Y \Rightarrow E = 0 \text{ ----- (3)}$$

Equation (3) shows that the Electric field will be zero in the region if the electric potential is constant in that region.

Case 02: In case of equipotential line, the electric potential at every point remains constant but the electric field is not zero in such a case.

Mathematically we can write

$$\Delta V = \vec{E} \cdot \vec{\Delta r}$$

$$\Rightarrow \Delta V = E \Delta r \cos\theta \text{ ----- (1)}$$

Angle “ θ ” of $\vec{E}$ with $\vec{\Delta r}$ on equipotential line is equal to 90° , so eq (1) we can write as

$$\Delta V = E \Delta r \cos 90^\circ$$

$$\Rightarrow \Delta V = 0$$

$$\Rightarrow V = \text{constant and } E \neq 0.$$

Thus in this case the potential remain constant and the Electric field is not zero.

Question 02: If a point charge “q” of mass “m” is released in a non uniform electric field with field lines pointing in the same direction, will it make a rectilinear motion?

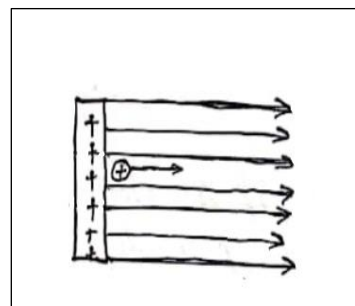
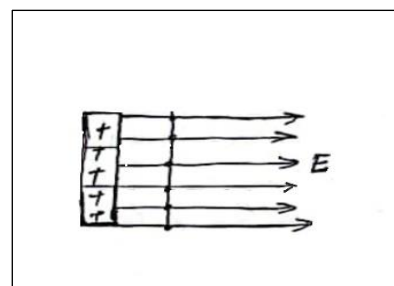
Answer: In the given statement it is clearly mentioned that all field lines of non uniform electric field pointing in the same direction. So it's mean that only magnitude of Electric field will be changing not direction. So whenever a charge “q” of mass “m” is released in such a non uniform electric field, then electric force will act on it and due to which a charge “q” will move in a straight line (Rectilinear motion) and will produce linear acceleration “a”. As we know that

$$E = F/q$$

$$\Rightarrow F = q/E \text{ ----- (1)}$$

Now this force also produces acceleration, so we can also write as

$$\Rightarrow F = ma \text{ ----- (2)}$$



Now compare ① and ②. We get

$$ma = q/E \Rightarrow a = q/E/m.$$

So charged particle will move in a straight line (Rectilinear Motion) with accelerated motion.

Question 03: What is the relationship between potential difference and electric potential energy?

Answer: Electric potential difference is define as “The work done on a test charge inside the electric field between two points against the electric field is called the electric potential difference. Mathematically It can be written as

$$\Delta V = W_{BA} / q_0 \text{ ----- (1)}$$

Now in equation (1) the work done on a test charge between the given points will change the electric potential energy of the test charge. So we can write as

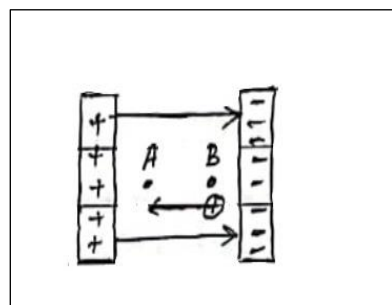
$$W_{BA} = \Delta U \text{ ----- (2)}$$

Now put eq (2) in eq(1). We get as

$$\Delta V = \Delta U / q_0 \Rightarrow \Delta U = q_0 \Delta V \text{ ----- (3)}$$

Now from eq (3) we can write as $\Delta U \propto \Delta V$ -----(4)

Now from relation (4) we can say that the electric potential energy of the test charge show directly proportionality to the potential difference between the two points.



Question 04: How are unit's volt and electron volt related? How do they differ?

Answer: First we explain differences between volt and electron volt.

Volt

- 1) It is the unit of potential difference.
- 2) 1 volt is the potential difference when 1 joule of work is done on 1C charge between the two points against the electric field.
- 3) $1V = 1J C^{-1}$
- 4) It is SI unit

Electron volt

- 1) It is the unit of energy.
- 2) 1 electron volt is the amount of energy gained or lost by an electron between the two points which having the potential difference of value equal to 1 volt.
- 3) $1eV = 1.6 \times 10^{-19}J$
- 4) It is non SI unit.

Now we explain relation between volt and electron volt.

As we know that the electric potential difference can be written as

$$\Delta V = \Delta U / q_0$$

$$\Rightarrow \Delta U = q_0 \Delta V \Rightarrow \Delta K.E = q_0 \Delta V \text{ ----- (1)}$$

Now equation (1) for simplicity we can write as

$$E_{(eV)} = V_{(V)} Q_{(e)} \text{ ----- (2)}$$

Now from eq (2) we can say that the electron volt energy show directly proportionality to the volt of Potential difference between the two points. If the volt of potential difference is greater between the two points, then electron volt energy will be also greater and vice versa. Now we find 1eV energy in joule unit. i.e

$$1\text{eV} = 1 \times 1.6 \times 10^{-19}\text{C} \times 1\text{V}$$

$$\Rightarrow 1\text{eV} = 1.6 \times 10^{-19} \text{ J} \quad (\text{CV} = \text{J})$$

Question 05: How can electric potential be high if electric potential energy is low?

Answer: The electric potential may be high when the electric potential energy is relatively low.

Explanation: In case of capacitor we know that the electric potential energy can be written as

$$U = Q^2/2C \text{ ----- (1)}$$

Now we also know that the electric potential can be written as

$$V = U/Q \text{ ----- (2)}$$

Now put eq (1) in eq (2). We get as

$$V = Q^2/2CQ$$

$$\Rightarrow V = Q/2C \text{ ----- (3)}$$

Now from eq (1) and eq (2) we can say that if the stored charge w.r.t the capacitance is low, then electric potential energy will be also low and electric potential will be greater or high. i.e

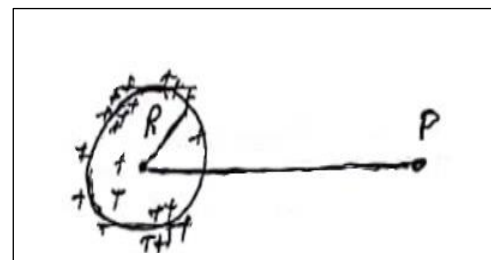
For low charge w.r.t capacitance we can write as

$$q/C > q^2/2C \quad \Rightarrow V > U \text{ ----- (4)} \quad (0.1/C) > (0.1)^2/2C$$

From (4) we can say that electric potential will be high, when electric potential energy is relatively low.

Question 06: Can the potential of a non uniformly charged sphere be the same as that of point charge?

Answer: The potential of a non uniformly charged sphere will be the same as that of a point charge at all points lying outside the sphere at a greater distance, but will be different at a points lying inside the non uniformly charged sphere.



Explanation: case 01: When point "P" lies outside the sphere ($r > R$)

Consider a non-uniformly charged sphere of radius "R" as shown in figure.

Let there is a point "P" situated outside the charged sphere at a greater distance "r". i.e ($r > R$). Then from point "P" the non uniformly charged sphere will become look like as a point charge due to a greater distance.

Then we can say that the electric potential at point "P" will be equal to the electric potential of the point charge. i.e

$$V_P = V_{\text{point charge}}$$

$$\Rightarrow V_P = KQ/r$$

Case 02: When point "P" lies inside the sphere ($r < R$)

When point "P" lies inside the sphere, then the potential due to a non uniformly charged sphere and potential due to a point charge will not be the same due to the following reasons.

- ① Non uniformly charged sphere will not be considered as a point charge.
- ② The charge density on the surface of the sphere is non uniform.
- ③ The electric field intensity at point "P" will be different due to a non-uniform charges on the sphere.

So in this case we can write as

$$V_P \neq V_{\text{point charge}}$$

$$\Rightarrow V_P \neq KQ/r$$

Question 07: will the energy stored in three capacitors be greater when they are connected in series or in parallels.

Ans: As we know that the expression for energy can be written as

$$U = \frac{1}{2} CV^2 \text{ ----- ①}$$

We also know that the expression for equivalent capacitance when capacitors are in series combination can be written as

$$1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3 \text{ ----- ②}$$

We also know that the expression for equivalent capacitance when capacitors are in parallel combination can be written as

$$C_{eq} = C_1 + C_2 + C_3 \text{ ----- ③}$$

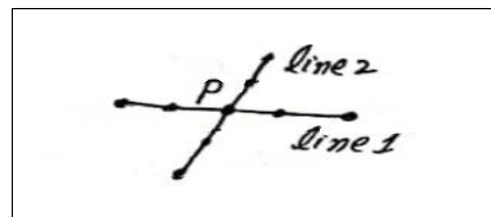
Now from eq ② and ③ we can say that the equivalent capacitance in parallel combination is greater than the equivalent capacitance of capacitors in series combination.

Now from equation ① we can say that the energy stored will be greater when the capacitors are in parallel combination.

Question 08: Can different equipotential lines cross each other.

Ans: Different equipotential lines can not cross each other.

Explanation: It is not possible for different equipotential lines to cross each other. It is because at each point of an equipotential line, there is a specific single value of electric field intensity and electric potential. Now if the equipotential lines cross each other, then we must have two values of electric fields and of the electric potentials at a single point "P" as shown in figure. Now two values of electric fields and of the electric potentials at a single point "P" physically are not possible. So now



we can say that different equipotential lines never cross each other.

Question 09: Water has a large dielectric constant, but it is rarely used in capacitors. Why?

Ans: Water has a large dielectric constant, but it is rarely used in the capacitors due to a following reasons.

- ① When water is polarized in the capacitor, then polarized water molecule will decrease the effect of electric field between the plates. i.e As we know that

$$C_{\text{med}} = \epsilon_r C_{\text{vac}}$$

$$\Rightarrow Q/V_{\text{med}} = \epsilon_r Q/V_{\text{vac}}$$

$$\Rightarrow V_{\text{med}} = V_{\text{vac}}/\epsilon_r$$

$$\Rightarrow E_{\text{med}} \times d = E_{\text{vac}}/\epsilon_r \times d$$

$$\Rightarrow E_{\text{med}} = E_{\text{vac}}/\epsilon_r$$

- ② Water molecules contain “H⁺” and “OH⁻” ions. Thus it can conduct electricity easily.
- ③ The ions of water molecule can release the charge between the plates of the capacitor. As a result, the discharging of the capacitor would take place.
- ④ Water is sensitive to temperature and impurities which effect its dielectric constant.
- ⑤ Water has the property of corrosion. It will corrode the plates of the capacitor and as a result the plates may become useless.

Note: Impurities: The materials which are solvable in water is called impurities for water.

Q.10: How can you increase the capacitance of a Capacitor?

Ans: As we know that the capacitance of a parallel plate capacitor can be written as

$$C_{\text{med}} = \epsilon_0 \epsilon_r A / d$$

Equation shows that the capacitance of a parallel plate capacitor depends on the following factors.

- ① Area of the plates: If area of the plates is greater then capacitance will be also greater, then capacitor will store more charges-and vice versa.
- ② Separation between the plates: If the separation or distance between the plates is less. Then stronger electrostatic induction will increase the capacitance and then capacitor will store more charges and vice versa.
- ③ Medium between the plates: If the permittivity of the medium or dielectric constant is greater then capacitance will be also greater and then capacitor will store more charges and vice versa.
- ④ Quality of the plates: Quality means if the plates are good conductors, then capacitance will be greater and then Capacitor will store more charges and vice versa.

"Numerical Questions"

Q.01: What is the magnitude of the force of attraction between an iron nucleus bearing charge $q_1 = 26e$ and its inner most electron, if the distance between them is $1 \times 10^{-12} \text{ m}$.

Given: $q_0 = 26e$, $q_1 = e$, $r = 10^{-12} \text{ m}$.

Required : $F = ?$

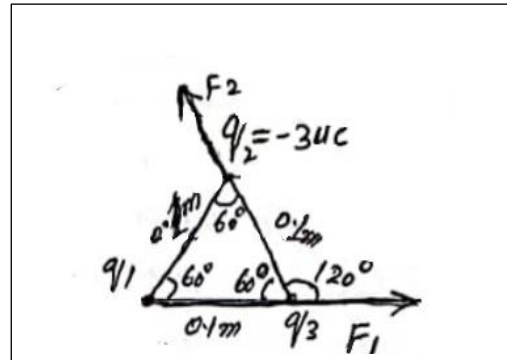
Sol: According to Coulomb's law the force between two charges is

$$F = K q_0 q_1 / r^2$$

$$\Rightarrow F = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 26 e \times e$$

$$\Rightarrow F = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 26 \times (1.6 \times 10^{-19} \text{ C})^2 / (10^{-12} \text{ m})^2$$

$$\Rightarrow F = 6 \times 10^{-3} \text{ N Ans.}$$



Q.2: Charges $2\mu\text{C}$, $-3\mu\text{C}$ and $4\mu\text{C}$ are placed in air at the vertices of an equilateral triangle of sides 10cm , what is magnitude of resulting force acting on $4\mu\text{C}$ charge.

Given: $q_1 = 2\mu\text{C} = 2 \times 10^{-6} \text{ C}$ $q_2 = -3 \times 10^{-6} \text{ C}$ $q_3 = 4 \times 10^{-6} \text{ C}$ $r = 10\text{cm} = 0.1\text{m}$

Required : Force on q_3 i.e $F_2 = ?$

Sol: First we find force " F_1 " due to " q_1 " charge on " q_3 " charge from Coulomb's law. i.e

$$F = K q_1 q_3 / r^2$$

$$\Rightarrow F = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 2 \times 10^{-6} \text{ C} \times 4 \times 10^{-6} \text{ C} / (0.1\text{m})^2$$

$$\Rightarrow F_1 = 7.2 \text{ N} \text{ ----- (1)}$$

Now we find " F_2 " force due to " q_2 " charge on " q_3 " charge. i.e

$$F_2 = K q_2 q_3 / r^2$$

$$\Rightarrow F_2 = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times (-3 \times 10^{-6}) \times (4 \times 10^{-6}) / (0.1\text{m})^2$$

$$\Rightarrow F_2 = -10.8 \text{ N} \text{ ----- (2)}$$

Now we find the resultant force. The resultant force of " F_1 " and " F_2 " we can write as

$$F = \sqrt{(F_1)^2 + (F_2)^2 + 2F_1 F_2 \cos \theta}$$

$$\Rightarrow F = \sqrt{(7.2\text{N})^2 + (-10.8\text{N})^2 - 2 \times 7.2\text{N} \times 10.8\text{N} \cos 120^\circ}$$

$$\Rightarrow F = \sqrt{246.24}$$

$$\Rightarrow F = 15.69\text{N (Ans)}$$

Q.3: Two equal and opposite charges of magnitude $2 \times 10^{-7} \text{ C}$ are placed 15cm apart. What is the magnitude and direction of electric intensity E at a point mid way between the charges? What force would act on a proton (charge $= +1.6 \times 10^{-19} \text{ C}$) placed there?

Given : $q_1 = q_2 = q = 2 \times 10^{-7} \text{ C}$ $D = 15\text{cm} \Rightarrow d = 0.15\text{m}$

$$r = d/2 \Rightarrow r = 0.15\text{m}/2 = 0.075\text{m}$$

Required : a) E at point " P " b) F at point " P "

Sol a) At point " P " there are two electric field intensities in the same direction.

There force for total Electric field intensity

We can write as $E = E_1 + E_2$

$$E = kq_1/r^2 + kq_2/r^2$$

$$\Rightarrow E = 2kq/r^2$$

$$\Rightarrow E = 2 \times 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 2 \times 10^{-7} \text{ C} / (0.075 \text{ m})^2$$

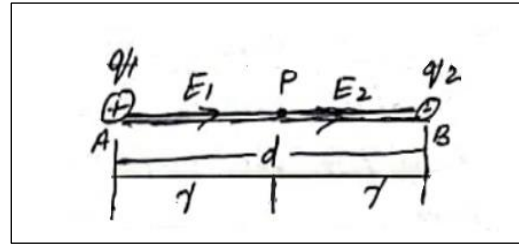
$$\Rightarrow E = 0.64 \times 10^6 \text{ N/C along AB. (Ans)}$$

Sol b) Now we find force on proton at point

“P”. For this we can write as $E = F/q \Rightarrow F = q/E \Rightarrow F = q_{\text{Proton}} E$

$$\Rightarrow F = 1.6 \times 10^{-19} \text{ C} \times 0.64 \times 10^6 \text{ N/C}$$

$$\Rightarrow F = 1.024 \times 10^{-13} \text{ N along AB (Ans)}$$



Q.4]: Two positive point charges of $15 \times 10^{-10} \text{ C}$ and $13 \times 10^{-10} \text{ C}$ are placed 12cm apart. Find the work done in bringing the two charges 4cm closer.

Given: $q_1 = 15 \times 10^{-10}$, $q_2 = 13 \times 10^{-10}$ $r_1 = 0.12 \text{ m}$,
 $r_2 = 0.08 \text{ m}$

Required: work done = $W = ?$

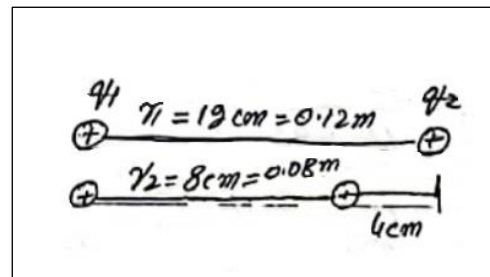
Sol: As we know that

$$W = q_1 q_2 / 4\pi\epsilon_0 (1/r_2 - 1/r_1)$$

$$\Rightarrow W = k q_1 q_2 (1/r_2 - 1/r_1)$$

$$\Rightarrow W = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 15 \times 10^{-10} \text{ C} \times 13 \times 10^{-10} \text{ C} (1/(0.08 \text{ m}) - 1/(0.12 \text{ m}))$$

$$\Rightarrow W = 7.9 \times 10^{-8} \text{ J (Ans).}$$



Q.5: A hollow sphere is charged to $14 \mu\text{C}$. Find the Potential (a) at its surface (b) inside the sphere (c) at a distance of 0.2m from the surface. The radius of the sphere is 0.3m.

Given: $q = 14 \times 10^{-6} \text{ C}$ radius = $r = 0.3 \text{ m}$ $r_1 = 0.2 \text{ m}$

Sol (a): First we find potential at surface.

As we know that $V_s = Kq/r$

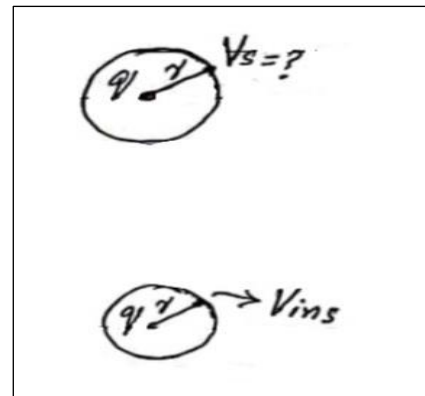
$$\Rightarrow V_s = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 14 \times 10^{-6} \text{ C} / 0.3 \text{ m}$$

$$\Rightarrow V_s = 4.2 \times 10^5 \text{ V (Ans)}$$

Sol (b): Now we find inside potential. Inside point have the same distance r from charge q . Therefore we can write as

$$V_{\text{ins}} = V_{\text{surface}}$$

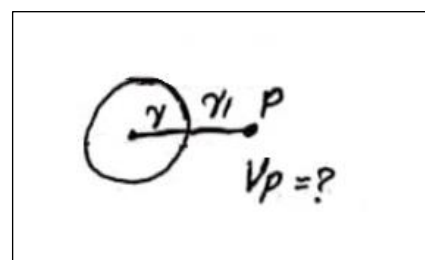
$$\Rightarrow V_{\text{ins}} = 4.2 \times 10^5 \text{ V (Ans)}$$



Sol ©: Now we find potential at a 0.2m distance away from the sphere Surface. In this case we can write as $V_p = Kq/(r + r_1)$

$$\Rightarrow V_p = 9 \times 10^9 \text{ Nm}^2/\text{C}^2 \times 14 \times 10^{-6} \text{ C} / (0.3 \text{ m} + 0.2 \text{ m})$$

$$\Rightarrow V_p = 25.2 \times 10^4 \text{ V (Ans)}$$



Q.6: If 280J of work is done in carrying a charge of 2C from a place where the potential is -12V to another place where potential is 'V', calculate the value of V.

Given: $W = 280\text{J}$ $q = 2\text{C}$ $V_1 = -12\text{V}$ $V_2 = V$. Req: potential = $V = ?$

Sol: As we know that $\Delta V = W/q \Rightarrow V_2 - V_1 = W/q$

$$\Rightarrow V - (-12\text{V}) = 280\text{J}/2\text{C}$$

$$\Rightarrow V + 12\text{V} = 140\text{V}$$

$$\Rightarrow V = 140\text{V} - 12\text{V}$$

$$\Rightarrow V = 128\text{V} \text{ Ans}$$

Q.7: The electric field at a point due to a point charge is 26N/c and the electric potential is 13 J/c. Calculate the distance of the point from the charge and magnitude of charge.

Given: $E = 26 \text{ N/C}$ $V = 13 \text{ J/c}$. Required: a) distance = $r = ?$ b) charge = $q = ?$

Sol a) we know that

$$V = kq/r \Rightarrow V = kq/r^2 \Rightarrow V = Er$$

$$\Rightarrow r = V/E \Rightarrow r = 13 \text{ J/c} / 26 \text{ N/C} \Rightarrow r = 0.5 \text{ m (Ans)}$$

Sol (b): Now we find charge "q". As we know that

$$V = Kq/r \Rightarrow q = rV/K$$

$$\Rightarrow q = 0.5 \text{ m} \times 13 \text{ J} / 9 \times 10^9 \text{ Nm}^2/\text{c}$$

$$\Rightarrow q = 0.722 \times 10^{-9} \text{ C (Ans)}$$

Q.8: Two point charges of $8\mu\text{C}$ and $-4\mu\text{C}$ are separated by a distance of 10cm in air. At what point on the line joining the two charges is the electric potential is Zero?

Given: $q_1 = 8\mu\text{C} = 8 \times 10^{-6}$ $q_2 = -4\mu\text{C} = -4 \times 10^{-6}$

Required: The distances of the point from the charges " q_1 " and " q_2 ".

Sol: consider at point "P" the electric potential is zero. i.e

$$V_{\text{net}} = 0$$

$$\Rightarrow V_1 + V_2 = 0$$

$$\Rightarrow V_1 = -V_2$$

$$\Rightarrow Kq_1/10-x = -Kq_2/x$$

$$\Rightarrow 8 \times 10^{-6} / 10-x = -(-4 \times 10^{-6}) / x$$

$$\Rightarrow 8 / 10-x = 4 / x$$

$$\Rightarrow 8x = 4(10 - x)$$

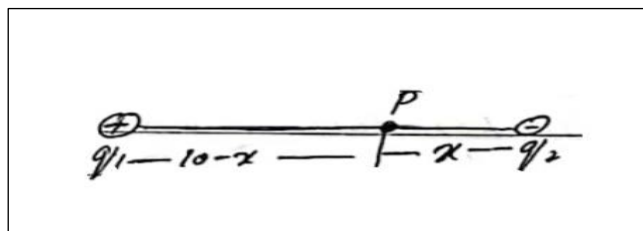
$$\Rightarrow 8x = 40 - 4x$$

$$\Rightarrow 12x = 40$$

$$\Rightarrow x = 40 / 12$$

$$\Rightarrow x = 3.3 \text{ cm from } -4\mu\text{C}$$

$$\Rightarrow 10-x \Rightarrow 10 - 3.3 \text{ cm} \Rightarrow 6.6 \text{ cm so 6.6cm from } q_1 \text{ charge.}$$



Q.9: An electron with an initial speed $29 \times 10^5 \text{ m/s}$ is fired in the same direction as a uniform electric field with a magnitude of 80 N/C . How

far does the electron travel before being brought to rest momentarily and turned back.

Given: $v = 29 \times 10^5 \text{ m/s}$ $E = 80 \text{ N/C}$. Required: $d = ?$

Sol: As we know that $\Delta V = \Delta U/q$ or $V = U/q$

$$\Rightarrow U = qV \Rightarrow K.E = qV \Rightarrow K.E = qEd \quad (V = Ed)$$

$$\Rightarrow \frac{1}{2}mv^2 = qEd$$

$$\Rightarrow d = mv^2/2qE$$

$$\Rightarrow d = 9.11 \times 10^{-31} \text{ kg} \times (29 \times 10^5 \text{ m/s})^2 / (2 \times 8.6 \times 10^{-19} \text{ C} \times 80 \text{ N/C})$$

$$\Rightarrow d = 0.299 \text{ m Ans.}$$

Q.10: Three capacitors of capacitance $4\mu\text{F}$, $6\mu\text{F}$ and $8\mu\text{F}$ respectively are connected in series to a 250V dc supply. Find (i) the total capacitance (ii) charge on each capacitor (iii) P.D across each capacitor.

Given: $C_1 = 4 \mu\text{F} = 4 \times 10^{-6} \text{ F}$, $C_2 = 6 \mu\text{F} = 6 \times 10^{-6} \text{ F}$, $C_3 = 8 \mu\text{F} = 8 \times 10^{-6} \text{ F}$

Req: (i) $C_{eq} = ?$ (ii) $Q_1 = ?$, $Q_2 = ?$, $Q_3 = ?$ (iii) $V_1 = ?$, $V_2 = ?$, $V_3 = ?$

$$\Rightarrow 1/C_{eq} = 1/C_1 + 1/C_2 + 1/C_3$$

$$\Rightarrow 1/C_{eq} = 1/4 + 1/6 + 1/8$$

$$\Rightarrow 1/C_{eq} = (6+4+3)/24$$

$$\Rightarrow 1/C_{eq} = 13/24$$

$$\Rightarrow C_{eq} = 24/13 = 1.84 \mu\text{F}$$

$$\Rightarrow Q_1 = Q_2 = Q_3 = Q$$

$$\Rightarrow Q = C_{eq} V$$

$$\Rightarrow Q = 1.84 \times 10^{-6} \times 250$$

$$\Rightarrow Q = 460 \times 10^{-6} \text{ C}$$

$$\Rightarrow Q_1 = Q_2 = Q_3 = 460 \times 10^{-6} \text{ C}$$

$$\Rightarrow V_1 = Q_1/C_1$$

$$\Rightarrow V_1 = 460 \times 10^{-6} / 4 \times 10^{-6}$$

$$\Rightarrow V_1 = 115 \text{ V}$$

$$\Rightarrow V_2 = Q_2/C_2$$

$$\Rightarrow V_2 = 460 \times 10^{-6} / 6 \times 10^{-6}$$

$$\Rightarrow V_2 = 77 \text{ V}$$

$$\Rightarrow V_3 = Q_3/C_3$$

$$\Rightarrow V_3 = 460 \times 10^{-6} / 8 \times 10^{-6}$$

$$\Rightarrow V_3 = 57.5 \text{ V}$$

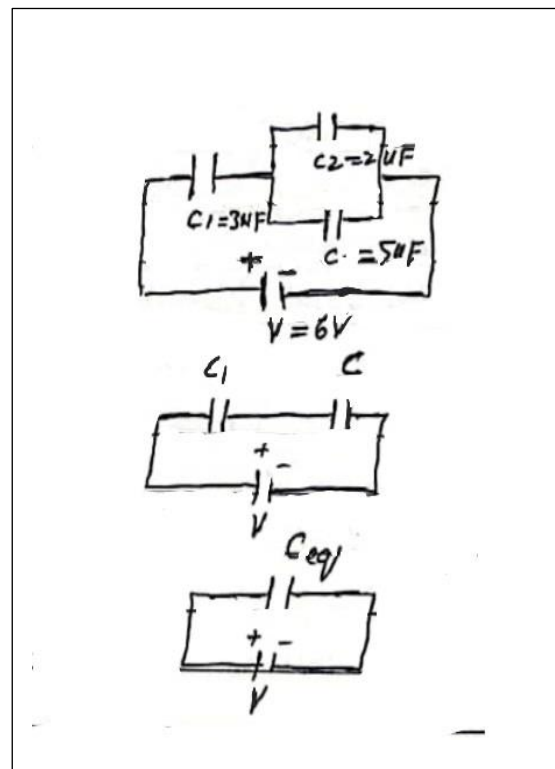
Q.11: Find the charge on $5\mu\text{F}$ Capacitor in the circuit shown in fig.

Given: $C_1 = 3 \times 10^{-6} \text{ F}$, $C_2 = 2 \times 10^{-6} \text{ F}$,

$C_3 = 5 \times 10^{-6} \text{ F}$, $V = 6 \text{ V}$

Req: $Q_3 = ?$

Sol : First we find equivalent



capacitance of capacitors C_2 and C_3 . They are in parallel so we can write as

$$\Rightarrow C = C_2 + C_3$$

$$\Rightarrow C = 2 + 5 = 7 \mu\text{F}$$

$$\Rightarrow 1/C_{\text{eq}} = 1/3 + 1/7$$

$$\Rightarrow C_{\text{eq}} = 2.1 \mu\text{F}$$

$$\Rightarrow Q = C_{\text{eq}} V$$

$$\Rightarrow Q = 2.1 \times 10^{-6} \times 6$$

$$\Rightarrow Q = 12.6 \times 10^{-6} \text{ C}$$

$$\Rightarrow V_3 = Q/C$$

$$\Rightarrow V_3 = 12.6 \times 10^{-6} / 7 \times 10^{-6}$$

$$\Rightarrow V_3 = 1.8 \text{ V}$$

$$\Rightarrow Q_3 = C_3 V_3$$

$$\Rightarrow Q_3 = 5 \times 10^{-6} \times 1.8$$

$$\Rightarrow Q_3 = 9 \times 10^{-6} \text{ C} = 9 \mu\text{C}$$

Q.12: A $6\mu\text{F}$ capacitor is charged to a P.D of 120V and then connected to an uncharged, $4\mu\text{F}$ capacitor in parallel. Calculate the P.D across the capacitor.

Given: $C_1 = 6 \mu\text{F} = 6 \times 10^{-6} \text{ F}$, $C_2 = 4 \mu\text{F} = 4 \times 10^{-6} \text{ F}$, $V_1 = 120 \text{ V}$

Req: $V = ?$

$$\Rightarrow Q = C_1 V_1$$

$$\Rightarrow Q = 6 \times 10^{-6} \times 120$$

$$\Rightarrow Q = 720 \times 10^{-6} \text{ C}$$

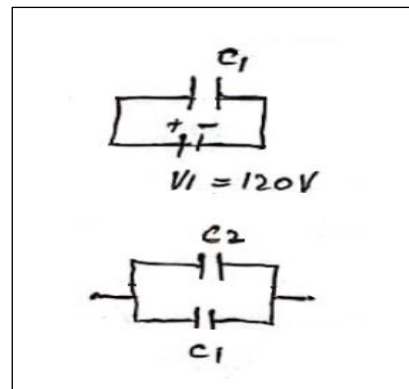
$$\Rightarrow C_{\text{eq}} = C_1 + C_2$$

$$\Rightarrow C_{\text{eq}} = 10 \mu\text{F} = 10 \times 10^{-6} \text{ F}$$

$$\Rightarrow V = Q / C_{\text{eq}}$$

$$\Rightarrow V = 720 \times 10^{-6} / 10 \times 10^{-6}$$

$$\Rightarrow V = 72 \text{ V}$$



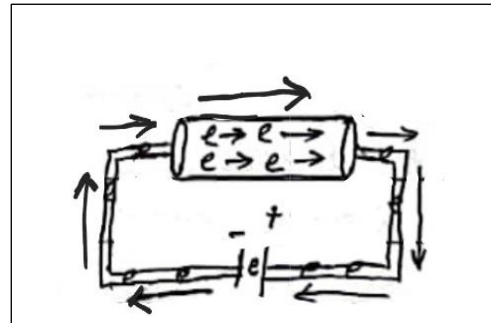
END OF CHAPTER 11

Chapter #12 “Current Electricity”

Steady current:

The continues flow of free electrons in a circuit is called steady current.

Explanation: A conductor which having many free electrons is connected with a battery of voltage “V” as shown in figure. After the connection of a conductor with a battery then electric field will be set up in the circuit from +ve terminal of a battery towards a -ve terminal. Then electric field produced by a battery voltage will exert an electric force on a free electrons in a circuit then due



to which such a free electron will star continuous flow from -ve terminal towards a +ve terminal. Now this continuous flow of electrons in a circuit is called steady electric current.

“Main points for the current in the circuits or materials”

① The actual direction of current in the circuit will be from -ve terminal of a battery towards a +ve terminal, through out that part of the circuit, which is external to the battery. And such current is said to be an electronic current. However, prior to an electron theory, it was assumed that current flows from +ve terminal of a battery towards a -ve terminal, through out that part of the circuit, which is external to the battery. Then physist said such a current will be due to a +ve charge mean protons. Then further they said the protons and electron has the same charge magnitude, so current magnitude will be the same in each direction if we took it from +ve terminal towards negative terminal due to a -ve charge or from -ve terminal towards +ve terminal due to a -ve charge.

Then further they said in any circuit we will always take current direction from +ve terminal towards a -ve terminal of a battery, and such a direction of current is said to be conventional direction, and the current is said to be a conventional current.

- ② In metallic conductors, the current is produced due to a flow of electrons which are negatively charged.
- ③ In electrolytes, the current is produced due to a flow of positive and negative ions.
- ④ In gasses, the current is produced due to a flow of electrons and positive ions.

Mathematical explanation of current

The flow of charges through a given cross-sectional area of a wire or conductor per unit time is called electric current. Mathematically it can be written as

$$I = Q/t$$

The unit of electric current is ampere (A).

In one second time if the flowing charges is one coulomb through a given cross-sectional area. Then current will be equal to one ampere.

$$1 \text{ A} = 1 \text{ C/1 s}$$

Ampere unit of current is bigger unit. Its smaller units are:

$$1 \text{ milliamphere} = 1 \text{ mA} = 10^{-3} \text{ A}$$

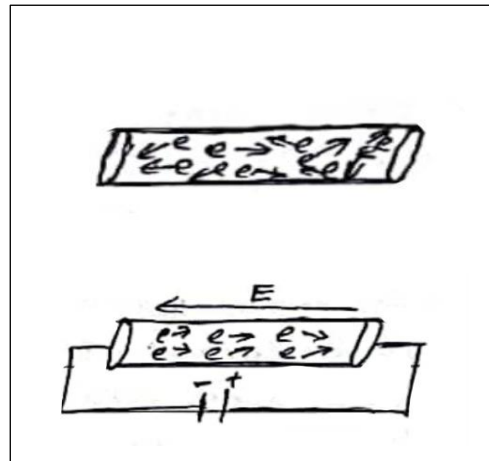
$$1 \text{ microampere} = 1 \mu\text{A} = 10^{-6} \text{ A.}$$

Drift velocity in conductor

The average velocity with which free electrons get drifted in a metallic conductor under the influence of electric field is known as drift velocity.

Explanations: Every conductor has many free electrons. These electrons move in random directions within the atoms of the conductor. Due to their random motion, the net or average velocity of these electrons is zero in the conductor.

Now a conductor which having many free electrons is connected with a battery of voltage "V" as shown in figure. After the connection of a conductor with a battery then electric field will be set up in the circuit from +ve terminal of a battery towards a -ve terminal.



Then electric field produced by a battery voltage will exert an electric force on free electrons in a circuit. Then due to which such a free electrons will start continuous flow with different velocities from -ve terminal towards a +ve terminal.

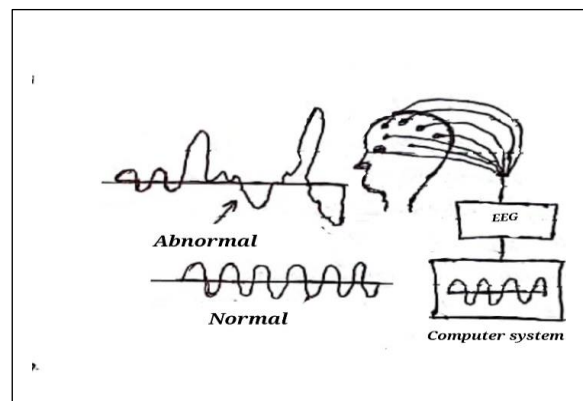
During flow of electrons its velocity will be different due to a successive collision of these electrons with the atoms of the conductor, and also will change the direction in each collision.

Now in all motion the average velocity of an electron in a metallic conductor under the influence of electric field is known as drift velocity. In the conductor the average velocity of electrons is equal to 10^{-5} m/s . It's very less velocity.

Electroencephalogram (EEG)

An electroencephalogram (EEG) is a test that uses metal discs or electrodes to find the electrical activity in the brain.

The brain constantly generates minute electrical currents even during sleep and in deep coma. The electrical currents are produced due to Na^+ , K^+ , Ca^{++} and Cl^- ions motion in a brain.



Then the electrodes with thin wires are attached around the head which detects the tiny electrical currents produced by the ions motion in a brain.

Then (EEG) device receives all the electrode currents. Then (EEG) device send information to computer system. Then computer System records the brain electrical currents on the screen or in graph paper as wavy line.

Then neurologist's doctors check the graphs for normal or abnormal function of the brain. Neurologist doctors refer the patient for (EEG) test in case of epilepsy, Brain tumors, stroke etc.

Ohm's Law

Statement(a): This law states that the magnitude of current in a conductor is directly proportional to the applied voltage while keeping the temperature of the conductor is constant. Mathematically we can write as

$$I \propto V$$

$$\Rightarrow I = KV \quad \text{put } K = 1/R$$

$$\Rightarrow I = V/R$$

$$\Rightarrow V = IR \text{ ----- (1)}$$

In eq (1) "R" is constant, which is known as the resistance of the conducting materials. Resistance depends upon the nature, dimension and physical state of the conductor.

Statement(b): The voltage across the conductor or the resistance of a conductor is directly proportional to the current which is passing through it while keeping the temperature of the conductor is constant. Mathematically we can write as

$$V \propto I$$

$$\Rightarrow V = KI \quad \text{put } k = R \rightarrow \text{constant}$$

$$\Rightarrow V = IR \text{ ----- (2)}$$

Ohmic Materials: Those materials in which current is directly proportional to the voltage is called ohmic materials.

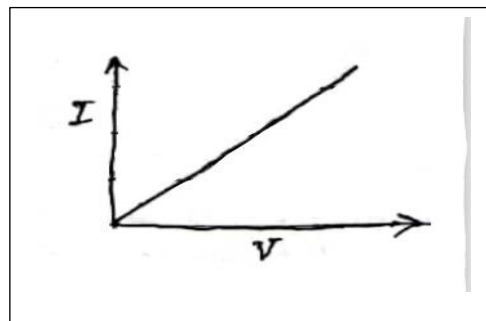
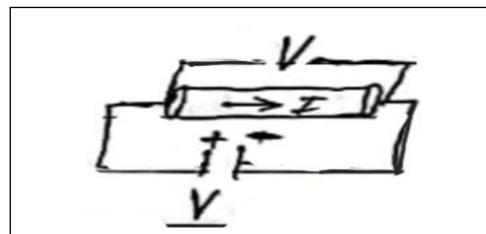
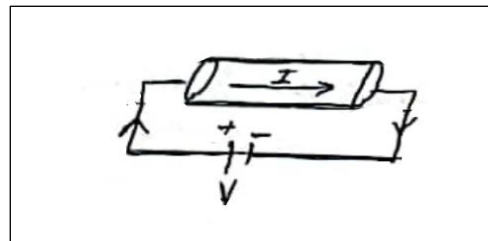
OR Those materials for which the I-V graph is a straight line then such materials are called Ohmic materials.

OR Those materials for which the slope of I-V graph is constant are called Ohmic materials.

In fig (a) the graph between I-V is a straight line in case of metals or conductors. So metals or conductors are called Ohmic materials. Now we find the slope of the graph from fig (a). i.e

$$\text{slope} = \Delta Y / \Delta X$$

$$\Rightarrow \text{slope} = I/V$$



$$\Rightarrow \text{slope} = 1/R \text{ ----- } \textcircled{3}$$

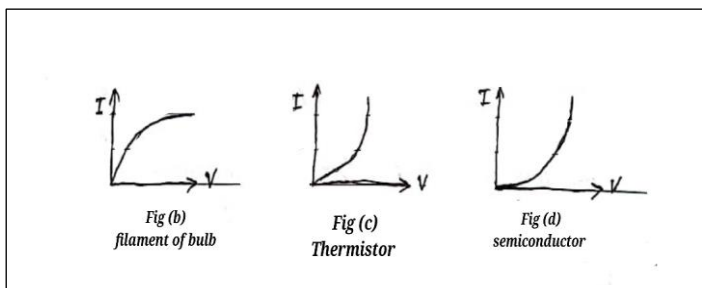
In eq $\textcircled{2}$ "R" is constant. So we can say that the slope of stright line graph will be constant.

Non Ohmic Materials: Those materials in which current is not directly proportional to the voltage is called Non ohmic Materials.

OR Those materials for which the I-V graph if not stright line, then such materials are called Non ohmic materials.

OR These materials for which the slope of I-V graph is not constant are called Non ohmic materials.

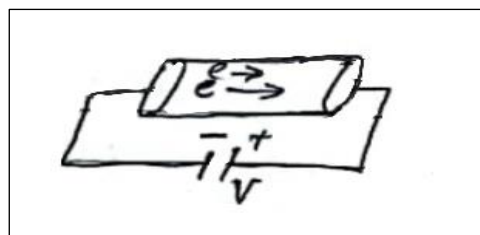
In case of Filament of bulb, thermistor and semiconductor diode the graphs between I-V are not stright line. So these materials are called Non ohmic materials. If the graphs are not stright line, then its slope will not be constant.



Electrical Resistance

The opposition offered by the molecules or atoms of a substance to the flow of free electrons is known as electrical Resistance.

Explanation: When a conductor or substance is connected to a battery. Then electrons will move from -ve terminal towards a +ve terminal of a battery. Now during motion of electrons it collides with atoms of molecules in the substance and due to which its flow opposes. Such



opposition to the flow of electrons due to a collision with atoms of molecules is called Resistance.

In each collision the electron will lost the speed as well as energy. And the lost energy will be converted to heat energy. Those materials which have low resistance to the flow electrons are said to be good conductors. Good conductors are (silver, copper and Aluminum). The unit of Resistance is called Ohm (Ω). It is define as "If applied voltage is one volt and due to which a current in a conductor if one ampere". Then the resistance of that conductor will be one ohm .i.e

$$R = V/I$$

$$\Rightarrow 1 \Omega = 1\text{volt}/1\text{ampere}$$

Factors upon which Resistance depends

The resistance of a conductor depends upon the following factors.

$\textcircled{1}$ The resistance is directly proportional to the length of the conductor. i.e

$$R \propto L \text{ ----- } \textcircled{1}$$

$\textcircled{2}$ The resistance is inversely proportional to the cross-sectional area of the

conductor.

$$R \propto 1/A \text{-----} \textcircled{2}$$

③ The resistance of a conductor depends upon its nature. Some materials offer high resistance while some conductor offer smaller resistance.

④ The resistance of a conductor also depends upon temperature. Its resistance increases with increase in temperature and vice versa.

Now by keeping the temperature constant, we can write as

$$R \propto L/A$$

$$\Rightarrow R = \rho L/A \text{-----} \textcircled{3}$$

In equation “ ρ ” is constant and is known as resistivity Or specific resistance of the material.

Resistivity or specific resistance: As we know that

$$R = \rho L/A$$

$$\Rightarrow \rho = RA/L$$

$$\Rightarrow \rho = 1\Omega \cdot 1\text{m}^2/1\text{m}$$

$$\Rightarrow \rho = R \times 1\text{m}^2/1\text{m}$$

Now resistivity or specific resistance we can define as

“The resistance of a wire which having length of 1m and cross sectional area of 1m² is called resistivity or specific resistance”.

Explanation: Different materials posse’s different values of resistivity. The resistivity of metals and alloys is very small. On the other hand resistivity of insulators is extremely large. So these materials hardly pass or conduct an electric current.

There is also an intermediate class of materials which are known as semiconductors. The resistivity of these materials lies between conductors and insulators.

Unit of resistivity: As we know that

$$\rho = RA/L$$

$$\Rightarrow \rho = 1\Omega \times 1\text{m}^2/1\text{m}$$

So the unit of resistivity is $\Omega \cdot \text{m}$.

Conductance: The reciprocal of resistance of a conductor is known as conductance. It is represented by “G”. If “R” represents the resistance of a conductor, then its conductance “G” can be written as $G = 1/R$.

The unit of conductance is “mho”. Another unit is “Siemen”. It is represented by “G”.

Conductivity: The reciprocal of resistivity of a conductor is called conductivity. It is represented by “ δ ”. If “ ρ ” represents the resistivity of a conductor, then its conductivity can be written as

$$\delta = 1/\rho$$

$$\Rightarrow \delta = L/RA$$

The unit of conductivity is (mho m) or Siemen meter (s.m)

Effect of Temperature on Resistance

The resistance of a material changes with the change in temperature. The change in resistance due to temperature depends upon the type of material. For example

① The resistance of pure metals like copper, aluminum increases with the increase in temperature. It is because when we increase the temperature across the metals, then at high temperature the atoms of metals then absorb high heat energy and due to which such atoms then start bigger vibration. Now due to a bigger vibration of atoms in each direction, then electron collision rate will be increased, and then due to which the electron will not flow easily. So by that reason we can say that the resistance of the metals will be increased with the increase in temperature and vice versa.

② The resistance of electrolytes, insulators (e.g. glass, mica, rubber) and semiconductor (e.g. germanium, Silicon etc) decreases with the increase in temperature. It is because a high temperature across these materials will produce more free electrons in them. And then such a more free electron flows easily in these materials. So by that reason we can say that the resistance of these materials will be decreased with the increase in temperature and vice versa.

Temperature coefficient of Resistance

Consider a conductor having resistance " R_0 " at temperature 0°C . And " R_T " at $T^\circ\text{C}$. Now let " $R_T - R_0$ " represents the change in resistance due to increase in temperature, then we can write as

① $(R_T - R_0)$ is directly proportional to initial resistance " R_0 ". i.e

$$R_T - R_0 \propto R_0 \text{ ----- ①}$$

② $(R_T - R_0)$ is directly proportional to the rise in temp " T ". i.e

$$R_T - R_0 \propto T \text{ ----- ②}$$

Now combining relation ① and ②. we get

$$R_T - R_0 \propto R_0 T$$

$$R_T - R_0 = \alpha R_0 T \text{ ----- ③}$$

In eq ③ " α " is constant and is known as temperature coefficient of resistance.

From eq ③ we can write as

$$R_T = R_0 + \alpha R_0 T$$

$$R_T = R_0 (1 + \alpha T) \text{ ----- ④}$$

Eq ④ represents the final resistance of the conductor.

Definition of temperature coefficient of Resistance " α "

From equation ③ we can write as

$$\alpha = (R_T - R_0) / R_0 T \text{ ----- ⑤}$$

Now from eq ⑤ the temperature coefficient of resistance " α " is defined as

"change in resistance per ohm initial resistance per degree rise in temperature is called temperature coefficient of resistance " α ". Its unit is K^{-1} . The

temperature coefficient of resistance " α " will be +ve for metals. Because with high temperature it increases the resistance. i.e $R_T > R_o$.

The temperature coefficient of resistance " α " will be negative for insulators and semiconductors, because with high temperature it decreases the resistance. i.e $R_T < R_o$.

Temperature coefficient of Resistivity

As we know that

$$R_T - R_o = \alpha R_o T \text{ ----- (1)}$$

Now eq(1) in term of " ρ " Resistivity we can write as.

$$\rho_T L / A - \rho_o L / A = \alpha \rho_o L / A T$$

$$\Rightarrow L/A(\rho_T - \rho_o) = \alpha \rho_o L / A T \text{ ----- (2)}$$

$$\Rightarrow (\rho_T - \rho_o) = \alpha \rho_o T$$

$$\Rightarrow \rho_T = \rho_o + \alpha \rho_o T \text{ ----- (3)}$$

Now from eq(2) we can write as

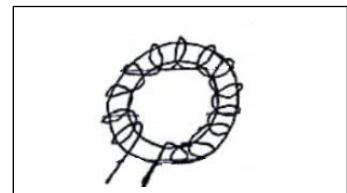
$$\alpha = (\rho_T - \rho_o) / \rho_o T \text{ ----- (4)}$$

Now in eq (4) " α " represents the temperature coefficient of resistivity and it is define as "change in resistivity Per unit initial resistivity per degree rise in temperature is called temperature coefficient of resistivity " α ". Its unit is also K^{-1} . The temperature coefficient of resistivity " α " will be +ve for metals, because with high temperature it increases the resistivity re $\rho_T > \rho_o$. The temperature coefficient of resistivity " α " will be -ve for insulators and semiconductors. Because with high temp these materials decreases the resistivity re $\rho_T < \rho_o$.

Wire wound Resistor and wire wound variable Resistor

Wire wound Resistor: A resistive wire which is wound around a ceramic or plastic core is called wire wound Resistor.

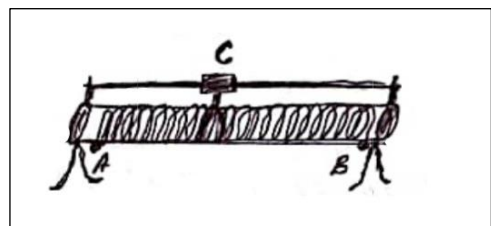
Explanation: A wire wound resistor is enclosed in an insulating cover. A resistive wire in it is made of nickel chromium, due to the following properties:



- ① It has very small temperature coefficient of resistance.
- ② It can safely operate at higher temperature than carbon type resistors.
- ③ It has high stability in the circuit.
- ④ It has high accuracy. Its mean it does not change the resistance value easily at high temperature.

A wire wound resistor is best type of resistor and it has a fixed value of Resistance.

wire wound variable Resistor: It is a type of wire wound resistor whose resistance can be varied according to our needs and desires. It consists of Nickel chromium wire which is wound on an insulated cylinder. The end of



the wire is connected to two fixed terminals "A" and "B" as shown in figure. There is 3rd terminal "C" which is sliding or moving over the wire. We can use wire wound variable resistor as a (a) Rheostat (b) as a potential divider.

a) use of wire wound variable resistor as Rheostat: When a wire wound variable resistor we can use in a circuit for a current control, then it is called Rheostat.

A wire wound variable resistor of rheostat consist of three terminals "A" "B" and "C" as shown in figure. The terminal "A" and "B" are fixed while the terminal "C" can slide over the wire. If the sliding contact "C" is moved away from terminal "A" then the length of the wire will be increasing and as a result the resistance of the wire will be also increasing. Now if the sliding contact "C" is moved towards terminal "A" then the length of the wire will be decreasing and as a result the resistance of the wire will be also decreasing.

Now by adjusting the rheostat in the circuit, then due to which we can change the current in that circuit to any desired value by changing the resistance of Rheostat.

b) use of wire wound variable resistor as a potential dividers: When

a wire wound variable resistor we can use in a circuit for the voltage or potential difference division, then it is called potential divider.

Explanation: In a figure the applied potential difference "V" of the battery, we want to divide it to any desired value between the terminals "A", "B" and "C". Now from Ohm's law the potential difference or voltage drop across the wire resistor between terminals "A" and "B" we can write as

$$V_{AB} = IR$$

such a voltage drop " V_{AB} " will be equal to battery voltage V. So we can write

$$V = IR \text{ ----- (1)}$$

In eq(1) "R" represents the resistance of the wire between "A" and "B" terminals and "I" represents the current to it. Now we can write as

$$I = V/R \text{ ----- (2)}$$

Now the potential difference or voltage drop across the wire resistor between terminals "B" and "C" we can write as

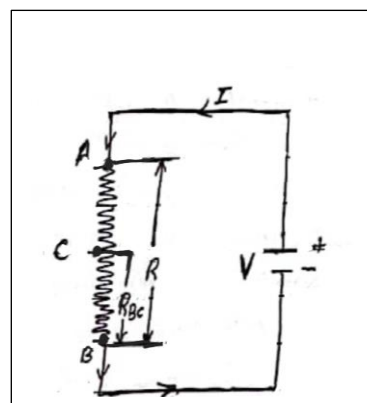
$$V_{BC} = I R_{BC} \text{ ----- (3)}$$

In eq(3) " R_{BC} " represents the resistance of the wire between "B" and "C" terminals and "I" represents the current to it.

Now put eq (2) in eq (3) we get as

$$V_{BC} = V(R_{BC}/R) \text{ ----- (4)}$$

Now from eq (4) we can say that the potential difference " V_{BC} " depends on the position of sliding terminal "C" when we move terminal "C" towards terminal "B" then the wire length and its resistance " R_{BC} " will be decreasing and as a



result V_{BC} will be also decreasing. On the other hand when we move terminal "C" towards terminal "A" then the wire length and its resistance " R_{BC} " will be increasing and as a result " V_{BC} " will be also increasing.

By this way we can get any value of " V_{BC} " by changing the " R_{BC} " resistance between terminals "B" and "C".

Note (1): The P.D or voltage drop across the resistor "R" by the ohm law we can write as

$$V_{AB} = IR$$

And such voltage drop will be equal to the applied battery voltage "V". i.e

$$V_{AB} = V.$$

Note(2): In the given figure the voltage or P.D drop across " R_1 " resistor will be

$$V_{AB} = IR_1 \text{ ----- } ①$$

And the voltage drop across " R_2 " resistor will be

$$V_{BC} = IR_2 \text{ ----- } ②$$

In this case the sum of voltages drops will be equal to the applied battery voltage, because the given circuit is series circuit.

$$V_{AB} + V_{BC} = V$$

V_{AB} Or V_{BC} either of them will be less than V.

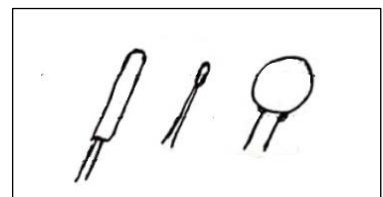
Note (3): The battery voltage we will call applied voltage and the voltage across the resistor we will call as the voltage drop because the resistor drops or lost the battery voltage.

All notes is just information they are not a part of topic.

Thermistor

Thermistor is a temperature sensing device which changes the resistance with change in temperature.

Explanation: Thermistor is made of metallic oxides and having shape like bead, disk or cylinder. It has an outer capsulated cover of impermeable material called epoxy or glass. Thermistor has the following important properties.



- ① A thermistor is a temperature or heat sensitive device.
- ② The resistance of thermistor changes very rapidly with change in temperature.
- ③ The temperature coefficient of resistance for a thermistor is very high.

The temperature coefficient of resistance for a thermistor can be +ve or -ve.

Types of thermistors

Negative Temperature Coefficient (NTC) Thermistor

The thermistor which decreases the resistance with increasing temperature conversely, which increase the resistance with decreasing temperature is called Negative Temperature coefficient (NTC) Thermistor. In such type

thermistors a rapid decrease occurs in resistance with small increase in temperature and vice versa.

Some appliances, such as washing machine, clothes dryers, refrigerators and freezers, as well as small appliances like hair dryers, ovens, toasters, thermostats air conditions and fire alarms all these have NTCs for temperature control. NTCs are ideal for accurate temperature measurement and control.

Positive Temperature coefficient (PTC) Thermistor

The thermistor which increases the resistance with increasing temperature conversely, which decrease the resistance with decreasing temperature is called positive Temperature coefficient (PTC) Thermistor.

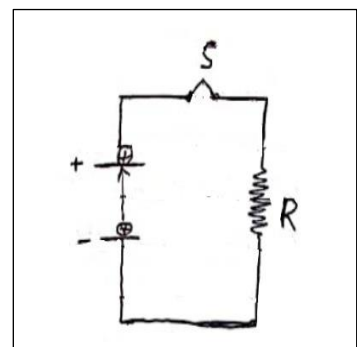
- ① This type thermistor is generally used for fuse purpose. When a high current flows through a (PTC) then heat generated raises the temperature of the (PTC). Then at high temperature the resistance of (PTC) rises very high and then such a high resistance decrease current in the circuit. By this way the positive Temperature coefficient (PTC) thermistor protect the circuits or devices from high currents.
- ② They are also used as a heater in automobile industry to provide heat to the people inside the cabin.
- ③ They are also used in diesel engine to provide heat to diesel in cold climate areas for the engine starting purpose.
- ④ They are also used in timing devices for the prevention of high heat effects produced in it.

Electromotive Force

The work done on a unit +ve charge by a battery from negative terminal (lowers Potential) to a positive terminal (higher Potential) is called electromotive force of a battery. i.e $\mathcal{E} = W/q$

Explanation: in a figure a battery, switch and external resistance "R" are shown in figure. Now a unit +ve charge on a negative terminal inside a battery is also shown in a figure.

Now when we close the switch then the chemicals in a battery start the work on a unit +ve charge from a -ve terminal and move It towards a +ve terminal, this work will be continuous until the charge move out from the battery. This work done on a unit +ve charge by a battery from negative terminal to a positive terminal is called electromotive force of a battery.



Then out side the battery the charge on a +ve terminal will gain electric potential energy due to the work done on it. Then such a +ve charge will move from +ve terminal towards a -ve terminal and will change electric potential energy to Kinetic energy, and will produce current in the external circuit. Then the external resistor will lost the +ve charge kinetic energy and then charge will entre with no kinetic energy inside to the battery.

After that, again the battery chemical will do the same work on the charge and will maintain the same current in the circuit. This process will be continuous by a battery until the battery chemical to stop the further work on a charge.

Now electromotive force we can also define as:

- 1) The work done on a unit +ve charge by a battery to maintain a continuous current in the circuit is called electromotive force.
- 2) It can be also defined as: The work done on a unit +ve charge by a battery to maintain a constant potential difference between any two points in the circuit is called electromotive force. The unit of emf is volt. i.e $1V = 1J/C$.

Sources of electromotive force (emf):

Those sources which have an ability of work on the charges and to maintain a current in the circuit is called sources of emf. Sources of the emf are the following:

- ① Batteries or cells convert chemical energy into electrical energy.
- ② Electric generators convert mechanical energy into electrical energy.
- ③ Thermocouples convert heat energy into electrical energy.
- ④ Radiant (a solar cell) converts sunlight into electrical energy.

Note (I): The work on a +ve charge starts from the negative terminal and it will be zero on that terminal. So electric potential of that terminal will be also zero. i.e $V_- = W/q \Rightarrow V_- = 0/q \Rightarrow V_- = 0$

That's why negative terminal is called lower electric potential terminal.

Note(II): The work on a +ve charge on the positive terminal from the negative terminal will not be zero. So electric potential of that terminal will not be also zero. i.e

$$V_+ = W/q \Rightarrow V_+ \neq 0$$

That's why positive terminal is called higher electric potential terminal.

All notes is just information it is not a part of topic

Internal Resistance of a Supply

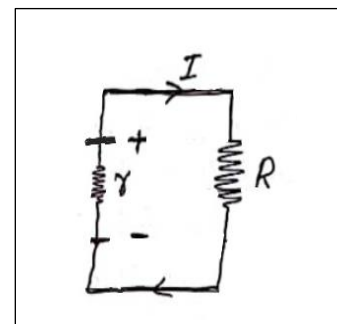
The resistance inside a cell or battery is called internal resistance of cell or battery. It is denoted by "r".

Terminal potential difference: In a figure a battery, internal resistance and external resistance are shown. A battery in such a figure supplies the current to external resistance "R".

Now in figure the potential difference across the terminals of a battery is called terminal potential difference or terminal voltage. Now in figure the voltage drop across the external resistor "R" will be equal to the terminal voltage or terminal potential difference of a battery. i.e

$$V_R = V_t$$

$$IR = V_t$$



Relation between electromotive force, Terminal P.D and internal resistance:

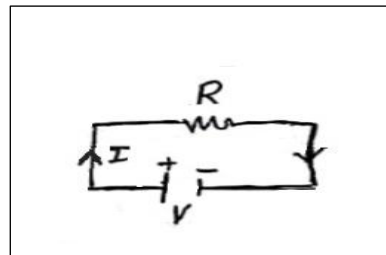
In a figure the battery emf will be equal to the sum of voltages drop across resistance “r” and external resistance “R”.

$$\mathcal{E} = V_r + V_R$$

$$\Rightarrow \mathcal{E} = Ir + IR$$

$$\Rightarrow \mathcal{E} = Ir + V_t$$

$$\Rightarrow V_t = \mathcal{E} - Ir \text{ ----- } \textcircled{1}$$



Now we explain special cases.

- ① If the circuit is open. i.e if $I = 0$ in the external circuit. Then eq ① we can write as

$$V_t = \mathcal{E} - 0 \times r$$

$$\Rightarrow V_t = \mathcal{E} \text{ ----- } \textcircled{2}$$

Now from ② we can say that the terminal Potential difference will be equal to emf of a battery.

- ② If current is not zero in the external circuit mean if the circuit is close, i.e. if $I \neq 0$ in the external circuit. Then we can write as

$$V_t = \mathcal{E} - Ir$$

$$\Rightarrow V_t < \mathcal{E} \text{ ----- } \textcircled{3}$$

Now from ③ we can say that terminal Potential difference will be less than the emf of a battery.

- ③ As we know that the reverse direction current when flows from a battery then the battery will get be charging. So for the battery charging we take $I = -I$. Now eq ① we can write as

$$V_t = \mathcal{E} - (-I)r$$

$$\Rightarrow V_t = \mathcal{E} + Ir$$

$$\Rightarrow V_t > \mathcal{E} \text{ ----- } \textcircled{4}$$

Now from ④ we can say that the terminal potential difference will be greater than the emf of a battery during charging process.

Electric power

In a figure a battery provides a current to a resistor “R”. Now electric power in that circuit we can define as

- 1) The work done on a charges by a battery w.r.t time is called power of the battery. i.e

$$P = W/t$$

- 2) The energy given to charges by a battery w.r.t time is also called power of the battery. i.e

$$P = E/t$$

The battery power is called applied power.

- 3) The work done on the charges by a resistor w.r.t time is called power of the

resistor. i.e. $P = W/t$

4) The energy lost of the charges by the resistor w.r.t time is called power of the resistor. i.e $P = E/t$

Power of the resistor is called power lost.

Other expressions for power

As we know that

$$P = W/t \text{ ----- } (1)$$

We also know that for the potential difference . i.e

$$V = W/Q$$

$$\Rightarrow W = QV \text{ ----- } (2)$$

Now put eq (2) in eq (1) we get as

$$P = QV/t \text{ ----- } (3)$$

Now we also know that for the current i.e

$$I = Q/t$$

$$\Rightarrow Q = It \text{ ----- } (4)$$

Now put eq (4) in eq (3). We get as

$$P = ItV/t$$

$$\Rightarrow P = IV \text{ ----- } (5)$$

Now we also know that from ohms law i.e

$$V = IR \text{ ----- } (6)$$

Put eq (6) in eq (5). We get as

$$P = IIV$$

$$\Rightarrow P = I^2R \text{ ----- } (7)$$

We also know that

$$V=IR$$

$$\Rightarrow I = V/R \text{ ----- } (8)$$

Put eq (8) in eq (5) . We get as

$$P = V^2R /R^2$$

$$\Rightarrow P = V^2/R \text{ ----- } (9)$$

The above three equation are equally valid for calculation of electric power in DC circuits.

Unit of electric power: The SI unit of power is watt. If current through a resistor is 1ampere and due to which if potential difference drop or lost is 1V across the Resistor. Then power lost or power consumed by a resistor will be equal to 1watt. i.e 1watt = 1V x 1A. The bigger units of power are kilowatt (KW) and megawatt (MW).

$$1KW = 1000 \text{ watts} = 10^3\text{watts}$$

$$1mw = 10^6\text{watts}.$$

Kilowatt-hour: It is the unit of electrical energy. It is define as

The energy lost in one hour by an appliance which having power equal to 1 kilowatt is called kilowatt-hour. We know that

$$P = E/t$$

$$\Rightarrow E = Pt$$

$$\Rightarrow E = \text{kilowatt-hour}$$

$$\Rightarrow E = 1000 \text{ watt} \times 3600 \text{ s } J=WS$$

$$\Rightarrow E = 3.6 \times 10^6 \text{ ws}$$

$$\Rightarrow E = 3.6 \times 10^6 J$$

Maximum power out put

The maximum power delivered to a load resistor “R” by a source is called Maximum power output.

Explanation: In a figure a battery, internal resistance and load resistance are shown in figure. In a figure a battery supplies a current to a load resistor “R”. Now we calculate the maximum Power out put through a load resistor in the circuit.

Now for this we can write as

$$P_{\text{out}} = I^2 R \text{ ----- (1)}$$

Now we can also write as

$$\mathcal{E} = V_r + V_R$$

$$\Rightarrow \mathcal{E} = Ir + IR$$

$$\Rightarrow \mathcal{E} = I (r + R)$$

$$\Rightarrow I = \mathcal{E} / (r + R)$$

$$\Rightarrow I = \mathcal{E} / (R + r) \text{ ----- (2)}$$

Now put eq (2) in eq (1). We get as

$$P_{\text{out}} = \mathcal{E}^2 R / (R + r)^2$$

$$\Rightarrow P_{\text{out}} = \mathcal{E}^2 R / (R^2 + r^2 + 2Rr)$$

$$\Rightarrow P_{\text{out}} = \mathcal{E}^2 R / (R^2 + r^2 + 2Rr + 2Rr - 2Rr)$$

$$\Rightarrow P_{\text{out}} = \mathcal{E}^2 R / (R^2 + r^2 - 2Rr + 4Rr)$$

$$\Rightarrow P_{\text{out}} = \mathcal{E}^2 R / ((R - r)^2 + 4R) \text{ ----- (3)}$$

In eq (3) if we put $R = r$ i.e. load Resistance = internal Resistance

then denominator will become minimum, and “ P_{out} ” will become maximum. So we can write as

$$(P_{\text{out}})_{\text{max}} = \mathcal{E}^2 R / ((r - r)^2 + 4r)$$

$$\Rightarrow (P_{\text{out}})_{\text{max}} = \mathcal{E}^2 / 4r \text{ or } (P_{\text{out}})_{\text{max}} = \mathcal{E}^2 / 4R$$

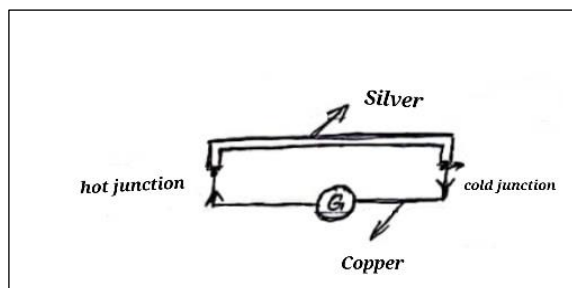
Maximum power transfer theorem: The power delivered to a load resistor by a source will be maximum if the load resistance value equal to the internal resistance of the source. Devices like Radio, Television, Antenna and speaker system operates on maximum power.

Thermocouples

Two different wire circuit which convert heat energy to electrical energy is called Thermocouple.

OR Two different wire circuit in which the emf and current is producing due to a temperature difference is called Thermocouple.

Explanation: In early 1821 Thomas seebeck searched experimentally for a relation between electricity and heat. He joined two wires of two dissimilar metals and formed a circuit. He discovered that if one junction is heated to a high temperature, and the other junction remained at cooled temperature, then the galvanometer connected at their ends shows a deflection. This is known as seebeck effect. The emf ($\mathcal{E} = W/q$) generated in the circuit due to a temperature difference is called thermoelectric emf and the resulting current is called Thermoelectric current.



then the galvanometer connected at their ends shows a deflection. This is known as seebeck effect. The emf ($\mathcal{E} = W/q$) generated in the circuit due to a temperature difference is called thermoelectric emf and the resulting current is called Thermoelectric current.

Uses of thermocouple: Thermocouples are the most widely used as temperature sensors in industry due to their low cost, Simplicity, size, and useable temperature range. The emf or current produced in it show the value of temperature.

Composition of thermocouple: The wires of thermocouple can be composed of Noble metals (such as platinum, indium, silver, osmium, gold, and Thorium) or base metals (such as copper, iron, nickel, copper, alloy lead and zinc).

Reversibility: The seebeck effect is reversible, if the hot and cold junctions are interchanged, the direction of emf (and hence current) reverses.

Mathematical form of thermoelectric emf: The thermoelectric emf is measured with respect to the temperature of hot Junction, when the cold Junction is maintained at 0°C . Now the emf produced in the two junctions w.r.t the temperature is given by

$$\mathcal{E} = \alpha T + \frac{1}{2} \beta T^2$$

Where “ α ” and “ β ” are constants (called thermoelectric coefficients), which depends on the nature of the metals. The greater the separation of the metals forming the thermocouple in greater will be the thermoemf produced.

Normally the thermoemf produced is very small, of the order of millivolt per every degree of temperature difference.

Variation in thermoelectric emf with temperature

Keeping the junction “B” at 0°C , the temperature of junction “A” is increased. When both the junctions are at the same temperature, there is no thermo emf.

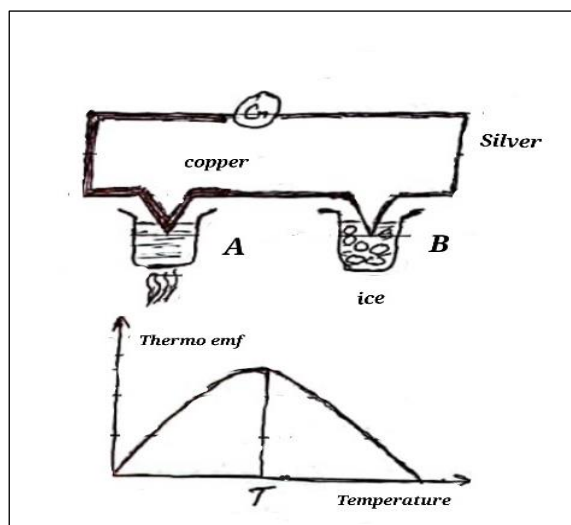
When the temperature at one junction is cold while at other end is increased the thermo emf also increases with temperature and reaches a maximum value at a certain temperature called the neutral temperature “T”.

The value of the neutral temperature is constant “T” Temperature for a thermocouple, depends on the nature of materials and is independent of the temperature of the cold junction.

As the temperature of the hot junction is increased, the thermo emf starts decreasing instead of increasing, it's because at high temperature the two metal wires will start increasing their resistances.

Then at very high temperature the emf is zero due to very high resistance in the wires. Now the particular temperature at which, the thermo emf becomes zero is called the inversion temperature.

The graph shows the variation of the thermo emf with the temperature of hot junction, with the cold junction.



Resistance Thermometers

Instruments which give us the temperature by its resistance are called Resistance thermometers. They are also called resistance temperature detectors (RTDS) or temperature sensors. They are made of platinum, copper or Nickel. Resistance thermometer has the following properties.

- ① They have a single value of resistance for each degree of Temperature.
- ② They have a long range a linear relationship between resistance and temperature.
- ③ They gives us accurate and repeatable temperature.

Resistance thermometer of platinum is the best thermometer because platinum has a long rang linear relationship between resistance and temperature. Its long range relationship between resistance and temperature ($R \propto T$) starts from -272.5°C and ends on 961.78°C . Its mean through such a thermometer we can easily find any temperature between the range -272.5°C to 961.78°C .

Uses of Resistance thermometers:

Due to higher accuracy and repeatability of resistance thermometer, They are uses food producing industry, hospitals, refiners, chemical plants, in research labs paramedical industries etc.

They are slowly replacing the use of thermocouples in many industries because thermocouples gives us the temperature up to 600°C only.

Complex Circuits

Those circuits in which the resistors are neither in series nor in parallel and having more batteries or sources of emf are called Complex circuits. Ohm's

law does not work in such circuits for the calculation of currents, resistances and voltages. In such circuits we will use Kirchhoff's laws. It has two parts.

- ① Kirchhoff's current law (KCL) ② Kirchhoff's voltage law (KVL).

Kirchhoff's current law (KCL)

Statement: It states that the algebraic sum of all the currents meeting at a junction of a circuit is equal to zero. i.e

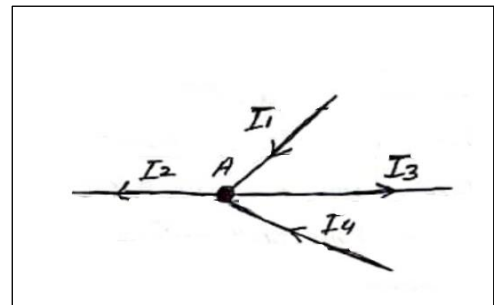
$$\Sigma I = 0$$

Explanation: Consider a circuit having a junction "A" as shown in the figure. Let I_1 and I_4 be the currents flowing into a junction and I_2, I_3 be the currents leaving the junction. Now the currents which are flowing into a junction are taken as positive, while the currents leaving the junction are taken as negative. Now according to Kirchhoff's current law, we have

$$\Sigma I = 0$$

$$\Rightarrow I_1 + I_4 + (-I_2) + (-I_3) = 0$$

$$\Rightarrow I_1 + I_4 = I_2 + I_3 \text{ ----- ①}$$



i.e Sum of incoming currents = Sum of out going currents.

Hence Kirchhoff's current law may also be stated as "The algebraic sum of currents flowing towards the junction in a circuit is equal to the algebraic sum of currents flowing away from that junction".

In case of KCL, no charge can accumulate at a Junction. So the total charge entering the junction per unit time must be equal to the total charge leaving the junction per unit time. Thus KCL is a manifestation of the conservation of electric charges. It can be written as

$$\Sigma I_{\text{entering}} = \Sigma I_{\text{leaving}}$$

$$\Rightarrow \Sigma Q_{\text{entering}} / t = \Sigma Q_{\text{leaving}} / t$$

$$\Rightarrow \Sigma Q_{\text{entering}} = \Sigma Q_{\text{leaving}}$$

Kirchhoff's voltage law (KVL)

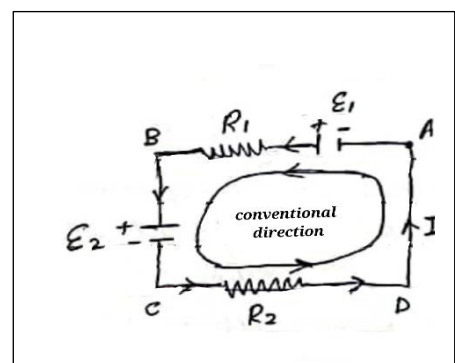
Statement: It states that, in any closed circuit, the algebraic sum of all the electromotive force (emf) and voltage drops at resistors is equal to zero. i.e

$$\Sigma \text{emf} + \Sigma V_R = 0$$

$$\text{Or } \Sigma \text{emf} + \Sigma IR = 0$$

Explanation: Consider a circuit which consists of two emf sources " $\mathcal{E}_1$ " and " $\mathcal{E}_2$ " and two resistors " R_1 " and " R_2 " as shown in figure. Now we explain some important rules for KVL.

- ① The circuit must be closed between the points.
- ② Then we will take the current direction either clock wise or anticlock wise in the circuit.



- ③ Then we will take conventional direction either clock wise or anticlock wise in the circuit.
- ④ Then we will move in the circuit according to a conventional direction from any point and then we will come back to this point.
- ⑤ In a circuit if we move from -ve terminal towards a +ve terminal then we will take the positive sign with its emf, conversely if we move from +ve terminal toward a -ve terminal then we will take a negative sign with its emf.
- ⑥ If the current and conventional directions are the same across the resistor then we will take negative sign with a voltage drop conversely If the current and conventional directions are the opposite across the resistor then we will take positive sign with a voltage drop.

Now applying these rules of Kirchhoff's law to the given circuit, then we get as

$$\mathcal{E}_1 + (-V_{R1}) + (\mathcal{E}_2) + (-V_{R2}) = 0$$

$$\Rightarrow \mathcal{E}_1 + (-\mathcal{E}_2) = V_{R1} + V_{R2}$$

$$\Rightarrow \mathcal{E}_1 + (-\mathcal{E}_2) = IR_1 + IR_2$$

$$\text{Or } \Sigma \mathcal{E} = \Sigma IR \text{ ----- (1)}$$

From eq (1) KVL can be also defined as "The sum of all the emfs in a circuit is equal to the sum of all the voltage drops in that closed circuit or loop."

The KVL is a manifestation of law of conservation of Energy. That is from eq(1) we can write as

$$\Sigma \mathcal{E} = \Sigma V_R$$

$$\Rightarrow \Sigma W_s / Q = \Sigma W_R / Q$$

$$\Rightarrow \Sigma E_s / Q = \Sigma E_R / Q \text{ ----- (2)}$$

From eq (2) we can say that for the charges "Q" the energy delivered by the sources must be equal to the energy drops by the resistors.

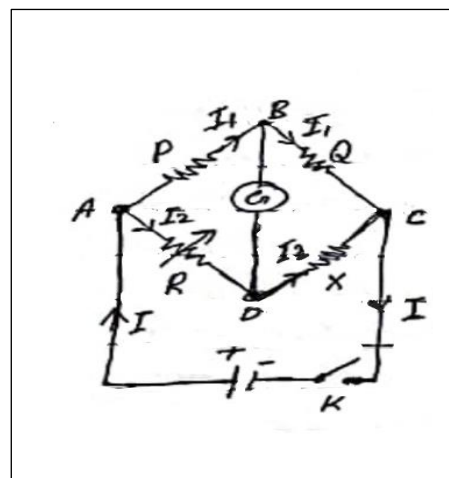
Wheatstone Bridge

An electric circuit which is used to determine the unknown resistance is called Wheatstone Bridge.

It was constructed by an English telegraph Engineer charless wheat stone and hence it is called wheat stone bridge as shown in figure.

Principle: To find the unknown resistance for this first we will shift the circuit to a balance condition. Balance condition mean that there is no current flow in the galvanometer. Balance condition also called null condition.

Construction: It has four resistors P, Q, R, and X. In these resistors "P" and "Q" is fixed resistors, "R" is variable resistor and "X" is unknown resistor. It has one key and one



battery connected on points "A" and "C" as shown in figure. And it has a galvanometer which is connected between the points "B" and "D".

Working: Generally when the key "k" is closed then the current "I" divides unequally at the point "A" between the two branches. Now due to a potential difference between the points "B" and "D", some current will flow through the galvanometer, which will show deflection. Then we will change the value of resistor "R" in such a way that there is no current flow through the galvanometer even when the key is closed. Then we will say the circuit or bridge is now in balance condition and the point "B" and "D" is now having the same electric potentials. Now when the bridge is balanced, let the current in the resistor "P" and "Q" is " I_1 " while the current in resistors "R" and "X" is " I_2 " as shown in figure. Now at balance condition the points "B" and "D" having the same electric potential, so potential difference between these two points will be equal to zero. Due to which now we can write as

P.D across P resistor = P.D across R resistor

$$V_P = V_R$$

$$\Rightarrow I_1 P = I_2 R \text{ ----- } \textcircled{1}$$

Similarly we can also write as

P.D across Q resistor = P.D across X resistor.

$$V_Q = V_X$$

$$\Rightarrow I_1 Q = I_2 X \text{ ----- } \textcircled{2}$$

Now divide eq $\textcircled{1}$ by eq $\textcircled{2}$ we get as

$$I_1 Q / I_1 P = I_2 X / I_2 R$$

$$\Rightarrow Q / P = X / R$$

$$\Rightarrow X = QR / P$$

if the values of three resistors are known, then we can easily find the unknown resistor X.

Potentiometer

A device through which we measure the unknown emf of the given source is called potentiometer.

Principle: To find the unknown emf of the source for this first, we will shift the potentiometer to a balance condition. Balance condition mean that there is no current flow in the galvanometer. Balance condition is called null condition.

Construction: In a figure AC represents a resistive wire. A battery of potential difference "V" is connected across the resistive wire ends. In a figure we also have a scale up to 100cm and two sources of emfs in which the emf of one source is unknown and of the other source is known. The two sources are connected with the two way switch, galvanometer, Jockey "K" and with a resistive wire as shown in figure.

Working: In a figure first we switch on the source " $\mathcal{E}_1$ " of known emf by two way switch. Then such a source will provide the current to a resistive wire, to

jockey "K" and to a galvanometer. Then we will move a jockey "K" on a resistive wire and will check the galvanometer current. Then on a resistive wire we will adjust the jockey on a certain point say it "B" such that on which the galvanometer current to become zero. Then of point "B" the battery Potential difference will balance the potential difference of " $\mathcal{E}_1$ " source because of zero current on that point. Now at balance condition we can say that the potential difference drop across the resistance " R_1 " of the wire between "A" and "B" points will be equal to the emf " $\mathcal{E}_1$ " of the source, if we ignore its internal resistance. i.e

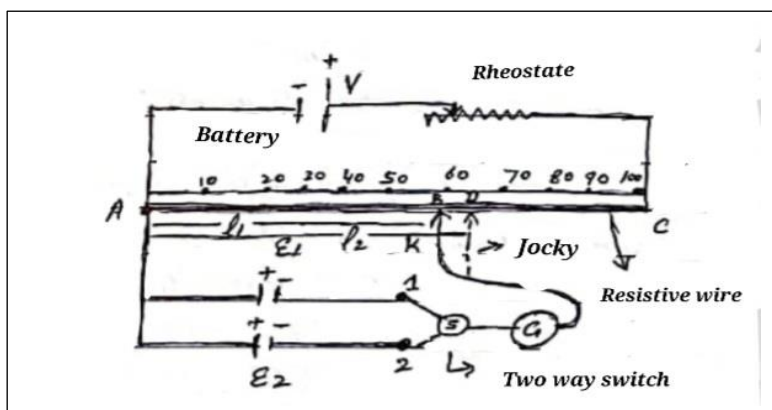
$$V_{R1} = \mathcal{E}_1 \text{ or}$$

$$\Rightarrow \mathcal{E}_1 = V_{R1}$$

$$\Rightarrow \mathcal{E}_1 = I R_1$$

$$\Rightarrow \mathcal{E}_1 = I \rho l_1 / A \text{ ----- (1)}$$

" l_1 " represents the length of the resistive wire between 'A' and 'B' points and " ρ " its density. After



that then we switch on the source " $\mathcal{E}_2$ " of unknown emf by two way switch. Then by the same procedure we will zero the galvanometer current by adjusting the jockey "K" on a certain point say "D" point on the resistive wire. Now at balance condition, we can say that the potential difference drop across the resistance " R_2 " of the wire between "A" and "D" points will be equal to the emf " $\mathcal{E}_2$ " of the source, if we ignore its internal resistance. i.e

$$V_{R2} = \mathcal{E}_2 \text{ or}$$

$$\Rightarrow \mathcal{E}_2 = V_{R2}$$

$$\Rightarrow \mathcal{E}_2 = I R_2$$

$$\Rightarrow \mathcal{E}_2 = I \rho l_2 / A \text{ ----- (2)}$$

" l_2 " represents the length of the wire between "A" and "D" points.

Now divide eq (2) by eq (1). We get as

$$\mathcal{E}_2 / \mathcal{E}_1 = (I \rho l_2 / A) / (I \rho l_1 / A)$$

$$\Rightarrow \mathcal{E}_2 / \mathcal{E}_1 = l_2 / l_1$$

$$\Rightarrow \mathcal{E}_2 = (l_2 / l_1) \times \mathcal{E}_1 \text{ ----- (3)}$$

From eq (3) we can easily find the unknown emf " $\mathcal{E}_2$ " of the Source.

Applications of potentiometer:

By comparison with the source of known emf following are the applications of potentiometer:

- ① Measurement of smaller emf (upto 2V) and larger emf (up to 250V).

- ② Comparison of emf of two cells.
- ③ Measurement of resistance
- ④ Measurement of current
- ⑤ Calibration of ammeter and calibration of voltmeter.

Examples and Assignments

Example 12.1: Each second 10^8 electrons flow from right to left across a cross-section of a wire attached to the two terminals of a battery. (a) Calculate the current in the wire (b) in which direction the current is flowing.

Given: $n = 10^8$ $t = 1\text{s}$ $e = 1.6 \times 10^{-19}\text{C}$

Required: a) $I = ?$ b) Direction of current = ?

Sol: As we know that

$$I = Q/t$$

$$\Rightarrow I = ne/t$$

$$\Rightarrow I = 10^8 \times 1.6 \times 10^{-19}/1\text{s}$$

$$\Rightarrow I = 0.16\text{A}$$

$$\Rightarrow I = 160\text{mA} \quad (\text{Ans})$$

b) The current is flowing from left to right i.e. in opposite direction of electrons flow.

Example 12.2: The high voltage in a TV receiver is 17KV. The maximum allowable current is $150\mu\text{A}$. What is the least permissible value of load resistance?

Given: $V = 17\text{KV} = 17 \times 10^3\text{V}$ $I = 150\mu\text{A} = 150 \times 10^{-6}\text{A}$

Req: Load Resistance = $R = ?$

Sol: As we know that from Ohm's law

$$V = IR$$

$$\Rightarrow R = V/I$$

$$\Rightarrow R = 17 \times 10^3\text{V} / 150 \times 10^{-6}\text{A}$$

$$\Rightarrow R = 113.3 \times 10^6\Omega \quad (\text{Ans})$$

Example: 12.3: A transmission line made of copper has a resistance of 100Ω at 0°C . Calculate the change in resistance between Summer and winter, knowing that temperature varies from $+35^\circ\text{C}$ to -30°C at 0°C . Assume the temperature coefficient of copper is $0.00427/^\circ\text{C}$ at 0°C .

Given: $R_o = 100\Omega$ $T_1 = 35^\circ\text{C}$ $T_2 = -30^\circ\text{C}$ $\alpha = 0.00427/^\circ\text{C}$

Required: Change in resistance = $\Delta R = ?$

Sol: As we know that

$$R_T = R_o (1 + \alpha T) \text{ ----- ①}$$

Now eq ① we can write as

$$R_{T1} = R_o (1 + \alpha T_1)$$

$$\Rightarrow R_{35^\circ C} = 100\Omega (1 + 0.00427 \times 35^\circ C)$$

$$\Rightarrow R_{35^\circ C} = 115\Omega$$

Now equation (1) we can write as

$$R_{T2} = R_o (1 + \alpha T_2)$$

$$\Rightarrow R_{-30^\circ C} = 100\Omega (1 + 0.00427 \times (-30^\circ C))$$

$$\Rightarrow R_{-30^\circ C} = 87.2\Omega$$

Now we find change in resistance " ΔR ". i.e

$$\Delta R = R_{35^\circ C} - R_{-30^\circ C}$$

$$\Rightarrow \Delta R = 115\Omega - 87.2\Omega$$

$$\Rightarrow \Delta R = 27.85\Omega$$

Assignment 12.1: (a) calculate the resistance per unit length of a 22- nichrome wire of radius 0.321 mm. (b) if a potential difference of 10V is maintained across 1m length of nichrome wire, what is the current in the wire. $\rho_{\text{nichrome}} = 1.5 \times 10^{-6} \Omega \cdot m$.

Given: $r = 0.321 \text{ mm} = 0.321 \times 10^{-3} m$, $V = 10V$, $L = 1m$, $\rho_{\text{nichrome}} = 1.5 \times 10^{-6} \Omega \cdot m$

Red: (a) $R/L = ?$ b) $I = ?$

Sol (a) As we know that

$$R = \rho L / A$$

$$\Rightarrow R/L = \rho / A \text{ ----- (1)}$$

Cross-sectional area of the wire = $A = \pi r^2$ ----- (2). Put (2) in eq(1) we get as

$$R/L = \rho / \pi r^2$$

$$\Rightarrow R/L = 1.5 \times 10^{-6} \Omega \cdot m / 3.14 \times (0.321 \times 10^{-3})^2$$

$$\Rightarrow R/L = 4.6 \Omega / m \text{ (Ans)}$$

Sol (b) As we know that

$$V = IR$$

$$\Rightarrow I = V/R \text{ ----- (3)}$$

Now we also know that

$$V = 10V \text{ ----- (4)}$$

We also know that

$$R/L = 4.6 \Omega / m$$

$$\Rightarrow R = L \times 4.6 \Omega / m$$

$$\Rightarrow R = 1m \times 4.6 \Omega / m$$

$$\Rightarrow R = 4.6 \Omega \text{ ----- (5)}$$

Now equation (3) we can write as

$$I = 10V / 4.6\Omega$$

$$\Rightarrow I = 2.2A \text{ Ans}$$

Example 12.4: Calculate the resistance of copper conductor having a length of 2km and a cross-section of 22 mm^2 . Assume the resistivity is $18 \times 10^{-9} \Omega \cdot m$.

Given: $L = 2 \text{ km} = 2 \times 10^3 m$ $A = 22 \text{ mm}^2$ $\rho = 18 \times 10^{-9} \Omega \cdot m$.

Req: Resistance of copper = $R = ?$

Sol: As we know that

$$R = \rho L/A$$

$$\Rightarrow R = 18 \times 10^{-9} \Omega \text{m} \times 2 \times 10^3 \text{m} / 22 \text{m}^2$$

$$\Rightarrow R = 1.64 \Omega \text{ Ans.}$$

Example 12.5: A heating coil has a resistance of 20Ω . It is designed to operate on 220V . What electric energy in Joules is supplied to the heater in 10s ?

Given: $R = 20 \Omega$ $V = 220 \text{V}$ $t = 10 \text{s}$

Req: Electric energy = $E = ?$

Sol: As we know that

$$P = E/t$$

$$\Rightarrow E = Pt \text{ ----- (1)}$$

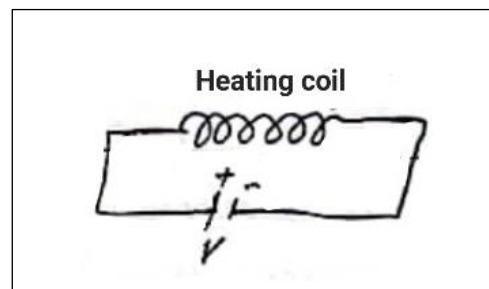
we also know that $P = V^2/R$

put eq (1) in eq (1). we get as

$$E = V^2 t / R$$

$$\Rightarrow E = (220 \text{V})^2 \times 10 \text{s} / 20 \Omega$$

$$\Rightarrow E = 24200 \text{J Ans.}$$



Example 12.6: The following are the details of load and a circuit connected through a supply meter.

- ① Six lamps of 40 watts each working for 4 hours per day.
- ② Two fluorescent tubes 125 watts each working for 2 hours per day.
- ③ 1000 watts heater working for 3 hours per day.

Find the cost of electrical energy consumed in 30 days for a month when the rate of electricity is 7Rs per unit.

Sol: Total watt for lamps = $40 \times 6 = 240 \text{ watts}$.

Total watt for tubes = $125 \times 2 = 250 \text{ watts}$

Heater = 1000 watts .

Energy consumed per day = $(240 \text{W} \times 4 \text{h}) + (250 \text{W} \times 2 \text{h}) + (1000 \text{W} \times 3 \text{h})$

Energy consumed per day = 4460 Wh

Energy consumed per day = 4.46 kWh

Now Total Energy consumed in the month of June = $4.46 \text{ kWh} \times 30$

Now Total Energy consumed in the month of June = 133.8 kWh

Now electric meter measures the electrical energy in kWh . Now when 1 kWh electrical energy is passed from the meter, then it changes the unit by one value. So now we can write as

Total energy consumed in the month of June = $133.8 \times 1 \text{ unit}$

Total energy consumed in the month of June = $133.8 \times 7 \text{ Rs}$

Total energy consumed in the month of June = 936.6 Rs

Example 12.7: Figure shows three resistors and two voltage Sources. Use Kirchhoff's voltage law to find the emf $\mathcal{E}_{ab}$.

Given: $\mathcal{E}_1 = 24 \text{V}$, $R_1 = 2 \text{k}\Omega$ $R_2 = 1 \text{k}\Omega$, $R_3 = 3 \text{k}\Omega$. Req: $\text{emf} = \mathcal{E}_{ab} = ??$

Sol: For the emf $\mathcal{E}_{ab}$ we will use Kirchhoff's law on the loop ABCDA.

So in this loop we take a start from "D" point and we will follow the convention direction. So we can write as

$$\mathcal{E}_1 - IR_1 - \mathcal{E}_{ab} = 0$$

$$\Rightarrow \mathcal{E}_{ab} = IR_1 - \mathcal{E}_1 \text{----- (1)}$$

Now we find the circuit current "I". For this we will

Use Ohm's law. i.e

$$V = IR_T$$

$$\Rightarrow I = V/R_T$$

$$\Rightarrow I = V/(R_1 + R_2 + R_3)$$

$$\Rightarrow I = 24V/(2k\Omega + 1k\Omega + 3k\Omega)$$

$$\Rightarrow I = 24V/6k\Omega$$

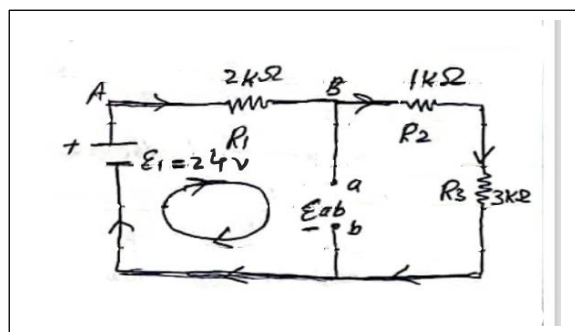
$$\Rightarrow I = 4mA$$

Now eq (1) we can write as

$$\mathcal{E}_{ab} = 24V - 4 \times 10^{-3} A \times 2 \times 10^3 \Omega$$

$$\Rightarrow \mathcal{E}_{ab} = 24V - 8V$$

$$\Rightarrow \mathcal{E}_{ab} = 16V \text{ Ans.}$$



Assignment 12.2: A heating coil is rated 100W, 220V. The coil is cut in half and two pieces are joined in parallel to the same source. Now what is the energy liberated or lost per second?

Given: $P = 100W$ $V = 220V$ Req: Energy liberated per second = $E/t = ?$

Sol: When we cut the coil in half, then its resistance will become also half for each piece as shown in figure. Now we find the equivalent resistance of parallel resistors. i.e

$$1/R_{eq} = 1/R + 1/R$$

$$\Rightarrow 1/R_{eq} = 2/R + 2/R$$

$$\Rightarrow 1/R_{eq} = 4/R$$

$$\Rightarrow R_{eq} = R/4 \text{----- (1)}$$

Now we find the equivalent power of these two resistors. i.e

$$P_{eq} = V^2/R_{eq}$$

$$\Rightarrow P_{eq} = V^2/(R/4)$$

$$\Rightarrow P_{eq} = 4V^2/R \text{----- (2)}$$

Now we can write as

$$P = V^2/R \quad \text{It's the power of heating coil before cutting.}$$

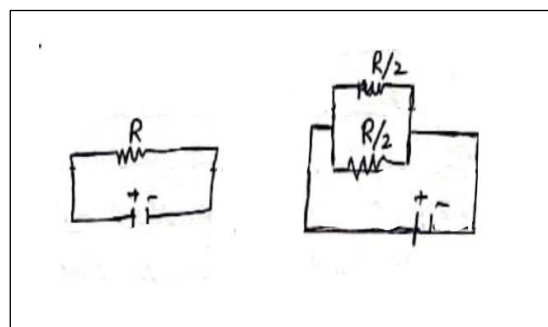
$$\Rightarrow R = V^2/P \text{----- (3)}$$

Now put eq (3) in eq (2). We get as

$$P_{eq} = 4V^2 \times P/V^2$$

$$\Rightarrow P_{eq} = 4P$$

$$\Rightarrow P_{eq} = 4 \times 100W \Rightarrow P_{eq} = 400 \text{ watt} \Rightarrow E/t = 400J/s \text{ Ans.}$$



Assignment 12.3: Circuit ABCDEF is shown in figure. Calculate the current in each branch of the circuit using Kirchhoff's voltage law.

Sol: Let we divide the given circuit into two loops i.e loop 'ABEFA' and loop 'BCDEB'

Now first we apply Kirchhoff's voltage law on loop "ABEFA" we get as

$$\begin{aligned}\varepsilon - I_1 R_1 - (I_1 - I_2) R_2 &= 0 \\ \Rightarrow 4 - 5I_1 - 20(I_1 - I_2) &= 0 \\ \Rightarrow 4 - 5I_1 - 20I_1 + 20I_2 &= 0 \\ \Rightarrow 4 - 25I_1 + 20I_2 &= 0 \quad \text{--- (1)}\end{aligned}$$

Now Applying KVL on loop 'BCDEB' we get

$$\begin{aligned}(I_1 - I_2) R_2 - I_2 R_3 - I_2 R_4 &= 0 \\ \Rightarrow 20(I_1 - I_2) - 10I_2 - 30I_2 &= 0 \\ \Rightarrow 20I_1 - 20I_2 - 10I_2 - 30I_2 &= 0 \\ \Rightarrow 20I_1 - 60I_2 &= 0 \\ \Rightarrow 20I_1 &= 60I_2 \\ \Rightarrow I_1 &= (60/20)I_2 \\ \Rightarrow I_1 &= 3I_2 \quad \text{--- (2)}\end{aligned}$$

Putting eq (2) in eq (1) we get

$$\begin{aligned}\Rightarrow 4 - 25 \times 3I_2 + 20I_2 &= 0 \\ \Rightarrow 4 - 75I_2 + 20I_2 &= 0 \\ \Rightarrow 4 - 55I_2 &= 0 \\ \Rightarrow 55I_2 &= 4 \\ \Rightarrow I_2 &= 4/55 \\ \Rightarrow I_2 &= 0.0727\text{A}\end{aligned}$$

Now putting the value of I_2 in eq (2)

$$\begin{aligned}\Rightarrow I_1 &= 3 \times 0.0727\text{A} \\ \Rightarrow I_1 &= 0.2181\text{A}\end{aligned}$$

a) Current through R_1

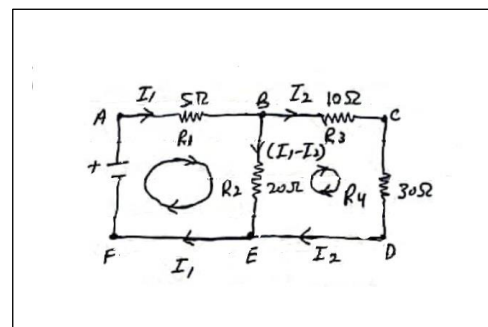
$$\Rightarrow I_1 = 0.2181\text{A}$$

b) Current through R_3 and R_4

$$\Rightarrow I_2 = 0.0727\text{A}$$

c) Current through R_2

$$\begin{aligned}\Rightarrow (I_1 - I_2) &= 0.2181 - 0.0727 \\ \Rightarrow (I_1 - I_2) &= 0.1454\text{A}\end{aligned}$$



Example 12.8: Using a Weston cadmium cell of 1.01834V and a standard resistance of 0.15Ω a potentiometer was adjusted so that 1.0183m was equivalent to the emf of the cell. When a certain direct current was flowing through the standard resistance, the voltage across it corresponds to 150cm. What was the value of current?

Given: $\varepsilon_1 = 1.0183\text{V}$ $l_1 = 1.0183\text{m}$ $l_2 = 150\text{cm} = 1.5\text{m}$ $R = 0.15\Omega$

Required: $I_2 = ?$

Sol: For " I_2 " we can write as

$$\Rightarrow I_2 = \varepsilon_2 / R \text{ ----- (1)}$$

$$\Rightarrow \varepsilon_2 / \varepsilon_1 = I_2 / I_1$$

$$\Rightarrow \varepsilon_2 = (I_2 / I_1) \times \varepsilon_1$$

$$\Rightarrow \varepsilon_2 = (1.5 / 1.0183) \times 1.0183V$$

$$\Rightarrow \varepsilon_2 = 1.5V \text{ ----- (2)}$$

Put (2) in (1) we get as

$$\Rightarrow I_2 = 1.5V / 0.15\Omega$$

$$\Rightarrow I_2 = 10A$$

Assignment 12.4: A potentiometer wire of length 1m is connected to a driven cell of emf 3V. When a cell of 1.5V emf is in secondary circuit the balance point is found to be at 60cm. On replacing this cell and using a cell of unknown emf the balance point shifts to 80cm. Calculate the unknown emf of the cell.

Given: $\varepsilon_1 = 1.5V$. $I_1 = 60cm$. $I_2 = 80cm$. Required: $\varepsilon_2 = ?$

Sol: As we know that

$$\Rightarrow \varepsilon_2 / \varepsilon_1 = I_2 / I_1$$

$$\Rightarrow \varepsilon_2 = (I_2 / I_1) \times \varepsilon_1$$

$$\Rightarrow \varepsilon_2 = (80 / 60) \times 1.5V$$

$$\Rightarrow \varepsilon_2 = 2V$$

Exercise MCQs

① Three resistors of resistance “R” each are combined in various ways.

Which of the following cannot be obtained?

(a) $3R\Omega$

(b) $R/4\Omega$ ✓

(c) $R/3\Omega$

(d) $2R/3\Omega$

Sol: Now we can write as

In Series: $R_e = R + R + R = 3R\Omega$

In Parallel: $1/R_e = 1/R + 1/R + 1/R \Rightarrow R_e = R/3$

Two parallel + one series: $R_e = R/2 + R = 3R/2$

Two series + one parallel: $1/R_e = 1/2R + 1/R \Rightarrow R_e = 2R/3$

$\Rightarrow 1/R_e = 1/2R + 1/R \Rightarrow R_e = 2R/3$

Possible: $3R, R/3, 3R/2, 2R/3 \Rightarrow$ Not possible: $R/4$

② When “n” resistances each of value “r” are connected in parallel then resultant resistance is “x”. When all are connected in series, the total resistance is?

(a) nx

(b) rx

(c) x/n

(d) n^2x ✓

Sol: We can write as

In Parallel: $1/R_e = 1/r + 1/r + \dots n \text{ times}$

$$\Rightarrow 1/R_e = n/r \Rightarrow R_e = r/n \Rightarrow x = r/n \Rightarrow r = xn \text{ ----- (1)}$$

In Series: $R_e = r + r + \dots n \text{ times} \Rightarrow R_e = nr \text{ ----- (2)}$

Now put (1) and (2) we get as

$$R_e = nxn \Rightarrow R_e = n^2x$$

- ③ Two bulbs marked 200W 250V and 100W 250V are joined in series to a 250V supply. Power consumed in the circuit is?

(a) 33W (b) 67W ✓ (c) 100W (d) 300W

Sol: For this we can write as

$$R_1 = V^2/P_1 = 250^2/200 = 312.5\Omega \text{ and}$$

$$R_2 = V^2/P_2 = 250^2/100 = 625\Omega$$

For Series we can write as

$$R_e = R_1 + R_2 = 312.5 + 625 = 937.5\Omega$$

Now for Power we can write as

$$P = V^2/R_e = 250^2/937.5 \approx 66.66W \approx 67W$$

- ④ Resistance of a wire is “r” ohms . Wire is stretched to double its length, then its resistance in ohms is?

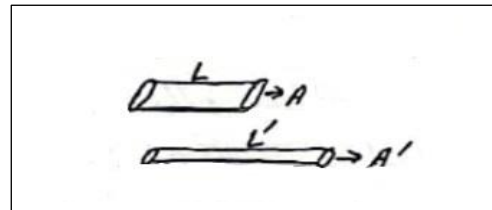
(a) $r/2$ (b) $4r$ ✓ (c) $2r$ (d) $r/4$

Sol: We can write as

$$r = \rho L/A \text{ ----- (1)}$$

It is given that

$$L' = 2L$$



Now volume will be constant. So we can write as

$$A'L' = AL \Rightarrow A' = A/2$$

Now for new resistance we can write as

$$R' = \rho L'/A' = \rho(2L)/(A/2) = 4\rho L/A = 4r$$

- ⑤ 10^6 electrons are moving through a wire per second. Current developed is?

(a) $1.6 \times 10^{-19}A$ (b) $2A$ (c) $1.6 \times 10^{-13}A$ ✓ (d) 10^6A

Sol: We can write as

$$I = ne/t = 10^6 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-13}A$$

- ⑥ When a wire is stretched so radius becomes $r/2$, then its resistance will be ?

(a) $16R$ ✓ (b) $4R$ (c) $2R$ (d) 0

Sol: We can write as

$$R = \rho L/A \text{ and}$$

$$A = \pi r^2$$

Now we can write as

$$A' = \pi r'^2 = \pi(r/2)^2 = A/4$$

Now Volume will be constant. So we can write as

$$A' L' = AL$$

$$\Rightarrow L' = AL/A' \Rightarrow L' = AL/(A/4) \Rightarrow L' = 4L$$

Now we can write as

$$\Rightarrow R' = \rho L'/A' = \rho(4L)/(A/4) = 16\rho L/A = 16R$$

- ⑦ Voltage required to make $2A$ flow through 8Ω ?

(a) $16V$ ✓ (b) $8V$ (c) $4V$ (d) $2V$

Sol: $\Rightarrow V = IR = 2 \times 8 = 16V$

- ⑧ Wire of uniform cross-section “A”, length “L”, and resistance “R” is cut into two equal parts. The resistivity of each part

- (a) Doubled (b) Halved (c) Same ✓ (d) One fourth

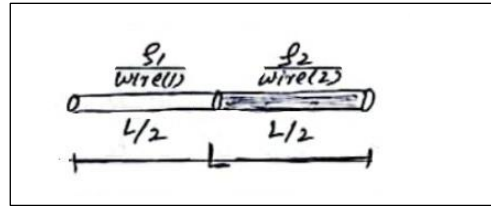
Sol: Now we can write as

$$\rho = RA/L \quad \text{After cutting: } R' = R/2, L' = L/2$$

$$\Rightarrow \rho' = R'A/L' = (R/2)A/(L/2) = RA/L = \rho$$

$$\Rightarrow \rho' = \rho. \text{ (same)}$$

- ⑨ The resistivity of two wires is ρ_1 and ρ_2 which are connected in series? If their dimensions are the same then the equivalent resistivity of the combination will be



- (a) $\rho_1 + \rho_2$ (b) $1/\rho_1 + 1/\rho_2$ (c) $(\rho_1 + \rho_2)/2$ ✓ (d) ρ_1/ρ_2

Sol: We can write as

$$\text{Series resistance: } R_e = R_1 + R_2 = \rho_1 L/2A + \rho_2 L/2A = (\rho_1 + \rho_2)L/2A$$

$$\text{So equivalent resistivity} = \rho_e = (\rho_1 + \rho_2)/2$$

- ⑩ Powers of two electric bulbs are 100W and 200W which are connected to power supply of 220V. The ratio of the resistance of their

- (a) 1:2 (b) 2:1 ✓ (c) 1:3 (d) 4:3

Sol: For this we can write as

$$R_1 = V^2 / P_1 \text{ and}$$

$$R_2 = V^2 / P_2$$

Now we can write as

$$R_1/R_2 = P_2/P_1 = 200/100 = 2:1$$

- ⑪ Thermocouple is an arrangement of two different metals.....?

- (a) To convert heat energy into electrical energy ✓
 (b) To produce more heat
 (c) To convert heat energy into chemical energy
 (d) To convert electrical energy into heat energy

- ⑫ In order to achieve high accuracy, slide wire of potentiometer should be:

- (a) As long as possible ✓ (b) As short as possible
 (c) Very thin (d) Very thick

Ans: Longer wire gives more resolution and accuracy

- ⑬ When a potentiometer is used for measurement of voltage of unknown source, the power consumed in the circuit of the unknown source under null condition?

- (a) High (b) Small (c) Ideally zero ✓ (d) Very high

Ans: No current flows at null condition. So Power ideally zero.

- ⑭ In a Wheatstone bridge method, the bridge is said to be balanced when galvanometer current is:

- (a) 1A (b) 0A ✓ (c) Maximum (d) Half

Sol: $\Rightarrow$ Balanced bridge $\Rightarrow$ Current through galvanometer = 0

Conceptual Questions

Question 01: A heavy duty battery of a truck maintaining a current of 3A for 24 hours. How much charge flows from the battery.

Given: Current = $I = 3A$ Time = $t = 24h = 24 \times 3600s = 86400s$

Required: Flowing charge = $Q = ?$

Sol. As we know that

$$I = Q/t$$

$$\Rightarrow Q = It \text{ ----- (1)}$$

putting the given values in equation (1) we get as

$$Q = 3A \times 86400s \Rightarrow Q = 259200 \Rightarrow Q = 2.592 \times 10^5 C$$

Thus $2.592 \times 10^5 C$ charges will flow from the battery during this time.

Question 02: While analyzing circuit the internal resistance of emf sources are ignored why?

Ans: The internal resistance of emf sources is very small. So we generally ignore it while analyzing a circuit.

Explanation: In the normal conditions, the load resistor “R” connected across on emf source are very large in magnitude whereas the resistance (r) of the emf source is very small and consequently the potential drop (Ir) across it also very small.

Therefore, we usually ignore this small resistance.

Example: consider a circuit as shown in fig.

Now in which we want to find current “I”.

So for this we can write as

$$\mathcal{E} = V_r + V_R$$

$$\Rightarrow \mathcal{E} = Ir + IR$$

$$\Rightarrow \mathcal{E} = I(r + R)$$

$$\Rightarrow I = \mathcal{E} / (r + R)$$

$$\Rightarrow I = \mathcal{E} / (R + r) \text{ ----- (1)}$$

Now putting the values in equation (1) then we get as

$$I = 10V / (0.001\Omega + 10\Omega)$$

$$\Rightarrow I = 0.99A \text{ ----- (2)}$$

Now if we ignore the internal resistance of the source.

Then eq (1) we can write as

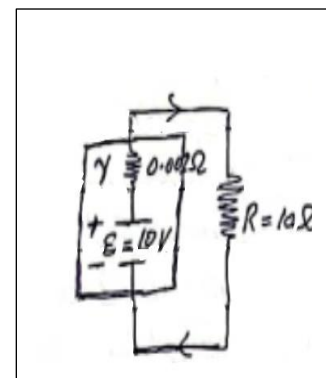
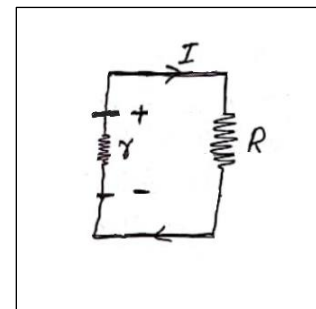
$$I = \mathcal{E} / R$$

$$\Rightarrow I = 10V / 10\Omega$$

$$\Rightarrow I = 1A \text{ ----- (3)}$$

Now from eq (2) and (3) we can say that the current

change is very small when we ignore the internal resistance. So now we can say that the internal resistance of a source can be ignored while analyzing the circuit.



Question 03: If aluminum and copper wires of the same length have the same resistance, which has the larger diameter? Why?

Ans: For same length and same resistance, the diameter of aluminum will be greater than that of copper.

Explanation: Now it is given that

$$L_A = L_C \text{ ----- } \textcircled{1} \quad \text{And}$$

$$R_A = R_C$$

$$\Rightarrow \rho_A L_A / A_A = \rho_C L_C / A_C \quad \text{Put eq } \textcircled{1} \text{ we get as}$$

$$\Rightarrow \rho_A L_C / A_A = \rho_C L_C / A_C$$

$$\Rightarrow \rho_A / A_A = \rho_C / A_C \text{ ----- } \textcircled{2}$$

Now generally we can write as

$$A = \pi r^2 \Rightarrow A = \pi (d/2)^2 \Rightarrow A = \pi d^2 / 4 \text{ ----- } \textcircled{3}$$

Now eq $\textcircled{2}$ we can write as

$$\Rightarrow \rho_A / \pi d_A^2 / 4 = \rho_C / \pi d_C^2 / 4$$

$$\Rightarrow \rho_A / d_A^2 = \rho_C / d_C^2$$

$$\Rightarrow \sqrt{d_A^2 / d_C^2} = \sqrt{\rho_A / \rho_C}$$

$$\Rightarrow d_A / d_C = \sqrt{\rho_A / \rho_C} \text{ ----- } \textcircled{4}$$

Now we know that

$$\rho_A = 0.0285 \times 10^6 \Omega \text{m} \quad \text{and} \quad \rho_C = 0.0178 \times 10^6 \Omega \text{m}$$

Now eq $\textcircled{4}$ we can write as

$$\Rightarrow d_A / d_C = \sqrt{0.0285 \times 10^6 \Omega \text{m} / 0.0178 \times 10^6 \Omega \text{m}}$$

$$\Rightarrow d_A / d_C = \sqrt{1.61}$$

$$\Rightarrow d_A = 1.27 d_C \text{ ----- } \textcircled{5}$$

Now from eq $\textcircled{5}$ we can say that the diameter of Aluminum is 1.27 times greater than the diameter of copper.

Question 04: Under what circumstances can terminal P.D of a battery exceed its emf?

Ans: The terminal potential difference of a battery exceeds its emf when the battery is being charged through a battery charger.

Explanation: As we know that the relation between terminal P.D " V_t " and emf of a battery is given by

$$V_t = \mathcal{E} - Ir \text{ ----- } \textcircled{1}$$

Now when a battery is being charged through a charger, then the current reverses its direction. So we replace " I " by " $-I$ " in equation $\textcircled{1}$ we get as

$$V_t = \mathcal{E} - (-I) r$$

$$\Rightarrow V_t = \mathcal{E} + Ir$$

$$\Rightarrow V_t > \mathcal{E}$$

Thus the terminal P.D will exceed the battery emf during charging of a battery.

Question 05: What is the difference between an emf and a P.D?

Electromotive force (emf)

$\textcircled{1}$ Emf of the cell is potential difference between terminals of a cell (or emf

source) when no current flows through it.

- ② The emf is regarded as "cause" because it causes the flow of current due to which P.D is developed across any two points in the circuit.
- ③ The emf of a battery is generated by the chemical actions occurs in the battery.
- ④ The emf of the cell is usually greater than the terminal P.D.
 $\epsilon = V_t + Ir$
 $\Rightarrow \epsilon > V_t$
- ⑤ if emf " ϵ " and current " I " are known. Then we can measure the total electric power of the source across the resistor.

Potential difference

- ① When circuit is closed and current " I " starts flowing through the load " R ", then the product of current and resistance " R " is called as terminal P.D " V_t ".
- ② The P.D is regarded as an "effect" because it is develops due to flow of current causes by the emf.
- ③ The P.D is generated due to an emf of a battery.
- ④ Usually terminal P.D " V_t " is lesser than emf of the cell due to the potential drop across the internal resistor (r) of the emf source.
 $V_t = \epsilon - Ir$
 $\Rightarrow V_t < \epsilon$
- ⑤ if P.D and current are known. Then we can measure power dissipated across this load resistor.

Question 06: How loop rule and Junction rule are based on conservation of energy and charge respectively?

Ans. The loop rule is based on the conservation of energy principle and Junction rule is based on the conservation of charge principle.

① Kirchhoff's loop rule or Kirchhoff's voltage law:

This law states that "the sum of all the emf's in a circuit is equal to the sum of all the voltage drops in that closed circuit or Loop".

$$\text{i.e } \Sigma \epsilon = \Sigma V_R$$

$$\Rightarrow \Sigma W_s / Q = \Sigma W_R / Q$$

$$\Rightarrow \Sigma E_s / Q = \Sigma E_R / Q \text{ ----- ①}$$

Eq ① represents the law of conservation of energy from which we can say that for the charges " Q " the energy delivered by the source is equal to the energy drop or lost by the resistors. Thus we can say that the loop rule or (KVL) is based on the conservation of energy principle.

② Kirchhoff's Junction rule or Kirchhoff's current law:

This law states that "the algebraic sum of all currents entering to a junction is equal to the algebraic sum of all currents leaving the junction". i.e

$$\Sigma I_{\text{entering}} = \Sigma I_{\text{leaving}}$$

$$\Rightarrow \Sigma Q_{\text{entering}}/t = \Sigma Q_{\text{leaving}}/t$$

Eq ② represents the law of conservation of charge, from which we can say that for the given time “t” the entering charges to the junction is equal to the leaving charges from the junction. Thus we can say that the Junction rule or (KCL) is based on the conservation of charge principle.

Question 07: Why rise in temperature of conductor is accompanied by rise in resistance?

Ans: When temperature of a conductor increases, its resistance also increases.

Explanation: Opposition offered by the atoms of the conductor is known as resistance “R”. The resistance of the conductor depends upon the temperature i.e if the temperature of the conductor increases, then its resistance also increases.

It is due to a fact that when temperature increases, the K.E of atoms of the conductor also increases. Due to increase in K.E of the atoms, its amplitude of vibration also increases. Now due to increase in amplitude of vibration, these atoms offer greater resistance to the flow of electrons passing through the conductor.

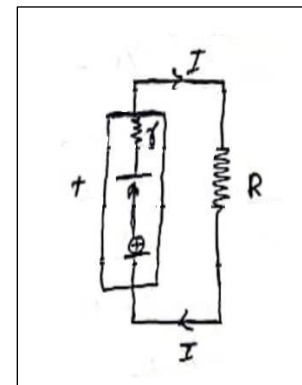
Thus the resistance of the conductor increases with increase in temperature.

Question 08: Does the direction of emf provided by a battery depends on the direction of current flow through the battery?

Ans: The direction of emf provided by a battery does not depend on the direction of current flowing through the battery.

Explanation: The emf of a battery is define as “The work done on the charges from low potential to a higher potential is called the emf of a battery.”

Now due to such an emf of a battery a current “I” is producing in the circuit. Such a current “I” is flowing from low potential to a higher potential inside the battery and is flowing from higher potential to a lower potential in the external part of the circuit. Now such a produced current “I” and its direction flowing through the battery or flowing in the external part of the circuit is depends on the direction of emf provided by a battery. So now we can say that the direction of emf provided by a battery does not depend on the direction of current flowing through the battery.



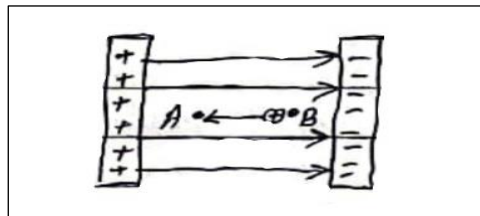
Question 09: Voltages are always measured between two points. Why?

Ans: Voltages of potential differences are always measured between two points.

Explanation: The voltage or potential difference is define as “The difference in electric potentials between the two points inside the electric field is called voltage or potential difference”.

i.e $\Delta V = V_A - V_B$ ----- ①

Now from eq ① we can say that the voltage or P.D can be calculated from the two different electric potential points, one is point “A” (higher electric potential point) and the second one is point “B” (lower Electric potential point). So now we can say that voltages or potential differences are always measured from two point.



Second reason: The voltage or potential difference is also define as “The work done on a unit positive test charge between the two points inside the electric field is also called the voltage or “P.D”.

i.e $\Delta V = W_{BA}/q$ ----- ②

Now from eq ② we can say that the voltage or potential difference can be calculated from the work done on a test charge between the two points, one is point ‘A’(higher electric potential point) and the second one is point “B” (lower electric potential point). So now we can say that voltages of Potential differences are always measured from two points.

Question 10: Is every emf a potential difference? Is every potential difference an emf?

Ans: In a figure the battery emf will be equal to the sum of voltages drop across resistance “r” and external resistance “R”.

$$\varepsilon = V_r + V_R$$

$$\Rightarrow \varepsilon = Ir + IR$$

$$\Rightarrow \varepsilon = Ir + V_t$$

$$\Rightarrow V_t = \varepsilon - Ir$$
 ----- ①

Now we explain cases.

① If the circuit is open. i.e if $I = 0$ in the external circuit.

Then eq① we can

write as

$$V_t = \varepsilon - 0 \times r$$

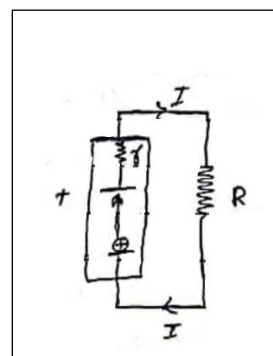
$$\Rightarrow V_t = \varepsilon$$
 ----- ② or $\varepsilon = V_t$ ----- ③

In this case from eq ③ and ② we can say that emf is equal to P.D or P.D is equal to emf.

② If current is not zero in the external circuit mean if the circuit is close, i.e. if $I \neq 0$ in the external circuit. Then we can write as

$$V_t = \varepsilon - Ir$$

$$\Rightarrow V_t < \varepsilon$$
 ----- ② or $\varepsilon > V_t$ ----- ③



Now from eq (3) and (2) we can say that the emf is greater than the P.D Or P.D is less than emf in this case.

(3) As we know that the reverse direction current when flows from a battery then the battery will get be charging. So for the battery charging we take $I = -I$. Now eq (1) we can write as

$$V_t = \varepsilon - (-I)r$$

$$\Rightarrow V_t = \varepsilon + Ir$$

$$\Rightarrow V_t > \varepsilon \text{ ----- (2) or } \varepsilon < V_t \text{ ----- (3)}$$

Now from (3) and (2) we can say that the emf is less than the P.D or P.D is greater than the emf. Now from these cases we can say that not every emf will be a P.D or every P.D not will be an emf except case first.

Question 11: How much charge flow in a pocket calculator each minute when the current is 0.0001A?

Given: Time = $t = 1$ minute $\Rightarrow t = 60s$, Current = $I = 0.0001$ A.

Required: Charges = $Q = ??$

Sol: As we know that

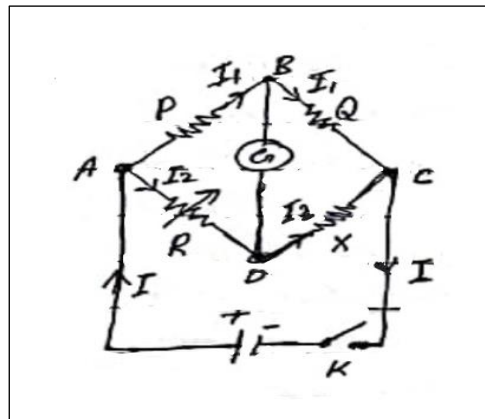
$$I = Q/t$$

$$\Rightarrow Q = It$$

$$\Rightarrow Q = 0.0001A \times 60s$$

$$\Rightarrow Q = 0.006C \text{ Ans}$$

Question 12: When the Wheatstone bridge is balanced, then no current flows through the galvanometer. Why?



Ans: When Wheatstone bridge is balanced, then no current flows through the galvanometer.

Explanation: The circuit arrangement of a Wheatstone bridge is shown in a figure. Now for a balance Wheatstone bridge, the variable resistance "R" is changed in such a way that the point "B" and "D" across the galvanometer have the same electric potentials.

Now due to a same electric potentials, the potential difference between these points will be equal to zero, and due to a zero potential difference between these points we can say that no current will flow through the galvanometer.

Numerical problems.

Q.01 A battery has an emf of 12.8V and supplies current of 3.2A. What is resistance of a circuit? How many coulombs do battery in 5 minute?

Given: $\mathcal{E} = 12.8\text{V}$, $I = 3.2\text{A}$ Req: a) $R=?$ b) $Q=?$

Sol (a): As we know that

$$V = IR$$

$$\Rightarrow \mathcal{E} = IR$$

$$\Rightarrow R = \mathcal{E} / I \Rightarrow R = 12.8\text{V}/3.2\text{A} \Rightarrow R = 4\Omega . \text{ Ans}$$

Sol (b): As we know that

$$I = Q/t$$

$$\Rightarrow Q = It$$

$$\Rightarrow Q = 3.2\text{A} \times 5 \times 60 \text{ s} \Rightarrow Q = 960\text{C} . \text{ Ans}$$

Q.02 Two light bulbs designed for 120V use are rated at 40W and 60W. Which bulb has greater filament resistance and why?

Given: $V = 120\text{V}$, $P_1 = 40\text{W}$, $P_2 = 60\text{W}$

Req: a) $R_1 = ?$ b) $R_2 = ?$

Sol (a): As we know that

$$P = V^2/R$$

$$\Rightarrow R = V^2/P$$

Now we can write as

$$R_1 = V^2 / P_1 \Rightarrow R_1 = (120\text{V})^2/40\text{W} \Rightarrow R_1 = 360\Omega . \text{ Ans}$$

Sol (b): Now we can write as

$$R_2 = V^2 / P_2 \Rightarrow R_2 = (120\text{V})^2/60\text{W} \Rightarrow R_2 = 240\Omega . \text{ Ans}$$

Q.03: A carbon electrode has a resistance of 0.125Ω at 20°C . The temperature co-efficient of carbon is $-0.0005/^\circ\text{C}$ at 20°C . What will be the resistance of the electrode at 85°C ?

Given: $R_0 = 0.125\Omega$, $T_1 = 20^\circ\text{C}$, $T_2 = 85^\circ\text{C}$, $\alpha = -0.0005/^\circ\text{C}$.

Req: $R_{85^\circ\text{C}} = ?$

Sol: As we know that

$$R_T = R_0 (1 + \alpha T)$$

$$\Rightarrow R_T = R_0 (1 + \alpha(T_2 - T_1))$$

$$\Rightarrow R_{85^\circ} = 0.125(1 + 0.0005(85^\circ - 20^\circ))$$

$$\Rightarrow R_{85^\circ} = 0.121\Omega \text{ Ans}$$

Q.04: Calculate the resistance of wire 10m long that has a diameter of 2mm and resistivity of $2.63 \times 10^{-2}\Omega\text{m}$.

Given: $L = 10\text{m}$, $d = 2\text{mm} \Rightarrow r = d/2 = 1\text{mm} \Rightarrow r = 10^{-3}\text{m}$, $\rho = 2.63 \times 10^{-2}\Omega\text{m}$.

Req: $R = ??$

Sol: As we know that

$$R = \rho L/A$$

$$\Rightarrow R = \rho L / \pi r^2$$

$$\Rightarrow R = 2.63 \times 10^{-2} \Omega \text{m} \times 10 \text{m} / 3.14 \times (10^{-3} \text{m})^2$$

$$\Rightarrow R = 0.083758 \Omega \text{ Ans}$$

Q.05 : A typical 12V automobile battery has a resistance of 0.012Ω.

What is terminal voltage of this battery when starter draws a current of 100A? Calculate “R”, “P_ε”, P_r and P_R.

Given: ε = 12V, r = 0.012Ω, I = 100A.

Req: (a) V_t = ? b) R = ? c) Power of source = P_ε = ?

d) Power lost across R = P_R = ? e) P_r = ?

Sol: (a) As we know that

$$\epsilon = Ir + V_t$$

$$\Rightarrow V_t = \epsilon - Ir$$

$$\Rightarrow V_t = 12\text{V} - 100\text{A} \times 0.012\Omega \Rightarrow V_t = 10.8\text{V Ans.}$$

(b) Now we know that

$$V_t = IR$$

$$\Rightarrow R = V_t / I$$

$$\Rightarrow R = 10.8\text{V} / 100\text{A} \Rightarrow R = 0.108\Omega \text{ Ans.}$$

Sol (c): As we know that P = IV

$$\text{Now } P_\epsilon = I \epsilon \Rightarrow P_\epsilon = 100\text{A} \times 12\text{V} \Rightarrow P_\epsilon = 1200\text{W} . \text{ Ans}$$

Sol (d): As we know that

$$P = I^2 R$$

$$\Rightarrow P_R = I^2 R \Rightarrow P_R = (100\text{A})^2 \times 0.108 \Omega \Rightarrow P_R = 1080\text{W Ans}$$

sol (e): Now we can write as

$$P_r = I^2 r$$

$$\Rightarrow P_r = (100\text{A})^2 \times 0.012 \Omega \Rightarrow P_r = 120\text{W Ans.}$$

Q-06. A 120 Ω resistor has a value of 120 Ω. What is the rated current through the resistor?

Given: P = 10W, R = 120 Ω. Req: I = ?

Sol: As we know that

$$P = I^2 R$$

$$\Rightarrow I^2 = P / R$$

$$\Rightarrow I = \sqrt{P / R} \Rightarrow I = \sqrt{(10\text{W} / 120 \Omega)} \Rightarrow I = 0.2886\text{A Ans}$$

Q-07: Resistor of 50 Ω has a P.D of 100V DC across 1h. Calculate

a) Power b) Energy.

Given: R = 50 Ω, V = 100V, t = 1h = 3600s. Req: a) P = ? b) E = ?

Sol (a) As we know that

$$P = V^2 / R \Rightarrow P = (100\text{V})^2 / 50 \Omega \Rightarrow P = 200\text{W Ans}$$

sol (b) As we know that

$$P = E / t$$

$$\Rightarrow E = Pt$$

$$\Rightarrow E = 200\text{W} \times 3600\text{s} \Rightarrow E = 0.72 \times 10^6 \text{ J} \Rightarrow E = 0.72 \text{ MJ Ans}$$

Q-08: Calculate the current through a single loop circuit if $\epsilon = 120\text{V}$
 $R = 100\ \Omega$ and internal resistance $r = 0.01\ \Omega$.

Given: $\epsilon = 120\text{V}$, $R = 100\ \Omega$, $r = 0.01\ \Omega$ Req: $I = ?$

Sol: As we know that

$$\epsilon = Ir + V_t$$

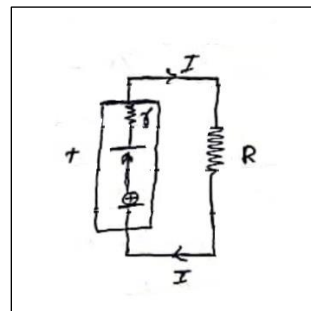
$$\Rightarrow \epsilon = Ir + IR$$

$$\Rightarrow \epsilon = I(r + R)$$

$$\Rightarrow I = \epsilon / (r + R)$$

$$\Rightarrow I = 120\text{V} / (0.01\Omega + 100\Omega)$$

$$\Rightarrow I = 0.1199\text{A} \text{ or } I = 120\text{mA. Ans}$$



Q-09: Find terminal P.D of each cell in the circuit of fig given.

Given: $r_1 = 0.15\Omega$, $r_2 = 0.9\Omega$, $R = 85\Omega$, $\epsilon_1 = 24\text{V}$, $\epsilon_2 = 6\text{V}$

Reap: a) $V_{t1} = ?$ b) $V_{t2} = ?$

Sol. As we know that

$$\epsilon = Ir + V_t$$

$$\text{Now for 1}^{\text{st}} \text{ cell: } V_{t1} = \epsilon_1 - Ir_1 \text{ ----- ①}$$

$$\text{For 2}^{\text{nd}} \text{ cell: } V_{t2} = \epsilon_2 - Ir_2 \text{ ----- ②}$$

Now we find the current "I" from Kirchhoff's voltage laws. i.e

$$\Rightarrow \epsilon_1 - Ir_1 - IR - Ir_2 - \epsilon_2 = 0$$

$$\Rightarrow \epsilon_1 - \epsilon_2 = Ir_1 + IR + Ir_2$$

$$\Rightarrow \epsilon_1 - \epsilon_2 = I(r_1 + R + r_2)$$

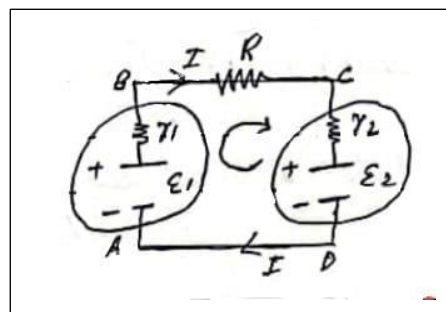
$$\Rightarrow I = (\epsilon_1 - \epsilon_2) / (r_1 + R + r_2)$$

$$\Rightarrow I = 24\text{V} - 6\text{V} / (0.1\Omega + 85\Omega + 0.9\Omega)$$

$$\Rightarrow I = 2\text{A} \text{ ----- ③}$$

$$\text{Now eq ① } V_{t1} = 24\text{V} - 2\text{A} \times 0.15\Omega \Rightarrow V_{t1} = 23.8\text{V. Ans}$$

$$\text{Now eq ② } V_{t2} = 6\text{V} - 2\text{A} \times 0.9\Omega \Rightarrow V_{t2} = 7.8\text{V. Ans}$$



Q-10: Voltmeter in circuit may be considered to be ideal values are $\epsilon = 15\text{V}$, internal resistance $r = 5\Omega$, $R_1 = 100\Omega$, $R_2 = 300\Omega$. Calculate current in R_1 .

Given: $\epsilon = 15\text{V}$, $r = 5\Omega$, $R_1 = 100\Omega$, $R_2 = 300\Omega$

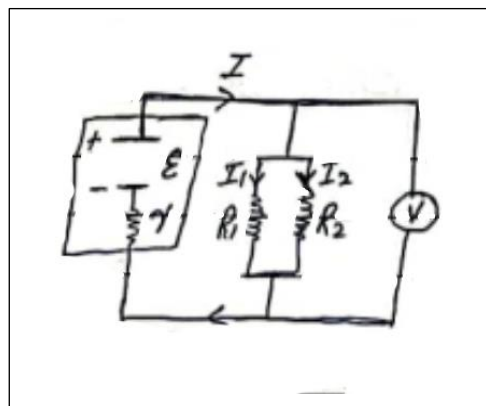
Req: current through $R_1 = I_1 = ?$

Solution: First we need to calculate equivalent resistance "Re" as the given circuit contains more than one resistors. Ideal voltmeter has infinite resistance, So there will be no current through it.

Now " R_1 ", and " R_2 " are in parallel so their equivalent resistance will be

$$\Rightarrow 1/R' = 1/R_1 + 1/R_2$$

$$\Rightarrow 1/R' = R_1 + R_2 / R_1 \times R_2$$



$$\Rightarrow R' = R_1 \times R_2 / R_1 + R_2$$

$$\Rightarrow R' = 100\Omega \times 300\Omega / 100\Omega + 300\Omega$$

$$\Rightarrow R' = 75\Omega$$

Now this “R’” is in series with “r” in the circuit.

So Now we find the total equivalent resistance. i.e

$$R_e = R' + r$$

$$\Rightarrow R_e = 75\Omega + 5\Omega$$

$$\Rightarrow R_{eq} = 80\Omega$$

Now we find total current “I” from Ohm’s law

$$\mathcal{E} = I R_e$$

$$\Rightarrow I = \mathcal{E} / R_e$$

$$\Rightarrow I = 15V / 80\Omega$$

$$\Rightarrow I = 0.1875A$$

Now this current “I” will pass through both “R’” and “r”. So potential drop across R’ is the terminal potential difference “V_t” is given by

$$V_t = I R'$$

$$\Rightarrow V_t = 0.1875A \times 75\Omega$$

$$\Rightarrow V_t = 14.00V$$

Now current through “R₁” will be $I_1 = V_t / R_1$

$$\Rightarrow I_1 = 14.0625V / 100\Omega$$

$$\Rightarrow I_1 = 0.1406A \text{ Ans.}$$

Q-12: Three arms of Wheatstone bridge are of 75Ω each. What is the resistance of its fourth arm.

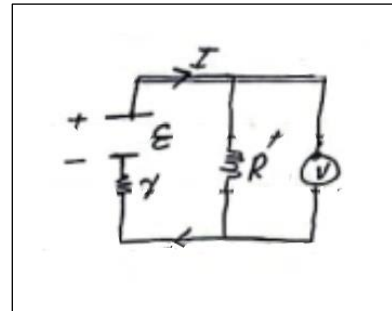
Given = P=75Ω, Q=75Ω, R=75Ω, Req: X=?

Sol: As we know that

$$X = QR/P$$

$$\Rightarrow X = 75\Omega \times 75\Omega / 75\Omega$$

$$\Rightarrow X = 75\Omega \text{ Ans.}$$

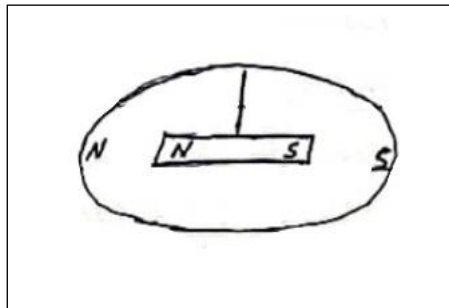


END OF CHAPTER 02

Chapter # 13 Electromagnetism

Electromagnetism: The study of magnetic properties due to an effect of electricity is called Electromagnetism.

Magnet: An object which attracts the small pieces of iron and which always stand in north-south direction, when freely suspended, is called Magnet. As we know that the earth has also a magnetic property due to which a bar magnet always stand in north-south direction, when it is freely suspended as shown in fig. Now the end of bar magnet pointing towards north is called north pole while the end pointing towards the south is called south pole. The like poles of two magnets repel while unlike poles attract each other.



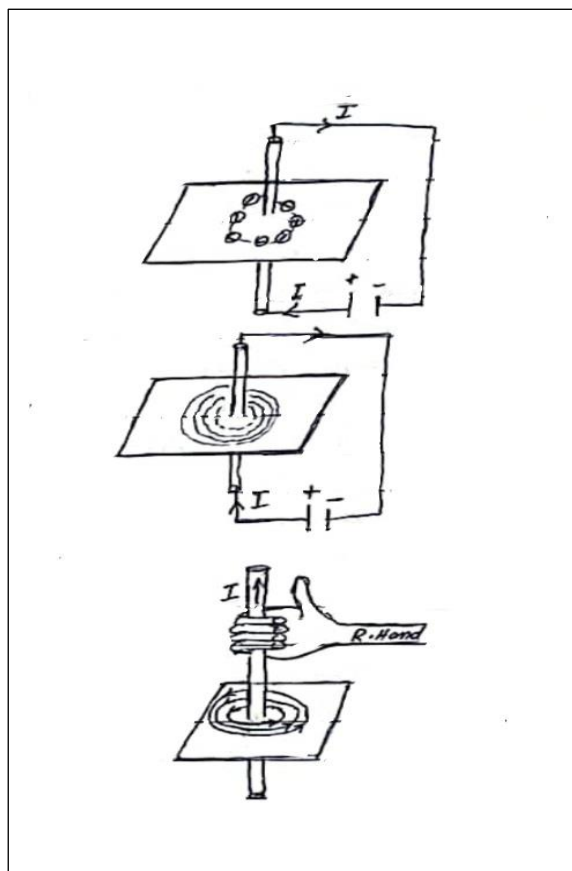
Magnetic field: The region of space around a big magnet or a current carrying conductor, where a test magnet is to experience a magnetic force is called magnetic field region, and the field in which existing is called magnetic field.

Explanation: The magnetic field is a vector field. It is represented by magnetic field lines or magnetic lines of force.

At any point, the magnetic field is specified by a vector $\vec{B}$ which is known as magnetic induction. The magnitude and direction of $\vec{B}$ at any point can be determined by magnetic force which is exerted by the field on a test magnet placed at that point.

Demonstration of magnetic field:

It has been found experimentally that moving charges can produce magnetic field around them. This fact can be explained with the help of following experiment (Demonstration). The magnetic field due to a current can be demonstrated by considering a conductor passed vertically through a horizontal card board. The iron filings are sprinkled over the board or compass needles are placed around the conducting wires as shown in Figures. When current "I" is allowed to pass through the conducting wires, the



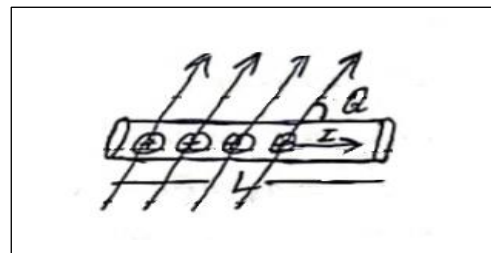
iron filings or compass needles arrange themselves in circles around the wire. Now from this demonstration we can say that the magnetic field produced around the wire due to a current (moving charges) will be circular in nature.

Direction of Magnetic field by Right hand rule:

The direction of magnetic field $\vec{B}$ due to current (moving charges) in a straight wire can be easily determined by Right hand rule, which is stated as, If the current carrying wire is grasped by right hand with the thumb in the direction of the current, then the remaining curl fingers will show us the direction of magnetic field around that wire.

Force on a current carrying conductor in a magnetic field

In a figure a current carrying conductor of length “L” is placed in an external uniform magnetic field as shown. A magnetic field makes an angle “ θ ” with the length of the conductor. Now due to an interaction of two magnetic fields one which is produced due to the current (moving charges) in the conductor and the second is an external magnetic field there will be a magnetic force on the conductor. Now we explain factors on which the magnetic force depends which are the following.



- ① The magnetic force on the conductor is directly proportional to the current “I” through the conductor i.e $F \propto I$ ----- ①
- ② The magnetic force on the conductor is directly proportional to the strength of an external magnetic field i.e $F \propto B$ ----- ②.
- ③ The magnetic force on the conductor is directly proportional to the length of the conductor i.e $F \propto L$ ----- ③
- ④ The magnetic force on the conductor is directly proportional to the $\sin \theta$. i.e $F \propto \sin \theta$ ----- ④ where “ θ ” is the angle between length “L” and external magnetic field.

Now combine all relations. Then we get as

$$F \propto ILB \sin \theta$$

$$\Rightarrow F = KILB \sin \theta \quad \text{put } k=1$$

$$\Rightarrow F = ILB \sin \theta \text{ ----- ⑤}$$

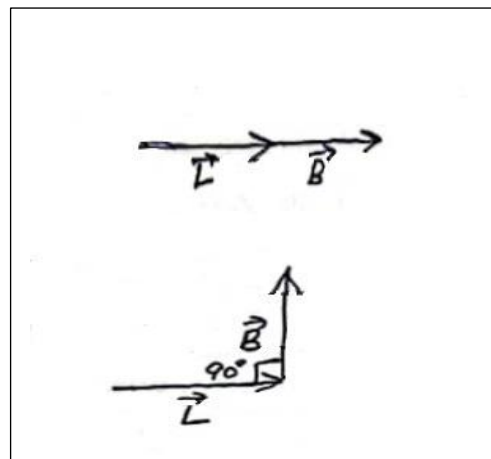
Eq ⑤ represents the magnitude form of magnetic force. Its vector form can be written as $\vec{F} = I (\vec{L} \times \vec{B})$ ----- ⑥

Special cases: ① if $\theta = 0^\circ$ between $\vec{L}$ and $\vec{B}$. Then eq ⑤ we can write as

$$F = ILB \sin 0^\circ$$

$$\Rightarrow F = 0 \quad \text{because } \sin 0 = 0$$

- ② if $\theta = 90^\circ$ between $\vec{L}$ and $\vec{B}$. Then eq ⑤ we can write as



$$F = ILB \sin 90^\circ$$

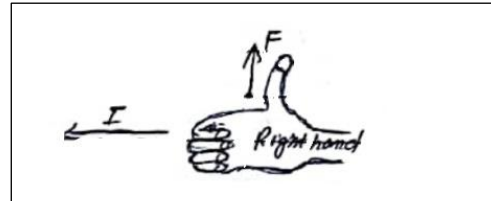
$$\Rightarrow F_{\max} = ILB \quad (\sin 90 = 1)$$

This shows when length of a conductor is perpendicular to the external magnetic field. Then magnetic force will be maximum on the conductor.

- ③ If $90^\circ < \theta < 0^\circ$. Then in this case magnetic force will depend on the value of " θ "

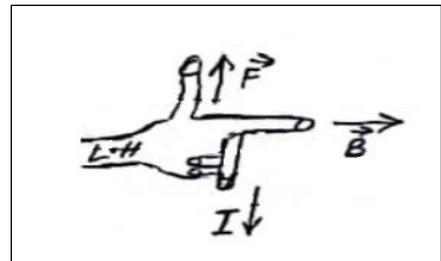
Direction of magnetic force by Right hand rule:

First we will open all the fingers of right hand in the direction of current " I " in the conductor. Then we will curl the four fingers toward " B ". Then the extended thumb will give us the direction of magnetic force as shown in figure.



Direction of magnetic force by Left hand rule:

First we will open the thumb, side finger and middle finger perpendicularly to each other as shown in figure. Then we set middle finger in the direction of current I and side finger in the direction of external magnetic field, then the extended thumb will give us the direction of magnetic force as shown in figure.



Unit of $\vec{B}$: The SI unit for magnetic induction is Tesla.

Now we know that $F = ILB$ when $\theta = 90^\circ$

$$\Rightarrow B = F/IL$$

$$\Rightarrow IT = IN/1A \times 1m$$

$$\Rightarrow IT = NA^{-1}m^{-1}$$

If we have 1m length of conductor and have a current in it is 1A, then due to which if magnetic force is 1N. Then magnetic induction " $\vec{B}$ " will be equal to 1 Tesla.

Other unit of $\vec{B}$: It can be written as

$$1 \text{ Gauss} = 10^{-4} \text{ Tesla}$$

$$\Rightarrow 1G = 10^{-4}T \quad \text{Or}$$

$$\Rightarrow IT = 10^4 G$$

Magnetic Flux

The measure of number of magnetic field lines that pass through a certain area is called magnetic flux.

Explanation: In a figure a number of magnetic field lines are passing through a certain area, so its measure from that area is called magnetic flux. Now magnetic flux " Φ " depends on the following factors.

- ① Surface area
- ② magnetic field intensity

③ On the angle between area vector " $\vec{A}$ " and magnetic field or magnetic induction " $\vec{B}$ ". Mathematically we can write as

$$\Phi \propto A \text{ ----- } ①$$

$$\Phi \propto B \text{ ----- } ②$$

$$\Phi \propto \cos\theta \text{ ----- } ③$$

From these relations we can say that if area is greater then more number of magnetic field lines will be passing and magnetic flux will be greater and vice versa. Similarly we can say that if magnetic induction is greater then more number of magnetic field lines will be passing through area and magnetic flux will be greater and vice versa.

Also we can say that if $\cos\theta$ value is greater then magnetic flux will be greater and vice versa. Now combine ① ② and ③ we will get

$$\Phi \propto BA \cos\theta$$

$$\Rightarrow \Phi = KBA \cos\theta$$

$$\Rightarrow \Phi = BA \cos\theta \quad \text{put } K=1$$

$$\Rightarrow \Phi = \vec{B} \cdot \vec{A} \text{ ----- } ④$$

In eq ④ from the dot product of " $\vec{B}$ " and " $\vec{A}$ " we can say that the magnetic flux is a scalar quantity.

Dependence of Magnetic flux at angle " θ ":

① **Maximum flux:** when area vector is in the direction of magnetic field of magnetic induction B , then $\theta = 0$.

Now we know that

$$\Phi = EA \cos\theta$$

$$\Rightarrow \Phi = EA \cos 0^\circ$$

$$\Rightarrow \Phi_{\max} = EA$$

So for $\theta = 0$ the magnetic flux will be maximum and it will be equal to EA .

② **Zero flux:** When area vector is perpendicular to the magnetic field, then $\theta = 90^\circ$.

Now we know that

$$\Phi = EA \cos\theta$$

$$\Phi = EA \cos 90^\circ$$

$$\Rightarrow \Phi = EA (0)$$

$$\Rightarrow \Phi = 0$$

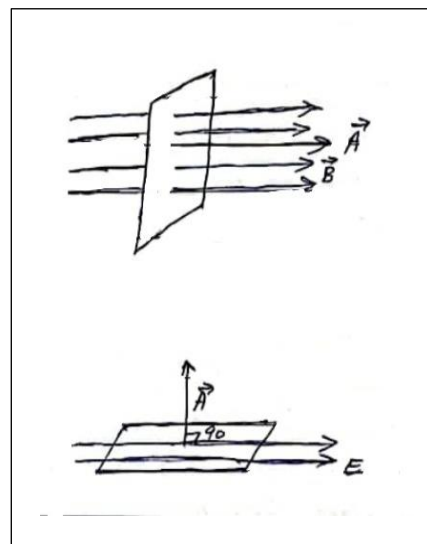
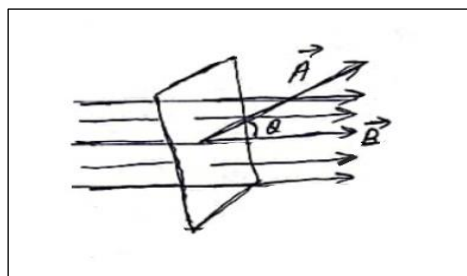
So for $\theta = 90^\circ$ the magnetic flux will be equal to zero.

③ **Flux at any angle:** When $90^\circ < \theta < 0^\circ$ between area vector and magnetic field. Then in this case we can say that the electric flux will depend on the value of " θ ".

Unit of magnetic flux: The SI unit of magnetic flux is weber.

One weber is given by

$$1 \text{ wb} = 1 \text{ T} \cdot 1 \text{ m}^2$$

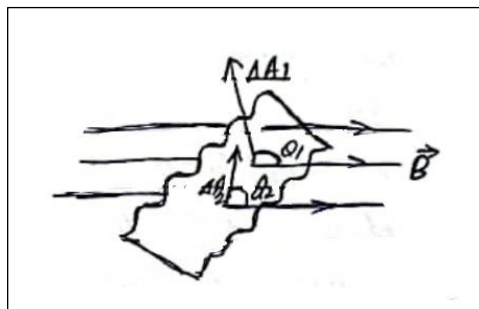


$$\Rightarrow 1wb = 1NA^{-1}m \quad (IT = N \cdot m^{-1}A^{-1})$$

Magnetic flux through curved or non-uniform magnetic field

A curved surface is shown in figure. Now we know that the whole surface area of the curved surface is a scalar quantity. Now for the magnetic flux we will convert the whole surface area of the curved surface to areas which are vector quantities.

So for this we will divide the whole surface area of the curved surface to very small flat areas as shown in figure.



Now each flat areas ($\Delta \vec{A}_1, \Delta \vec{A}_2, \Delta \vec{A}_3, \dots, \Delta \vec{A}_n$)

is a vector quantity and for each flat area the magnetic field intensity is uniform. Now for total magnetic flux from this curved surface we can write as

$$\begin{aligned} \Phi_t &= \Phi_1 + \Phi_2 + \Phi_3 + \dots + \Phi_n \\ \Rightarrow \Phi_t &= \vec{B} \cdot \Delta \vec{A}_1 + \vec{B} \cdot \Delta \vec{A}_2 + \vec{B} \cdot \Delta \vec{A}_3 + \dots + \vec{B} \cdot \Delta \vec{A}_n \\ \Rightarrow \Phi_t &= B \Delta A_1 \cos \theta_1 + B \Delta A_2 \cos \theta_2 + B \Delta A_3 \cos \theta_3 + \dots + B \Delta A_n \cos \theta_n \\ \Rightarrow \Phi_t &= B \sum_{i=1,2,3} \Delta A_i \cos \theta_i \end{aligned}$$

Now when area is uniform and magnetic field is non uniform then we can write

$$\Phi_t = A \sum_{i=1,2,3} B_i \cos \theta_i$$

Now when both area and magnetic field are non uniform, then we can write as

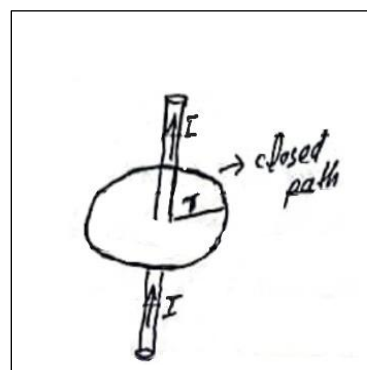
$$\Phi_t = \sum_{i=1,2,3} B_i \Delta A_i \cos \theta_i$$

Ampere circuital law

This law states that "The sum of dot products of magnetic field " $\vec{B}$ " and length element " $\Delta \vec{\ell}$ " of a closed loop is equal to μ_0 time of the total current which is passing in a conductor enclosed in a loop or path".

$$\text{i.e. } \sum \vec{B} \cdot \Delta \vec{\ell} = \mu_0 I$$

Proof: Consider a closed path around a current carrying wire in the form of circle as shown in figure. Now we want to find the total magnetic field on this closed path produced by current in that wire. For this we can write as



- ① The total magnetic field on the closed path is directly proportional to the current flowing through the wire i.e $B \propto I$ ----- ①
- ② The total magnetic field on the closed path is inversely proportional to the radius of the closed path. i.e $B \propto 1/r$ ----- ②

Now combine ① and ②, we get as

$$B \propto I / r$$

$\Rightarrow B = \mu_0 I / 2 \pi r$ ----- (3) where $(\mu_0 / 2 \pi)$ is constant.

In eq (3) μ_0 is constant of proportionality. In which " μ_0 " is called permeability of free space or vacuum and its value is equal to $4\pi \times 10^{-7} \text{ Tm/A}$ or $4\pi \times 10^{-7} \text{ Wb/Am}$. Now permeability of an medium is define as "The permission of a medium to the flow of magnetic field lines is called permeability (μ). If a medium give more permission to the flow of a magnetic field lines then its permeability will be greater and vice versa. Now equation (3) we can write as

$$B (2\pi r) = \mu_0 I$$

$$\Rightarrow B \ell = \mu_0 I$$
 ----- (4)

Now in eq (4), $\ell = 2\pi r$ represents total length or circumference of the closed path and 'B' represents total magnetic field on it.

Now we want to write eq (4) in vector form. For this we will divide the total length " ℓ " into small elements (i.e $\Delta \vec{\ell}_1, \Delta \vec{\ell}_2, \Delta \vec{\ell}_3, \dots \Delta \vec{\ell}_n$) because total length is a scalar quantity.

Then the small elements (i.e $\Delta \vec{\ell}_1, \Delta \vec{\ell}_2, \Delta \vec{\ell}_3, \dots \Delta \vec{\ell}_n$) will be a vector quantities. Now for each small element we will also divide the total magnetic field into its components (i.e $\vec{B}_1, \vec{B}_2, \vec{B}_3, \dots \vec{B}_n$) as shown in figure. Now in figure each element of length " ℓ " and each component of "B" are in parallel. So we will take dot in its product. i.e

$$\vec{B}_1 \cdot \Delta \vec{\ell}_1 + \vec{B}_2 \cdot \Delta \vec{\ell}_2 + \vec{B}_3 \cdot \Delta \vec{\ell}_3 + \dots \vec{B}_n \cdot \Delta \vec{\ell}_n = \mu_0 I$$

$$\Rightarrow \sum \vec{B}_i \cdot \Delta \vec{\ell}_i = \mu_0 I$$
 ----- (5) (proved)

Equation (5) represents the mathematical form of Ampere's law. From eq (5) we can also say that the Ampere's law can be applied to any closed path (Amperian loop) around a current carrying conductor as shown in figure. Because this law does not depends on the shape of closed path.

For any other medium apart from vacuum we will take permeability (μ) in equation. And then we will express permeability (μ) in term of relative permeability " μ_r ". i.e

$$\mu = \mu_0 \mu_r$$

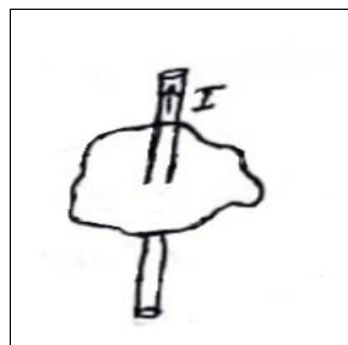
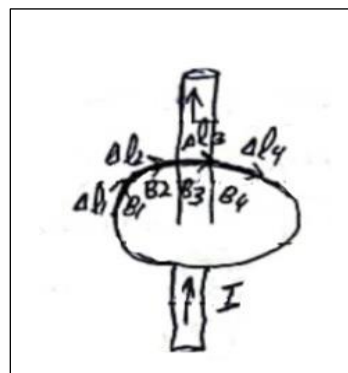
Now relative permeability of any medium is define as "The ratio of permeability of any medium to the permeability of free space or vacuum is called relative permeability i.e.

$$\mu_r = \mu / \mu_0$$

It has no unit.

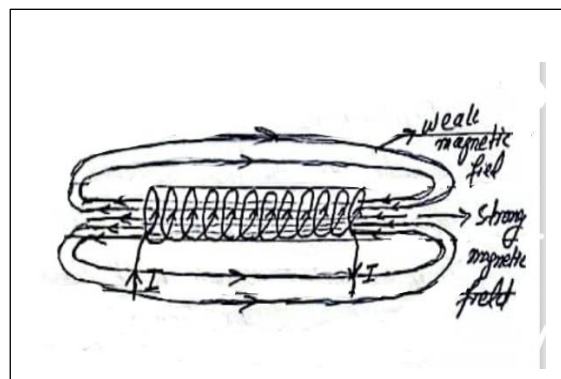
Note: Before using Ampere's law we must take a closed path called Amperian loop for calculation of magnetic field.

Magnetic field due to a current carrying solenoid



A material which is made of a conductor and has a spring like structure is called solenoid. Now we want to find the expression for magnetic field due to a current carrying solenoid. For this we consider a long solenoid with many closed turns as shown in figure.

Now when current is allowed to flow through this long solenoid then the magnetic fields will be producing, one inside the solenoid, and the second outside the solenoid. Now the inside magnetic field along the axis of solenoid will be strong and uniform due to a closer turns and due to its longer length and the out side magnetic field will be weak so we will neglect it.

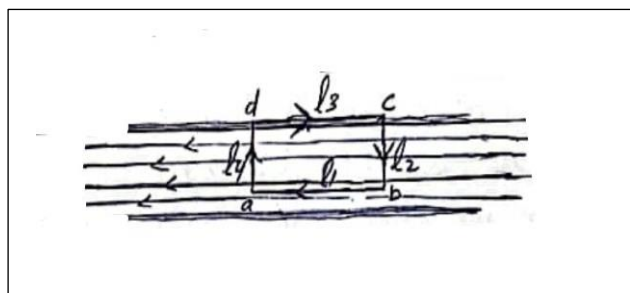


Now we apply Ampere's Law to find the expression for inside the magnetic field. For this first we consider a rectangular closed path between the points "ABCD" as shown in figure. Now we know that from Ampere's law

$$\sum_{i=1,2,3} \vec{B}_i \cdot \Delta \vec{\ell}_i = \mu_0 I \text{ ----- (1)}$$

Now in figure the inside magnetic field is uniform. So eq (1) we can write as

$$\begin{aligned} \vec{B} \sum_{i=1,2,3} \Delta \vec{\ell}_i &= \mu_0 I \\ \Rightarrow \vec{B} \cdot (\Delta \vec{\ell}_1 + \Delta \vec{\ell}_2 + \Delta \vec{\ell}_3 + \Delta \vec{\ell}_4) &= \mu_0 I \\ \Rightarrow \vec{B} \cdot \Delta \vec{\ell}_1 + \vec{B} \cdot \Delta \vec{\ell}_2 + \vec{B} \cdot \Delta \vec{\ell}_3 + \vec{B} \cdot \Delta \vec{\ell}_4 &= \mu_0 I \\ \Rightarrow B \Delta \ell_1 \cos \theta_1 + B \Delta \ell_2 \cos \theta_2 + B \Delta \ell_3 \cos \theta_3 + B \Delta \ell_4 \cos \theta_4 &= \mu_0 I \\ \Rightarrow B \Delta \ell_1 \cos 0^\circ + B \Delta \ell_2 \cos 90^\circ + B \Delta \ell_3 \cos 180^\circ + B \Delta \ell_4 \cos 90^\circ &= \mu_0 I \\ \Rightarrow B \Delta \ell_1 + 0 - B \Delta \ell_3 + 0 &= \mu_0 I \text{ ----- (2)} \end{aligned}$$



Now in figure for " $\Delta \ell_3$ " the inside field effect is zero. So we can write as

$$B \Delta \ell_3 = 0 \times \Delta \ell_3 = 0$$

Now Eq (2) we can write as

$$\begin{aligned} B \Delta \ell_1 + 0 + 0 + 0 &= \mu_0 I \\ \Rightarrow B \Delta \ell_1 &= \mu_0 I \text{ ----- (3)} \end{aligned}$$

For "N" number of turns in solenoid, eq (3) can be written as

$$B \Delta \ell_1 = N \mu_0 I$$

Now take $\Delta \ell_1 = \ell$ = Full length of solenoid.

$$\begin{aligned} B \ell &= N \mu_0 I \\ \Rightarrow B &= N \mu_0 I / \ell \text{ ----- (4)} \end{aligned}$$

Now take "n" represents number of turns per unit length i.e. $n = N/\ell$

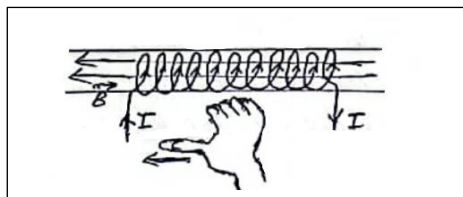
So we can write as

$$B = n \mu_0 I \text{ ----- (5)}$$

Eq (5) represents the inside magnetic field of solenoid.

Direction of magnetic field by Right hand rule:

We will curl the four fingers in the direction of current in solenoid. Then extended thumb will show us the direction of inside Magnetic field as shown in figure.



Magnetic force on a point charge

In a figure a point charge is moving with a velocity "v" in an external magnetic field as shown. An external magnetic field makes an angle "θ" with the velocity of the point charge. Now due to an interaction of two magnetic fields, one which is produced due to a moving point charge, and the second is an external magnetic field, there will be a magnetic force on the point charge. Now we explain factors on which the magnetic force depends which are the following.

- ① The magnetic force on the moving point charge is directly proportional to its velocity. i.e.

$$F \propto v \text{ ----- ①}$$

- ② The magnetic force on the moving point charge is directly proportional to its charge. i.e.

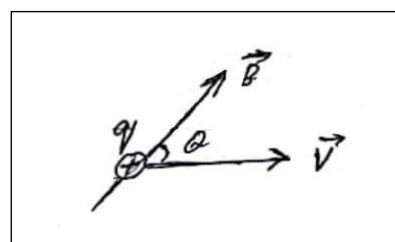
$$F \propto q \text{ ----- ②}$$

- ③ The magnetic force on the moving point charge is directly proportional to the strength of an external magnetic field . i.e

$$F \propto B \text{ ----- ③}$$

- ④ The magnetic force on the moving point charge is directly proportional to the sinθ. i.e

$$F \propto \sin\theta \text{ ----- ④}$$



where "θ" is the angle between velocity " $\vec{v}$ " and external magnetic field.

Now combine all relations, then we get as

$$F \propto q v B \sin\theta$$

$$\Rightarrow F = k q v B \sin\theta \quad \text{put } k=1$$

$$\Rightarrow F = q v B \sin\theta \text{ ----- ⑤}$$

Eq ⑤ represents the magnitude form of magnetic force. Its vector form can be written as $\vec{F} = q (\vec{v} \times \vec{B})$ ----- ⑥

Now we explain special cases.

- ① if $\theta=0^\circ$ between " $\vec{v}$ " and " $\vec{B}$ ", Then eq ⑤ becomes.

$$F = q v B \sin 0^\circ$$

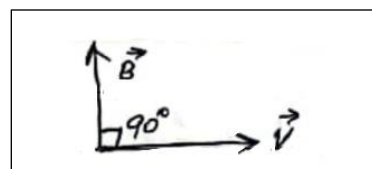
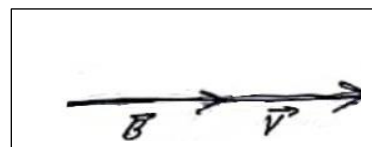
$$\Rightarrow F = 0 \quad \text{because } \sin 0^\circ = 0$$

- ② If $\theta=90^\circ$ between " $\vec{v}$ " and " $\vec{B}$ " Then eq ⑤ becomes

$$F = q v B \sin 90^\circ$$

$$\Rightarrow F_{\max} = q v B \quad \text{because } \sin 90^\circ = 1$$

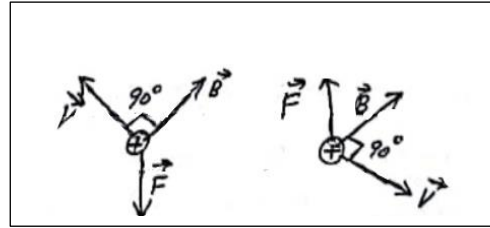
This shows that when velocity of a point charge is perpendicular to the external magnetic field. Then magnetic force will be maximum on the point charge.



③ If $90^\circ < \theta < 0^\circ$ then in this case magnetic force will depend on the value of “ θ ”.

Direction of magnetic force by Right hand rule:

First we will open all the fingers of right hand in the direction of velocity of the point charge. Then we will curl the four fingers towards B . Then the extended thumb will gives us the direction of magnetic force.



Direction of magnetic force by Left hand rules:

First we will open the thumb, side finger and middle finger perpendicularly to each other. Then we set the middle finger in the direction of velocity “ v ” and side finger in the direction of external magnetic field. Then the extended thumb will gives us the direction of magnetic force.

Prove that $F_B = ILB\sin\theta = qvB\sin\theta$

As we know that the force on a current carrying conductor can be written as

$$F_B = ILB\sin\theta \text{ ----- (1)}$$

Now current in a conductor we can write as

$$I = Q/t$$

$$\Rightarrow I = Nq \text{ ----- (2)}$$

Now the charge velocity in at conductor we can write as

$$v = L/t$$

$$\Rightarrow L = vt \text{ ----- (3)}$$

Now put eq (2) and (3) in eq(1). We get as

$$F_B = N(q/t)vB \times vt \times \sin\theta$$

$$\Rightarrow F_B = NqvB\sin\theta \text{ ----- (4)}$$

Now for a single charge. We put $N=1$ in eq(2). We get as

$$F_B = qvB\sin\theta.$$

Now we can write as

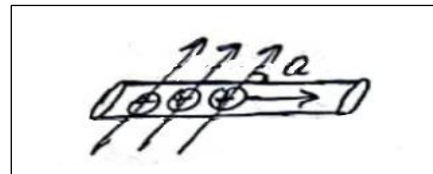
$$F_B = ILB\sin\theta = qvB\sin\theta \text{ ----- (5) (proved).}$$

From eq (5) we can say that the magnetic force “ F_B ” in case of current carrying conductor and in case of point charge has the same nature. Eq (5) in vector form we can write as

$$F_B = I(\vec{L} \times \vec{B}) = q(\vec{v} \times \vec{B}).$$

Circular motion of charge particle in uniform magnetic field

A uniform magnetic field is shown in a figure. Its direction is into the page and it is represented by “ x ”. Now a charge particle is entering to this magnetic field with velocity “ v ”. Now due to an interaction of two magnetic fields there will be a magnetic force on the charge and due to which it will change the direction. Now due to a continuous magnetic force on the charge particle it will change a direction continuously and as a result it will start a circular motion as shown in figure. Now at every point of the circle the direction of magnetic force is



towards a center and such a force will behave as centripetal force. So now we can write as

$$F_C = F_B$$

$$\Rightarrow mv^2/r = qvB \quad \text{where 'm' is the mass of particle.}$$

$$\Rightarrow mv/r = qB$$

$$\Rightarrow v = rqB/m \text{ ----- (1)}$$

Eq(1) represent the linear velocity of charge particle.

Radius of the circle: As we know that

$$v = rqB/m$$

$$\Rightarrow r = mv/qB \text{ ----- (2)}$$

Angular velocity of charge particle: As we know that

$$v = rqB/m \text{ ----- (3)}$$

we also know that $v = r \omega$ ----- (4)

Now put eq(4) in eq(3) we get as

$$r \omega = rqB/m$$

$$\Rightarrow \omega = qB/m \text{ ----- (5)}$$

Time period of charge particle: As we know that

$$T = 2\pi / \omega \text{ ----- (6)}$$

Now put eq(5) in eq (6). We get as

$$T = 2\pi / (qB/m)$$

$$\Rightarrow T = 2\pi m/qB \text{ ----- (7)}$$

Frequency of charged particle: As we know that

$$f = 1/T \text{ ----- (8)}$$

Now put eq(7) in eq (8). We get as

$$f = 1/(2\pi m/qB)$$

$$\Rightarrow f = qB/2\pi m \text{ ----- (9)}$$

In eq (9) the frequency is called the cyclotron frequency of a particle in a field because this is the frequency at which particles revolve in a cyclotron device. Cyclotron is a device which accelerates the charged particle with certain speed and frequency

Determination of charge to mass ratio for an electron

In a figure the electrons are produced from the filament "F". Then these electrons are accelerated towards a glass tube by the applied voltage "V" across the plates "A" and "B". Then in a glass tube the electrons will start circular motion due to a magnetic force produced by an applied magnetic field as shown in figure. Now such a magnetic force will behave as a centripetal force. So now we can write as

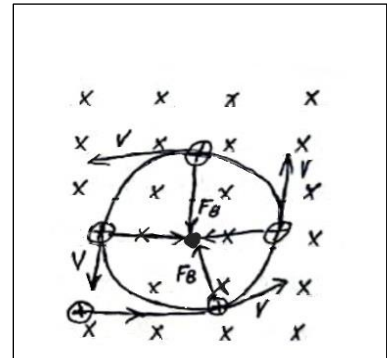
$$F_C = F_B$$

$$\Rightarrow m_e v^2/r = q_e v B$$

$$\Rightarrow m_e v/r = q_e B$$

$$\Rightarrow q_e / m_e = v B/r \text{ ----- (1)}$$

Now we know that the voltage or P.D can be written as



$$\Delta V = \Delta U / q_e$$

Now we write it in simple form

$$\text{i.e } V = U / q_e$$

$$\Rightarrow U = q_e V$$

$$\Rightarrow \text{K.E.} = q_e V$$

$$\Rightarrow \frac{1}{2} m v^2 = q_e V$$

$$\Rightarrow v^2 = 2 q_e V / m_e \text{ ----- (2)}$$

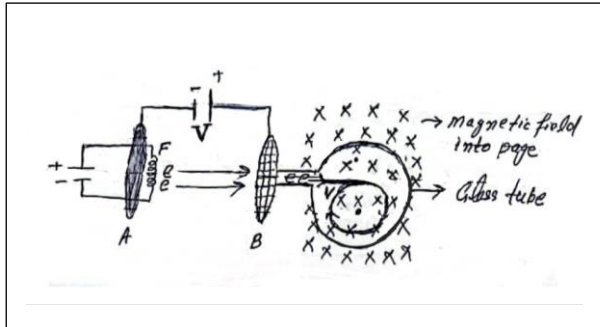
Now taking square of eq (1). we get as

$$q_e^2 / m_e^2 = v^2 / B^2 r^2 \text{ ----- (3)}$$

Now put eq (2) in eq (3). we get as

$$q_e^2 / m_e^2 = 2 q_e V / m_e B^2 r^2$$

$$\Rightarrow q_e / m_e = 2V / B^2 r^2 \text{ ----- (4)}$$



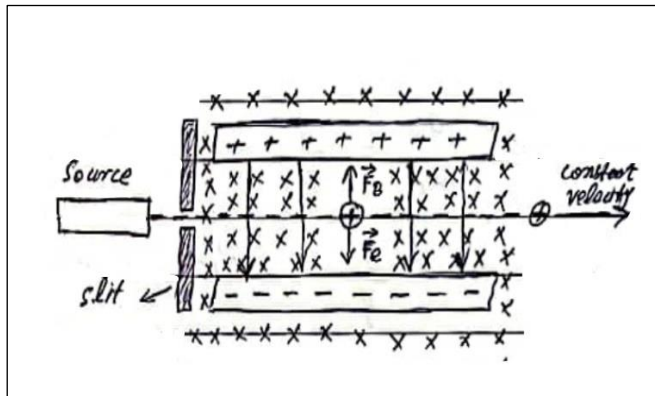
Measurement of Radius: When the electrons enter to the glass tube. Then during circular motion these electrons collide with helium gas atoms inside the tube. Then excitation and de-excitation will occur in helium gas atoms due to which a blue light will emit. Then due to this blue light the electrons circular path will become visible and then we will find its diameter. Then from diameter we can easily find its radius.

Thus by knowing the value of applied voltage “V” magnetic field “B” and radius “r” from the apparatus, then we can easily find charge to mass ratio for an electron by using equation (4).

Velocity selector

The device which allows only those charged particles which has a constant velocity is called velocity selector.

Explanation: In a figure a charged particles enters with different velocities to the two fields one is called electric field and the second is called magnetic field. Now on these charged particles there will be two forces one is called electric force ($F_e = qE$) in downward direction and the second one is called magnetic force ($F_B = qvB$) in upward direction. Now those charges



which have less velocities will be deflected in downward direction due to a greater effect of electric force. For those charges the magnetic force effect will be less.

Now those charges which have greater velocities will be deflected in upward direction due to a greater effect of magnetic force. For those charges the electric force effect will be less. Now there will be some charges, which will be moving straight with constant velocity between the two fields and then such

aparticles will go out with the same constant velocity from the exit of this device as shown in figure. Now we find the expression for such a constant velocity. Now the electric and magnetic forces will be equal when charged particle is moving straight between the fields. i.e.

$$F_e = F_m$$

$$\Rightarrow qE = qvB \quad \theta = 90^\circ \text{ between } \vec{v} \text{ and } \vec{B}.$$

$$\Rightarrow v = E/B$$

Equation (1) represents the constant velocity of charged particles.

Torque on a current carrying coil

A current carrying rectangular coil inside a uniform magnetic field is shown in figure. Now due to a current flow in that rectangular coil there will be magnetic forces on that coil. Which we explain as follows.

Force on (1) side: As we know that

$$F = ILB\sin\theta$$

Now for (1) side let its length $L = b$ and $\theta = 0^\circ$ between its length and magnetic field.

$$\Rightarrow F_1 = IbB\sin 0^\circ$$

$$\Rightarrow F_1 = IbB \times 0$$

$$\Rightarrow F_1 = 0 \text{ ----- (1)}$$

Force on (2) side: As we know that

$$F = ILB\sin\theta$$

Now for (2) side let $L = a$ and $\theta = 90^\circ$ between its length and magnetic field.

$$\Rightarrow F_2 = IaB\sin 90^\circ$$

$$\Rightarrow F_2 = IaB \text{ ----- (2) "F}_2\text{" force will be in upward direction}$$

Force on (3) side: As we know that

$$F = ILB\sin\theta$$

Now for (3) side let $L = b$ and $\theta = 0^\circ$ between its length and magnetic field.

$$\Rightarrow F_3 = IbB\sin 0^\circ$$

$$\Rightarrow F_3 = 0 \text{ ----- (3)}$$

Force on (4) side: As we know that

$$F = ILB\sin\theta$$

Now for (4) side let $L = a$ and $\theta = 90^\circ$ between its length and magnetic field.

$$\Rightarrow F_4 = IaB \sin 90^\circ$$

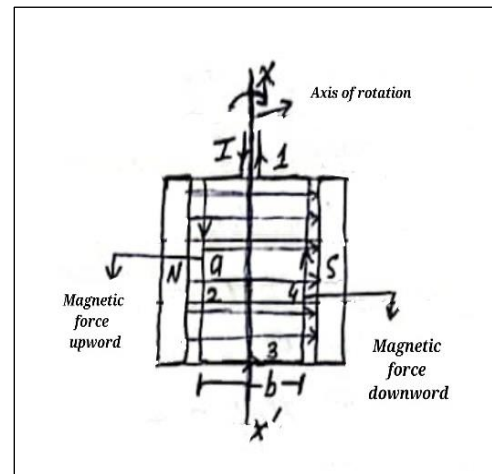
$$\Rightarrow F_4 = IaB \text{ ----- (4) "F}_4\text{" force will be in downward direction.}$$

Now from the above explanation we can say that there will be only two magnetic forces, one is " F_2 " on side (2) in upward direction, and the second force " F_4 " on side (4) in downward direction on rectangular coil. Now such a two forces will produce a torque in that coil.

Now for total torque we can write as

$$\tau = \tau_2 + \tau_4$$

$$\Rightarrow \tau = F_2 b / 2 + F_4 b / 2$$



$$\Rightarrow \tau = IabB / 2 + IabB / 2$$

$$\Rightarrow \tau = 2IabB / 2$$

$$\Rightarrow \tau = IabB \text{ ----- (5)}$$

Now in fig. "a" is the length and "b" is the width of the coil. So therefore $ab = A$ ----- (6)

Now put eq (6) in eq (5). We get as

$$\Rightarrow \tau = IAB \text{ ----- (7)}$$

Now if we have "N" turns in the coil. then eq (7) we can write as

$$\Rightarrow \tau = NIBA \text{ ----- (8)}$$

Eq (8) represents the total torque produced in the coil and such a torque will be clock wise. Now we find torque expression when the magnetic field makes angle $\theta < 90^\circ$ with the area vector of the rectangular coil.

In a figure $\frac{b}{2} \sin \theta$ are the moment arms on each side for forces F_2 and F_4 . Now for total torque we can write as

$$\tau = \tau_2 + \tau_4$$

$$\Rightarrow \tau = F_2 \frac{b}{2} \sin \theta + F_4 \frac{b}{2} \sin \theta$$

$$\Rightarrow \tau = IabB \frac{b}{2} \sin \theta + IabB \frac{b}{2} \sin \theta$$

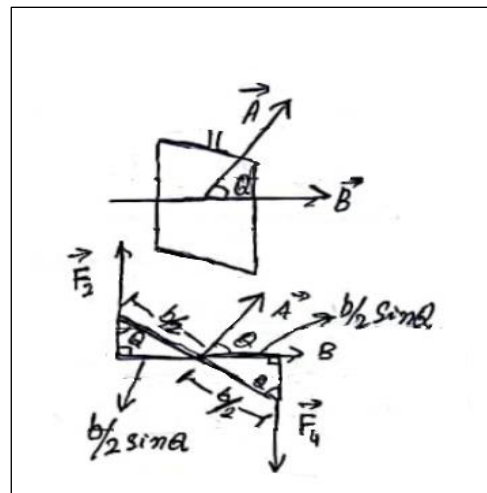
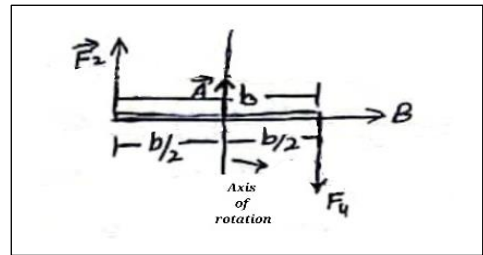
$$\Rightarrow \tau = 2IabB \sin \theta / 2$$

$$\Rightarrow \tau = IabB \sin \theta$$

$$\Rightarrow \tau = IAB \sin \theta \text{ ----- (9)}$$

If we have 'N' turns in the coil, then eq (9) can be written as

$$\tau = NIB \sin \theta$$

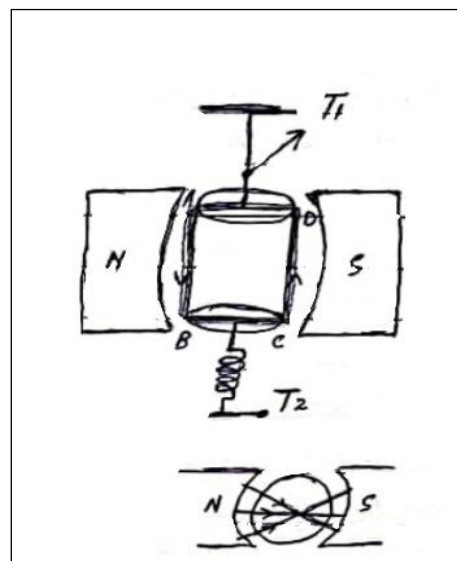


Galvanometer

A device used for detection and measurement of small electric currents is called galvanometer.

Principle: The principle of moving coil galvanometer is the torque produced in current carrying coil in the magnetic field.

Construction: The construction of a moving coil galvanometer is shown in figure. It has a rectangular coil "ABCD" which is wrapped around an aluminum frame, then it is suspended between the concave poles by means of a metallic wire, and its other end is connected with a loosely coil spring as shown in figure. It has also an iron cylinder in between the coil and has the following advantages.



① It makes the magnetic field radial and strong as shown in figure II. Radial magnetic field means the magnetic field which passes from the radius of the iron cylinder.

② It increases the moment of inertia of the suspending coil. It has also a pointer which will move on the scale and will give us the current.

It has also two terminals one is "T₁" and the other is "T₂" as shown in figure. "T₁" is the terminal for entering current and "T₂" for leaving current working of galvanometer: When a current "I" enters to the coil ABCD" by terminal "T₁". Then there will be two magnetic forces on the sides "AB" and "DC" of the coil due to radial magnetic field. The magnetic force on side "AB" will be in upward direction and the magnetic force on side "DC" will be in downward direction. Now such a two forces will produce a torque in that coil and it can be written as

$$\tau = NIBA \sin \theta \text{ ----- } \textcircled{1}$$

For $\theta = 90^\circ$ between $\vec{A}$ and $\vec{B}$. Eq ① can be written as

$$\tau = NIBA \text{ ----- } \textcircled{2}$$

Now due to a torque in the coil, a twist will be also producing in a spring and due to which a torque will be also restoring in a spring in opposite direction. Now such a restoring torque in the spring will show directly proportionality to the angle of deflection of rectangular coil. So now we can write as

$$\tau_{\text{res}} \propto \theta$$

$$\Rightarrow \tau_{\text{res}} \propto C\theta \text{ ----- } \textcircled{3}$$

In eq ③ "C" is spring constant and it depends on the nature of the spring.

Now when the suspending coil system is reached to equilibrium after deflection due to a current flow in it, then we can write as

Torque produced in the coil = Torque produced in the spring. i.e

$$\tau = \tau_{\text{res}}$$

$$\Rightarrow NIAB = C\theta$$

$$\Rightarrow I = C\theta/NAB \text{ ----- } \textcircled{4}$$

$$\Rightarrow I \propto \theta \text{ ----- } \textcircled{5} \text{ where } C/NAB \text{ is constant.}$$

Now from relation ⑤ we can say that the current measurement of the galvanometer depends on the angle of deflection of the coil. Now by knowing the angle of deflection by pointer and the constant C/NAB, then we can easily find the magnitude of current by using eq ④. Now when the current is stopped to the galvanometer, then spring will come back a suspended coil system to its initial position.

Ammeter" OR "Conversion of galvanometer into Ammeter

It is a device which is used for the measurement of large current in a circuit.

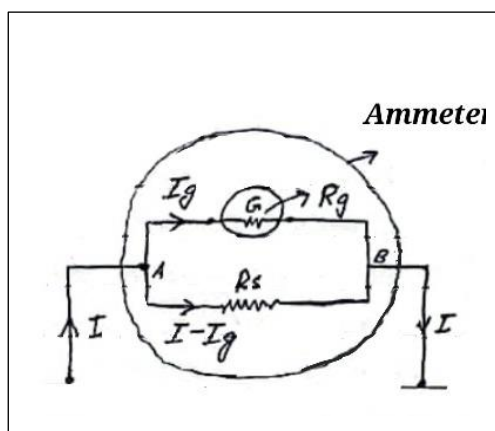
Explanation: A sensitive galvanometer can measure the current of few milliamperes only. Thus in case of large current, we can not use such a sensitive galvanometer.

For measuring large currents, the galvanometer is modified by shunt resistor, then such modified galvanometer is said to be an Ammeter.

Modification or conversion of Galvanometer into Ammeter:

For converting the galvanometer into ammeter, a low resistance called shunt resistance is connected in parallel with galvanometer "G" in the circuit as shown in the figure. Now when large current flows in the circuit, it divides at point "A".

Then some current " I_g " will pass through galvanometer and due to which a galvanometer will show a full scale deflection. Now the remaining current $(I - I_g)$ will pass from the shunt resistor (R_s).



Expression for shunt Resistance:

As the galvanometer or its resistor " R_g " and shunt resistor " R_s " are in parallel in the circuit. So we can write as

Potential difference drop across " R_g " = Potential difference drop across " R_s ".

i.e $V_{R_g} = V_{R_s}$

$$\Rightarrow I_g R_g = (I - I_g) R_s$$

$$\Rightarrow R_s = I_g R_g / (I - I_g) \text{ ----- (1)}$$

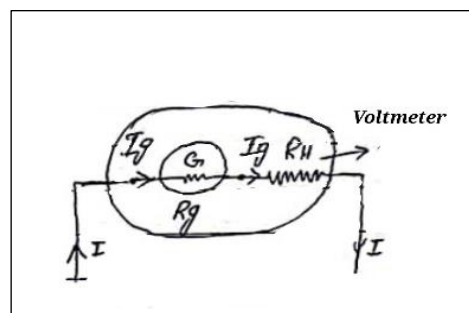
Eq (1) represents the expression for shunt resistance. It is also called conversion equation. Now for any desired current " I " say it up to 100A, first we will find the shunt Resistance " R_s " from eq (1) by knowing the values of I_g and R_g from the galvanometer, then we will adjust such a shunt Resistance " R_s " in parallel with galvanometer, then such a given galvanometer will be converted to ammeter, and then it will shows us the current " I " up to 100 A. Ammeter must be connected in series in the circuit, and ideal Ammeter have zero resistance.

Voltmeter OR Conversion of galvanometer into voltmeter

A voltmeter is a device which measure the voltage or potential difference between any two points in the circuit. For measuring voltages, the galvanometer is modified by high resistor, then such modified galvanometer is said to be a voltmeter.

Modification or conversion of galvanometer into voltmeter:

For converting a galvanometer into voltmeter, a high resistance " R_H " is connected in a series with galvanometer "G" in the circuit as shown in figure. Now when current flows in the circuit then " I_g " will be the current through " R_g " and the same current " I_g " will also pass from " R_H ". So for both resistors the current will be same, and it will be equal to " I_g ".



Expression for high Resistance: As the galvanometer or its resistor " R_g " and high resistor " R_H " are in Series in the circuit. So we can say that the total voltage drop will be equal to the sum of voltage drop across the " R_H " and the voltage drop across " R_g ". i.e

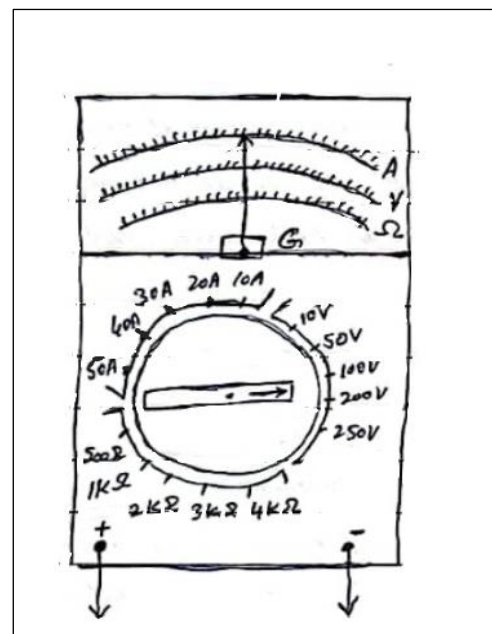
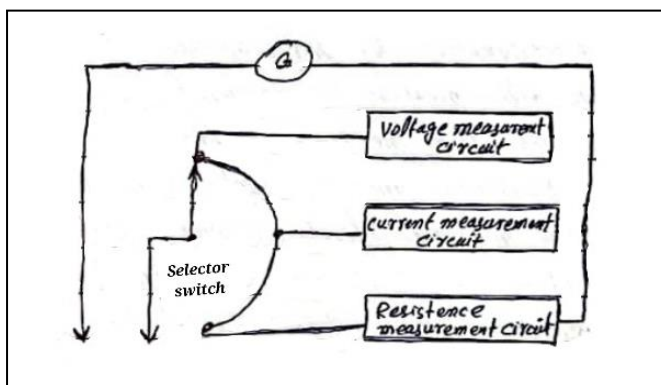
$$\begin{aligned} V &= V_{R_g} + V_{R_H} \\ \Rightarrow V &= I_g R_g + I_g R_H \\ \Rightarrow I_g R_H &= V - I_g R_g \\ \Rightarrow R_H &= \frac{V - I_g R_g}{I_g} \text{----- (1)} \end{aligned}$$

Eq (1) represents the expression for high resistance. It is also called conversion equation. Now for any desired voltage " V " say it up to 250V first we will find the high resistance " R_H " from eq (1) by knowing the values of " I_g " and " R_g " from the galvanometer. Then we will adjust such a high resistance in series with galvanometer. Then such a gives galvanometer will be converted to voltmeter, and then it will show us the voltage up to 250V. Voltmeter must be connected in parallel in the circuit and Ideal voltmeter has an infinite resistance.

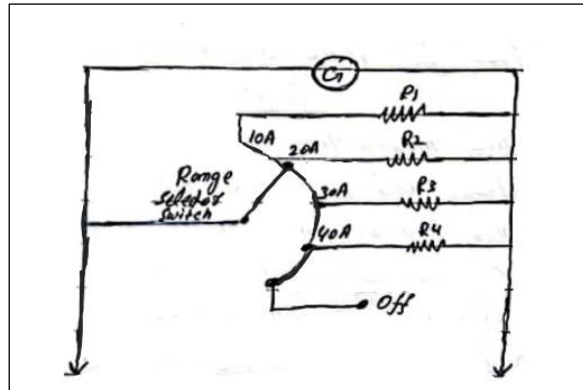
Avometer

It is an instrument which is used to measure current, voltage, and resistance. It is also known as multimeter. It can measure the direct as well as alternating current and voltage. It is a galvanometer and take a connection with inside combination of resistors by a selector switch for a required measurement of current, voltage and resistor. It has different scales graduated.

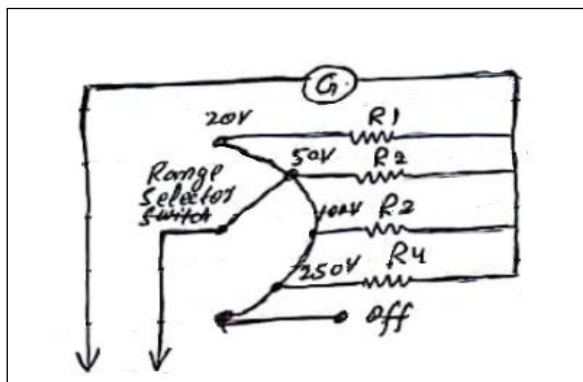
In such a manner that all the three quantities can be measured. It has its own battery for its function and for the operating the electrical circuits. The required quantity to be measured, for this we will adjust the selector switch on some particular range value.



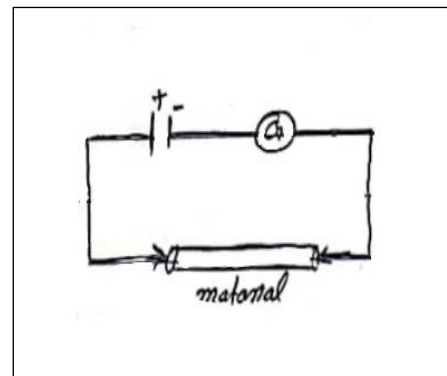
(a) Current measurement: For the current measurement we will adjust the selector switch on some particular range value of current. Then by adjusting the selector switch, the galvanometer will take a connection with one of the particular resistor in parallel inside the construction. By this way then we will measure the required current from the scale according to a selected range value of current.



(b) Voltage measurement: For the voltage measurement we will adjust the selector switch on some particular range value of voltage. Then by adjusting the selector switch, the galvanometer will take a connection with one of the particular resistor in series inside the construction. By this way then we will measure the required voltage from the scale according to a selected range value of current.



C) Resistance measurement: For the measurement of resistance of the material first we will insert the material in between the leads of the Avometer as shown in figure. Then we will adjust the selector Switch on some particular range value of resistance. Then by adjusting the selector switch, the galvanometer will take a connection with a battery of Avometer and with a material.



Then the battery will pass the current through material and as well as from galvanometer.

Then with respect to current flow from the material we will find the resistance of that material from the scale according to a selected range value of resistance.

Digital Multimeter

Digital multimeters are Avometers, these also measures the current, voltage and resistance but gives the answers in digits with concern units. Digital multimeters have additional digital circuits instead of galvanometer. Such digital circuits work on electrical signals and give us the result in digits with concern units. The results of digital multimeters are accurate, reliable and

durable. Digital multimeters prevent the parallax error as it is existing in Avometer. Digital multimeters have additional functional as well, due to which its scope of use is very high.

Magnetic Resonance imaging (MRI)

Magnetic Resonance imaging (MRI) is one of the most advance diagnostic tool available today. MRI uses a combination of strong magnetic field and radio waves (Electromagnetic waves) to produce high resolution images of the interior part of the body.

Working: The human body is composed of fats and water, which are made up mostly of hydrogen atoms (protons). Now when a radio waves are applied to a specific part of the body from radio frequency coil, then resonance occurs to protons (Hydrogen atoms) and will start motion in the magnetic field. Then the strong magnetic field will exert a force on protons and push them back, Now when a protons comes back to the initial position, then these protons will also emit the radio waves as charges does in antenna. Now emitted radio waves by protons are then received by another coil of MRI machine. Then MRI machine processes these waves and get information about the image from these waves. Then MRI machine send their information to computer set up. Then computer set up gives us the high resolution images of the interior part of the body.

Examples and Assignments

Example 13.1: A wire carrying a current of 2A and having length 10cm between the poles of a magnet is kept at an angle of 30° to a uniform magnetic field of 0.6T. Find the force acting on the wire.

Given:

$$I = 2A, \quad L = 10\text{cm} = 0.1\text{m}, \quad B = 0.6\text{T}, \quad \theta = 30^\circ \quad \text{Req: Magnetic force } F_B = ?$$

Sol: As we know that

$$F_B = ILB \sin\theta$$

$$\Rightarrow F_B = 2 \times 0.1 \times 0.6 \times \sin 30^\circ$$

$$\Rightarrow F_B = 2 \times 0.1 \times 0.6 \times 0.5$$

$$\Rightarrow F_B = 0.06\text{N}$$

Example 13.2: Two long parallel wires 6cm apart carry currents of 8A and 2A in the same direction. What is the magnitude of magnetic field midway between them?

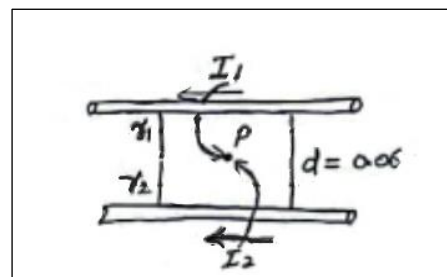
Given:

$$d = 6\text{cm} = 0.06\text{m} \quad r_1 = r_2 = r = d/2 = 0.03\text{m}.$$

$$I_1 = 8A, \quad I_2 = 2A$$

Req: Magnetic field $B_{\text{net}} = ?$

Sol: At midpoint P, the magnetic field can be written as



$$\begin{aligned}
 B_{\text{net}} &= B_1 - B_2. \text{ Negative sign is due to opposite direction of magnetic field} \\
 \Rightarrow B_{\text{net}} &= \mu_0 I_1 / 2\pi r - \mu_0 I_2 / 2\pi r \\
 \Rightarrow B_{\text{net}} &= \mu_0 / 2\pi r (I_1 - I_2) \\
 \Rightarrow B_{\text{net}} &= 4\pi \times 10^{-7} / 2\pi \times 0.03 \times (8 - 2) \\
 \Rightarrow B_{\text{net}} &= 2 \times 10^{-7} / 0.03 \times (6) \\
 \Rightarrow B_{\text{net}} &= 4 \times 10^{-5} \text{ T}
 \end{aligned}$$

Assignment 13-1: A long straight wire carries a current of 1A. Find the magnitude of the magnetic field at a distance of 1m from it.

Given: $I = 1\text{A}$, $r = 1\text{m}$. Req: Magnetic field $B = ?$

Sol: As we know that

$$\begin{aligned}
 B &= \mu_0 I / 2\pi r \\
 \Rightarrow B &= 4\pi \times 10^{-7} \times 1 / 2\pi \times 1 \\
 \Rightarrow B &= 2 \times 10^{-7} \text{ T}
 \end{aligned}$$

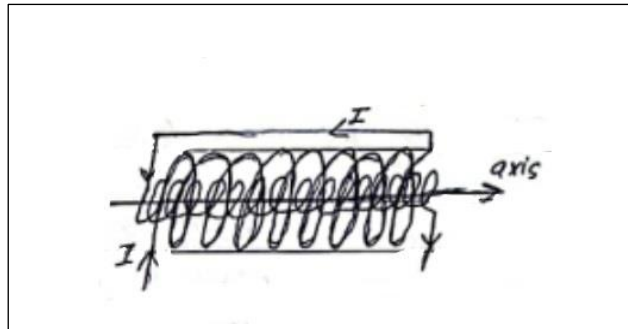
Example 13.3: A solenoid is 10cm long and is wound with two layers of wire. The inner layer has 50 turns and the outer layer has 40 turns. A current of 3A flows in both layers in the same direction. Find the magnetic flux density along the axis of the solenoid.

Given: $L = 10\text{cm} = 0.1\text{m}$. $N_1 = 50$ turns, $N_2 = 40$ turns. $I_1 = I_2 = I = 3\text{A}$

Req: Magnetic flux density $B_{\text{net}} = ?$

Sol: Since both layers carry current in same direction

$$\begin{aligned}
 \Rightarrow B_{\text{net}} &= B_1 + B_2 \\
 \Rightarrow B_{\text{net}} &= \mu_0 N_1 I / L + \mu_0 N_2 I / L \\
 \Rightarrow B_{\text{net}} &= \mu_0 I / L (N_1 + N_2) \\
 \Rightarrow B_{\text{net}} &= 4\pi \times 10^{-7} \times 3 / 0.1 (50 + 40) \\
 \Rightarrow B_{\text{net}} &= 3.4 \times 10^{-3} \text{ T}
 \end{aligned}$$



Assignment 13.2: You are asked to design a solenoid that will produce a magnetic field of 0.10T, while the current must not exceed 10A. Find the number of turns per unit length that the solenoid should have.

Given: $B = 0.10\text{T}$, $I = 10\text{A}$. Req: $n = ?$

Sol: As we know that

$$\begin{aligned}
 B &= \mu_0 n I \\
 \Rightarrow n &= B / \mu_0 I \\
 \Rightarrow n &= 0.10 / 4\pi \times 10^{-7} \times 10 \\
 \Rightarrow n &= 7.96 \times 10^3 \text{ turns/m}
 \end{aligned}$$

Example 13.4: The path of an electron in a uniform magnetic field of flux density $1 \times 10^{-2}\text{T}$ in vacuum is circular with radius 1cm. Calculate the time period of its orbit.

Given: $B = 1 \times 10^{-2} \text{T}$, $r = 0.01 \text{m}$. Req: Time period $T = ?$

Sol: As we know that

$$T = 2\pi m/qB$$

$$\Rightarrow T = 2 \times 3.14 \times 9.1 \times 10^{-31} / 1.6 \times 10^{-19} \times 1 \times 10^{-2}$$

$$\Rightarrow T = 3.57 \times 10^{-9} \text{s}$$

Assignment 13.3: A velocity selector in a mass spectrometer uses a magnetic field of 0.100T . What electric field strength is needed to select a speed of $4 \times 10^6 \text{m/s}$?

Given: $B = 0.100 \text{T}$, $v = 4 \times 10^6 \text{m/s}$. Req: Electric field $E = ?$

Sol: As we know that

$$v = E/B$$

$$\Rightarrow E = vB$$

$$\Rightarrow E = 4 \times 10^6 \times 0.100$$

$$\Rightarrow E = 4 \times 10^5 \text{N/C}$$

Example 13.5: A square loop of wire of side 4cm carries a current of 3A . A uniform magnetic field of 0.6T makes an angle of 30° with the plane of the loop.

(a) What is the magnitude of the torque on the loop?

(b) What is the net magnetic force on the loop?

Given: $N = 1$, $I = 3 \text{A}$. $A = 0.0016 \text{m}^2$ $B = 0.6 \text{T}$, $\theta = 60^\circ$, $\phi = 30^\circ$

Req: a) Torque = ? b) Net force = $F_{\text{net}} = ?$

Sol (a) : As we know that

$$\tau = NIAB \sin \theta$$

$$\Rightarrow \tau = 1 \times 3 \times 0.0016 \times 0.6 \times \sin 60^\circ$$

$$\Rightarrow \tau = 2.5 \times 10^{-3} \text{Nm}$$

Sol (b) : Now we find net force on the loop. So we can write as

$$F_{\text{net}} = F_{\text{Left}} + F_{\text{Right}} + F_{\text{Top}} + F_{\text{Bottom}} \text{ ----- (1)}$$

Now we can write as

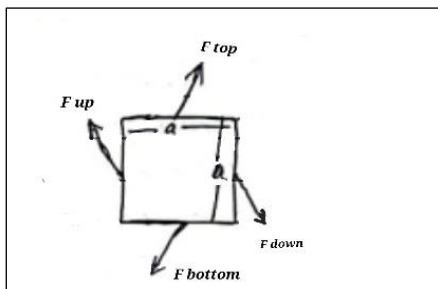
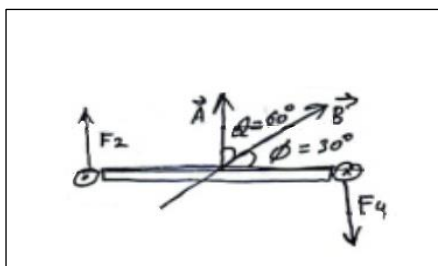
$$\Rightarrow F_{\text{Left}} = -F_{\text{Right}} \quad \text{and}$$

$$\Rightarrow F_{\text{Top}} = -F_{\text{Bottom}}$$

Now eq (1) we can write as

$$F_{\text{net}} = -F_{\text{Right}} + F_{\text{Right}} - F_{\text{Bottom}} + F_{\text{Bottom}}$$

$$\Rightarrow F_{\text{net}} = 0 \text{ N}$$



Example 13.6: A galvanometer has resistance 100Ω and gives full-scale deflection at 1mA . How it can be converted into (a) an ammeter of range 10A . (b) voltmeter of range 10V .

Given: $R_g = 100 \Omega$. $I_g = 10^{-3} \text{A}$ $I = 10 \text{A}$. $V = 10 \text{V}$

Conversion to Ammeter: For this we find " R_s " first. i.e

$$R_s = R_g / I - I_g$$

$$\Rightarrow R_s = 10^{-3} \times 100 / 10 = 10^{-3}$$

$$\Rightarrow R_s = 0.01\Omega$$

By connecting such a shunt resistance in parallel with galvanometer, then galvanometer will be converted to ammeter and will show us the current up to 10A range.

Conversion to Voltmeter: For this first we find " R_h ". i.e

$$R_h = V / I_g - R_g$$

$$\Rightarrow R_h = 10 / 10^{-3} - 100$$

$$\Rightarrow R_h = 10000 - 100$$

$$\Rightarrow R_h = 9900\Omega$$

By connecting such a high resistance in series with galvanometer, then galvanometer will be converted to voltmeter and will show us the voltage up to 10V range.

Exercise MCQs

- ① A current is flowing north in a magnetic field that points west. It experiences
(a) A force down (b) A force up ✓ (c) A force west (d) No force

- ② Two long, straight wires have currents flowing in them in the same direction. The force between the wires is

- (a) Attractive ✓ (b) Repulsive (c) Zero (d) Infinite

Sol: See figure on page 150 in example 13.2. The two fields will be in opposite directions, and opposite fields will produce attractive force.

- ③ One Weber is equal to

- (a) NA^2/m (b) Nm^2/A (c) NA/m (d) Nm/A ✓

Sol: As we know that

$$\Phi = BA \text{ ----- (1)}$$

We also know that

$$F = ILB \Rightarrow B = F/IL \text{ ----- (2) Put (2) in (1) we get as}$$

$$\Phi = FA/IL \Rightarrow \Phi = \text{Nm}^2/\text{Am} \Rightarrow \Phi = \text{Nm}/\text{A} \Rightarrow 1 \text{ weber} = \text{Nm}/\text{A} \text{ (ANS)}$$

- ④ The magnetic flux will be maximum if the angle between magnetic field strength and vector area is

- (a) 0° ✓ (b) 45° (c) 60° (d) 90°

Sol: As we know that

$$\Phi = BA \cos\theta \Rightarrow \Phi_{\max} = BA \text{ when } \theta = 0^\circ$$

- ⑤ Ampere's law is applied over any

- (a) Surface (b) Closed surface (c) Path (d) Closed path ✓

- ⑥ When the number of turns in a solenoid is doubled without any change in the length of solenoid, its self-induction will be

- (a) Four times ✓ (b) Double (c) Halved (d) One fourth

Sol: From next chapter we can write

$$L = N\Phi / I$$

$$\Rightarrow L = NBA / I \text{ ----- (1)}$$

We also know that

$$B = \mu_0 (N/\ell) I \text{ ----- (2)}$$

Now put (2) in eq (1) we get as

$$\Rightarrow L = \mu_0 N^2 A / \ell$$

If N becomes 2N. Then we can write as

$$L' = \mu_0 (2N)^2 / \ell \Rightarrow L' = 4\mu_0 N^2 / \ell \Rightarrow L' = 4L$$

⑦ If the charge is at rest in a magnetic field, then the force on the charge is

- (a) qvB (b) $qvB\cos\theta$ (c) $qvB\sin\theta$ (d) Zero ✓

Sol: As we know that $F = qvB \sin\theta$. Now put $v = 0$. Then we get as $\Rightarrow F = 0$

⑧ Work done on a charged particle moving in a uniform field along a closed path is

- (a) Maximum (b) Minimum (c) Zero ✓ (d) Infinite

⑨ The Earth's northern magnetic pole acts like

- (a) The north pole of a magnet (b) The south pole of a magnet ✓
(c) It has a positive charge (d) It has a negative charge

⑩ The pole pieces of the magnet in a galvanometer are made concave to make the field

- (a) Radial (b) Stronger (c) Weaker (d) Both A and B ✓

⑪ When a small resistance is connected parallel to a galvanometer, the resulting circuit behaves as

- (a) Voltmeter (b) Ammeter ✓ (c) Velocity selector (d) Avometer

⑫ To measure voltage, the voltmeter is connected with the circuit in

- (a) Series (b) Parallel ✓ (c) Perpendicular (d) Straight

⑬ The resistance of an ideal voltmeter is

- (a) Small (b) Large (c) Zero (d) Infinite ✓

⑭ The torque in the coil can be increased by increasing

- (a) Number of turns
(b) Current and magnetic field
(c) Area of coil
(d) All of these ✓

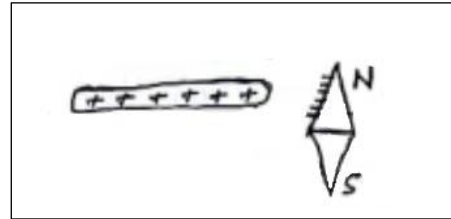
Conceptual Questions

Question 01: A compass needle is deflected when a charged plastic rod is held near it. What is the origin of the force that produces the deflection?

Ans: The origin of the force that produces the deflection is electrostatic induction.

Explanation: A compass needle is deflected by a charged plastic rod due to an electrostatic induction. A compass needle is made of conductor, which means that it has free electrons.

Now if we bring a charged plastic rod close to the needle, then electrostatic induction will occur to the compass needle, and due to which



the electrons will move in the needle towards a charged plastic rod. Then Charged plastic rod will exert an attractive force on the induced electrons in the needle and then due to such a force a compass needle will be deflected towards a charged plastic rod.

Question 02: What is the difference between permeability and permittivity?

Ans: Permittivity

- ① The opposition of the medium to the flow of electric field lines is called permittivity.
- ② It is a property related with Electric field.
- ③ The permittivity is because of polarization.
- ④ Its unit is C^2/Nm^2 .
- ⑤ Due to which we can construct's good capacitors
- ⑥ Its value is $\epsilon_0 = 8.85 \times 10^{-12} C^2/Nm^2$.

Permeability

- ① The permission of the medium to the flow of magnetic field lines is called. permeability.
- ② It is a property related with magnetic field.
- ③ The permeability is because of magnetization.
- ④ Its unit is Wb/Am
- ⑤ Due to which we can constructs a good transformers and good electric motors etc.
- ⑥ Its value is $\mu_0 = 4\pi \times 10^{-7} Wb/Am$.

Question 03: Electron and proton are projected with the same velocity normal to the magnetic field. Which one will suffer greater deflection. why?

Ans: When electron and proton are projected with the same velocity normal to the magnetic field, then these particles will start circular motion due to an effect of magnetic force on these two particles as shown in figure. Now such a magnetic force will behave as a centripetal force. So we can write as

$$F_C = F_B$$

$$\Rightarrow mv^2/r = qvB$$

$$\Rightarrow mv/r = qB$$

$$\Rightarrow r = mv/qB$$

$\Rightarrow r \propto m$ ----- ① (By keeping “q” and “B” are constant).

Now from relation (1) we can say that the radius of an electron will be less due to its less mass as compared with proton. So now we can say that, due to less radius, the electron will suffer greater deflection as compared with proton.

Question 04: A charged particle moves in a straight line through a particular region of space. Could there be non-zero magnetic field in this region explain.

Ans: When a charged particle moves in a straight line through a particular region of space. Then magnetic field may or may not be zero.

Case 1: When magnetic field is zero in the region:

When magnetic field is zero in the region. Then the magnetic force on the charge particle will be also zero and due to which a charged particle will move in a straight line in this region. Now we can write as

$$F_m = q v B \sin\theta \Rightarrow F_m = q v \times 0 \times \sin\theta \Rightarrow F_m = 0$$

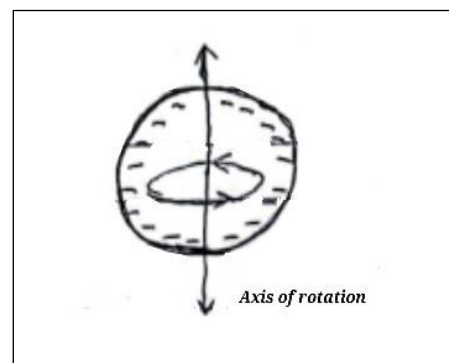
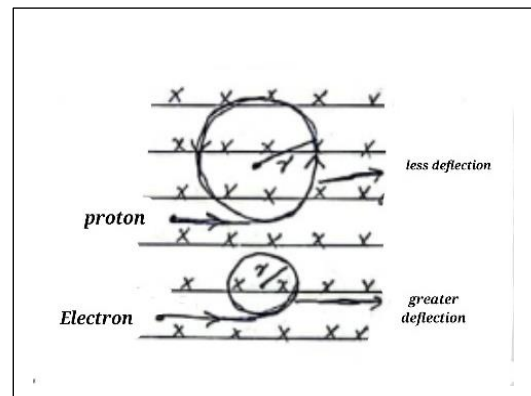
Case 2: when magnetic field is non zero and $\theta = 0^\circ$ or $\theta = 180^\circ$.

When magnetic field is non zero in the region and angle “ θ ” is zero between the charged particle velocity and this magnetic field. Then due to which the magnetic force will be zero on the charged particle and then the charged particle will move in a straight line in this region. Now we can write as

$$F_m = q v B \sin\theta \Rightarrow F_m = q v B \sin 0^\circ \Rightarrow F_m = 0$$

Question 05: Can the objects in rotational motion create a magnetic field? Provide an example.

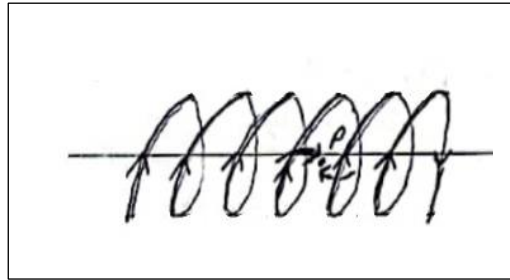
Ans: A rotating conducting sphere is shown in figure. Due to its rotation there will be a centripetal force on the free electrons in it. Then due to such a force the electrons will move towards the surface. Then due to these electron rotation on the surface with the rotation of conducting sphere, there will be producing the magnetic fields because we know that the magnetic fields are producing



due to a charges motion. The strength of such magnetic fields depends on the angular velocity of a rotating conducting sphere. If angular velocity is greater the strength of magnetic fields will be also greater and vice versa. Thus the objects in rotational motion create magnetic fields.

Question 06: If a current is passed through an unstratched spring, will the spring contract or expand? Explain.

Ans: A current carrying unstratched spring is shown in figure in which a current is flowing in upward direction. Now the magnetic fields produced by a current flow between the two adjacent coils are also shown in figure at point "P". Now at point "P" these magnetic fields are in opposite direction and due to which an attractive force will be producing between the two coils. Then due to such attractive force the two adjacent coils will attract each other. Now everywhere between the two coils there will be an attractive force and due to which all coils of the spring will attract each other and due to which a spring will be contracting.



Question 07: How can neutrons be accelerated in a cyclotron?

Explain.

Ans: A neutrons can not be accelerated in cyclotron.

Explanation: Cyclotron is a type of particle accelerator in which a charged particle is accelerated towards a target under the effect of electric and magnetic fields. Now we know that the neutrons are chargless particles, so in cyclotron there will be no effect of electric and magnetic fields on the neutrons. So by that reason we can say that the neutrons can not be accelerated in cyclotron.

Question 08: A current carrying loop, free to turn, is placed in a uniform field 'B'. What will be its orientation relative to "B" in the equilibrium state?

Ans: A current carrying loop is shown in magnetic field "B" with current "I" turns 'N' and area "A" in a figure. Now such a current carrying loop is said to be in equilibrium state for which when the torque effect is zero. i.e

$$\tau = 0 \text{ ----- (1)}$$

Now we know that Torque in a loop can be written as

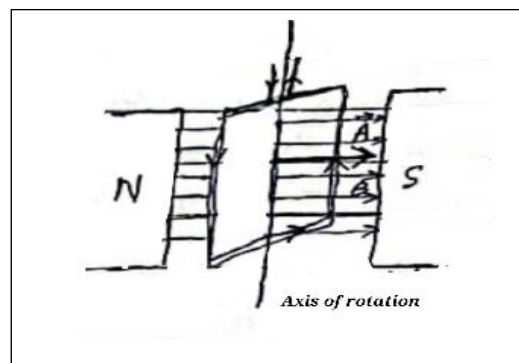
$$\tau = NIAB \sin\theta \text{ ----- (2)}$$

Now put eq (1) in eq (2) we get as

$$NIAB \sin\theta = 0$$

$$\Rightarrow \sin\theta = 0/NIAB$$

$$\Rightarrow \sin\theta = 0$$



$$\Rightarrow \theta = \sin^{-1} 0$$

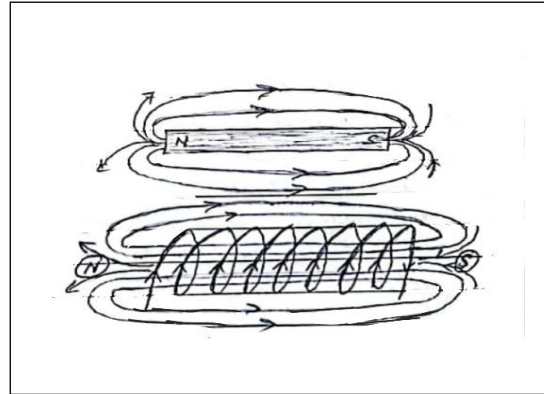
$$\Rightarrow \theta = 0^\circ$$

Now from equation we can say that the current carrying loop will be in equilibrium state when the angle “ θ ” is zero between its Area vector (A) and magnetic field (B).

Question 09: How does a current carrying coil behave like a bar magnet?

Ans: A bar magnet has a definite north and south poles as shown in figure. In which a magnetic field lines starts from North Pole and enters on South Pole.

Now a current carrying coils is also shown in figure in which a current is flowing in upward direction. Now the magnetic fields produced inside and out side of the coils are also shown in a figure. Now externally of the coils, the magnetic field lines start from LHS and enter on RHS. So LHS of the coil behave as a North Pole of a bar magnet and RHS behave as a South Pole of a bar magnet.



Numerical Questions

Q-01: At what distance from a long straight wire carrying a current of 10A is the magnetic field is equal to the earth's magnetic field of $5 \times 10^{-5} \text{T}$?

Given: $I = 10\text{A}$ $B = 5 \times 10^{-5}\text{T}$ Req: Distance from wire = $r = ?$

Sol: As we know that from amperes law

$$B = \mu_0 I / 2\pi r$$

$$\Rightarrow r = \mu_0 I / 2\pi B$$

$$\Rightarrow r = 4\pi \times 10^{-7} \text{Tm A}^{-1} \times 10\text{A} / 2\pi \times 5 \times 10^{-5}\text{T} = 0.04\text{m}$$

Q.02: A long solenoid having 1000 turns uniformly distributed over a length of 0.5m produces a magnetic field of $2.5 \times 10^{-3}\text{T}$ at the center. Find the current in the solenoid.

Given: $N = 1000$ $L = 0.5\text{m}$ $B = 2.5 \times 10^{-3}\text{T}$. Required: current = $I = ?$

Sol: As we know that in case of solenoid

$$B = \mu_0 n I$$

$$\Rightarrow B = \mu_0 N / L \times I$$

$$\Rightarrow I = BL / \mu_0 N$$

$$\Rightarrow I = 2.5 \times 10^{-3}\text{T} \times 0.5\text{m} / 4\pi \times 10^{-7} \text{Tm A}^{-1} \times 1000$$

$$\Rightarrow I = 1\text{A Ans}$$

Q.03: A proton moving at right angles to a magnetic field of 0.1T experiences a force of $2 \times 10^{-12}\text{N}$. What is the speed of the proton?

Given: $B = 0.1\text{T}$ $q = 1.6 \times 10^{-19}\text{C}$. $F = 2 \times 10^{-12}\text{N}$. $\theta = 90^\circ$ Req: $v = ?$

Sol: As we know that magnetic force on a moving charge is

$$F = qvB \sin\theta$$

$$\Rightarrow v = F / qB \sin\theta$$

$$\Rightarrow v = 2 \times 10^{-12}\text{N} / 1.6 \times 10^{-19}\text{C} \times 0.1\text{T} \sin 90^\circ$$

$$\Rightarrow v = 1.25 \times 10^8 \text{ m/s Ans}$$

Q.4: An 8MeV proton enters perpendicularly into a uniform magnetic field of 2.5T. Find (a) the force on the proton (b) what will be the radius of the path of proton?

Given: K.E = 8 MeV = $8 \times 10^6 \times 1.6 \times 10^{-19} \text{ J} = 12.8 \times 10^{-13} \text{ J}$. $\theta = 90^\circ$ $B = 2.5\text{T}$

mass of proton = $m = 1.67 \times 10^{-27} \text{ kg}$ Req: (a) $F = ?$ (b) Radius = $r = ?$

Sol:(a) As we know that the magnetic force on moving charge is

$$F = qvB \sin\theta \text{ ----- (1)}$$

Now 1st we find "v". It is given that

$$\text{K.E} = 12.8 \times 10^{-13} \text{ J}$$

$$\Rightarrow \frac{1}{2} mv^2 = 12.8 \times 10^{-13} \text{ J} \Rightarrow v^2 = 2 \times 12.8 \times 10^{-13} \text{ J/m}$$

$$\Rightarrow v = \sqrt{(2 \times 12.8 \times 10^{-13} / 1.67 \times 10^{-27} \text{ kg})}$$

$$\Rightarrow v = 3.9 \times 10^7 \text{ m/s. Now eq (1) we can write as}$$

$$\Rightarrow F = 1.6 \times 10^{-19} \text{ C} \times 3.9 \times 10^7 \text{ m/s} \times 2.5\text{T} \sin 90^\circ$$

$$\Rightarrow F = 1.6 \times 10^{-11} \text{ N Ans.}$$

Sol (b) Now such a magnetic force will behave as a centripetal force so we can write as

$$F_c = F_m$$

$$\Rightarrow mv^2 / r = F_m$$

$$\Rightarrow r = mv^2 / F_m$$

$$\Rightarrow r = 1.67 \times 10^{-27} \text{ kg} \times (3.9 \times 10^7 \text{ m/s})^2 / (1.6 \times 10^{-11} \text{ N})$$

$$\Rightarrow r = 0.16 \text{ m Ans.}$$

Q.5: What is the time period of an electron projected into a uniform magnetic field of 10mT and moves in a circle of radius 6cm?

Given:

$B = 10 \times 10^{-3} \text{ T}$ $r = 6\text{cm} = 6 \times 10^{-2} \text{ m}$ Req : $T = ?$

Sol. As we know that

$$T = 2\pi m / qB$$

$$\Rightarrow T = 2 \times 3.14 \times 9.11 \times 10^{-31} / ((1.6 \times 10^{-19} \text{ C}) (10 \times 10^{-3})) \text{ kg}$$

$$\Rightarrow T = 3.6 \times 10^{-9} \text{ s}$$

$$\Rightarrow T = 3.6 \text{ ns Ans.}$$

Q.6: A rectangular coil with 250 turns is 6cm long and 4cm wide. When the coil is placed in a magnetic field of 0.25T its maximum torque is 0.20Nm. What is the current in the coil?

Given $N = 250$. $L = 6\text{cm} = 0.06\text{m}$. $W = 4\text{cm} = 0.04\text{m}$

$$A = l \times W = 0.06\text{m} \times 0.04\text{m} = 2.4 \times 10^{-3} \text{ m}^2 \quad B = 0.25\text{T} \quad \tau_{\max} = 0.20\text{Nm}$$

Req: Current $I = ?$

Sol: As we know that about Torque in a coil is

$$\tau = NIAB \sin \theta$$

$$\Rightarrow \tau_{\max} = NIAB \sin 90^\circ$$

$$\Rightarrow \tau_{\max} = NIAB$$

$$\Rightarrow I = \tau_{\max} / NAB$$

$$\Rightarrow I = 0.20 \text{ Nm} / 250 \times 2.4 \times 10^{-3} \text{ m} \times 0.25 \text{ T}$$

$$\Rightarrow I = 1.33 \text{ A Ans}$$

Q.7 Cross electric and magnetic fields are established over a certain region. The magnetic field is 0.105T and the electric field is $2 \times 10^5 \text{ V/m}$. An electron experiences zero net force from these fields and so continuous moving in a straight line. What is the electron's speed.

Given $B = 0.105 \text{ T}$. $E = 2 \times 10^5 \text{ V/m}$. Req : $v = ?$

Sol: As we know that in the velocity selector

$$v = E/B$$

$$\Rightarrow v = 2 \times 10^5 \text{ V/m} / 0.105 \text{ T}$$

$$\Rightarrow v = 1.905 \times 10^6 \text{ m/s} / \text{Ans.}$$

Q.8 : Full scale deflection for galvanometer is 10 mA. Its resistance is 100Ω . How it be converted into ammeter of range 100A?

Given: $I_g = 10 \text{ mA} = 0.01 \text{ A}$ $R_g = 100 \Omega$ $I = 100 \text{ A}$

Sol: First we find shunt resistance for the conversion of galvanometer to ammeter. For this we know that

$$R_s = (I_g R_g) / (I - I_g)$$

$$\Rightarrow R_s = (0.01 \text{ A} \times 100 \Omega) / (100 \text{ A} - 0.01 \text{ A})$$

$$\Rightarrow R_s = 0.01 \Omega$$

By connecting such a shunt resistance in parallel with galvanometer, then the galvanometer will be converted to ammeter and will shows us the current up to 100A range.

Q.9: How 5mA, 100Ω galvanometer converted to 20V voltmeter?

Given: $I_g = 5 \text{ mA} = 5 \times 10^{-3} \text{ A}$ $R_g = 100 \Omega$ $V = 20 \text{ V}$

Sol: First we find high resistance for the conversion of galvanometer to voltmeter. For this we know that

$$R_h = V / I_g - R_g$$

$$\Rightarrow R_h = 20 \text{ V} / 5 \times 10^{-3} \text{ A} - 100 \Omega$$

$$\Rightarrow R_h = 3900 \Omega$$

By connecting such a high resistance in series with galvanometer, then the galvanometer will be converted to voltmeter and will show us the voltage up to 20V.

Q.10: Two parallel wires 10cm apart carry current in opposite direction of 8A. What is the magnetic field halfway between them.

Given: $d=10\text{cm} \Rightarrow r_1 = r_2 = 10\text{cm}/2 = 5\text{cm} = 0.05\text{m}$. $I_1 = I_2 = 8\text{A}$, $r_2 = 0.05\text{m}$

Req: Magnetic field = $B = ??$ at mid point

Sol: Now we find " B_{net} " at point 'P'. Now at Point "P" " B_1 " and " B_2 " are in parallels. So we can write as

$$B_{\text{net}} = B_1 + B_2$$

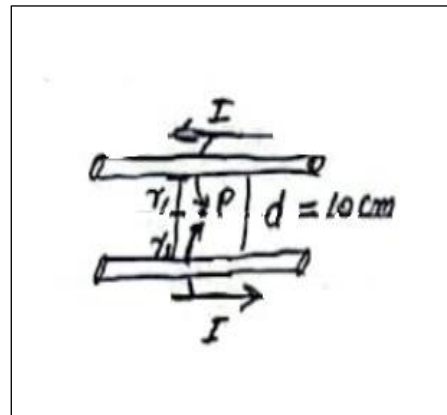
$$\Rightarrow B_{\text{net}} = \mu_0 I_1 / 2\pi r_1 + \mu_0 I_2 / 2\pi r_2$$

$$\Rightarrow B_{\text{net}} = 4\pi \times 10^{-7} \text{ Tm/A} \times 8\text{A} / 2\pi \times 0.05 \text{ m} + 4\pi \times 10^{-7} \text{ Tm/A} \times 8\text{A} / 2\pi \times 0.05\text{m}$$

$$\Rightarrow B_{\text{net}} = 320 \times 10^{-7} \text{ T} + 320 \times 10^{-7} \text{ T}$$

$$\Rightarrow B_{\text{net}} = 640 \times 10^{-7} \text{ T}$$

$$\Rightarrow B_{\text{net}} = 6.4 \times 10^{-5} \text{ T}$$



END OF CHAPTER 03

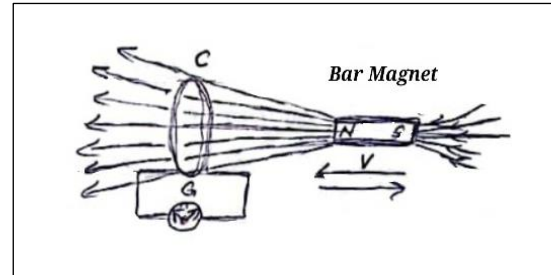
Chapter # 04 Electromagnetic induction

Electromagnetic Induction: The phenomenon in which the emf is induced (produced) due to a magnetic flux change linking with coil or conductor is called electromagnetic induction.

Explanation of induced emf due to a magnetic flux change;

The induced emf due to a magnetic flux change we explain with the help of following experiments.

Experiment-1: Consider a coil which is connected to galvanometer as shown in the figure. When a bar magnet is held stationary near the coil "C" the galvanometer shows no current this means that the emf is zero in the circuit.



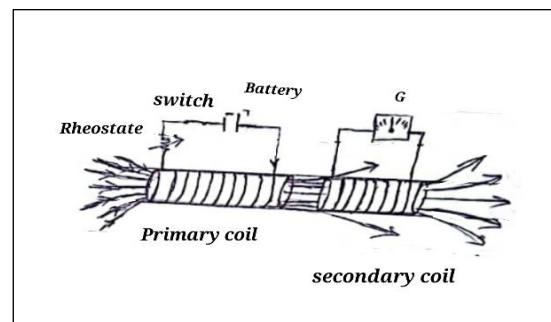
Now if we move the bar magnet towards the coil "C" the galvanometer shows deflection in one direction. This is because when we move the bar magnet towards the coil, then the magnetic flux changes in the coil and due to which an emf is produced. Then such emf produces the current in the circuit and due to which the galvanometer shows deflection in one direction.

Similarly if we move the bar magnet away from the coil, the galvanometer shows deflection in opposite direction. This is because when we move the bar magnet away from the coil, then the magnetic flux changes in the coil and due to which an emf is produced in opposite direction. Then such emf produces the current in the circuit and due to which the galvanometer shows deflection in opposite direction.

The production of an emf and current in the circuit is associated with the magnetic flux change in the coil. Now if the movement of magnet is stopped then there will be no change in the magnetic flux. As a result, there will be no induced emf and current in the circuit and thus the galvanometer will show zero deflection.

Experiment 02: In a figure a two coils are shown in figure. One coil is primary coil in which a battery, switch and rheostat are connected with coil and the second one is secondary coil in which a galvanometer is connected with the coil.

Now by decreasing the resistance of rheostat, the current in the primary coil is increasing, due to which a magnetic flux changes in primary coil as well as in secondary coil. Then due to such a magnetic flux change in the secondary coil, the emf is produced. Then such emf produces the current in the secondary coil and due to which the galvanometer shows deflection in one direction.



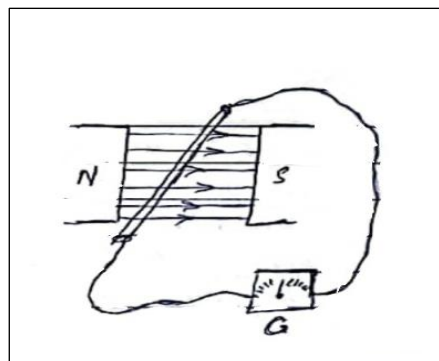
Now by increasing the resistance of rheostat, the current in the primary coil is decreasing, due to which a magnetic flux changes (decreasing) in primary coil as well as in secondary coil. Then due to such a magnetic flux change in the secondary coil, the emf is produced. Then such emf produces the current in the secondary coil and due to which the galvanometer shows deflection in opposite direction.

The production of an emf and current is associated with the magnetic flux change in the secondary coil. Now if changes of resistance are stopped, then there will be no change in magnetic flux due to a constant current in the primary coil. As a result, there will be no induced emf and current in the coil and thus the galvanometer will show zero deflection.

Experiment 03: In a figure a conductor of length "L" connected with galvanometer by wire in the form of loop is shown in horizontal magnetic field. Now when the loop is held stationary in the field, the galvanometer indicates no deflection, it means that there is no emf and current in the loop.

Now when the loop is moved upwards, the galvanometer deflects to one side, it means that the emf and current is produced in the loop due to magnetic flux changes (increasing) in the loop.

Now when the loop is moved downwards, the galvanometer deflects in opposite direction, it means that the emf and current is produced in opposite direction in the loop due to a magnetic flux changes (decreasing) in the loop.



Thus from the above experiments we can say that due to a magnetic flux changes in increasing or decreasing form in the coils or loops; the emf will be producing in it. Then such emf will produce the current in it, and such phenomenon is called electromagnetic induction.

Faraday's Law of electromagnetic Induction

This law states that

- ① The emf is directly proportional to the rate of change of magnetic flux $\Delta\Phi$ in time Δt . Greater the rate of change in magnetic flux through the coil, greater is induced emf. i.e.

$$\varepsilon \propto \Delta\Phi/\Delta t \text{ ----- ①}$$

- ② The emf is directly proportional to the number of turns 'N' in the coil. Greater the number of turns, greater is induced emf. i.e

$$\varepsilon \propto N \text{ ----- ②}$$

Now combine relation ① and ②. we get as

$$\varepsilon \propto N\Phi/\Delta t$$

$\Rightarrow \varepsilon = KN\Phi/\Delta t$. Where "K" is constant and take $k=1$.

$$\Rightarrow \varepsilon = N\Phi/\Delta t \text{ ----- ③}$$

Now according to Lenz's law the induced emf always opposes the change in

magnetic flux. Therefore eq (3) can be written as

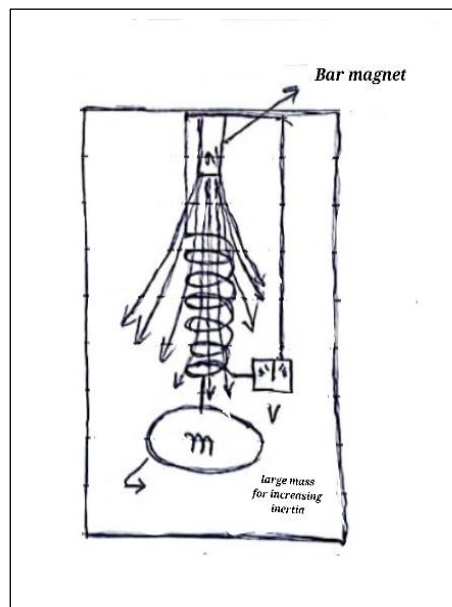
$$\varepsilon = -N\Phi/\Delta t$$

Application of electromagnetic induction

Electromagnetic seismometer: A device through which we measure the earthquakes is called seismometer. The shaking of earth surface is called earth quake.

Working: In seismometer figure the bar magnet, voltage and suspended mass spring system are shown in figure.

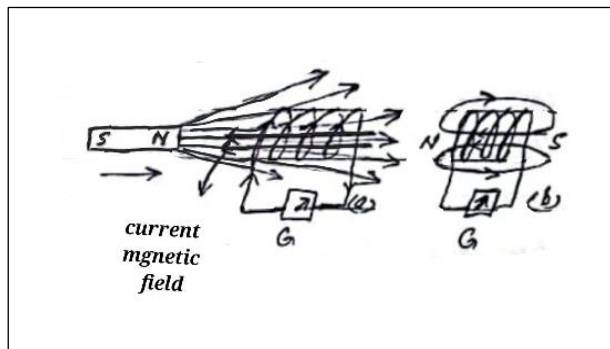
Now during earth quake, the whole seismometer along with bar magnet vibrates up down and the suspended mass spring system is remain stationary due to its greater inertia. Now due to a bar magnet vibration, the magnetic flux changes in the stationary spring mass system, and due to which the emf is produced. Then voltmeter measures such an induced emf. Then from the value of induced emf the earthquake is measured.



Lenz's Law

Statement: The induced current or emf in the coils we will take in such a direction, such that to oppose the change of magnetic flux in the coils.

Explanation: (1) In a figure, a North Pole of a magnet is moving towards the coil, then due to which a magnetic flux will be increasing in the coil. Then Lenz's law says that take a direction of induced current or emf in such a direction, such that, then the current magnetic field is to decrease, the increase of magnetic flux as shown in fig. So from this law, we can say that the induced



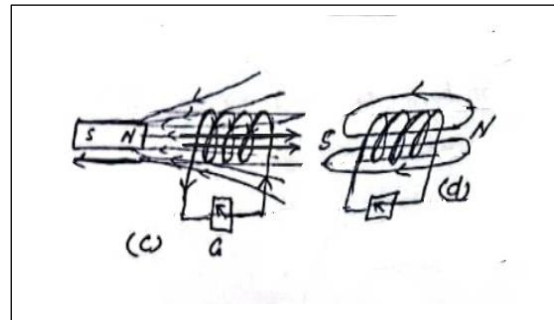
current or emf in the coil will be clock wise. Now the magnetic field lines produced by induced current as shown in fig (b). According to the direction of these field lines the L.H.S of the coil will behave as North and R.H.S will behave as South.

Now from this we can say that when we move a bar magnet towards the coil, then the coil L.H.S face will behave as a North and due to which there will be a repulsive force on the magnet.

Now more work will be needed on a magnet to move it towards the coil and such a work then changes to electrical energy in the coil. So in Lenz's law,

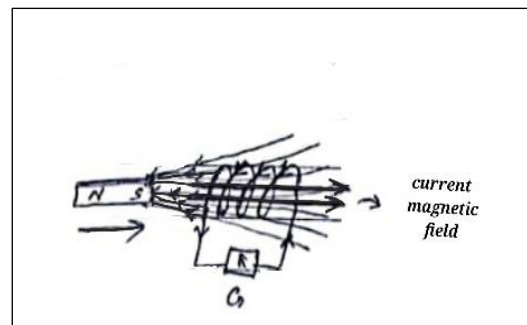
conservation of energy is also take place and by that reason, it is also called law of conservation of energy.

② In a figure a North pole of magnet is moving away from the coil. Then due to which a magnetic flux will be decreasing in the coil. The Lenz's law says that take a direction of induced current or emf in such a direction, Such that, then the current magnetic field is to increase, the decrease of magnetic flux as shown in figure. So from this law we can say that the induced current or emf in the coil will be anticlock wise. Now the magnetic field lines produced by an induced current as shown in figure . According to the direction of these field lines the L.H.S of the coil will behave as a south and R.H.S will behave as North. Now from this we can say that when we move a bar magnet away from the coils, then the coil L.H.S face will behave as a south and due to which there will be an attractive force on the magnet.

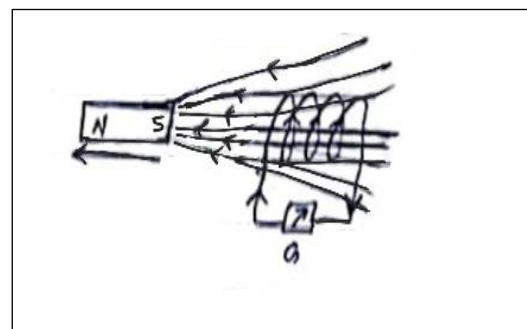


Now more work will be needed on a magnet to move it away from the coil and such a work then changes to electrical energy in the coil.

③ in case of South Pole of a magnet when it is moving towards the coil, the induced current or emf will be anticlock wise as shown in figure. And there will be repulsive force on the magnet.



④ In case of South Pole of a magnet when it is moving away from the coil, the induced current or emf will be clock wise as shown in figure. And there will be an attractive force on the magnet.



In both last two cases the work on a magnet will be converted to electrical energy.

Motional emf

The emf produced due to a motion of a conductor in a uniform magnetic field is called motional emf.

Explanation: Consider a uniform magnetic field in which a conductor "NM" is placed on U-shaped conductor as shown in figure. Now when a conductor "NM" is moving from position " x_1 " to position " x_2 " with certain velocity " v " by an applied force " F_{app} ", then due to which a magnetic flux will be changing (decreasing) in the loop "NMSR". Then according to Lenz's law the induced current or emf will be clockwise in the loop by opposition of such a magnetic

flux. Now we find mathematical expression for produced emf. As we know that from Faraday's law i.e.

$$\epsilon = -N\Phi/\Delta t \text{ ----- (1)}$$

we also know that $\Phi = BA\cos\theta$ ----- (2)

Now put eq (1) in eq (2). we get as

$$\epsilon = NAB\cos\theta/\Delta t.$$

As $\theta = 0^\circ$ b/w $\vec{B}$ and $\vec{A}$ of the loop. So we can write as

$$\epsilon = NABA\cos 0^\circ/\Delta t$$

$$\Rightarrow \epsilon = NABA/\Delta t$$

$$\Rightarrow \epsilon = NB\Delta A/\Delta t \text{ ----- (3)}$$

Now put $N = 1$ and $\Delta A = \text{Area of the loop} = \ell\Delta X$

$$\epsilon = B\ell\Delta X/\Delta t$$

$$\Rightarrow \epsilon = B\ell v \text{ ----- (4)}$$

Eq(4) will be valid when B , ℓ and v are mutually perpendicular as shown in figure. If they are not perpendicular then we will use

$$\epsilon = B\ell v \sin\theta \text{ ----- (5)}$$

Expression for current due to motional emf: From Ohm's law we can write as

$$\epsilon = IR$$

$$\Rightarrow I = \epsilon/R$$

$$\Rightarrow I = B\ell v/R \text{ ----- (6) where "R" is the resistance of the conductor.}$$

Expression for applied power on a conductor "NM"

$$P_{app} = Fv \text{ ----- (7)}$$

Expression for power dissipated across the conductor Resistance "R"

We can write as $P = I^2R$ ----- (8)

Now put eq (6) in eq (8) we get as

$$P_{dis} = (B\ell v)^2 R/R^2$$

$$\Rightarrow P_{dis} = B^2 \ell^2 v^2/R \text{ ----- (9)}$$

Now dissipated power is produced due to on applied power. So we can write

$$P_{dis} = P_{app}$$

Explanation of magnetic force on a conductor "NM":

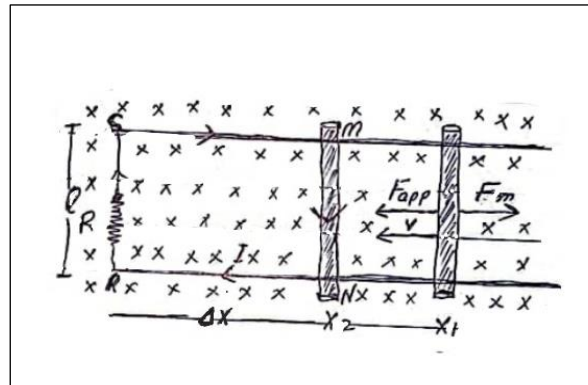
When a current is induced in the conductor "NM". Then conductor "NM" is called current carrying conductor. Now the magnetic force on current carrying conductor "NM" due to the applied magnetic field can be written as

$$F_m = I\ell B\sin\theta$$

Now put $\theta=90$ b/w I and B . Then we get as

$$F_m = I\ell B \text{ ----- (10)}$$

By using L.H Fleming rules the direction of such a magnetic force will be opposite to the direction of applied force as shown in fig. For non accelerated slow motion we can write as $F_m = F_{app}$



Self Induction and Self induced emf

The phenomenon in which an induced emf is produced in a coil due to its own changing current is known as self Induction. The emf induced in this case is known as self induced emf.

Explanation: consider a coil which is connected to rheostat and a battery as shown in a figure. Now if we vary the resistance of rheostat, then the current in the coil changes.

Due to change in current, the magnetic flux also changes. Due to change in magnetic flux, an induced emf is produced in the coil. Such induced emf is called self induced emf. Now in the coil such an induced emf is directly proportional to the rate of change of current. i.e

$$\varepsilon \propto \Delta I / \Delta t$$

$$\Rightarrow \varepsilon = L \Delta I / \Delta t$$

$$\Rightarrow \varepsilon = - L \Delta I / \Delta t \text{ ----- } \textcircled{1}$$

In eq $\textcircled{1}$ "L" is constant and is known as self inductance of the coil. The negative sign shows that the induced emf always opposes the cause due to which it is produced.

Self Inductance "L": The property of a coil that opposes any change in the amount of current flowing through it is known as self inductance or Inductance.

Explanation: Such a property of opposition to the change in the amount of current in the coil is produced due to an induced emf. Now from Faraday's law we know that

$$\varepsilon = - N \Delta \Phi / \Delta t \text{ ----- } \textcircled{2}$$

We also know that in this case

$$\varepsilon = - L \Delta I / \Delta t \text{ ----- } \textcircled{3}$$

Now compare eq $\textcircled{2}$ and eq $\textcircled{3}$. we get as

$$- L \Delta I / \Delta t = - N \Delta \Phi / \Delta t$$

$$\Rightarrow L = N \Delta \Phi / \Delta I$$

Factors affecting Self inductance: The following factors affecting the Self inductance

$\textcircled{1}$ The shape and Number of turns of the coil. i.e $L \propto N$.

$\textcircled{2}$ The time rate of change of magnetic flux w.r.t current change in the coil. i.e $L \propto \Delta \Phi / \Delta I$.

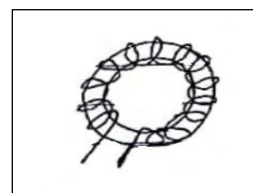
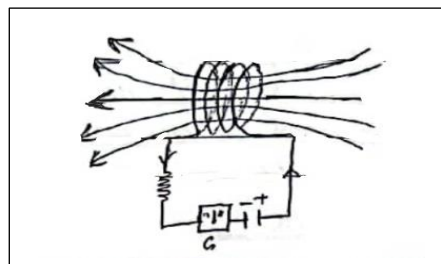
$\textcircled{3}$ Self Inductance also depends on the permeability of the material on which the coil is wound. As we know that

$$L = N \Delta \Phi / \Delta I$$

$$\Rightarrow L = N \Delta B A / \Delta I \text{ Or}$$

$$\Rightarrow L = N B A / I \text{ ----- } \textcircled{1}$$

In case of solenoid we know that



$$B = \mu_0 n I \text{ ----- (2)}$$

Now Put (2) in (1) we get as

$$L = N A n \mu_0 l$$

$$\Rightarrow L \propto \mu_0$$

Unit of self inductance: The unit of self inductance is henry (H).

Now we know that

$$\varepsilon = L \Delta I / \Delta t$$

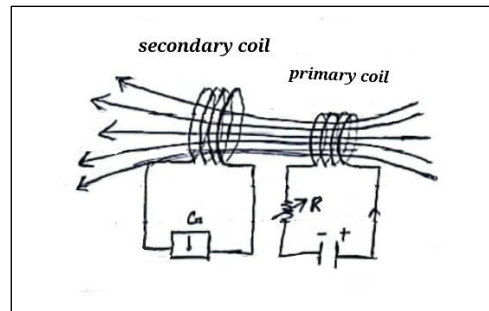
$$\Rightarrow L = \varepsilon / (\Delta I / \Delta t)$$

$$\Rightarrow 1H = 1V / (1A/1s)$$

Mutual Induction and Mutual induced emf

The phenomenon in which an induced emf is produced in a secondary coil due to a changing current in the primary coil is known as mutual induction. The emf induced in the secondary coil in this case is known as mutual induced emf.

Explanation: Consider a primary coil which is connected to rheostat and a battery as shown in figure. Now if we vary the resistance of rheostat, then the current in the primary coil changes. Due to change in current, the magnetic flux also changes in primary coil as well as in secondary coil. Now due to change in magnetic flux in the secondary coil, an induced emf is produced in that coil. Such induced emf is called mutual induced emf.



Now in the secondary coil such an induced emf is directly proportional to the rate of change of current in the primary coil. i.e

$$\varepsilon \propto \Delta I_p / \Delta t$$

$$\Rightarrow \varepsilon = M \Delta I_p / \Delta t \text{ or}$$

$$\Rightarrow \varepsilon = - M \Delta I_p / \Delta t \text{ ----- (1)}$$

In eq (1) "M" is constant and is known as mutual inductance of the two coils. The negative sign shows that the induced emf always opposes the cause due to which it is produced.

Mutual Inductance "M": The property of the two coils that opposes any change in the amount of current flowing in the primary coil is known as Mutual Inductance.

Explanation: Such a property of opposition to the change in the amount of current in the primary coil is produced due to an emfs induced in that two coils. Now from Faradays law we know that

$$\varepsilon = - N_s \Delta \Phi / \Delta t \text{ ----- (2)}$$

we also know that in this case

$$\varepsilon = - M \Delta I_p / \Delta t \text{ ----- (3)}$$

Now compare (2) and (3). we get as

$$-M \Delta I_p / \Delta t = -N_s \Delta \Phi / \Delta t$$

$$M = N_s \Delta \Phi / \Delta I_p$$

Factors affecting the Mutual Inductance: The following factors affecting the mutual inductance:

- ① The shape and number of turns in the secondary coil i.e $M \propto N_s$
- ② The time rate of change of magnetic flux w.r.t current change in the primary coil. i.e
 $M \propto \Delta \Phi$.
- ③ On the permeability of the materials on which they are wound.
- ④ Distance between the coils.
- ⑤ Orientation of the two coils.

Unit of mutual Inductance: Its unit is also henry (H).

Now we know that

$$\varepsilon = -M \Delta I_p / \Delta t$$

$$\Rightarrow M = \varepsilon / \Delta I_p / \Delta t$$

$$\Rightarrow H = IV / (1A/1s)$$

If 1V emf is produced in the secondary coil due to 1ampere current change in it then the mutual inductance will be equal to 1 henry.

Dynamically induced emf: The emf induced in a coil due to a motion of a conductor in a uniform magnetic field is called dynamically induced emf.

Examples: Motional emf in an AC generator and DC generator.

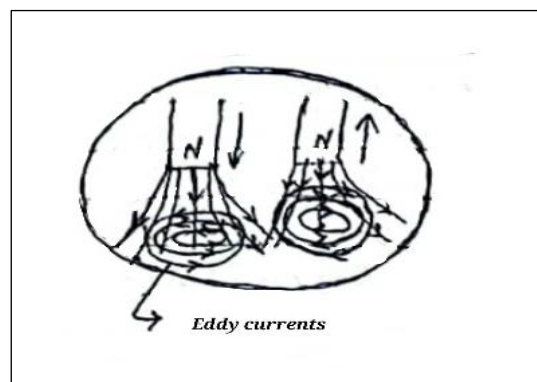
Statically induced emf: The emf produced in a stationary conductor due to a changing magnetic field is called statically induced emf.

Examples: Self induced emf in a coil and mutually induced emf in transformer.

Eddy currents

The current produced due to induce emf in metals moving in a magnetic field or exposed to a changing magnetic flux are called Eddy currents.

Explanation: (1) In a figure a north pole of bar magnet is moving towards the conductor in downward direction. Due to which a magnetic flux is changing (increasing) in particular area of the conductor. Then due to flux change the emf is induced in this area of the conductor. Then due to which the current is induced in this area and such a current is called Eddy current. Then according to Lenz's law the eddy current or emf will be anticlock wise in this area by the opposition of such a magnetic flux change. Also according to Lenz's law the magnetic field produced by eddy current will oppose the applied magnetic field of a bar magnet. Then there will be a repulsive force on the bar magnet



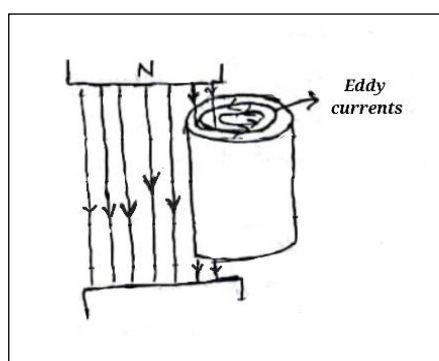
and due to which its motion will be opposes.

2) In a figure a north pole of a bar magnet is moving away from the conductor in upward direction. Due to which a magnetic flux is changing (decreasing) in particular area of the conductor. Then due to flux change the emf is induced in this area of the conductor. Then due to which the current is induced in this area and such a current is called eddy current. Then according to Lenz's law the eddy current or emf will be clockwise in this area by the opposition of such a magnetic flux change.

Also according to Lenz's law the magnetic field produced by eddy current will oppose the magnetic field change of a bar magnet. In this case the eddy current magnetic field will take a direction, in the direction of bar magnet magnetic field, and due to which there will be an attractive force on the bar magnet, and due to which its motion will be also opposes in the upward direction.

3) In a figure a conductor is moving from RHS to LHS in the uniform magnetic field as shown in figure. Now due to a magnetic flux change there will be producing an emf and eddy currents in that conductor as shown.

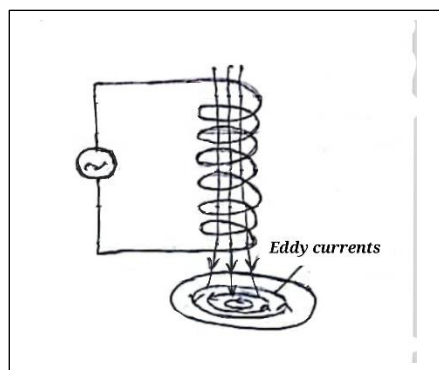
Now according to Lenz's law the direction of such a produced eddy currents will be anticlockwise due to opposition of magnetic flux change.



Metal detector

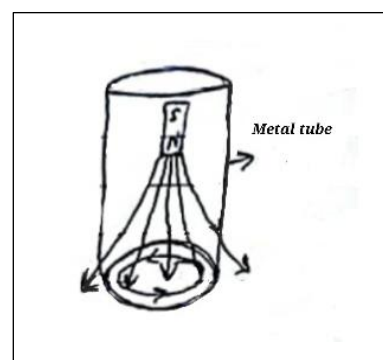
The metal detector consists of the oscillator or solenoid or inductor and alternating current source as shown in figure. In a figure the alternating current source passes current through a coils of oscillator and producing the changing magnetic field.

Now if the piece of metal is close to the coil, eddy currents will be induced in it due to a changing magnetic field. Now eddy currents will produce it own magnetic field due to which alarm will produce a sound and notify us the metal.



Projects

A bar magnet is dropped inside a long vertical tube. If the tube is made of metal, the magnet quickly approaches a terminal speed, but if the tube is made of cardboard, the magnet falls with constant acceleration. Explain why the magnet falls differently in the metal tube than it does in the cardboard tube?



Explanation: ① When a bar magnet is dropped inside a metal long vertical tube. Then in the base of this metal tube the magnetic flux will be changing and due to which an eddy current will be produced as shown in figure. Now the magnetic fields of eddy currents will oppose the motion of a bar magnet due to which it will gain quickly the terminal speed and the bar magnet will move slowly towards the base of the tube.

② When a bar magnet is dropped inside a cardboard (wood) tube, then there will be no eddy currents in the base end. Also there will be no repulsive force on the bar magnet, so due to which the bar magnet will gain slowly the terminal speed (constant speed) and also the bar magnet will get move quickly towards the base of this tube.

So from this explanation we can say that the bar magnet in case of metal tube will reaches slowly towards the base due to a produced eddy currents in the base.

Alternating current generator

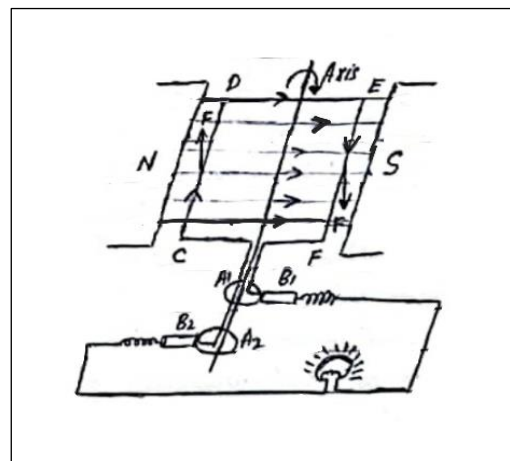
A device which converts mechanical energy into electrical energy is called Alternating current generator.

Principle: Its principle is based on Faraday's law of electromagnetic induction. Due to changing magnetic flux in coil the emf is produced.

Construction: The AC generator consists of the following parts.

1. **Armature:** It is a rectangular coil (CDEF) which is placed between the poles of a magnet. It can rotate about its axis of rotation as shown in figure.
2. **Field Magnets:** The field magnets are used to produce a uniform magnetic field around the armature.
3. **Slip rings:** The slip rings " A_1 " and " A_2 " are in contact with the coil. It can rotate with armature.
4. **Carbon Brushes:** The carbon brushes are in contact with slip rings and external circuit.

Working: In the figure when the coil (CDEF) rotates, then magnetic flux changes in the coil due to which the emf and current is produced in the coil. Now by using right hand rules, the current flows from 'C' to 'D' and "E" to "F". Thus the current enters the external circuit at " B_1 " and leaves at " B_2 ". After half revolution the side "FE" will take the position of "CD" and the current direction is reversed, so the current flows "F" to "E" and "D" to "C". Now the current enters the external circuit at ' B_2 ' and leaves it at " B_1 ". Thus in figure the induced emf and the current are changes in every half revolution in magnitude and direction.



Mathematical expressions for induced emf and for induced current

As we know that from Faraday's law

$$\varepsilon = - N \Delta \Phi / \Delta t \text{ ----- (1)}$$

$$\varepsilon = - N \Delta \Phi / \Delta t \text{ ----- (1)}$$

we also know that

$$\Phi = B A \cos \theta \text{ ----- (2)}$$

Now put (2) in eq (1) we get as

$$\varepsilon = - N \Delta B A \cos \theta / \Delta t$$

$$\Rightarrow \varepsilon = - N B A \Delta \cos \theta / \Delta t \text{ ----- (3)}$$

Now we also know that

$$\omega = \theta / t \Rightarrow \theta = \omega t \text{ ----- (4)}$$

Now put eq (4) in eq (3) we get as

$$\varepsilon = - N B A \Delta \cos \omega t / \Delta t \text{ ----- (5)}$$

Now from math's we can write as

$$\cos \omega t / \Delta t = - \omega \sin \omega t \text{ ----- (6)}$$

Now put eq (6) in eq (5) we get as

$$\varepsilon = - N B A (- \omega \sin \omega t)$$

$$\Rightarrow \varepsilon = N B A \omega \sin \omega t \text{ ----- (7)}$$

Now for max emf we put $\sin \omega t = 1$. So we get as

$$\varepsilon_{\max} = N B A \omega \text{ ----- (8)}$$

Now put eq (8) in eq (7) we get as

$$\varepsilon = \varepsilon_{\max} \sin \omega t \text{ ----- (9)}$$

Now we also know that

$$\omega = 2\pi f \text{ ----- (10)}$$

Now put eq (10) in eq (9) we get as

$$\varepsilon = \varepsilon_{\max} \sin(2\pi f t) \text{ ----- (10)}$$

In eq (10) "f" is the frequency of coil. Now we find expression for current.

For this eq (10) we can write as

$$I R = I_{\max} R \sin(2\pi f t)$$

$$\Rightarrow I = I_{\max} \sin(2\pi f t) \text{ ----- (11)}$$

Curves or graph for produced emf and current w.r.t angle "θ" between $\vec{A}$ and $\vec{B}$.

As we know that

$$\varepsilon = \varepsilon_{\max} \sin \omega t \text{ or } \varepsilon = \varepsilon_{\max} \sin \theta \text{ ----- (12) and}$$

$$I = I_{\max} \sin \omega t \text{ or } I = I_{\max} \sin \theta \text{ ----- (13)}$$

Case 1: When $\theta = 0^\circ$ in eq (12) and (13) we get

$$\varepsilon = \varepsilon_{\max} \sin 0^\circ \Rightarrow \varepsilon = \varepsilon_{\max} \times 0 \Rightarrow \varepsilon = 0 \text{ and}$$

$$I = I_{\max} \sin 0^\circ \Rightarrow I = I_{\max} \times 0 \Rightarrow I = 0$$

Case 2: When $\theta = 90^\circ$ in eq (12) and (13) we get

$$\varepsilon = \varepsilon_{\max} \sin 90^\circ \Rightarrow \varepsilon = \varepsilon_{\max} \times 1 \Rightarrow \varepsilon = \varepsilon_{\max} \text{ and}$$

$$I = I_{\max} \sin 90^\circ \Rightarrow I = I_{\max} \times 1 \Rightarrow I = I_{\max}$$

Case 3: When $\theta = 180^\circ$ in eq (12) and (13) we get

$$\varepsilon = \varepsilon_{\max} \sin 180^\circ \Rightarrow \varepsilon = \varepsilon_{\max} \times 0 \Rightarrow \varepsilon = 0 \text{ and}$$

$$I = I_{\max} \sin 180^\circ \Rightarrow I = I_{\max} \times 0 \Rightarrow I = 0$$

Case 4: When $\theta = 270^\circ$ in eq (12) and (13) we get

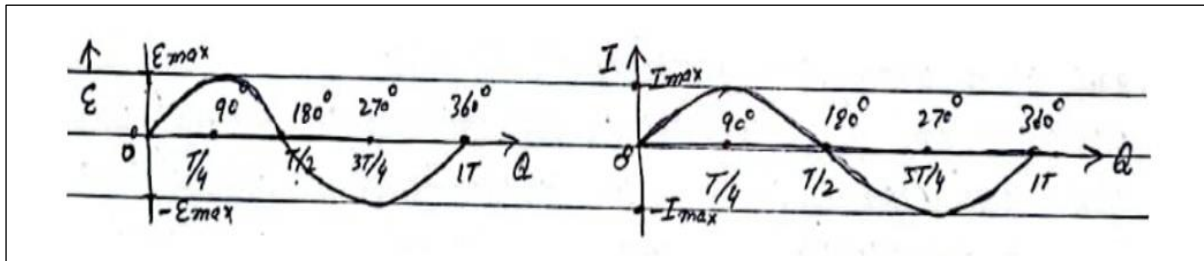
$$\epsilon = \epsilon_{\max} \sin 270^\circ \Rightarrow \epsilon = \epsilon_{\max} \times (-1) \Rightarrow \epsilon = -\epsilon_{\max} \text{ and}$$

$$I = I_{\max} \sin 270^\circ \Rightarrow I = I_{\max} \times (-1) \Rightarrow I = -I_{\max}$$

Case 5: When $\theta = 360^\circ$ in eq (12) and (13) we get

$$\epsilon = \epsilon_{\max} \sin 360^\circ \Rightarrow \epsilon = \epsilon_{\max} \times 0 \Rightarrow \epsilon = 0 \text{ and}$$

$$I = I_{\max} \sin 360^\circ \Rightarrow I = I_{\max} \times 0 \Rightarrow I = 0$$



Fleming Right hand rules for the induced current

First we will open the thumb, side finger and middle finger perpendicularly to each other. Then we set thumb in the direction of applied force on the conductor and side finger in the direction of external magnetic field, then the middle finger will gives us the direction of induced current.

AC Motor

A device which converts electrical energy into mechanical energy is called AC motor.

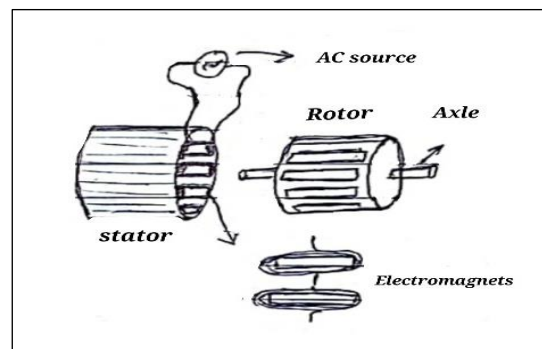
Principle: Its principle is based on Faraday's law of electromagnetic induction.

Construction: An AC motor has two basic electrical parts as shown.

① **Stator:** A stator is the stationery part of the motor. It has the group of individual electromagnets. These electromagnets are arranged inside the surface of hollow cylinder and which are connected with alternating current source. Due to alternating current each electromagnet will produce the changing magnetic field.

② **Rotor:** A rotor is the rotating part of the motor. It has a cylindrical shape and having rectangular conducting plates on its surface in separated (laminated) form as shown in figure. It has also an axle at its center. It is located inside the stator.

Working: Due to alternating current each electromagnet in the stator will produce the changing magnetic field. Then due to changing magnetic flux the emf and eddy currents will be produced in the rectangular conducting plates. Then according to Lenz's law the magnetic fields produced due to eddy currents in



the conducting plates will oppose the changing magnetic field or changing magnetic flux of each electromagnet. Then due to such opposition between the magnetic fields the rotor then starts rotation inside the stator. The rotating speed of the rotor is called synchronous speed and it is equal to the speed of magnetic flux change across the conducting plates. If the speed of magnetic flux change is greater then synchronous speed of the rotor will be also greater and vice versa.

Back emf

The emf which opposes the applied voltage is called back emf.

Explanation: In an AC motor due to the applied alternating voltage, the alternating current will pass from each electromagnet. Then due to which each electromagnet in the stator will produce the changing magnetic field. Then due to changing magnetic field or changing magnetic flux the emf will be produced in the conducting plates on the rotor. Then according to Lenz's law such a produced emf will oppose the applied alternating voltage. So such a produced emf in the rotor part is called back emf.

Now we can say that the applied voltage will produce a back emf in the rotor and remaining voltage will drop across the wire resistance in electromagnets.

So we can write as

$$V = \varepsilon + VR$$

$$\Rightarrow V = \varepsilon + IR$$

$$\Rightarrow \varepsilon = V - IR \text{ ----- } \textcircled{1}$$

In eq $\textcircled{1}$ 'R' is the resistance of the wiring in electromagnets and 'I' is the current through it.

See For your information on page 147. It is very important.

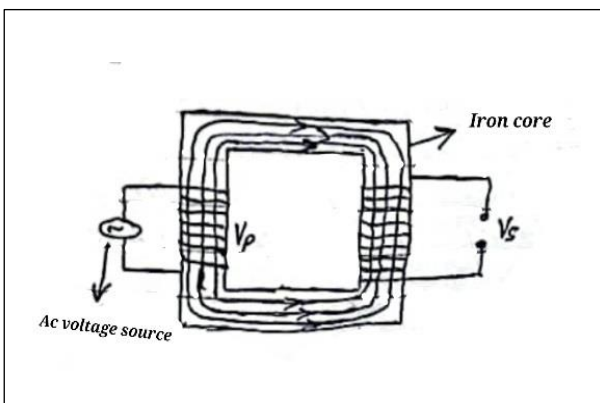
Transformer

A device which increase or decrease the value of a given AC voltage is called Transformer.

Principle: Transformer works on the principle of mutual induction.

Construction: It consists of laminated soft iron core upon which two coils are wound as shown in figure. One coil is called primary coil and the other is called secondary coil.

Working: When an alternating voltage is applied across the primary coil, then an alternating current flows in the primary coil. Then this current produces a continuously changing magnetic flux. Then the changing magnetic flux flows in an iron core from primary coil as well as in from secondary coil. Now due to changing magnetic flux the emf produced in a



primary coil is said to be self induced emf and the emf produced in the secondary coil is said to be mutual induced emf. Now the self induced emf in the primary coil can be written as

$$V_p = -N_p \Delta\Phi/\Delta t \text{ ----- (1)}$$

“ N_p ” represents the No of turns in the primary coil.

And now the mutual induced emf in the secondary coil can be written as

$$V_s = -N_s \Delta\Phi/\Delta t \text{ ----- (2)}$$

“ N_s ” represents the No of turns in the secondary coil.

Now divide eq (2) by eq (1). We get as

$$\begin{aligned} V_s / V_p &= -N_s \Delta\Phi/\Delta t / -N_p \Delta\Phi/\Delta t \\ \Rightarrow V_s / V_p &= N_s / N_p \text{ ----- (3)} \end{aligned}$$

Eq (3) is called Transformer equation.

Types of Transformer

Step up Transformer: The transformer in which the voltage across the secondary is greater than the voltage across the primary coil is called step up transformer.

If $N_s > N_p$ then $V_s > V_p$.

Step down transformer: The transformer in which the voltage across the secondary is smaller than the voltage across the primary coil is called step down transformer.

If $N_s < N_p$ then $V_s < V_p$.

Power of Transformer

For an ideal Transformer we can write as

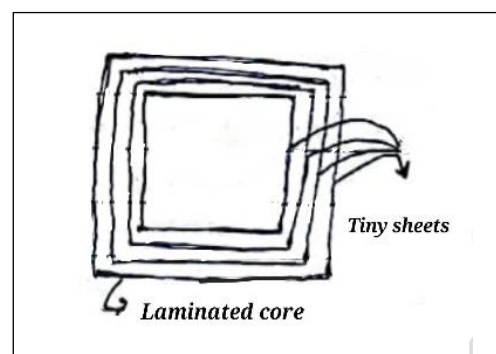
input power = output power

$$\begin{aligned} P_p &= P_s \\ \Rightarrow I_p V_p &= I_s V_s \\ \Rightarrow V_s / V_p &= I_p / I_s \text{ ----- (1)} \end{aligned}$$

From eq (1) we can also write as $I_s \propto 1/V_s$ and $I_p \propto 1/V_p$.

From these relations we can say that if the voltage is stepped up, then the current is stepped down and vice versa. If a high voltage is required, then for this purpose, a step up transformer is used. While if high current is required, then for this purpose a step down transformer is used.

Transformer core is made of laminated iron that is, made of many thin sheet of iron covered by insulating layers. This limits the over all effect of eddy current and reduces the large heating effect due to an insulating layers.



Examples and Assignments

Example 14.1: At what rate would it be necessary for a single conductor to cut the flux in order that a current of 1.2mA flows through it when 10Ω resistor is connected across its ends?

Given: $N=1$ $R=10\Omega$ $I = 1.2\text{mA} = 1.2 \times 10^{-3}\text{A}$.

Req: The rate of change of flux = $\Delta\Phi = ?$

Sol: As we know that

$$\epsilon = N\Delta\Phi/\Delta t \text{ ----- (1)}$$

Now we can also write as

$$\epsilon = IR \text{ ----- (2)}$$

Compare ① and ② we get as

$$IR = N\Delta\Phi/\Delta t$$

$$\Rightarrow \Delta\Phi/\Delta t = IR/N$$

$$\Rightarrow \Delta\Phi/\Delta t = 1.2 \times 10^{-3} \text{ A} \times 10^5 \Omega$$

$$\Rightarrow \Delta\Phi/\Delta t = 0.012 \text{ Wb/sec} \quad \text{Ans}$$

Example 14.2: A coil of mean area 500cm^2 and having 1000 turns is held perpendicular to a uniform field of 0.4 Gauss. The coil is turned through 180° in $2/10\text{s}$. Calculate the average induced emf.

Given: $A = 500 \text{ cm}^2 = 500 \times 10^{-4} \text{ m}^2$ $N = 1000$.

Initial Angle b/w area vector of the coil and magnetic field = $\theta_1 = 0$

After turning the coil then $\theta_2 = 180^\circ$ $t = 1/5 \text{ s} = 0.1\text{s}$. $B = 0.4\text{G} = 0.4 \times 10^{-4} \text{ T}$

Req: Average induced emf = $\epsilon = ?$

Sol: As we know that

$$\epsilon = -N \Delta\Phi/\Delta t$$

$$\Rightarrow \epsilon = -N(\Phi_2 - \Phi_1)/\Delta t$$

$$\Rightarrow \epsilon = -N(BA \cos\theta_2 - BA \cos\theta_1)/\Delta t$$

$$\Rightarrow \epsilon = -NBA (\cos 180^\circ - \cos 0^\circ)/\Delta t$$

$$\Rightarrow \epsilon = -NBA (-1 - 1)/\Delta t$$

$$\Rightarrow \epsilon = 2 NBA/\Delta t$$

$$\Rightarrow \epsilon = 2 \times 1000 \times 0.4 \times 10^{-4} \text{ T} \times 500 \times 10^{-4} \text{ m}^2 / 0.1 \text{ s}$$

$$\Rightarrow \epsilon = 0.04 \text{ V} \quad \text{Ans}$$

Assignment 14.1: In the initial state (1), the magnetic flux is changed by 5Wb in 2 seconds in a coil with 20 loops. In the final state (2), the same flux changed in 8 seconds. Determine the ratio of the initially induced emf and the final induced emf.

Given: In the initial state $\Delta\Phi_1 = 5\text{Wb}$ $\Delta t_1 = 2\text{s}$ $N=20$

in the final state $\Delta\Phi_2 = 5\text{Wb}$ $\Delta t_2 = 8\text{s}$ $N=20$

Ref: Ratio of initial to final induced emf = $\epsilon_1/\epsilon_2 = ?$.

Sal: As we know that

$$\epsilon = -N\Delta\Phi/\Delta t$$

Now we can write as

$$\begin{aligned}\epsilon_1 &= (-N\Delta\Phi_1/\Delta t_1) \\ \Rightarrow \epsilon_1 &= (-20 \times 5\text{Wb}/25). \\ \Rightarrow \epsilon_1 &= -50\text{V}\end{aligned}$$

Now we find " ϵ_2 "

$$\begin{aligned}\epsilon_2 &= (-N\Delta\Phi_2/\Delta t_2) \\ \Rightarrow \epsilon_2 &= (-20 \times 5\text{Wb}/85) \\ \Rightarrow \epsilon_2 &= -12.5\text{V}.\end{aligned}$$

Now we can write as

$$\begin{aligned}\Rightarrow \epsilon_1/\epsilon_2 &= -50\text{V}/-12.5\text{V} \\ \Rightarrow \epsilon_1/\epsilon_2 &= 4/1 \\ \Rightarrow \epsilon_1/\epsilon_2 &= 4:1 \text{ Ans.}\end{aligned}$$

Example 14.3: A loop of resistance 0.15Ω is placed in a magnetic field of 2T . If a conductor of length 0.2m is sliding along a loop with a velocity of 0.2 m/s . Find (a) the emf produced in the conductor if the motion of a conductor is perpendicular to the field (b) the current through the loop (c) the electric power generated (d) the input mechanical power.

Given: $R = 0.15\Omega$ $B = 2\text{T}$ $l = 0.2\text{m}$ $V = 0.2\text{m/s}$

Req: a) emf = ϵ = ? (b) current = I = ? (c) power generated = P = ?

(d) input power = p' = ?

Sol (a) As we know that

$$\begin{aligned}\epsilon &= B \ell v \\ \Rightarrow \epsilon &= 2\text{T} \times 0.2\text{m} \times 0.2\text{m/s} = 0.08\text{V}\end{aligned}$$

Sol (b) We also know that

$$\begin{aligned}\epsilon &= IR \\ \Rightarrow I &= \epsilon/R \Rightarrow I = 0.08\text{V}/0.15\Omega = 0.8\text{A}\end{aligned}$$

Sol (c) We also know that the power generated across the resistor

$$\begin{aligned}P &= B^2 \ell^2 V^2 / R \\ \Rightarrow P &= (2\text{T})^2 (0.2\text{m})^2 \times (0.2\text{m/s})^2 / 0.15\Omega \\ \Rightarrow P &= 0.064 \text{ Watt Ans}\end{aligned}$$

sol (d) we also know that the input power or applied powers. i.e

$$\begin{aligned}P' &= F_{\text{app}} v \\ \Rightarrow P' &= F_m v \\ \Rightarrow P' &= I \ell B v \\ \Rightarrow P' &= (0.8\text{A}) \times (0.2\text{m}) (2\text{T}) (0.2\text{m/s}) \\ \Rightarrow P' &= 0.064 \text{ watt Ans.}\end{aligned}$$

Assignment 14.2: The current flowing in the first coil changes from 2A to 10A in 0.45 . Find the mutual inductance between two coils if an emf of 60mV is induced in the second coil. Also determine the induced emf in the second coil if the current in the first coil is

changed from 4A to 16A in 0.035. Consider only the magnitude of induced emf.

Given: a) Initial current = $I_1 = 2A$ Final current = $I_2 = 10A$

Time = $\Delta t = 0.4s$ Induced emf = $\epsilon = 60mV = 60 \times 10^{-3}V$

Req: (a) Mutual inductance = $M = ?$ (b) Induced emf in 2nd coil = $\epsilon_s = ?$

Sol: As we know that

$$\epsilon = M \Delta I / \Delta t$$

$$\Rightarrow \epsilon = M(I_2 - I_1) / \Delta t$$

$$\Rightarrow M = \epsilon \Delta t / (I_2 - I_1)$$

$$\Rightarrow M = 60 \times 10^{-3} \times 0.4s / (10A - 2A)$$

$$\Rightarrow M = 3 \times 10^{-3}H \text{ Ans.}$$

b) Initial current = $I_1 = 4A$ Final current = $I_2 = 16A$

Time = $\Delta t = 0.03s$, $M = 3 \times 10^{-3}H$

Sol (b) : Induced emf in 2nd coil = $\epsilon_s = ?$

As we know that

$$\epsilon_s = M \Delta I / \Delta t$$

$$\Rightarrow \epsilon_s = M(I_2 - I_1) / \Delta t$$

$$\Rightarrow \epsilon_s = 3 \times 10^{-3}H (16A - 4A) / 0.03s$$

$$\Rightarrow \epsilon_s = 1.2V \text{ Ans}$$

Example 14.4: An AC motor has an internal resistance of 4Ω . When powered by a 50Hz-AC Supply of peak voltage 120V it draws a peak current of 5A. What is the peak back emf produced by the motor? (2) What is the peak power delivered to the motor by the AC supply? (3) What is the peak rate of energy loss as heat in the motor?

Given: $R = 4\Omega$. $f = 50Hz$. $I = 5A$. Peak voltage = $V = 120V$

Req: (1) Peak back emf = $\epsilon = ?$ (2) Peak power = $P_{\text{peak}} = ?$

(3) peak rate of energy loss = $E_{\text{loss}}/t = P_{\text{loss}} = ?$

Sol (1): As we know that the back emf in AC motor can be written as

$$\epsilon = V - IR \Rightarrow \epsilon = 120V - 5A \times 4\Omega \Rightarrow \epsilon = 100V \text{ Ans}$$

(2): The peak power delivered by the AC supply can be written as

$$P_{\text{peak}} = IV \Rightarrow P_{\text{peak}} = 5A \times 120V \Rightarrow P_{\text{peak}} = 600W$$

(3): Now the peak rate of energy loss across the resistance can be written as

$$E_{\text{loss}}/t = P_{\text{loss}}$$

$$\Rightarrow E_{\text{loss}}/t = I^2 R \Rightarrow E_{\text{loss}}/t = (5A)^2 \times 4\Omega \Rightarrow E_{\text{loss}}/t = 100W \text{ Ans}$$

Assignment 14.3: An electric generated consist of 100 turns of wire formed into a rectangular loop 50cm by 30cm, placed entirely in a uniform magnetic field of 3.50T. What is the maximum value of the emf produced when the loop is spun at 1000 rev/min about an axis perpendicular to "B"?

Given: $N = 100$

$L = 50cm = 0.5m$

$W = 30cm = 0.3m$

Area of rectangular loop = $A = L \times W = 0.5\text{m} \times 0.3\text{m} = 0.15\text{m}^2$

Magnetic field = $B = 3.50\text{T}$

Angular speed of rectangular loop = $\omega = 1000 \text{ rev / min}$

$$\Rightarrow \omega = 1000 \times 2\pi \text{ rad / } 60\text{s}$$

$$\Rightarrow \omega = 1000 \times 2 \times 3.14 \text{ rad / } 60\text{s} \quad \Rightarrow \omega = 104.7 \text{ rad/s}$$

Req: Maximum emf = $\epsilon_{\text{max}} = ?$

Sol: As we know that in AC generator

$$\epsilon_{\text{max}} = NBA\omega$$

$$\Rightarrow \epsilon_{\text{max}} = 100 \times 3.50\text{T} \times 0.15\text{m}^2 \times 104.7 \text{ rad/s}$$

$$\Rightarrow \epsilon_{\text{max}} = 5497\text{V} \quad \text{Ans}$$

Exercise MCQS

① For inducing emf in a coil, the basic requirement is that:

- (a) Flux should link the coil
- (b) Change in flux should link the coil ✓
- (c) Coil should form a closed loop
- (d) Both (b) and (c) are true

② When a single turn coil rotates in a uniform magnetic field at uniform speed, the induced emf will be:

- (a) Alternating ✓
- (b) Steady
- (c) Pulsating
- (d) Zero

③ The principle of dynamically induced emf is used in a:

- (a) Choke
- (b) Transformer
- (c) Generator ✓
- (d) Thermocouple

④ In Fleming's left-hand rule, the thumb always represents the direction of:

- (a) Current flow
- (b) Induced emf
- (c) Magnetic field
- (d) Mechanical force ✓

⑤ The magnetic field at any point on the axis of a current-carrying coil will be:

- (a) Perpendicular to the axis
- (b) Parallel to the axis ✓
- (c) At an angle of 45° with the axis
- (d) Zero

⑥ When a coil rotates in a magnetic field, the induced current in it:

- (a) Continuously changes ✓
- (b) Remains the same

- (c) Becomes zero
(d) Becomes maximum
- ⑦ In an AC generator, increasing the number of turns in the coil:
- (a) Decreases the electromotive force
(b) Electromotive force remains the same
(c) Increases the electromotive force ✓
(d) Electromotive force becomes zero
- ⑧ The device in which induced emf is statically induced emf is:
- (a) Transformer ✓
(b) AC generator
(c) Alternator
(d) Dynamo
- ⑨ What is the coefficient of mutual inductance when the magnetic flux changes by 2×10^{-2} Wb and the change in current is 0.01 A?
- (a) 2 H
(b) 3 H
(c) 2 H ✓
(d) Zero
- sol: we know that $M = N\Delta\Phi/\Delta I \Rightarrow M = 1 \times 2 \times 10^{-2} \text{ Wb}/0.01\text{A} \Rightarrow M = 2\text{H Ans}$
- ⑩ The induced emf in a coil is proportional to:
- (a) Magnetic flux through the coil
(b) Rate of change of magnetic flux through the coil ✓
(c) Area of the coil
(d) Product of magnetic flux and area of the coil
- ⑪ In a coil, current changes from 2 A to 4 A in 0.05 s. If the average induced emf is 8 V, then the coefficient of self-inductance is:
- (a) 0.2 henry ✓
(b) 0.1 henry
(c) 0.8 henry
(d) 0.04 henry
- Sol: As we know that $\epsilon = L\Delta I/\Delta t$
 $\Rightarrow L = \epsilon\Delta t/\Delta I$
 $\Rightarrow L = \epsilon\Delta t/(I_2 - I_1) \Rightarrow L = 8\text{V} \times 0.05\text{s}/(4\text{A} - 2\text{A}) \Rightarrow L = 0.2 \text{ henry Ans.}$
- ⑫ Which of the following quantities remains constant in a step up transformer?
- (a) Current
(b) Voltage
(c) Power ✓
(d) Heat
- ⑬ A step-up transformer has a transformation ratio of 3:2. What is the voltage in the secondary, if the voltage in the primary is 30 V?
- (a) 45 V ✓

(b) 15 V

(c) 90 V

(d) 300 V

Sol: As we know that

$$V_s / V_p = N_s / N_p$$

$$\Rightarrow V_s = N_s / N_p \times V_p$$

$$\Rightarrow V_s = 3/2 \times 30V \Rightarrow V_s = 45V \text{ Ans}$$

(14) Eddy current is produced when:

(a) A metal is kept in a varying magnetic field ✓

(b) A metal is kept in a steady magnetic field

(c) A circular coil is placed in a steady magnetic field

(d) Current is passed through a circular coil

General Topic for exercise

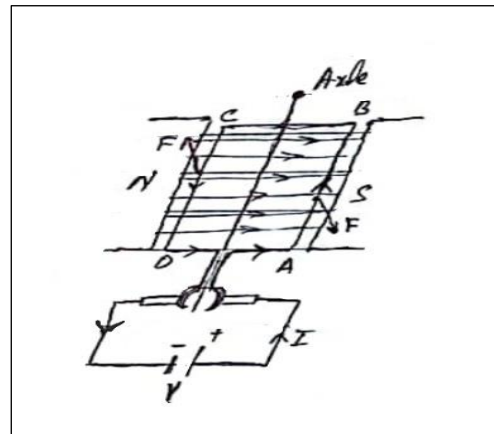
DC Motor

A device which work on direct current source and convert electrical energy into mechanical energy is called DC motor.

Construction: It has permanent magnets, Rectangular coil, split ring, Brushes, axle and Direct current source (Battery).

Working: In figure a direct current source supplies a current to Rectangular coil as shown. Now due to a current flow in Rectangular coil there are magnetic forces on sides "AB" and "CD" due to a magnetic field. Now due to a magnetic forces as shown, the rectangular coil starts rotation and converts electrical energy into mechanical energy.

Note: The rotating part in DC motor is called armature.



Back emf in DC Motor

The emf produced in a rectangular coil such that to oppose the applied voltage is called back emf in DC motion.

Explanation: When a rectangular coil rotates then magnetic field is changing in figure. Due to which an emf produced in the coil and such emf is called back emf. The Back emf in DC motor depends on the following factors.

- ① Number of turns in the coil.
- ② Magnetic field produced by permanent magnets.
- ③ Angular speed of rectangular coil or armature.
- ④ The permeability of a core on which the rectangular coil is wounded. The back emf in DC motor can be written as

$$\varepsilon = V - IR$$

Conceptual Questions

Question 01: What factors limit the size of the back emf?

Ans: The emf produced in motors such that to oppose the applied voltage is called back emf. Now we explain factors.

Factors which limit the size of the back emf in DC motor are the following.

- ① Number of turns in the coil.
- ② Magnetic field produced by permanent magnets.
- ③ Angular Speed of armature.
- ④ The permeability of a core on which the rectangular coil is wound.
- ⑤ Resistance of the rectangular coil.

Factors which limit the size of the back emf in AC motor are the following.

- ① The rate of changing current in electromagnets.
- ② The rate of change of magnetic flux produced by electromagnets.
- ③ The eddy currents produced in the rotor.

Generally the external loads connected with motors also limit the size of back emf. If the external connected load is greater, then angular speed of the rotating part will be less and then back emf will be also less and vice versa.

Question 02: Why does back emf tend to decrease as the rate of doing work increases?

Ans: As we know that for the back emf in the motor operations. i.e

$$\varepsilon = V - IR \text{ ----- (1)}$$

In eq ① “ ε ” represents the back emf, “V” represents the applied voltage and “R” is the resistance of the winding.

Now multiply eq ① by “I” to both side. Then we get as

$$I \varepsilon = IV - I^2 R$$

$$\Rightarrow I \varepsilon = P_v - P_R$$

$$\Rightarrow P_v = I \varepsilon + P_R$$

$$\Rightarrow P_v = I \varepsilon + W_{R/t}$$

$$\Rightarrow P_v = I \varepsilon + \text{rate of doing work by a resistance "R"} \text{ ----- (2)}$$

Now from eq(2) we can say that when rate of doing work is greater, then back emf will be less.

Example: A greater external loads connected with motors decreases its angular speed, then due to which the back emf will be also less and then the rate of doing work by a resistance will be greater, and then there will be a chance of burning of the motor.

Question 03: In Faraday's experiments, what would be the advantage of using coils with many turns?

Ans: As we know that from Faraday's law for induced emf is

$$\varepsilon = - N \Delta \Phi / \Delta t \text{ ----- (1)}$$

Now from eq (1) we can say that the induced emf in the coil show directly proportionality to the number of turns N. Thus by increasing the number of turns in the coil, the induced emf will be increasing in that coil.

Question 04: Show that the relationship $\epsilon = - \Delta\Phi/\Delta t$ is dimensionally correct.

Ans: Given equation is $\epsilon = - \Delta\Phi/\Delta t$ ----- (1)

Now first we find the dimension of L.H.S. of eq (1). For this we write as

$$\epsilon = W/q$$

We also know that $W = Fd$ and $q = It$

$$\Rightarrow \epsilon = Fd/It$$

We also know that $F = ma = mv/t = ms/t^2$

$$\Rightarrow \epsilon = msd/It^2$$

$$\Rightarrow \epsilon = [ML^2T^{-3}A^{-1}] \text{ ----- (2)}$$

Now we find the dimension of R.H.S of eq (1).

$$\Rightarrow - \Delta\Phi / \Delta t = \Phi/t . \quad \text{We know that } \Phi = BA$$

$$\Rightarrow - \Delta\Phi / \Delta t = BA/t \text{ ----- (3)}$$

We also know that $F = ILB$

$$\Rightarrow B = F/IL = ma/IL = mv/ILt = ms/ILt^2 \text{ ----- (4)}$$

we also know that $A = L \times W$ ----- (5)

$$\Rightarrow - \Delta\Phi / \Delta t = msLW/ILt^3$$

$$\Rightarrow - \Delta\Phi / \Delta t = msW/It^3$$

$$\Rightarrow - \Delta\Phi / \Delta t = [ML^2T^{-3}A^{-1}] \text{ ----- (6)}$$

Now from (2) and (3) we can say that eq (1) is dimensionally correct equation.

Question 05: Give the formulae for the flux linkage in terms of angular orientation.

Ans: The flux linkage is define as “The magnetic flux through number of turns of the coil is called flux linkage. i.e

$$\text{Flux linkage} = N\Phi \text{ ----- (1)}$$

Now we want to express eq (1) in angular orientation. i.e

$$\Rightarrow \text{Flux linkage} = NB.A$$

$$\Rightarrow \text{Flux linkage} = NBA \cos\theta$$

$$\Rightarrow \text{Flux linkage} = NBA \cos\omega t \text{ ----- (2)}$$

Eq (3) represents the flux linkage in term of angular orientation.

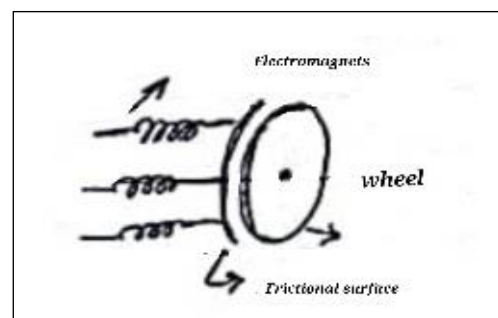
Question 06: How electromagnet brake works? Explain.

Ans: An electromagnetic brake is a type of brake that uses electromagnetic forces to slow or stop the rotation of a shaft or wheel. The components of an electromagnet are the following.

① **Electromagnet's:** It is a coil of wire wrapped around a core, which generates a magnetic field when an electric current flows through it.

② **Armature:** It is a rotating shaft or wheel made of metal.

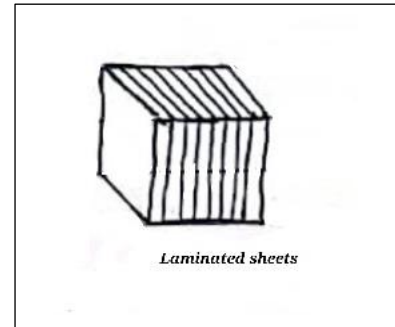
③ **Friction Surface:** It is a stationary surface, which is used to slow or stop the rotating shaft or wheel due to its friction, when brake is applied.



Working of brake: When current flows through an electromagnet then they produce the magnetic fields. Then such a magnetic field attracts the shaft or wheel towards the friction surface. Then friction surface slows or stops the wheel due to its friction. After stopping the current, the spring attached in it then push the wheel back and then the wheel goes to free state.

Question 07: How eddy currents can be minimized in transformer?

Ans: In order to minimize the eddy currents in transformer for which the solid iron core is replaced by thin lamination sheets. Then such lamination sheets are stacked together and each sheet is insulated from the other one. Now due to laminated sheets the produced Eddy currents are divided in each sheet. Then due to an individual greater resistance of each laminated sheet, the divided eddy currents are minimized.



Question 08: How electromagnetic induction is used in cook tops in electric ranges?

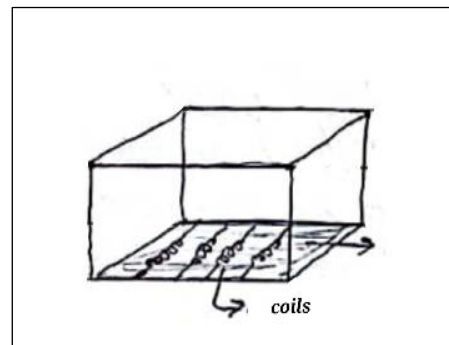
Ans: Induction cook top is an electrical device used for cooking and heating of food.

Principle: Induction cook top works on the Faraday's law of electromagnetic induction.

Components of induction cook top:

- 1) Coil: It is a spiral shaped wire and is located beneath the cook top surface..
- 2) Electronic controls: It is a control unit which regulates the frequency and power of the alternating current through the coil.
- 3) Cookware: It is a metal plate made of ferromagnetic materials.

Working of induction cook top: When an alternating current flows through the coil, then it produces the changing magnetic field. Then changing magnetic field or changing magnetic flux produces the eddy currents in the cookware. Then due to a high resistance of cookware the eddy currents produce heat in the cookware. Then heat is transferred to the food and then it is cooked.



Question 09: What is meant by the term back emf in any electric motor operation?

Ans: When the armature of a motor starts to rotate, the magnetic flux linking the coil changes. Thus an emf is induced in the coil due to such a magnetic flux change. Such emf is called back emf " ϵ " and acts in opposite direction to the applied voltage " V " according to Lenz's law. The expression for back emf can be written as

$$\varepsilon = V - IR$$

Question 10: Why it is an advantage for the armature to rotate in a radial magnetic field rather than a uniform one?

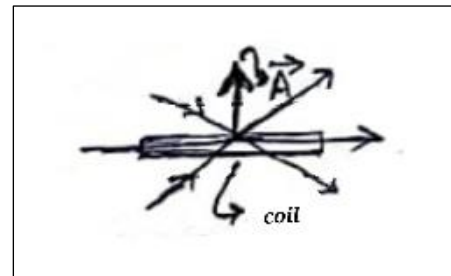
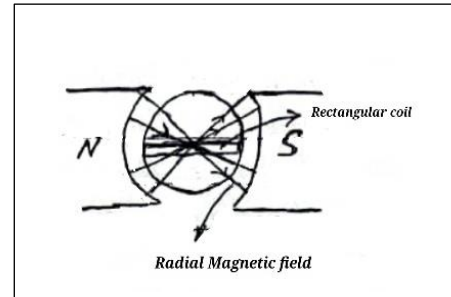
Ans: In a magnetic field a rotating rectangular coil wound around an iron cylinder is called armature. Radial magnetic field means a field which passes along a radius of iron cylinder is called radial magnetic field. Now there are two advantages for the armature to rotate in radial magnetic field rather than a uniform one.

① The radial magnetic field produces an extra magnetic field in the iron cylinder. Due to which the over all effect of magnetic field is greater around the armature, then the produce torque i.e $\tau = NIAB\sin\theta$ in the armature will be greater due to greater magnetic field.

② Radial magnetic field makes an angle $\theta = 90^\circ$ with the area vector A of the coil. So due to $\theta = 90^\circ$ the produce torque in the armature will be maximum and constant. i.e

$$\tau = NIAB \sin 90^\circ$$

$$\Rightarrow \tau_{\max} = NIAB.$$



Question 11: (a) Which materials would you prefer while making Transformer core; iron, solid soft iron, laminated soft iron, aluminum?

(b) What is the reason for using this material?

Ans: The following features of the material can make a transformer to be more efficient.

Performance of Iron: Which has high permeability less energy loss or hysteresis loss and minimum effect of eddy currents.

Performance of iron core: Which has high permeability, high hysteresis loss and high effect of eddy currents.

Performance of solid Soft iron : Which has high permeability, low hysteresis loss or energy loss and high effect of eddy currents.

Performance of Laminated Soft iron: Which has high permeability, low hysteresis loss or energy loss and minimum effect of eddy currents.

Performance of Aluminum: Aluminum is non-magnetic material so it is not suitable for core of transformer.

Conclusion: From the above discussion it is clear that laminated soft iron is the more efficient material for making the Transformer core as compared with the other materials.

Numerical problems

Question 01: A wheel with 12 metals spokes each 0.6m long is rotated with a speed of 180 r.p.m in a plane normal to carter's magnetic field at a place. If the magnitude of the field is 0.6 T, what is the magnitude of induced emf between the axle and rim of the wheel?

Given: $l = r = 0.6\text{m}$ $B = 0.6\text{T} = 0.6 \times 10^{-4}\text{T}$

$\omega = 180 \text{ r.p.m} = 180 \times 2\pi \text{ rad/min} = (180 \times 2\pi \text{ rad/s})/60\text{s} = 6\pi \text{ rad/s}$

Req: Emf produced in 12 triangles between the axle and rim.

Sol: First we find ϵ_1 in triangle 1 between axle and rim. As we know that

$$\epsilon_1 = N \Delta \Phi / \Delta t \quad \text{put } N=1 \text{ and } \Phi = BA$$

$$\Rightarrow \epsilon_1 = \Delta BA / \Delta t$$

$$\Rightarrow \epsilon_1 = B \Delta A / \Delta t \quad \text{----- (1)}$$

Area of triangle 1 = $A = \frac{1}{2} l \times w$

$$\Rightarrow A = \frac{1}{2} r \times s \quad \text{put } s = r \theta$$

$$\Rightarrow A = \frac{1}{2} r \times r \theta$$

$$\Rightarrow A = \frac{1}{2} r^2 \theta \quad \text{----- (2) put eq (2) in (1). we get as}$$

$$\Rightarrow \epsilon_1 = \Delta B r^2 \theta / 2 \Delta t .$$

$$\Rightarrow \epsilon_1 = B r^2 \Delta \theta / 2 \Delta t . \quad \text{Put } \Delta \theta / \Delta t = \omega$$

$$\Rightarrow \epsilon_1 = B r^2 \omega / 2$$

$$\Rightarrow \epsilon_1 = (0.6 \times 10^{-4} \text{T} \times (0.6\text{m})^2 \times 6 \times 3.14 \text{ rad/s}) / 2$$

$$\Rightarrow \epsilon_1 = 2.035 \times 10^{-4} \text{V} \quad \text{Book Ans.}$$

Now we find total emf in 12 triangles.

$$\epsilon = 12 \times \epsilon_1$$

$$\Rightarrow \epsilon = 12 \times 2.035 \times 10^{-4} \text{V.} \quad \text{Complete Ans.}$$

Question 02: A circuit has 1000 turns enclosing a magnetic circuit 20cm^2 in section with 4A current, the flux density is 1Wb/m^2 and with 9A current, it is 2.03Wb/m^2 . Find the mean value of the inductance between these current limits and the induced emf, if the current falls from 9A to 4A in 0.05s.

Given: $N = 1000$ $B_1 = 1\text{Wb/m}^2$ $B_2 = 1.4\text{wb/m}^2$ $I_1 = 4\text{A}$

$I_2 = 9\text{A}$ $\Delta t = 0.05\text{s}$ $A = 20\text{cm}^2 = 20 \times 10^{-4} \text{m}^2$

Req: a) Self inductance = $L = ?$

b) emf = $\epsilon = ??$

Sol: As we know that

$$L = N \Phi / \Delta I$$

$$\Rightarrow L = N(\Phi_2 - \Phi_1) / I_2 - I_1$$

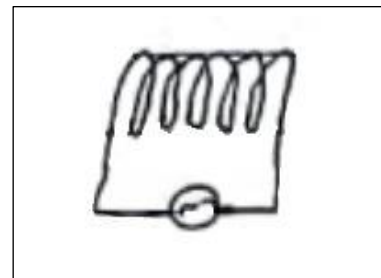
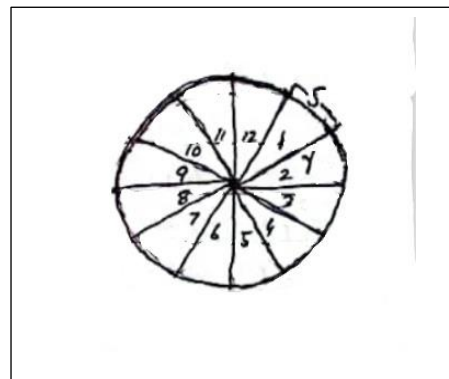
$$\Rightarrow L = N(B_2 - B_1)A$$

$$\Rightarrow L = 1000 (1.4\text{Wb/m}^2 - 1\text{Wb/m}^2) 20 \times 10^{-4} \text{m}^2 / (9\text{A} - 4\text{A})$$

$$\Rightarrow L = 0.16\text{H} \text{ Ans}$$

b) Now we also know that

$$\epsilon = L \Delta I / \Delta t$$



$$\Rightarrow \varepsilon = L(I_2 - I_1)/\Delta t$$

$$\Rightarrow \varepsilon = 0.16H (9A-4A)/0.05 s$$

$$\Rightarrow \varepsilon = 16V \text{ Ans}$$

Question 03: A coil of resistance 100Ω is placed in a magnetic field of $1m \text{ wb}$. The coil has 100 turns and a galvanometer of 400Ω resistance is connected in series with it. Find the average emf and the current if the coil is moved $1/10 s$ from the given field to a field of $0.2 mWb$.

$$\text{Given: } R = 100\Omega \quad \Phi_1 = 1 mWb = 1 \times 10^{-3} Wb \quad \Phi_2 = 0.2 \times 10^{-3} Wb$$

$$R_g = 400\Omega. \quad N = 100 \quad \Delta t = 1/10 s = 0.1s$$

Req: a) emf = ? b) Current = I = ?

Sol: a) As we know that

$$\varepsilon = -N\Delta\Phi/\Delta t$$

$$\Rightarrow \varepsilon = -N(\Phi_2 - \Phi_1)/\Delta t$$

$$\Rightarrow \varepsilon = -100(0.2 \times 10^{-3} Wb - 1 \times 10^{-3} Wb)/0.1s$$

$$\Rightarrow \varepsilon = 0.8V \quad \text{Ans}$$

b) For current we can write as

$$\varepsilon = IR_T$$

$$\Rightarrow \varepsilon = I(R + R_g)$$

$$\Rightarrow I = \varepsilon/(R + R_g)$$

$$\Rightarrow I = 0.8V/(100\Omega + 400\Omega)$$

$$\Rightarrow I = 1.6 \times 10^{-3} A \quad \text{Ans.}$$

Question 04: A horizontal straight wire $10m$ long extending from east to west is falling with a speed of $5.0 m/s$, at right angles to the horizontal component of the earth's magnetic field, $0.30 \times 10^{-4} Wb/m^2$. (a) What is the instantaneous value of the emf induced in the wire? (b) What is the direction of the emf? (c) Which end of the wire is at the higher electric potential?

$$\text{Given: } \ell = 10m \quad v = 5m/s. \quad B = 0.30 \times 10^{-4} Wb/m^2 \quad \theta = 90^\circ$$

Req: a) emf = ε = ? b) Direction of emf = ? c) Which end is at high potential?

Sol(a): As we know that

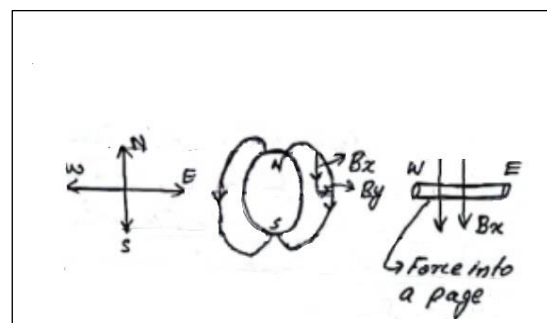
$$\varepsilon = B\ell v \sin\theta$$

$$\Rightarrow \varepsilon = 0.30 \times 10^{-4} Wb/m^2 \times 10m \times 5m/s \sin 90^\circ$$

$$\Rightarrow \varepsilon = 1.5 \times 10^{-3} V. \text{ Ans}$$

(b). Applying Fleming Right hand rule shows that the direction of this induced emf is from east to west.

c). The end towards west will be at higher electric potential because all the charges accumulate at west end.



Question 05: Current in a circuit falls from 5.0A to 0A in 0.1s. If an average emf of 200V induced, give an estimate of the self inductance of the circuit.

Given: $I_1 = 5A$ $I_2 = 0A$ $\Delta t = 0.1s$ $\epsilon = 200V$

Req: Self inductance = $L = ?$

Sol: As we know that

$$\epsilon = -L \Delta I / \Delta t$$

$$\Rightarrow L = -\epsilon \Delta t / \Delta I$$

$$\Rightarrow L = -\epsilon \Delta t / (I_2 - I_1)$$

$$\Rightarrow L = -200V \times 0.1s / (0A - 5A)$$

$$\Rightarrow L = 6H \text{ Ans}$$

Question 06: A long solenoid with 15 turns per cm has a small loop of area 2.0 cm^2 placed inside the solenoid normal to its axis. If the current carried by the solenoid changes steadily from 2.0 A to 4.0 A in 0.1s, what is the induced emf in the loop while the current is changing?

Given: $n = 15/\text{cm} = 15 \times 100/\text{m} = 1500/\text{m}$ $\Delta t = 0.1s$ $I_1 = 2A$ $I_2 = 4A$

$$A = 2\text{cm}^2 = 2 \times (10^{-2}\text{m})^2 = 2 \times 10^{-4} \text{ m}^2$$

Req: emf = $\epsilon = ?$

Sol: As we know that

$$\epsilon = N \Delta \Phi / \Delta t \quad \text{put } N=1 \text{ due to one loop}$$

$$\Rightarrow \epsilon = \Delta \Phi / \Delta t$$

$$\Rightarrow \epsilon = \Delta BA / \Delta t$$

$$\Rightarrow \epsilon = A \Delta B / \Delta t \quad \text{put } B = n\mu_0 I \text{ for solenoid.}$$

$$\Rightarrow \epsilon = A n \mu_0 \Delta I / \Delta t$$

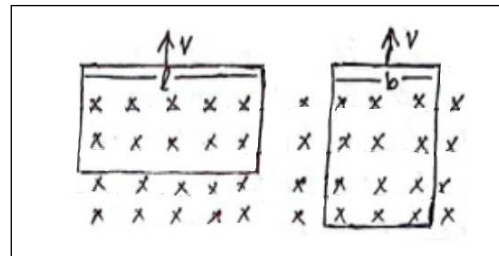
$$\Rightarrow \epsilon = n\mu_0 A \Delta I / \Delta t$$

$$\Rightarrow \epsilon = n\mu_0 A (I_2 - I_1) / \Delta t$$

$$\Rightarrow \epsilon = 1500/\text{m} \times 4\pi \times 10^{-7} \text{ Tm/A} \times 2 \times 10^{-4} \text{ m}^2 \times (4A - 2A) / 0.1s$$

$$\Rightarrow \epsilon = 7.54 \times 10^{-6} \text{ V Ans}$$

Question 07: A rectangular wire loop of sides 8cm and 2cm with a small cut is moving out of a region of uniform magnetic field of magnitude 0.3T directed normal to the loop. What is the emf developed across the cut if the velocity of the loop is 1cm/s in a direction normal to the (a) longer side to shorter side of the loop? For how long does the induced voltage last in each case?



Given: length = $\ell = 8\text{cm} = 0.08\text{m}$ width = $b = 2\text{cm} = 0.02\text{m}$ $N=1$

$$B = 0.3T$$

$$\theta = 90^\circ$$

$$v = 1\text{cm/s} = 0.01\text{m/s.}$$

Req: a) $\epsilon_1 = ?$ $\Delta t_1 = ?$ b) $\epsilon_2 = ?$ $\Delta t_2 = ?$

Sol: a) we find emf (ϵ_1) when a rectangular loop moves length wise as shown in fig. For this we can write as

$$\epsilon_1 = B \ell v \sin\theta$$

$$\Rightarrow \epsilon_1 = 0.3T \times 0.08m \times 0.01m/s \times \sin 90^\circ$$

$$\Rightarrow \epsilon_1 = 2.4 \times 10^{-4} \text{ V Ans}$$

Now we find time Δt_1 for emf (ϵ_1) in rectangular coil.

For this we can write as

$$\epsilon_1 = N \Delta\Phi / \Delta t_1$$

Now we can write as $\Delta\Phi = BA \Rightarrow B(\ell b)$. Now we can write as

$$\epsilon_1 = N B \ell b / \Delta t_1$$

$$\Rightarrow \Delta t_1 = N B \ell b / \epsilon_1$$

$$\Rightarrow \Delta t_1 = 1 \times 0.3T \times 0.08m \times 0.02m / 2.4 \times 10^{-4} \text{ V}$$

$$\Rightarrow \Delta t_1 = 2s \quad \text{Ans}$$

Part (b): Now we find emf (ϵ_2) when a rectangular loop moves width wise as shown in figure. For this we can write as

$$\epsilon_2 = Bbv \sin\theta$$

$$\Rightarrow \epsilon_2 = 0.3T \times 0.02m \times 0.01m/s \times \sin 90^\circ$$

$$\Rightarrow \epsilon_2 = 0.6 \times 10^{-4} \text{ V Ans}$$

Now we find Δt_2 for emf (ϵ_2) in rectangular coil. For this we can write as

$$\Delta t_2 = N B \ell b / \epsilon_2$$

$$\Rightarrow \Delta t_2 = 1 \times 0.3T \times 0.08m \times 0.02m / 0.6 \times 10^{-4} \text{ V}$$

$$\Rightarrow \Delta t_2 = 8s \quad \text{Ans.}$$

Question 08: A 90mm length of wire moves with an upward velocity of 35 m/s between the poles of a magnet. The magnetic field is 80mT directed to the right. If the resistance in the wire is 5.00m Ω , what are the magnitude and direction of the induced current?

$$\text{Given: } \ell = 90\text{mm} = 90 \times 10^{-3}\text{m} \quad V = 35\text{m/s} \quad B = 80\text{mT} = 80 \times 10^{-3} \text{ T} \quad \theta = 90^\circ$$

$$R = 5\text{m}\Omega = 5 \times 10^{-3}\Omega$$

$$\text{Req: Magnitude and direction of current} = I = ?$$

Sol: First we find an emf (ϵ) in a wire. i.e

$$\epsilon = B \ell v \sin\theta$$

$$\Rightarrow \epsilon = 80 \times 10^{-3}\text{T} \times 90 \times 10^{-3}\text{m} \times 35\text{m/s} \sin 90^\circ$$

$$\Rightarrow \epsilon = 0.252 \text{ V}$$

Now for current we can write as

$$I = \epsilon / R$$

$$\Rightarrow I = 0.252\text{V} / 5 \times 10^{-3}\Omega = 50.4\text{A}$$

Direction of current, according to Fleming Right hand rule, is into the plane of page. (See figure).

Question 09: A pair of adjacent coils has a mutual inductance of 1.5H. If the current in one coil changes from 0 to 20A in 0.5s, what is the change of flux linkage with the other coil ?

$$\text{Given: } M = 1.5\text{H} \quad I_1 = 0\text{A} \quad I_2 = 20\text{A} \quad \Delta t = 0.5\text{s} \quad \text{Req: Flux linkage} = N \Delta\Phi = ?$$

Sol: As we know that

$$M = N \Delta\Phi / \Delta I$$

$$\Rightarrow N \Delta\Phi = M \Delta I$$

$$\Rightarrow N \Delta\Phi = M(I_2 - I_1)$$

$$\Rightarrow N \Delta\Phi = 1.5H(20A - 0A)$$

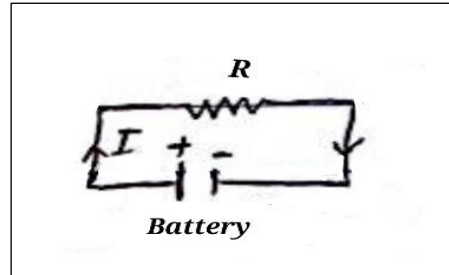
$$\Rightarrow N \Delta\Phi = 30Wb.$$

END OF CHAPTER 04

Chapter #05 Alternating current

Direct voltage or constant voltage: The voltage which does not change its polarity as well as magnitude with respect to time is called direct voltage or constant voltage. Direct voltage Sources are Batteries and cells produces a direct voltage or constant voltage.

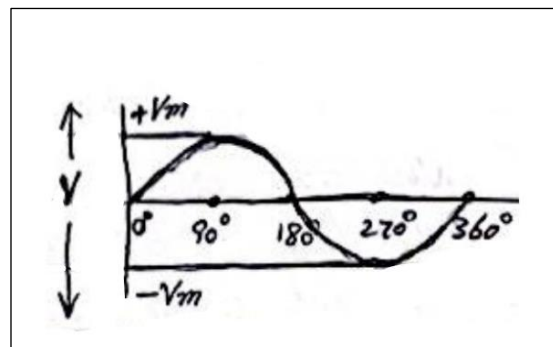
Direct current or constant current: The current which does not change its direction as well as magnitude with respect to time is called direct current or constant current. In direct current, the electrons flow continuously in one direction from the source through a conductor then to a load and then back to the source. Due to direct current or constant current the voltage drop across the load is always constant.



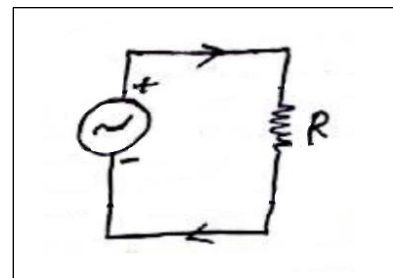
Alternating voltage or sinusoidal voltage: The voltage which changes its polarity as well as magnitude with respect to time is called Alternating voltage. It is also called sinusoidal voltage because its wave form is like as sine wave form. The source of alternating voltage is AC generator. The alternating voltage produced by an AC generator can be written as

$$V = V_m \sin \omega t \text{ ----- ① or } V = V_m \sin \theta \text{ ----- ②}$$

Where “V” is the instantaneous value of alternating voltage, V_m is the maximum value of alternating voltage and “ ω ” is the angular velocity of the coil. Now with the help of equation the waveform of alternating voltage is shown in figure. Now from the waveform we can say that the alternating voltage is +ve and changes the magnitude in the first half cycle of the coil and then it is negative and changes its magnitude in next half cycle of the coil. From the waveform we can also say that the alternating voltage changes with rapid rate at zero point and changes more slowly at peak points.

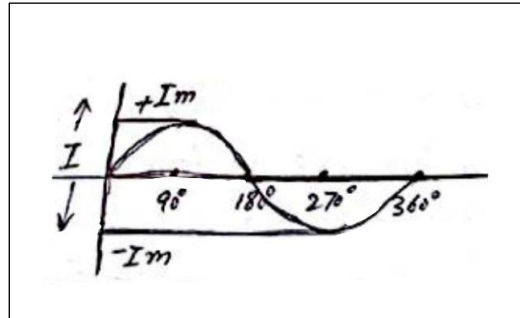


Alternating current of sinusoidal current: The current which changes its direction as well as magnitude with respect to time is called Alternating current. It is also called sinusoidal current because its wave form is like as sine wave form. The source of alternating current is AC generator. The alternating current produced by an AC generator can be written as



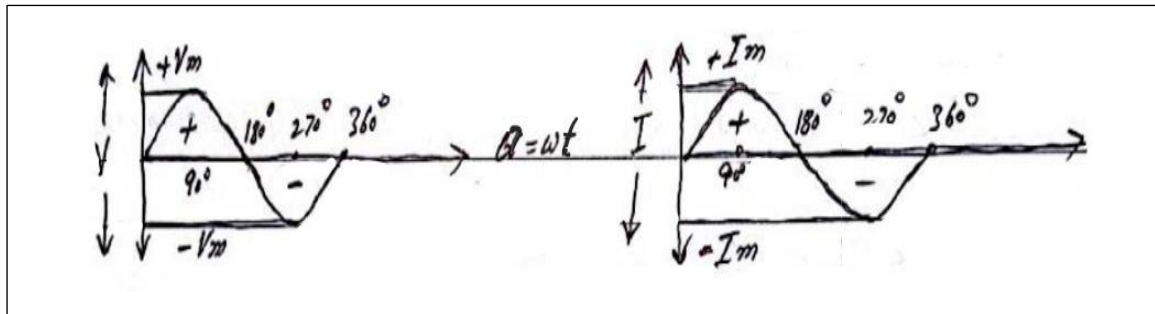
$$I = I_m \sin \omega t \text{ ----- ① or } I = I_m \sin \theta \text{ ----- ②}$$

Where “I” is the instantaneous value of alternating current, “ I_m ” the maximum value of alternating current and “ ω ” is the angular velocity of the coil. Now with the help of equations the wave form of alternating current is shown in figure. Now from the wave form we can say that the alternating current is + ve and changes the magnitude in the first half cycle and then it changes the direction and then become negative with changing magnitude in next half cycle of the coil. From the wave form we can also say that the alternating current changes with rapid rate at zero point and changes more slowly at peak points. Now due to alternating current the voltage drop across the load in the circuit is also alternating.



AC Terminologies

Alternating voltage or current changes continuously in magnitude and alternates in direction at regular intervals of time. It uses from zero to maximum positive value, falls to zero, increase to a maximum in the reverse direction and falls back to zero again. The important AC terminology is defined below.



- ① **Cycle:** One complete set of positive and negative values of an alternating voltage or current is known as a cycle as shown in figures. One cycle corresponds to 360° or 2π radians.
- ② **Time period:** The time required for one cycle of alternating voltage or current is called Time period. It is represented by “T”.
- ③ **Frequency:** The number of cycles of alternating voltage or current in one second is called frequency (f). It is measured in cycle per second or hertz. The frequency of alternating voltage or current or of power in Pakistan is 50 Hz. It means that alternating voltage or current completes 50 cycles in one second.
- ④ **Instantaneous value:** The value of an alternating voltage or current at any instant of time is called instantaneous value. The instantaneous values of alternating voltage and current are represented by “V” and “I” respectively.

As an example the instantaneous values of voltage at 0° , 90° and 270° are “0”, $+V_m$ and $-V_m$ respectively as shown in figure.

⑤ **Peak values:** The maximum values at the peak of AC wave forms is called peak values. It is also known as the maximum value or amplitude. The peak values of alternating voltage and current is represented by “ $+V_m$ ”, “ $-V_m$ ” and “ I_m ”, “ $-I_m$ ” respectively.

⑥ **Average value:** The average value of a waveform is the average of all its values over a period of time. For one cycle the average value of alternating voltage or current is equal to zero. i.e

$$\langle V \rangle = (+ \text{A.values in first half cycle} - \text{A.values in next half cycle}) / \text{Time period}$$

$$\Rightarrow \langle V \rangle = 0/T$$

$$\Rightarrow \langle V \rangle = 0$$

Similarly we can write as

$$\langle I \rangle = (+ \text{A.values in first half cycle} - \text{A.values in next half cycle}) / \text{Time period}$$

$$\Rightarrow \langle I \rangle = 0/T$$

$$\Rightarrow \langle I \rangle = 0$$

Average value can be also define as “The sum of positive and negative areas of wave form over a period of time.” For one cycle the average value of alternating voltage and current w.r.t second definition are calculated as i.e

$$\langle V \rangle = (+ \text{area in first half cycle} - \text{area in next half cycle}) / \text{Time period}$$

$$\Rightarrow \langle V \rangle = (+A - A)/T$$

$$\Rightarrow \langle V \rangle = 0/T$$

$$\Rightarrow \langle V \rangle = 0$$

Similarly we can write as

$$\langle I \rangle = (+ \text{area in first half cycle} - \text{area in next half cycle}) / \text{Time period}$$

$$\Rightarrow \langle I \rangle = (+A - A)/T$$

$$\Rightarrow \langle I \rangle = 0/T$$

$$\Rightarrow \langle I \rangle = 0$$

R.M.S OR EFFECTIVE VALUE

The value of alternating voltage or current whose behavior is equivalent to direct voltage or direct current is called RMS or effective value.

Explanation: when we find the RMS value of alternating voltage of current. Then the effect produced from such a values in a given circuit will be same of effect produced by the DC values of same magnitude.

For example we have $V_{rms} = 5V$ and $I_{rms} = 2A$. Then

$$P = V_{rms} I_{rms} \Rightarrow P_{rms} = 5V \times 2A \Rightarrow P_{rms} = 10W \text{ ----- } \textcircled{1}$$

Now we have $V_{D.C} = 5V$ and $I_{D.C} = 2A$ Then

$$P = V_{D.C} \times I_{D.C} \Rightarrow P_{D.C} = 5V \times 2A \Rightarrow P_{D.C} = 10W \text{ ----- } \textcircled{2}$$

Now from ① and ② we can say that the effect of power produced due to rms values in a given circuit is same of the effect produced by the DC values of same magnitude.

Calculation of RMS value for Alternating current:

As we know that the alternating current can be written as

$$I = I_m \sin \omega t \quad \text{or} \quad I = I_m \sin \theta \quad \text{----- (1)}$$

Now taking square of eq (1) . We get as

$$\Rightarrow I^2 = I_m^2 (\sin \theta)^2 \quad \text{----- (2)}$$

Now for one cycle we have the following angles ($0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$). Then

$$\begin{aligned} \sin \theta &= 0, \quad +1, \quad 0, \quad -1, \quad 0 \\ (\sin \theta)^2 &= 0, \quad +1, \quad 0, \quad +1, \quad 0 \end{aligned}$$

Now we find mean of $(\sin \theta)^2$. For this we can write as i.e

$$\langle (\sin \theta)^2 \rangle = (\text{minimum value} + \text{maximum value})/2$$

$$\Rightarrow \langle (\sin \theta)^2 \rangle = (0 + 1)/2 = 1/2 \quad \text{----- (3)}$$

Now we taking mean of equation (3).

Then we get as i.e

$$\langle I^2 \rangle = \langle I_m^2 (\sin \theta)^2 \rangle$$

$$\Rightarrow \langle I^2 \rangle = I_m^2 \langle (\sin \theta)^2 \rangle \quad \text{----- (4)}$$

$$\langle I_m^2 \rangle = I_m^2 \quad \text{because it is constant.}$$

Now put eq (3) in eq (4). We get as

$$\Rightarrow \langle I^2 \rangle = I_m^2 \times 1/2$$

$$\Rightarrow \langle I^2 \rangle = I_m^2 / 2 \quad \text{----- (5)}$$

Now we taking Root of equation (5).i.e

$$\Rightarrow \sqrt{\langle I^2 \rangle} = \sqrt{I_m^2 / 2}$$

$$\Rightarrow I_{\text{rms}} = I_m / \sqrt{2}$$

$$\Rightarrow I_{\text{rms}} = 0.707 I_m \quad \text{----- (6)}$$

From eq (6) we can easily find RMS value of current if we know the maximum value of alternating current.

Calculation for RMS value of Alternating voltage:

As we know that

$$I_{\text{rms}} = 0.707 I_m \quad \text{----- (1)}$$

Now multiply both side by "R" of equation (1). We get as

$$I_{\text{rms}} R = 0.707 I_m R$$

$$\Rightarrow V_{\text{rms}} = 0.707 V_m \quad \text{----- (2)}$$

From eq (2) we can easily find RMS value of voltage if we know the maximum value of alternating voltage.

Calculation for average value of power:

As we know that

$$P = I^2 R$$

Now put $I = I_m \sin \theta$. Then we get as

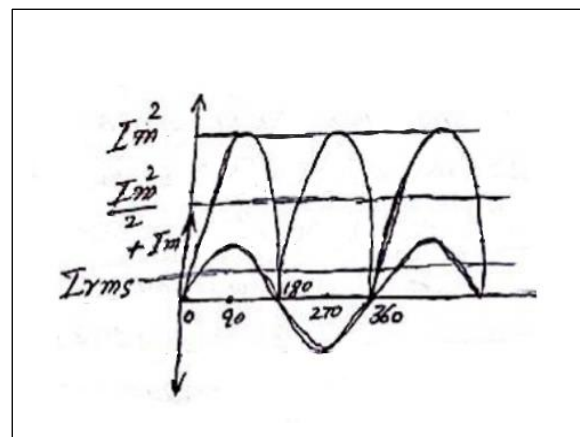
$$P = (I_m \sin \theta)^2 R$$

$$\Rightarrow P = I_m^2 (\sin \theta)^2 R \quad \text{----- (1)}$$

Now taking average of eq (1). We get as

$$\langle P \rangle = \langle I_m^2 (\sin \theta)^2 R \rangle$$

$$\Rightarrow \langle P \rangle = I_m^2 R \langle (\sin \theta)^2 \rangle$$



$$\Rightarrow \langle P \rangle = I_m^2 R \times 1/2$$

$$\Rightarrow \langle P \rangle = I_m^2 R / 2 \text{ ----- (2)}$$

Now we know that

$$I_{rms} = I_m / \sqrt{2} \quad \Rightarrow I_m = \sqrt{2} I_{rms} \quad \Rightarrow I_m^2 = 2 I_{rms}^2 \text{ ----- (4)}$$

Put eq (4) in eq (3) we get as

$$\langle P \rangle = 2 I_{rms}^2 R / 2$$

$$\Rightarrow \langle P \rangle = I_{rms}^2 R \text{ ----- (5)}$$

Eq (5) can be written as

$$\langle P \rangle = I_{rms} \times I_{rms} R$$

$$\Rightarrow \langle P \rangle = I_{rms} V_{rms} \text{ ----- (6)}$$

From eq (5) or (6) we can easily find average power if we know the RMS value of alternating current and voltage.

Note 1: If the average value of a define quantity is zero. Then we need to find its RMS value. Because RMS value is never zero.

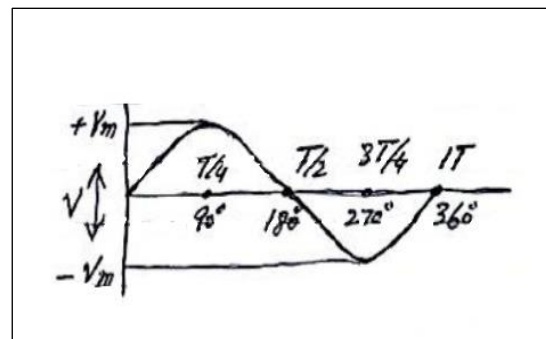
Note 2: The average value of power over one cycle is not equal to zero. So we don't need to find its RMS value.

Note 3: If examiner wants to find the RMS value of voltage. Then you first calculate the RMS value of current, and then find RMS value of voltage.

Phase of AC

The angle or fraction of time period which specifies the instantaneous value or state of alternating voltage or current is called its phase.

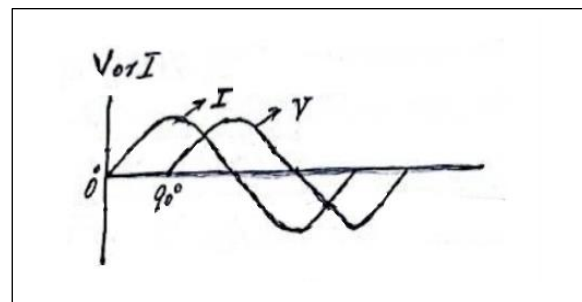
Explanation: The wave form of alternating voltage is shown in figure. Now in figure the angle (90°) or fraction of time period ($T/4$) represents the maximum value of alternating voltage i.e. ($+V_m$). So angle (90°) or fraction of time period ($T/4$) is called phase. Similarly angle (270°) or fraction of time period ($3T/4$) represents the minimum value of alternating voltage i.e. ($-V_m$). So angle (270°) or fraction of time period ($3T/4$) is called phase.



Phase difference

When two alternating quantities (wave forms) of the same frequency have different zero points, then they are said to have a phase difference.

Explanation: Consider a figure in which the wave form of AC voltage and current are shown. In a figure, the wave forms take a start from different position, so we can say that these two wave forms have phase difference.



Phase difference of two waveforms is represented by an angle " Φ ". For the given wave form the phase difference angle " Φ " can be calculated as

$$\Phi = 90^\circ - 0^\circ \Rightarrow \Phi = 90^\circ.$$

Now the wave form which starts earlier for which we will use leading word and the wave form which starts later for which we will use lagging word. Now from figure we can say that the wave form of current is in leading from wave form of voltage with phase difference angle 90° .

For this the equations of currents and voltage we can write as

$$I = I_m \sin(\omega t + 90^\circ) \text{ and } V = V_m \sin \omega t \text{ ----- (A)}$$

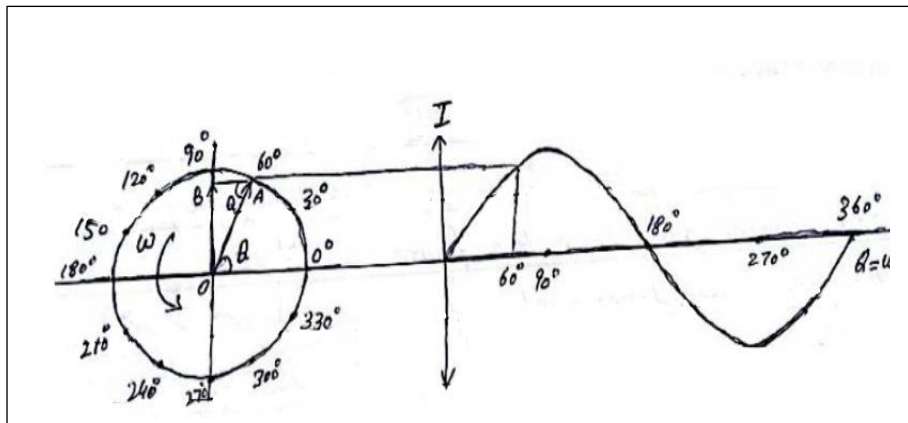
Now from figure we can also say that the wave form of voltage is in lagging from wave form of current with phase angle 90° . For this the equations of current and voltage we can write as

$$I = I_m \sin \omega t \text{ and } V = V_m \sin(\omega t + 90^\circ) \text{ ----- (B)}$$

Both set up equations (A) and (B) are correct for the given wave forms.

Phasor diagram

An equivalent diagram for AC wave form is called Phasor diagram. In a Phasor diagram we will take a vector which is said to be Phasor or Phase vector. Such a vector we will rotate with angular velocity " ω " equal to the wave form angular velocity in anti clock wise direction. We will take its length equal to maximum value in wave form.



And we will take its projection on Y-axis and such a projection will represent the instantaneous value in waveform. And its angle " θ " from x-axis will represent the angle in waveform as shown by two lines. Now from the figure we can write as

$$\sin \theta = OB/OA$$

$$\Rightarrow OB = OA \sin \theta \text{ ----- (1)} \quad I = I_m \sin \theta \text{ ----- (2)}$$

In eq (1) "OB" is projection on y-axis, it will represent the instantaneous value in wave form. OA is the length of a vector, it will represent the maximum value in wave form. Angle " θ " from x-axis will represent the angle in wave form. Now we explain some cases.

Case 1: When a phase vector on +ve x-axis:

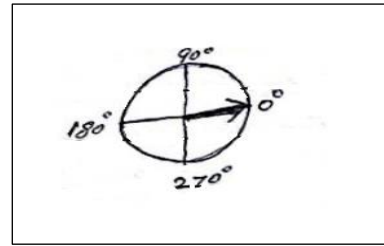
When a phase vector on +ve x-axis in a Phasor diagram, then we find its projection. In this case the angle $\theta = 0$.

Now we know that $OB = OA \sin \theta$

$$\Rightarrow OB = I_m \sin \theta$$

$$\Rightarrow OB = 0$$

Now the projection "OB" of a phase vector is zero. So we can say that the instantaneous value of current will be also zero. So now we can also say that when a phase vector is on +ve x-axis. Then it will represent the zero value of instantaneous current. i.e ($I = 0$).



Case 2: When a phase vector on +ve y-axis:

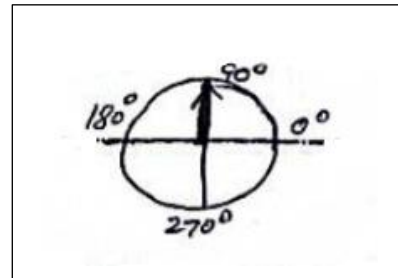
When a phase vector on +ve y-axis in a Phasor diagram, then we find its projection. In this case the angle $\theta = 90^\circ$.

Now we know that $OB = OA \sin \theta$

$$\Rightarrow OB = I_m \sin 90^\circ \quad (\sin 90^\circ = 1)$$

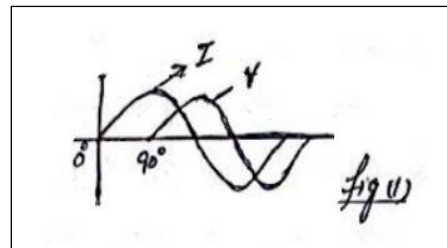
$$\Rightarrow OB = I_m$$

Now the projection "OB" represents the maximum value of instantaneous current. So now we can say that when a phase vector is on +ve y-axis. Then it will represent the maximum value of instantaneous current. i.e $I = I_m$.



Phasor diagram when two waveforms have a phase difference:

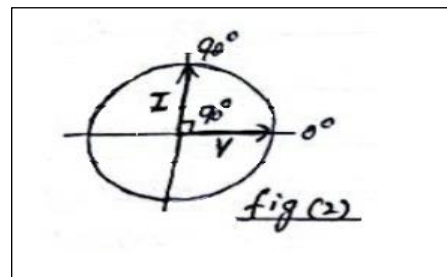
Two waveforms having a phase difference are shown in figure. Now we can say that the wave form of current is in leading from wave form of voltage with phase difference angle 90° . For this the equation of current and voltage we can write as.



$$I = I_m \sin(\omega t + 90^\circ) \text{ ----- } (1)$$

$$V = V_m \sin \omega t \text{ ----- } (2)$$

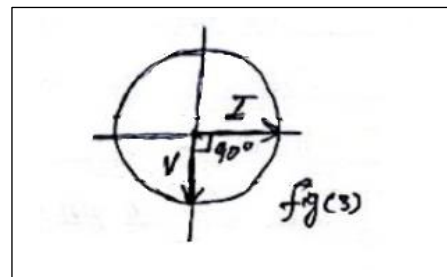
Now from eq (1) and (2) we draw a Phasor diagram. In a figure the phase vector of current is in leading from phase vector of voltage with phase angle 90° . From figure we can also say that the wave form of voltage is in lagging from wave form of current with phase difference angle 90° . For this the equation of current and voltage we can write as



$$I = I_m \sin \omega t \text{ ----- } (3)$$

$$V = V_m \sin(\omega t - 90^\circ) \text{ ----- } (4)$$

Now from (3) and (4) we draw a Phasor diagram. In a figure the phase vector of voltage is in lagging from phase vector of current with phase angle 90° . Fig (2) and (3) both are correct Phasor diagrams for a given waveforms.



AC through Resistor

AC through resistor means to study the behavior of alternating voltage and current across the resistor. For this we consider a circuit which has alternating voltage source and one resistor. The alternating current in the circuit is also shown.

Now in the given circuit the voltage across the resistor is "V". And such a voltage can be written as

$$V = V_m \sin \omega t \text{ ----- } (1)$$

Now due to such a voltage "V" the current "I" is flowing in the resistor. And such a current "I" we find from ohms law. i.e

$$V = IR$$

$$\Rightarrow I = V/R \text{ ----- } (2)$$

Now put eq (1) in eq (2) we get as

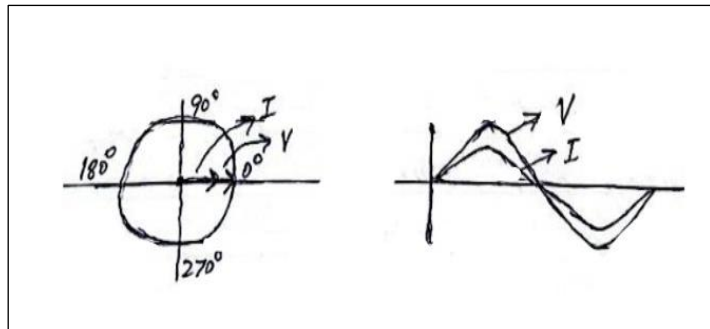
$$I = V_m \sin \omega t / R . \text{ OR}$$

$$\Rightarrow I = V_m / R \sin \omega t$$

$$\Rightarrow I = I_m \sin \omega t \text{ ----- } (3) \quad (\text{where } V_m/R = I_m)$$

Waveforms for alternating voltage "V" and current "I" in a resistor:

As we know that $V = V_m \sin \omega t$ and $I = I_m \sin \omega t$. By using these equations the waveforms are shown in figure. From the given waveforms we can say that both alternating voltage "V" and current "I" are in phase in the resistor. The Phasor diagram for the given wave form is also shown in figure.



Power loss in a resistor:

The energy dissipated per second in a resistor in the form of heat is called power loss. Now we find the average power dissipated in resistor "R" for one complete cycle of alternating voltage "V_R" and current "I". For this we can write as

$$\langle P \rangle = \langle VI \rangle$$

$$\Rightarrow \langle P \rangle = \langle V_m \sin \omega t \times I_m \sin \omega t \rangle$$

$$\Rightarrow \langle P \rangle = V_m I_m \langle (\sin \omega t)^2 \rangle$$

$$\Rightarrow \langle P \rangle = V_m I_m \langle (\sin \theta)^2 \rangle$$

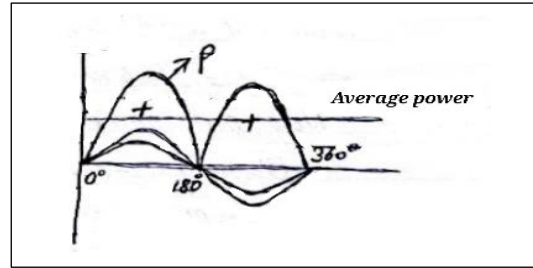
$$\Rightarrow \langle P \rangle = V_m I_m / 2$$

$$\Rightarrow \langle P \rangle = V_{rms} I_{rms} \text{ ----- } (4)$$

From equation (4) we can say that power will be dissipated in a resistor.

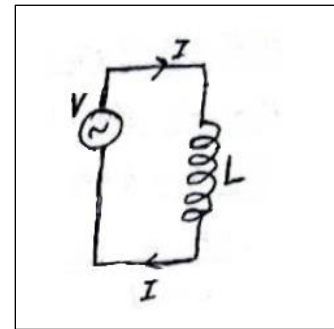
Power loss in a resistor from wave forms:

As we know that $P = VI$. So in the wave forms we multiply the regions of alternating voltage "V" and of current "I" every where, then we will get the wave form for power dissipated as shown in figure. The wave form of power represents the power dissipated is always positive and it is zero at 0° , 180° and 360° . Now in figure the average power is define, So we can say that power will be dissipated in a resistor and its average value will be positive.



AC through inductor

An inductor is a coil shaped conductor which resists the change in electric current passing through it. Now AC through inductor mean to study the behaviors of alternating voltage and current across the inductor. For this we consider a circuit which has alternating voltage source and one inductor. The alternating current in the circuit is also shown. Now in the given circuit the voltage across the inductor is "V". And such a voltage can be written as



$$V = \varepsilon \text{ ----- (1) (where } \varepsilon \text{ is called Back emf in the inductor).}$$

Now we know that the back emf in the inductor can be written as

$$\varepsilon = L \Delta I / \Delta t \text{ ----- (2) Now put eq (2) in eq (1). We get as}$$

$$\Rightarrow V = L \Delta I / \Delta t \text{ ----- (3)}$$

In eq (3) "I" is the alternating current in the inductor. And it can be written as

$$I = I_m \sin \omega t \text{ ----- (4) Now put eq (4) in eq (3). We get as}$$

$$\Rightarrow V = L \Delta I_m \sin \omega t / \Delta t$$

$$\Rightarrow V = L I_m \Delta \sin \omega t / \Delta t \text{ ----- (5)}$$

Now from math's we know that

$$\Delta \sin \omega t / \Delta t = \omega \cos \omega t \text{ ----- (6)}$$

Now put eq (6) in eq (5). We get as

$$V = L I_m \omega \cos \omega t \text{ ----- (7)}$$

Now "V" will be maximum, if we put maximum value of $\cos \omega t = 1$. i.e

$$V_m = L I_m \omega \text{ ----- (8). Now put eq (8) in eq (7). we get as}$$

$$V = V_m \cos \omega t \text{ ----- (9)}$$

Now from math's we know that $\cos \omega t = \sin(\omega t + \pi/2)$ ----- (10)

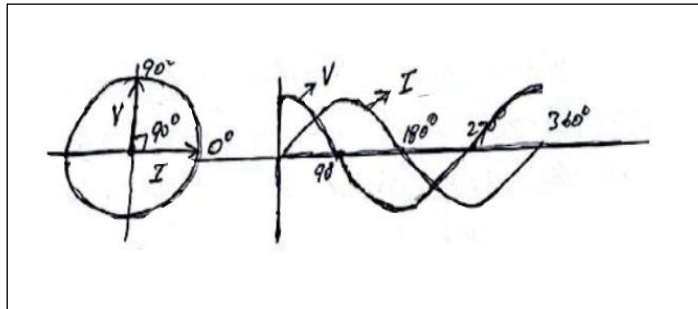
Now put eq (10) in eq (9). We get as

$$V = V_m \sin(\omega t + \pi/2) \text{ ----- (11)}$$

Now eq (11) and (4) represents the behavior of alternating voltage and current in an inductor.

Wave forms for alternating voltage “V” and current “I” in an inductor:

As we know that $V = V_m \sin(\omega t + \pi/2)$ and $I = I_m \sin \omega t$. By using these equations the wave form are shown in figure. From the given wave forms we can say that the current is in lagging from alternating voltage with phase difference angle 90° . OR we can say that the voltage is in leading from current with phase difference angle 90° or $\pi/2$. The Phasor diagram is also shown in figure.



Inductive Reactance:

The opposition offered by an inductor to the flow of alternating current is known as inductive reactance of an inductor. Inductive reactance is represented by “ X_L ”. Now we know that from Ohm’s law

$V = IR$ ----- (12) . Now replace “R” by “ X_L ” in eq(12) we get as

$$V = I X_L$$

$$\Rightarrow V_m = I_m X_L$$

$$\Rightarrow X_L = V_m / I_m$$
 ----- (13) Put eq(8) in eq(13). We get as

$$X_L = L I_m \omega / I_m$$

$$\Rightarrow X_L = L \omega$$

$$\Rightarrow X_L = 2\pi f L$$
 ----- (14) (where $\omega = 2\pi f$)

Equation (14) shows that the inductive reactance depends on inductance “L” and frequency of AC source.

Power loss in an inductor

The energy dissipated per second in an inductor in the form of heat is called power loss. Now we find the average power dissipated in an inductor for one complete cycle of alternating voltage and current. For this we can write as

$$\langle P \rangle = \langle V \rangle \langle I \rangle$$

$$\Rightarrow \langle P \rangle = \langle V_m \cos \omega t \rangle \langle I_m \sin \omega t \rangle$$

$$\Rightarrow \langle P \rangle = V_m I_m \langle \cos \omega t \rangle \langle \sin \omega t \rangle$$
 ----- (15)

Now we can write as

$$\theta = \omega t = 0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$$

$$\sin \omega t = 0 \quad 1 \quad 0 \quad -1 \quad 0$$

$$\cos \omega t = 1 \quad 0 \quad -1 \quad 0 \quad 1$$

$$\text{Now } \langle \sin \omega t \rangle = (\text{min value} + \text{max value})/2$$

$$\langle \sin \omega t \rangle = (-1+1)/2 = 0$$

$$\text{Now } \langle \cos \omega t \rangle = (\text{min value} + \text{max value})/2$$

$$\langle \cos \omega t \rangle = (-1+1)/2 = 0$$

Now eq (15) we can write as

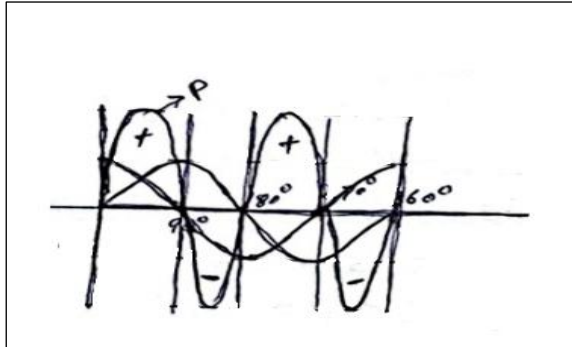
$$\langle P \rangle = V_m I_m \times 0 \times 0$$

$$\Rightarrow \langle P \rangle = 0 \text{ ----- (16)}$$

From eq (16) we can say that the dissipated power is equal to zero in an inductor over one cycle.

Power loss in an inductor from wave forms:

As we know that $P=VI$. So in the wave forms we multiply the regions of alternating voltage “V” and of current “I” every where, then we will get the waveform for power dissipated as shown in figure. Now the wave form of power represents the dissipated power is zero, because of same +ve and -ve regions in this wave form. Now from wave form we can say that



- ① During first 90° of the cycle, the power is +ve. It means that the power is flowing from the source to the inductor.
- ② During next 90° of the cycle, the power is negative. It means that the power is flowing from the inductor to the source.
- ③ During next 90° of the cycle it is again +ve.
- ④ During the last 90° of the cycle it is -ve.

Note: As we have explained the dissipated power in an inductor is zero. It means the inductor does not dissipate the electrical energy but the inductor stores the electrical energy in the form of back emf. In an inductive circuit the energy supplied from the source is equal to the energy stored in the inductors. i.e we know that

$$V = \mathcal{E}$$

Electrical Energy Supplied/Q = Electrical Energy stored/Q

Choke coil

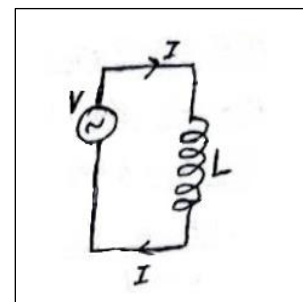
A choke is an inductor which is used in AC circuits. The reactance of inductor can be written as

$$X_L = 2\pi fL$$

$$\Rightarrow X_L \propto f \text{ ----- (1)}$$

So from relation we can say that a choke offers high reactance to frequencies above a certain frequency range without appreciable limiting the flow of current.

A choke can prevent electric signals along undesired paths. It is used as filter in power supply to prevent undesired signals. It is used as radio frequency choke (RFC) to prevent radio frequency signals from entering audio frequency circuits.



AC through Capacitor

Capacitor is a device which stores the electric charges or electric energy.

AC through capacitor means to study the behaviors of alternating voltage and current across the Capacitor. For this we consider a circuit which has alternating voltage source and one resistor. The alternating current in the circuit is also shown.

Now in the given circuit the voltage across the capacitor is "V". And such a voltage can be written as

$$V = V_m \sin \omega t \text{ ----- (1)}$$

Now due to such a voltage the current "I" in the capacitor can be written as

$$I = \Delta q / \Delta t \text{ ----- (2)}$$

The store charge "q" in the capacitor will be changing w.r.t time because of alternating voltage. Now we know that the charge on the capacitor can be written as

$$q = CV \text{ ----- (3) put (3) in eq (2). We get as}$$

$$I = \Delta CV / \Delta t$$

$$\Rightarrow I = C \Delta(V_m \sin \omega t) / \Delta t$$

$$\Rightarrow I = CV_m \Delta \sin \omega t / \Delta t \text{ ----- (4)}$$

Now from math's we know that

$$\Delta \sin \omega t / \Delta t = \omega \cos \omega t \text{ ----- (5)}$$

Now put eq (5) in eq (4). We get as

$$\Delta \sin \omega t / \Delta t = \omega \cos \omega t \text{ ----- (5) . Now put eq (5) in eq (4). We get as}$$

$$\Rightarrow I = CV_m \omega \cos \omega t \text{ ----- (6)}$$

Now "I" will be maximum, if we put max value of $\cos \omega t = 1$. i.e

$$I_m = C V_m \omega \text{ ----- (7) . Now put eq (6) in eq (5). We get as}$$

$$\Rightarrow I = I_m \cos \omega t \text{ ----- (8)}$$

Now from math's we know that

$$\cos \omega t = \sin (\omega t + \pi/2) \text{ ----- (9)}$$

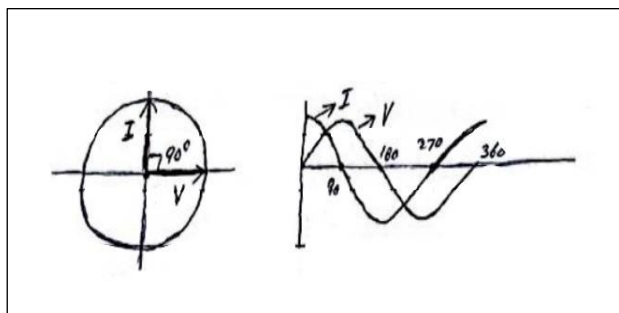
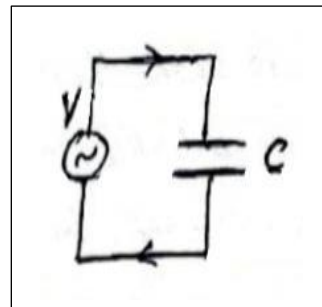
Now put eq (9) in eq (8). We get as

$$\Rightarrow I = I_m \sin (\omega t + \pi/2) \text{ ----- (10)}$$

Now eq (1) and (10) represents the behavior of alternating voltage and current in the capacitor.

Wave forms for alternating voltage "V" and current "I" in a capacitor:

As we know that $V = V_m \sin \omega t$ and $I = I_m \sin (\omega t + \pi/2)$. By using these equations the wave forms are shown in figure. From the given wave forms we can say that the current is in leading from alternating voltage with phase



difference angle “90°” or $\pi/2$.

OR we can say that the voltage is in lagging from current with phase angle 90° or $\pi/2$. The Phasor diagram is also shown in figure.

Capacitive Reactance

The opposition offered by a capacitor to the flow of alternating current in a circuit is known as capacitive reactance “ X_c ”.

Now we know that from Ohm's law i.e

$$V = IR \text{ ----- (1) Replace "R" by "Xc" in eq (1) we get as}$$

$$\Rightarrow V = I X_c$$

$$\Rightarrow V_m = I_m X_c$$

$$\Rightarrow X_c = V_m / I_m \text{ ----- (2)}$$

Now put eq (2) in eq (1) we get as

$$X_c = V_m / C V_m \omega$$

$$\Rightarrow X_c = 1 / C \omega$$

$$\Rightarrow X_c = 1 / 2\pi f C \text{ ----- (3)}$$

Eq (3) show that the capacitive reactance “ X_c ” depends on capacitance and frequency of AC source.

Power loss in a capacitor:

The energy dissipated per second in a capacitor in the form of heat is called power loss. Now we find the average power dissipated in a capacitor for one complete cycle of alternating voltage and current. For this we can write as:

$$\langle P \rangle = \langle V I \rangle$$

$$\Rightarrow \langle P \rangle = \langle V_m \sin \omega t \rangle \langle I_m \cos \omega t \rangle$$

$$\Rightarrow \langle P \rangle = V_m I_m \langle \sin \omega t \rangle \langle \cos \omega t \rangle . \quad \text{Now } \langle \sin \omega t \rangle = 0 \quad \text{and} \quad \langle \cos \omega t \rangle = 0$$

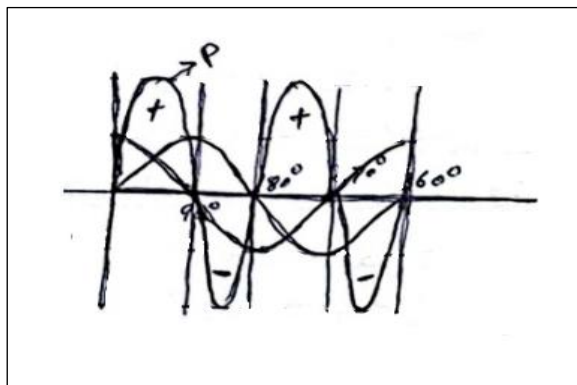
$$\Rightarrow \langle P \rangle = V_m I_m \times 0 \times 0$$

$$\Rightarrow \langle P \rangle = 0 \text{ ----- (4)}$$

From eq (4) we can say that the dissipated power is equal to zero in a capacitor over one cycle. It mean capacitor does not dissipate energy but it store the energy.

Power loss in a capacitor from wave forms:

As we know that $P=VI$. So in the wave forms we multiply the regions of alternating voltage “ V ” and of current “ I ” every where, then we will get the waveform for power dissipated as shown in figure. Now the wave form of power represents the dissipated power is zero, because of same +ve and -ve regions in this wave form.



RL Series A.C Circuit

A circuit which contains Inductor (L) and resistor connected in series to an AC is called RL series A.C circuit as shown in figure. Now we explain the behaviors of resulting alternating voltage and of current in this circuit. For this we know that in case of resistor the equations for current and voltage can be written as

$$I = I_m \sin \omega t \text{ ----- (1) and } V_R = V_m \sin \omega t \text{ ----- (2)}$$

The Phasor diagram from equations is shown in figure.

Now we also know that in case of inductor the equations for current and voltage can be written as

$$I = I_m \sin \omega t \text{ ----- (1) and } V = V_m \sin(\omega t + \pi/2) \text{ ----- (4)}$$

The Phasor diagram from equations is shown in fig (2) and (3). Now to find the resultant voltage in the given RL series AC circuit, for this we combine fig (2) and (3) as shown in fig (4). In figure (4) we will take one phase vector for current, because it is common for both resistor "R" and inductor "L".

Now from Pythagoras theorem we write as

$$\begin{aligned} V^2 &= V_R^2 + V_L^2 \\ \Rightarrow V &= \sqrt{V_R^2 + V_L^2} \\ \Rightarrow V &= \sqrt{(IR)^2 + (IX_L)^2} \\ \Rightarrow V &= \sqrt{I^2(R^2 + X_L^2)} \\ \Rightarrow V &= I \sqrt{R^2 + X_L^2} \\ \Rightarrow V &= I Z \text{ ----- (5)} \end{aligned}$$

In eq (4) $Z = \sqrt{R^2 + X_L^2}$ is called impedance. It is define as the combine opposition offered to the current flow by resister "R" and inductor "L" is called impedance. Now from fig (4) the expression for resultant voltage and current can be written as

$$V = V_m \sin(\omega t + \Phi) \quad \text{and} \quad I = I_m \sin \omega t$$

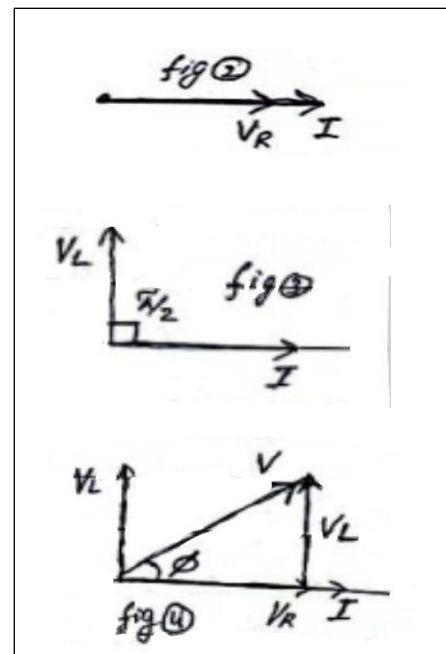
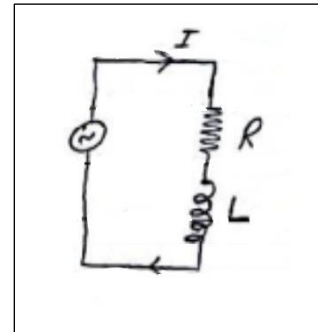
OR $V = V_m \sin \omega t \text{ ----- (5) and } I = I_m \sin(\omega t - \Phi) \text{ ----- (6)}$

Phase angle " Φ ": To find the phase angle " Φ " for this we use tan trigonometric ratio in fig (3). So we can write as

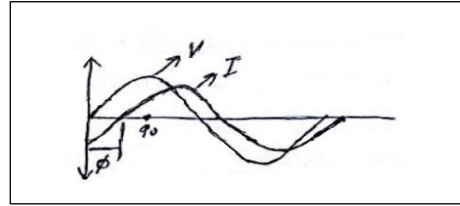
$$\begin{aligned} \tan \Phi &= V_L / V_R \\ \Rightarrow \Phi &= \tan^{-1}(V_L / V_R) \quad \Rightarrow \Phi = \tan^{-1}(IX_L / IR) \quad \Rightarrow \Phi = \tan^{-1}(X_L / R) \end{aligned}$$

Wave forms for resultant voltage and current in RL series A.C circuit:

As we know that $V = V_m \sin \omega t$ and $I = I_m \sin(\omega t - \Phi)$. By using these equations the wave forms are shown in figure. From the given wave forms we can say that the current is in lagging from voltage with phase difference angle $\Phi < 90^\circ$.



OR we can say that the voltage is in leading from current with phase difference angle $\Phi < 90^\circ$.

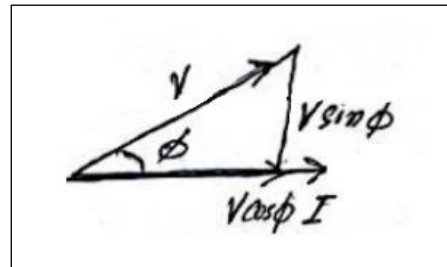


Power loss in RL series AC circuit

As we know that in case of inductor the power loss is equal to zero. But in case of resistor the power loss is not equal to zero. In case of resistor the power loss can be written as

$$\langle P \rangle = V_{\text{rms}} I_{\text{rms}} \text{ ----- (7)}$$

Eq(7) will be valid when the phase vector of " V_R " is in phase with phase vector of current " I " as shown in fig (2). Now in fig (4) the phase vector " V " is not in phase with phase vector of current " I ". But its component $V \cos \Phi$ is in phase with current I . So for power loss in RL circuit eq (7) can be written as

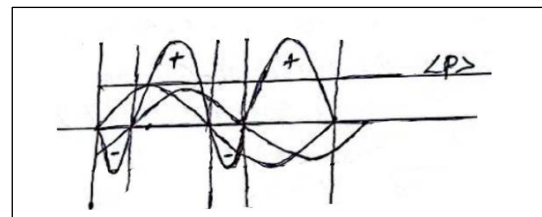


$$\langle P \rangle = V_{\text{rms}} \cos \Phi I_{\text{rms}}$$

$$\Rightarrow \langle P \rangle = V_{\text{rms}} I_{\text{rms}} \cos \Phi \text{ ----- (8). (Here } \cos \Phi \text{ is called power factor).}$$

Power loss in RL Series circuit from wave forms:

Now in figure the magnitude of +ve regions is greater than the magnitude of -ve regions in power wave form. So we can say that the average power is define and it is +ve.



Power factor: The term $\cos \Phi$ in eq (8) is called power factor.

The power factor for resistive circuit is equal to 1. i.e $\cos \Phi = \cos 0 = 1$

The power factor for inductive circuit is equal to 0. i.e $\cos \Phi = \cos 90^\circ = 0$

The power factor for capacitive circuit is also 0. i.e $\cos \Phi = \cos 90^\circ = 0$

Impedance triangle: The triangle which we draw from an equation

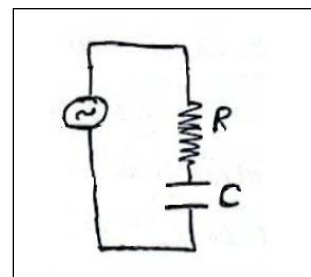
$$Z = \sqrt{R^2 + X_L^2} \text{ or } z^2 = R^2 + X_L^2 \text{ is called Impedance triangle.}$$

Hyp B Perp

The advantage of this triangle is that through which we can easily find " Z " or " X_L " or " R " or " Φ ".

RC Series AC circuit

A circuit which contains capacitor " C " and resistor " R " connected in series to an AC is called RC series AC circuit as shown in fig. Now we explain the behavior of resulting voltage and of current in this circuit. For this now we know that in case of resistor the equations for current and voltage can be written as



$$I = I_m \sin \omega t \text{ ----- (1) and } V_R = V_m \sin \omega t \text{ ----- (2)}$$

The phasor diagram from eq (1) and (2) is shown in figure (2). Now we also know that in case of capacitor the equation for current and voltage can be written as

$$I = I_m \sin \omega t \text{ ----- (3) and } V = V_m \sin(\omega t - \pi/2) \text{ ----- (4)}$$

The phasor diagram from eq (3) and (4) is shown in figure (3). Now to find the resultant voltage in the given RC series AC circuit, for this we combine fig (2) and (3) as shown in figure (4). In figure (4) we will take one phase vector for currents because it is common for both resistor "R" and capacitor "C".

Now from Pythagoras theorem we write as

$$\begin{aligned} V^2 &= V_R^2 + (-V_C)^2 \\ \Rightarrow V &= \sqrt{(V_R^2 + (-V_C)^2)} \\ \Rightarrow V &= \sqrt{(IR)^2 + (-IX_C)^2} \\ \Rightarrow V &= \sqrt{I^2(R^2 + X_C^2)} \\ \Rightarrow V &= I\sqrt{R^2 + X_C^2} \\ \Rightarrow V &= IZ \text{ ----- (5)} \end{aligned}$$

In eq (5) $Z = \sqrt{R^2 + X_C^2}$ is called impedance. It is defined as the combined opposition offered to the current flow by resistor "R" and capacitor "C" is called impedance. Now from figure (4) the expression for resultant voltage and current can be written as

$$\begin{aligned} V &= V_m \sin(\omega t - \Phi) \quad \text{and} \quad I = I_m \sin \omega t \\ \text{OR } V &= V_m \sin \omega t \text{ ----- (5) and } I = I_m \sin(\omega t + \Phi) \text{ ----- (6)} \end{aligned}$$

Phase angle " Φ ": To find the phase angle " Φ " for this we will use the trigonometric ratio in fig (4). So we can write as

$$\begin{aligned} \tan \Phi &= -V_C / V_R \\ \Rightarrow \Phi &= \tan^{-1}(-V_C / V_R) \quad \Rightarrow \Phi = \tan^{-1}(-IX_C / IR) \quad \Rightarrow \Phi = \tan^{-1}(-X_C / R) \end{aligned}$$

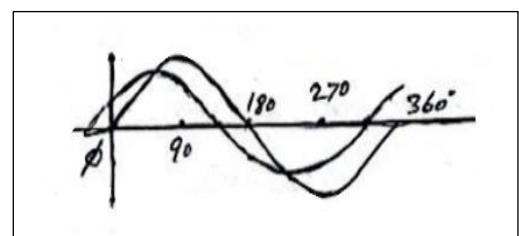
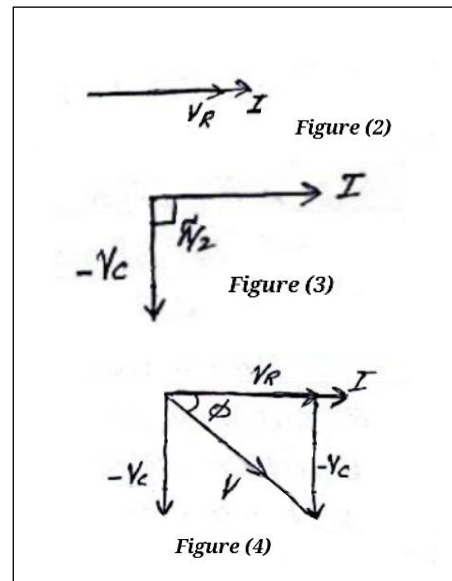
Wave forms for resultant voltage and current in RC series AC circuit:

As we know that $V = V_m \sin \omega t$ and $I = I_m \sin(\omega t + \Phi)$. By using these equations the wave forms are shown in figure. From the given wave forms we can say that the current is leading from voltage with phase difference angle ($\Phi = 90^\circ$). OR we can say that the voltage is lagging from current with phase difference angle ($\Phi = 90^\circ$).

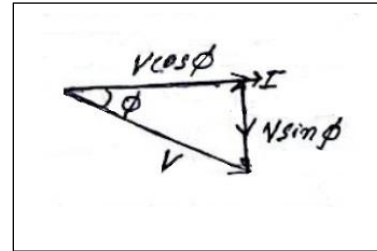
Power loss in RC series AC circuit:

As we know that in case of capacitor the power loss is equal to zero. But in case of resistor the power loss is not equal to zero. In case of resistor the power loss can be written as

$$\langle P \rangle = V_{rms} I_{rms} \text{ ----- (7)}$$



Eq (7) will be valid when the phase vector of " V_R " is in phase with phase vector of current " I " as shown in fig(2). Now in fig (4) the phase vector " V " is not in phase with phase vector of current " I ". But its component $V \cos \phi$ is in phase with current I . So for power loss in RL circuit eq (7) can be written as

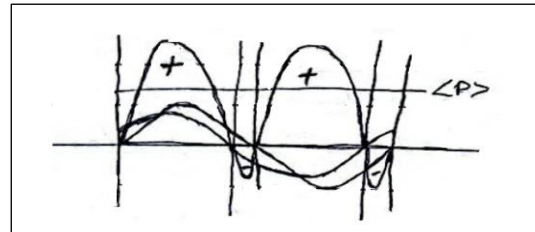


$$\langle P \rangle = V_{\text{rms}} \cos \Phi I_{\text{rms}}$$

$\Rightarrow \langle P \rangle = V_{\text{rms}} I_{\text{rms}} \cos \Phi$ ----- (8). (Here $\cos \phi$ is called power factor).

Power loss in RC series AC circuit from wave forms:

Now in figure the magnitude of +ve regions is greater than the magnitude of -ve regions in power wave form. So we can say that the average power is define and it is + ve.

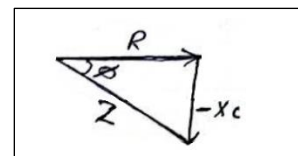


Impedance triangle:

$Z = \sqrt{R^2 + X_C^2}$ or $z^2 = R^2 + X_C^2$ is called Impedance triangle .

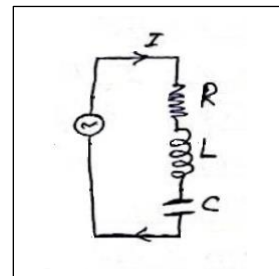
Hyp Base Perp

The advantage of this triangle is that through which we can easily find " Z " or " X_L " or " R " or " Φ ".



RLC Series AC Circuit

A circuit which contains resistor (R) inductor (L) and capacitor (C) connected in series to an AC is called RLC Series AC circuit as shown in figured. Now we explain the behavior of resulting voltage and of current in this circuit. For this now we know that in case of resistor the equations for current and voltage can be written as



$$I = I_m \sin \omega t \text{ -----(1) and } V_R = V_m \sin \omega t \text{ ----- (2)}$$

The Phasor diagram from eq (1) and (2) is shown in figure

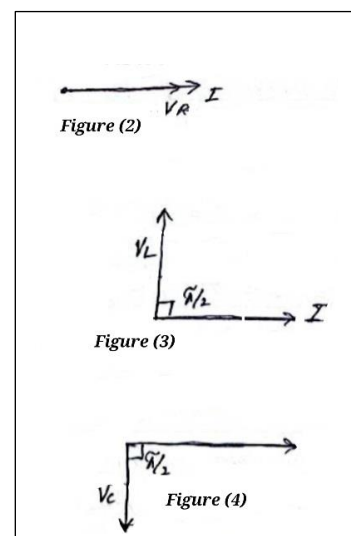
(2). Now we also know that in case of inductor the equation for current and voltage can be written as

$$I = I_m \sin \omega t \text{ -----(3) and } V = V_m \sin(\omega t + \pi/2) \text{ -- (4)}$$

The Phasor diagram from eq (3) and (4) is shown in figure (3). Now we also know that in case of capacitor the equation for current and voltage can be written as

$$I = I_m \sin \omega t \text{ -----(5) and } V = V_m \sin(\omega t - \pi/2) \text{ --- (6)}$$

The Phasor diagram from eq (5) and (6) is shown in figure (4). Now to find the resultant voltage in the given RLC Series AC circuit, for this we combine figures 2, 3, and 4 as shown in fig (5). In fig (5) we will take one phase vector for current, because it is common for all circuit element, and for simplicity we will take $V_L > V_C$.



Now from Pythagoras theorem we write as

$$\begin{aligned} V^2 &= V_R^2 + (V_L - V_C)^2 \\ \Rightarrow V &= \sqrt{(V_R^2 + (V_L - V_C)^2)} \\ \Rightarrow V &= (IR)^2 + (IX_L - IX_C)^2 \\ \Rightarrow V &= \sqrt{I^2(R^2 + (X_L^2 - X_C^2))} \\ \Rightarrow V &= I\sqrt{R^2 + (X_L^2 - X_C^2)} \\ \Rightarrow V &= IZ \text{ ----- (7)} \end{aligned}$$

In eq(7) $Z = \sqrt{R^2 + (X_L^2 - X_C^2)}$

is called impedance. It is define as the combine opposition offered to the current flow by all circuit elements is called impedance. Now from fig (5) the expression for resultant voltage and current can be written as

$$V = V_m \sin(\omega t + \Phi) \quad \text{and} \quad I = I_m \sin \omega t$$

OR $V = V_m \sin \omega t$ ----- (8) and $I = I_m \sin(\omega t - \Phi)$ ----- (9)

Phase angle “ Φ ” : Phase angle can be calculated as

$$\tan \Phi = (V_L - V_C)/V_R$$

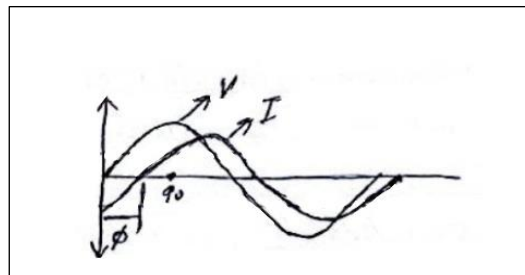
$$\Rightarrow \Phi = \tan^{-1}(V_L - V_C)/V_R \Rightarrow \Phi = \tan^{-1}(IX_L - IX_C)/IR) \Rightarrow \Phi = \tan^{-1}(X_L - X_C)/R$$

Power factor: $\cos \Phi$ is called power factor. It can be written as

$$\cos \Phi = V_R / V = IR / IZ \Rightarrow \cos \Phi = R / \sqrt{R^2 + (X_L^2 - X_C^2)}$$

Wave forms for resultant voltage and current in RLC series AC circuit:

The wave forms are shown in figure. From wave forms we can say that the current is in lagging from the voltage with phase difference angle $\Phi < 90^\circ$.



Power loss in RLC series AC circuit:

In RLC series AC circuit only resistor “R” will dissipate the power. And such power loss can be written as

$$\langle P \rangle = V_{rms} I_{rms} \cos \Phi$$

Impedance triangle: The triangle which we draw from an equation

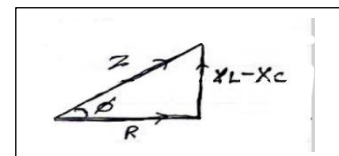
$$Z = \sqrt{R^2 + (X_L^2 - X_C^2)}$$

or $Z^2 = R^2 + (X_L^2 - X_C^2)$ is called impedance triangle.

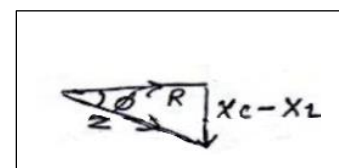
Special cases: As we know that $Z = \sqrt{R^2 + (X_L^2 - X_C^2)}$.

From this equation we explain special cases.

Case (1): When $X_L - X_C$ is positive (i.e $X_L > X_C$), then phase angle Φ will be positive and then the circuit will be inductive.



Case(2): When $X_L - X_C$ is negative (i.e $X_C > X_L$), then phase angle Φ will be negative and the circuit will be capacitive.



Case (3): When $X_L - X_C = 0$, then the circuit will be purely resistive and the phase angle Φ will be zero. Then power

factor $\cos\Phi = \cos 0 = 1$

From the cases the equation for voltage and current generally we can write

$$V = V_m \sin \omega t \text{ and } I = I_m \sin(\omega t \pm \Phi).$$

Resonance in RLC Series AC circuits

The RLC series AC circuit is said to be in resonance when in which the inductive reactance " X_L " is equal to the inductive capacitance X_C . i.e $X_L = X_C$.

Now we explain some important points when the RLC circuit at resonance.

1) As we know that the impedance of RLC series AC circuit can be written as

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \text{-----} \textcircled{1}$$

Now at resonance we also know that i.e $X_L = X_C$ ----- $\textcircled{2}$.

Now put $\textcircled{2}$ in $\textcircled{1}$ we get as

$$Z = \sqrt{R^2 + (X_C - X_C)^2} \Rightarrow Z = R \text{-----} \textcircled{3}$$

Now from $\textcircled{3}$ we can say that at resonance the impedance become small and it is equal to the resistance " R " only.

2) At resonance when the impedance becomes small, then large current will flow in the RLC series AC circuit.

3) At resonance we know that i.e. $Z = R$ -----(4). Then from eq(4) we can say that the RLC circuit will be resistive circuit only. Then we know that in case of resistive circuit the phase angle $\Phi = 0^\circ$ due to same phase of voltage and current. Then at resonance the power factor will be unity, i.e. $\cos\Phi = \cos 0^\circ = 1$.

Expression for resonance frequency: For resonance we know that

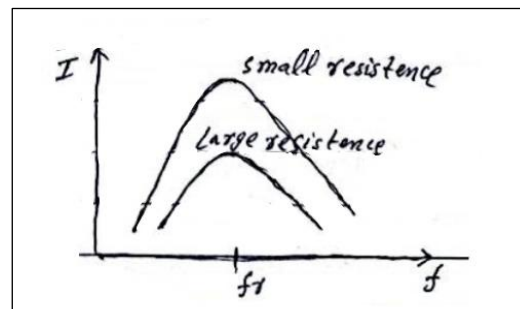
$$X_L = X_C$$

$$\Rightarrow 2\pi f_r L = 1/2\pi f_r C$$

$$\Rightarrow f_r^2 = 1/4\pi^2 L C . \text{ Taking square root to both side then we get as}$$

$$\Rightarrow f_r = 1/2\pi\sqrt{LC} \text{-----} \textcircled{5}$$

From eq (5) it is clear that on increasing either the inductance or the capacitance causes the resonant frequency to decrease. From eq (5) we can also say that for a given value of inductance " L " and capacitance " C " there will be one value of resonance frequency.



Resonance curve b/w current " I " and frequency " f " of power supply:

The magnitude of the current as a function of frequency is shown in figure which shows that the current is maximum at resonance frequency. The current is minimum at low frequency because of greater impedance. That is at low frequency the $X_L = 2\pi f L$ is small and $X_C = 1/2\pi f C$ is large. Then in impedance the term $(X_L - X_C)$ will not be equal to zero. So impedance will be greater.

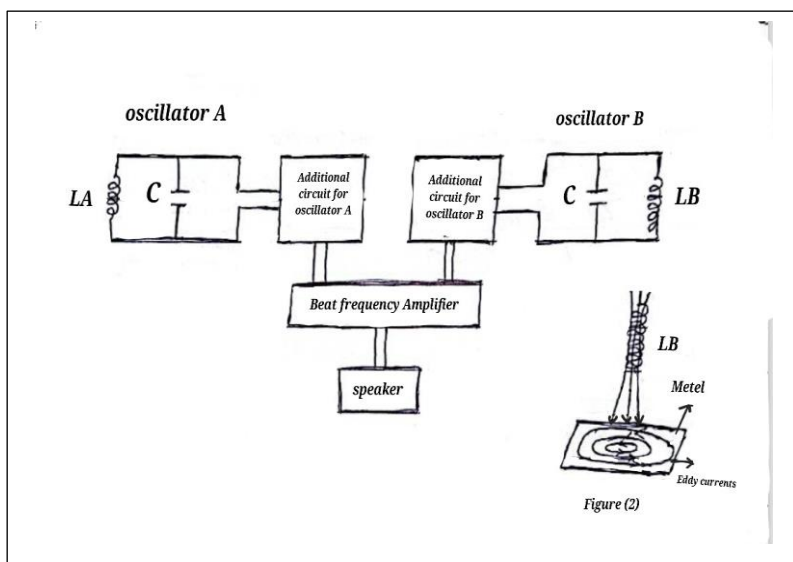
The current is also minimum at greater frequency because of greater impedance. That is at greater frequency the $X_L = 2\pi f L$ is large and $X_C = 1/2\pi f$

C is small. Then in impedance the term $(X_L - X_C)$ will not be equal to zero. So impedance will be greater.

Principle of metal Detectors:

An LC circuit is a circuit in which current or electrical energy oscillates between a capacitor and an inductor. Two such oscillators "A" and "B" are used for the operation of common type of metal detector as shown in figure. In the absence of any nearby metal object, the inductance L_A and L_B are the same and hence the resonance frequency $f = 1/2\pi\sqrt{LC}$ of the two circuits is also same.

Now when inductor "B" called the search coil comes near a metal object as shown in figure.



Then due to magnetic flux change in inductor "B" the eddy current is produced in the metal. Then due to such an eddy current the opposition or inductance L_B is increases. Then due to which the resonance frequency of circuit "B" is decreasing, then in the Beat frequency amplifier the two current signals of different frequencies will not cancel each other. Then the net current of particular frequency will pass from the speaker. Then the sound of the speaker will shows us the presence of the metal near the detector. Such detectors are not only used for security checks but also used to locate burned metal objects.

Maximum power Transfer

The maximum power delivered to a load resistor " R_L " of impedance " Z_L " by a source is called Maximum power transfer.

Explanation: In a figure a source of AC, its internal resistance " R_i " or internal impedance " Z_g " and load resistance " R_L " or impedance " Z_L " are shown in figure. Now we calculate the maximum power out put through a load resistor " R_L " or impedance " Z_L " in the circuit. Now for this we can write as

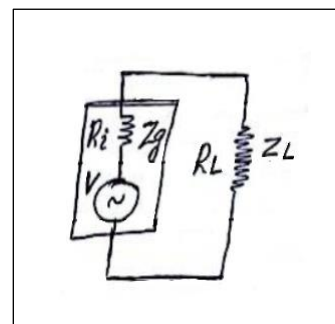
$$P_{out} = I^2 R_L \text{ ----- (1)}$$

Now we can also write as

$$\mathcal{E} = V_{R_i} + V_{R_L}$$

$$\Rightarrow \mathcal{E} = I R_i + I R_L$$

$$\Rightarrow \mathcal{E} = I (R_i + R_L)$$



$$\Rightarrow I = \mathcal{E} / (R_i + R_L)$$

$$\Rightarrow I = \mathcal{E} / (R_L + R_i) \text{ ----- (2)}$$

Now put eq (2) in eq (1). We get as

$$P_{out} = \mathcal{E}^2 R_L / (R_L + R_i)^2$$

$$\Rightarrow P_{out} = \mathcal{E}^2 R_L / (R_L^2 + R_i^2 + 2 R_L R_i)$$

$$\Rightarrow P_{out} = \mathcal{E}^2 R_L / (R_L^2 + R_i^2 + 2 R_L R_i + 2 R_L r - 2 R_L R_i)$$

$$\Rightarrow P_{out} = \mathcal{E}^2 R_L / (R_L^2 + R_i^2 - 2 R_L R_i + 4 R_L R_i)$$

$$\Rightarrow P_{out} = \mathcal{E}^2 R_L / (R_L - R_i)^2 + 4 R_L \text{ ----- (3)}$$

In eq (3) if we put $R_L = R_i$ or $Z_g = Z_i$ i.e. load Resistance = internal Resistance then denominator will become minimum, and " P_{out} " will become maximum. So we can write

$$(P_{out})_{max} = \mathcal{E}^2 R_i / (R_i - R_i)^2 + 4 R_i^2$$

$$\Rightarrow (P_{out})_{max} = \mathcal{E}^2 / 4 R_i \text{ or } (P_{out})_{max} = \mathcal{E}^2 / 4 R_L$$

Maximum power transfer theorem: The power delivered to a load resistor " R_L " or impedance " Z_g " by a source of AC will be maximum if the load resistance value equal to the internal resistance of " R_i " or internal impedance " Z_i " of the Source. Devices like Radio, Television, Antenna and Speaker system operate on maximum power.

Maxwell's equations

Maxwell's equations are a set of equations that describe how electric and magnetic fields propagate, interact, and how they are influenced by objects. Maxwell used the following laws for derivation of his equation.

- ① Gauss's law ② Ampere's law ③ Faraday's law.

(1) Maxwell 1st equation

A changing magnetic field produces an electric field:

Derivation: Consider the figure in which a bar magnet is moving toward a conducting loop with its north pole down, as shown in figure.

Now by Faraday's law changing magnetic flux induces an induced emf in a loop and it can be written as

$$\mathcal{E} = N \Delta \Phi / \Delta t. \quad \text{Put } N = 1 \quad \text{and} \quad \Phi = BA$$

$$\Rightarrow \mathcal{E} = A \Delta B / \Delta t$$

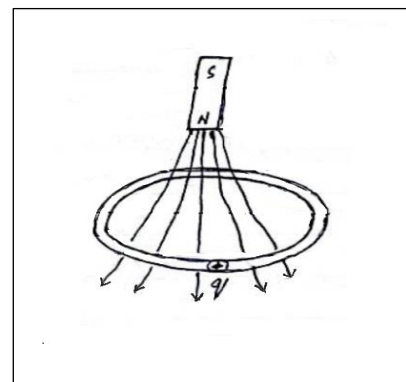
$$\Rightarrow \mathcal{E} = A \Delta B / \Delta t \text{ ----- (1)}$$

Now such an emf will start work on the charge " q " and due to which a charge will move in the loop and current will be producing. So now we can write as

$$\mathcal{E} = W / q$$

$$\Rightarrow \mathcal{E} = Fd / q \text{ ----- (2)}$$

Now the force on the charge will be electric force. So we take



$$F = Fe$$

$$\Rightarrow F = q/E \text{ ----- (3)}$$

Also we take $d = 2\pi r$ for complete rotation of charge "q". Now eq (2) we can write as

$$\varepsilon = q/E \times 2\pi r / q$$

$$\Rightarrow \varepsilon = 2\pi r E \text{ ----- (4)}$$

Now compare eq (1) and (4) we get as

$$2\pi r E = A \Delta B / \Delta t$$

$$\Rightarrow E = \Delta B A / 2\pi r \Delta t \text{ ----- (5)}$$

$$\Rightarrow E = 1 / 2\pi r \Delta \Phi / \Delta t \text{ ----- (6)}$$

From eq (5) and (6) we can say that the changing magnetic field or flux will gives us the electric field.

(2) Maxwell 2nd equation

A changing electric field can produce a magnetic field:

Derivation: Consider a capacitor is connected to an alternating voltage source as shown in the figure. In this case the store charge in the capacitor will be changing w.r.t time due to an alternating voltage. Also the sign of the store charge will depend on the polarity of alternating voltage. Also in this case the current in the circuit will be continuous due to an alternating voltage.

Now the current across the capacitor can be written as

$$I = \Delta Q / \Delta t \text{ ----- (1)}$$

$$\text{Now we know that } Q = CV \text{ ----- (2)}$$

Put eq (2) in eq(1) we get as

$$I = \Delta CV / \Delta t \text{ ----- (3)}$$

$$\text{Now we also know that } C = A\varepsilon_0 / d \text{ ----- (4)}$$

Now put eq (4) in eq(3) we get as

$$I = A\varepsilon_0 \Delta V / \Delta t \text{ ----- (5)}$$

$$\text{Now we also know that } V = Ed \text{ ----- (6)}$$

Now put eq (6) in eq(5). We get as

$$I = A\varepsilon_0 \Delta E d / \Delta t$$

$$\Rightarrow I = \varepsilon_0 A \Delta E / \Delta t \text{ ----- (7)}$$

$$\Rightarrow I = \varepsilon_0 \Delta E A / \Delta t$$

$$\Rightarrow I = \varepsilon_0 \Delta \Phi / \Delta t \text{ ----- (8)}$$

Now from Ampere's law in case of solenoid we know that

$$B = \mu_0 n I \text{ ----- (9)}$$

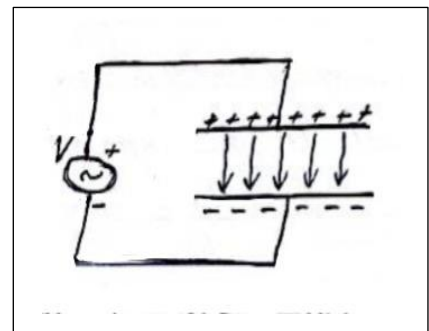
Now put eq(7) in (9) then we get as

$$B = \mu_0 n \varepsilon_0 A \Delta E / \Delta t \text{ ----- (10)}$$

Now put eq(8) in (9) then we get as

$$B = \mu_0 n \varepsilon_0 \Delta \Phi / \Delta t \text{ ----- (11)}$$

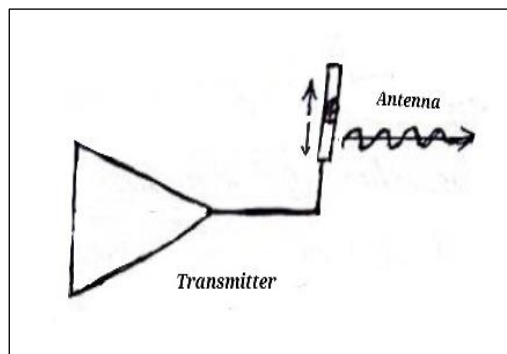
From eq (10) and (11) we can say that changing electric field or changing magnetic flux creates a magnetic field.



(3) Generating of Electromagnetic waves

An antenna generates electromagnetic waves through a process called radiation.

Explanation: In a figure an antenna is connected with a transmitter as shown. A transmitter gives electric signals to antenna. Then these electric signals causes the electrons in the antenna to accelerate back and forth creating an electric current. As the electrons accelerate, they generate a magnetic field around the antenna. Then



the changing magnetic field, in turn, induces an electric field. Then the oscillating electric and magnetic fields combine to form electromagnetic waves, which then radiates from an antenna. Then the electromagnetic waves propagates with two alternating field perpendicular to each other.

(4) Speed of electromagnetic waves equal to the speed of light:

Light is a type of electromagnetic waves. Maxwell predicted theoretically that the speed of electromagnetic waves in free space is given by

$$C = 1/\sqrt{(\epsilon_0\mu_0)} \text{ ----- (1)}$$

Where $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N m}^2$ is the permittivity of free space. And $\mu_0 = 4\pi \times 10^{-7} \text{ Wb/Am}$ is the permeability of free space. Now put these values in eq ① we get as

$$C = 1 / \sqrt{(8.85 \times 10^{-12} \times 4\pi \times 10^{-7})}$$

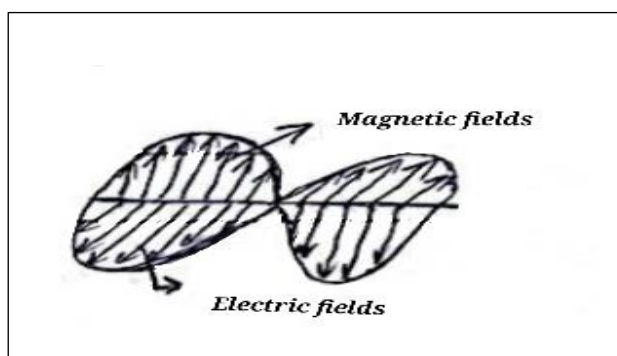
$$C = 3 \times 10^8 \text{ m/s} \text{ ----- (2)}$$

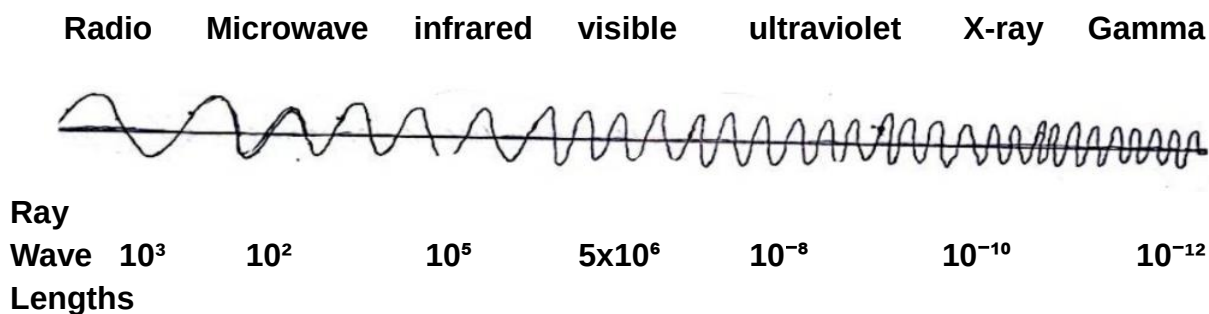
From ② He showed that the speed of electromagnetic waves is equal to the speed of light. Then he said light is one type of electromagnetic waves.

Electromagnetic waves

Those waves which travels through vacuum and having two perpendicular alternating fields with the direction of travelling is called E.M waves.

Electromagnetic waves such as infrared, ultraviolet, etc are different from each other due to their properties. But they have some features in common such as electric and magnetic fields. Therefore it can be described in terms of electric and magnetic fields, and they all travel through vacuum with the same speed of light. Fundamentally these radiations are different only in wavelength or frequency.





Now we explain different type of electromagnetic waves in detail.

① **Light waves:** Light waves are electromagnetic radiations which are emitted by the sun. The wave length of light waves ranges from 400nm to 700nm. Light is often emitted when the outermost electrons in atoms change their state of motion. The color of light tells us something about the atoms of the objects from which it was emitted. The study of the light emitted from the sun and other stars gives information about their composition.

② **Infrared waves:** There are electromagnetic radiations whose wave length ranges from 0.7 micrometer to about 1mm. These radiations are commonly emitted by atoms or molecules when they change their rotational or vibrational motion.

The warmth which your hand feels near a glowing light bulb is initially a result of the infrared radiations emitted from the bulb. Objects of temperature ranges from 3K to 3000K emit their most intense thermal radiation in the infrared region of the spectrum.

The infrared radiations are used in hand remote control for operating TV, DVD set etc in daily life. The signal between a remote control handset and the device it control consists of pulses of infrared light which is invisible to human eye. The infrared radiation is also used cooking the surface of food.

③ **Microwaves:** The microwaves are also called short radio waves whose wavelength ranges from 1mm to 1m. These radiations are commonly produced in electric circuits like in microwave ovens. These radiations are used to transmit telephone conversation. These radiations are also emitted by a neutral hydrogen atom present between the stars of our galaxy.

④ **Radio waves:** These are electromagnetic waves whose wavelengths is longer than 1m. These radiations produced from terrestrial sources through electrons oscillating in wire of electric circuits. These radiations are also emitted from extraterrestrial sources called radio astronomy. These radiations give us useful information about the universe.

⑤ **Ultraviolet waves:** These are electromagnetic radiation whose wavelength ranges from 1nm to 400nm. These radiations can be produced due to atomic transitions of electrons. These radiations are also emitted from the sun. But due to ozone layer; very little amount of these radiations can reach our earth. Brief exposure to ultraviolet radiation can cause common sun burn

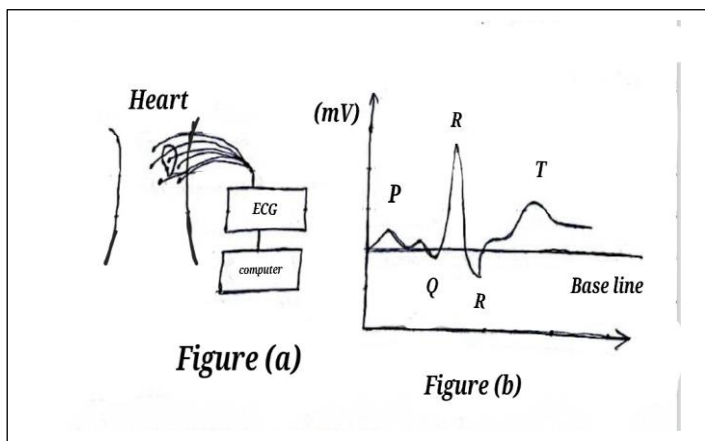
but long term exposure can cause serious effects like skin cancer. These radiations can be used for destroying bacteria's, viruses, molds and the other harmful organisms.

⑥ **X-rays:** These are electromagnetic radiations whose wavelength ranges from 0.01nm to 10nm. These radiations can be produced due to inner shell transition of electrons in an atom. These radiations are also produced when a high speed charged particle becomes slow down due to collision with certain target. These radiations are used in health, industry, agriculture and detective departments for various purposes.

⑦ **Gamma Rays (X-rays):** These are electromagnetic radiations whose wavelength is less than 10 Pico meter. These radiations can be produced during the transitions of an atomic nucleolus from one state to another. These radiations can also be emitted during the decay processes of certain radioactive element. These radiations possess high penetrating power and can damage human body.

ELECTROCARDIOGRAM or (E.C.G)

The electrocardiogram is worldwide used for diagnosing heart conditions. The electric waves in the heart are recorded in millivolts by the electrocardiograph. Our heart produces time varying voltages as it beats. These heart voltages produce small voltage differences between points on your skin that can be measured and used to diagnose the condition of your heart as shown in figure (4). Some of the characteristics of ECG curve are shown in figure (b). When the curve falls below the base line it shows a negative deflection and when it rises above the base line it shows a positive deflection.



A negative deflection indicates that the recorded wave has travelled away from the electrode and a positive deflection means it has travelled towards it. A typical plot of voltage difference between two points on the human body vs time is shown in figure (b). The "P" deflection corresponds to the contraction of the atria at the start of the heart beat. The QRS group corresponds to the contraction of the ventricles.

The "T" deflection corresponds to a recovery of the heart cells in preparation for the next beat. The voltage effect on the skin is due to the charge motion in the heart with the heart beats.

Note: In examples and assignments “V” represents “ V_{rms} ” and “I” represents “ I_{rms} ”.

Examples and Assignments

Example 15.1: An AC circuit consists of a pure resistance of 20Ω and is connected across A.C supply of 220V, 50 Hz. Calculate (a) current (b) Power consumed and (c) equation for voltage and current.

Given: $f = 50\text{Hz}$. $R = 20\Omega$ $V = 220\text{V}$.

Req: $I = ?$ (b) $P = ?$ (c) Equation for voltage and current.

Sol (a): As we know that

$$V = IR$$

$$\Rightarrow I = V/R \quad \Rightarrow I = 220\text{V}/20\Omega \quad \Rightarrow I = 11\text{A} \quad \text{Ans}$$

$$(b): \text{ We also know that } P = VI \quad \Rightarrow P = 220\text{V} \times 11\text{A} \quad \Rightarrow P = 2420\text{W} \quad \text{Ans}$$

(C): First we find equation for voltage. For this we know that

$$V = V_m \sin \omega t \text{ ----- (1)}$$

Now we also know that

$$V_{rms} = V_m / \sqrt{2}$$

$$\Rightarrow V_m = \sqrt{2} V_{rms} \quad \Rightarrow V_m = \sqrt{2} V \quad \Rightarrow V_m = \sqrt{2} \times 220\text{V} \quad \Rightarrow V_m = 311.1\text{V} \text{ --- (2)}$$

We also know that

$$\omega = 2\pi f \quad \Rightarrow \omega = 2 \times 3.14 \times 50 \text{ Hz} \quad \Rightarrow \omega = 314 \text{ rad/s} \text{ ----- (3)}$$

Now put (2) and (3) in eq (1) we get as

$$V = 311.1 \sin (314t) \quad (\text{Ans})$$

Now we find equation for current. For this we know that

$$I = I_m \sin (\omega t) \text{ ----- (4)}$$

Now we also know that

$$I_{rms} = I_m / \sqrt{2}$$

$$\Rightarrow I_m = \sqrt{2} I_{rms} \quad \Rightarrow I_m = \sqrt{2} I \quad \Rightarrow I_m = \sqrt{2} \times 11\text{A} \quad \Rightarrow I_m = 15.55\text{A} \text{ ----- (5)}$$

Now put eq (5) and (3) in eq (4) we get as

$$I = 15.55 \sin (314t). \quad (\text{Ans})$$

Assignment 15.1: A 60Ω resistor is connected across AC supply of 220V. Calculate the current drawn from the Supply and the power consumed by the heating element.

Given: $R = 60\Omega$ $V = 220\text{V}$ Req: (a) $I = ?$ (b) $P = ?$

Sol (a) As we know that

$$V = IR$$

$$\Rightarrow I = V/R \quad \Rightarrow I = 220\text{V}/60\Omega \quad \Rightarrow I = 3.66\text{A} . \text{ Ans}$$

(b) we know that

$$P = IV$$

$$\Rightarrow P = 3.66\text{A} \times 220\text{V} \quad \Rightarrow P = 805\text{W}. \text{ Ans}$$

Example 15.2: A pure inductive coil allows a current of 20A to flow from a 220V, 50Hz Supply. Find (1) inductive reactance (2) Inductance of the coil (3) Power absorbed. Write down the equation for voltage and current.

Given: $I = 20\text{A}$ $V = 220\text{V}$ $f = 50\text{Hz}$

Req: ① $X_L = ?$ ② $L = ?$ ③ $P = ?$ ④ Equation for “V” and “I”

Sol ①: we know that

$$V = I X_L$$

$$\Rightarrow X_L = V/I = 220\text{V}/20\text{A} \quad \Rightarrow X_L = 11\Omega \text{ (Ans)}$$

② we also know that

$$X_L = 2\pi f L$$

$$\Rightarrow L = X_L / 2\pi f \quad \Rightarrow L = 11\Omega / 2 \times 3.14 \times 50 \text{ Hz} \quad \Rightarrow L = 0.035\text{H} \text{ (Ans)}$$

③ Power absorbed or power loss = $P = 0$ (In inductive circuit).

④ Now first we find equation for voltage. For this we know that

$$V = V_m \sin \omega t \text{ ----- (1)}$$

Now we also know that

$$V_{\text{rms}} = V_m / \sqrt{2}$$

$$\Rightarrow V_m = \sqrt{2} V_{\text{rms}} \quad \Rightarrow V_m = \sqrt{2} V \quad \Rightarrow V_m = \sqrt{2} \times 220\text{V} \quad \Rightarrow V_m = 311.1 \text{ V} \text{ -- (2)}$$

We also know that

$$\omega = 2\pi f \quad \Rightarrow \omega = 2 \times 3.14 \times 50 \text{ Hz} \quad \Rightarrow \omega = 314 \text{ rad/s} \text{ ----- (3)}$$

Now put (2) and (3) in eq (1) we get as

$$V = 311.1 \sin (314t) \text{ (Ans)}$$

Now we find equation for current. For this we know that

$$I = I_m \sin (\omega t - 90^\circ) \text{ (In inductive circuit). ----- (4)}$$

Now we also know that

$$I_{\text{rms}} = I_m / \sqrt{2}$$

$$\Rightarrow I_m = \sqrt{2} I_{\text{rms}} \quad \Rightarrow I_m = \sqrt{2} I \quad \Rightarrow I_m = \sqrt{2} \times 20\text{A} \quad \Rightarrow I_m = 28.3\text{A} \text{ ----- (5)}$$

Now put eq (5) and (3) in eq (4) we get as

$$I = 28.3 \sin (314t - 90^\circ). \text{ (Ans)}$$

Example 15-3: The current through a 60mH inductor is

$0.2\sin(377t - 25^\circ)\text{A}$. Write the mathematical expression for the voltage across it.

Given: $L = 60\text{mH} = 60 \times 10^{-3}\text{H}$ $I = 0 \sin (377t - 25^\circ)$.

Req: Expression for voltage “V”

Sol: For voltage we can write as

$$V = V_m \sin (\omega t + 90^\circ - 25^\circ) \quad \Rightarrow V = V_m \sin (\omega t + 65^\circ) \text{ ----- ①}$$

Now it is given that $I = 0.2\sin (377t - 25^\circ) \text{ ----- ②}$

Now from eq ② we can write as

$$\omega = 377 \text{ rad/s} \quad \Rightarrow 2\pi f = 377 \text{ rad/s} \quad \Rightarrow f = 377 \text{ rad/s} / 2 \times 3.14 \quad \Rightarrow f = 60 \text{ Hz} \text{ -- ③}$$

Now we know that $X_L = 2\pi f L$

$$\Rightarrow X_L = 2 \times 3.14 \times 60 \text{ Hz} \times 60 \times 10^{-3}\text{H} \quad \Rightarrow X_L = 22.652 \text{ ----- ④}$$

Now we also know that

$$V_m = I_m X_L \Rightarrow V_m = 0.2A \times 22.652 \Rightarrow V_m = 4.5V \text{ ----- } (5)$$

Now eq(1) can be written as $V = 4.5 \sin(377t + 65^\circ)$ Ans.

Assignment 15.2: A pure inductor is connected across a 10V, 200 Hz supply, and the current flowing through it is measured as 0.4A. Determine the value of its inductance.

Given: $I = 0.4A$ $V = 10V$ $f = 200 \text{ Hz}$ Req : Inductance = $L = ?$

Sol: As we know that

$$X_L = 2\pi f L \Rightarrow L = X_L / 2\pi f \text{ ----- } (1). \text{ We also know that}$$

$$V = I X_L \Rightarrow X_L = V/I \text{ ----- } (2) \text{ put (2) in (1) we get as}$$

$$L = V / 2\pi f I \Rightarrow L = 10V / 2 \times 3.14 \times 200\text{Hz} \times 0.4A \Rightarrow L = 19.9 \times 10^{-3} \text{ H Ans.}$$

Example 15.4: A $318\mu\text{F}$ capacitor is connected across a 220V, 50Hz system. Determine (a) the capacitive reactance (b) RMS value of current and (c) equation for voltage and current.

Given: $C = 318\mu\text{F} = 318 \times 10^{-6}\text{F}$ $V = 220V$ $f = 50 \text{ Hz}$

Req: (a) $X_c = ?$ (b) I_{rms} (c) Equations for “V” and “I”.

Sol. (a) As we know that

$$X_c = 1 / 2\pi f C$$

$$\Rightarrow X_c = 1 / 2 \times 3.14 \times 50\text{Hz} \times 318 \times 10^{-6}\text{F}$$

$$\Rightarrow X_c = 10\Omega \text{ Ans.}$$

(b) We also know that

$$V_{\text{rms}} = I_{\text{rms}} X_c$$

$$\Rightarrow I_{\text{rms}} = V_{\text{rms}} / X_c$$

$$\Rightarrow I_{\text{rms}} = 220V / 10\Omega = 22A \text{ Ans.}$$

© Now we find equation for voltage. For this we can write as

$$V = V_m \sin(\omega t) \text{ ----- } (1)$$

Now we know that

$$V_{\text{rms}} = V_m / \sqrt{2}$$

$$\Rightarrow V_m = \sqrt{2} V_{\text{rms}} \Rightarrow V_m = \sqrt{2} \times 220V \Rightarrow V_m = 311V \text{ ----- } (2)$$

We also know that

$$\omega = 2\pi f \Rightarrow \omega = 2 \times 3.14 \times 50 \text{ Hz} \Rightarrow \omega = 314 \text{ rad/s} \text{ ----- } (3)$$

Now put (2) and (3) in eq (1). We get as

$$V = 311 \sin(314t). \text{ (Ans)}$$

Now for current we can write as

$$I = I_m \sin(\omega t + 90^\circ) \text{ ----- } (4)$$

Now we know that

$$I_{\text{rms}} = I_m / \sqrt{2} \Rightarrow I_m = \sqrt{2} I_{\text{rms}} \Rightarrow I_m = \sqrt{2} \times 22 \text{ A} \Rightarrow I_m = 31.2A \text{ ----- } (5)$$

Now put (5) and (3) in eq (4) we get as

$$I = 31.1 \sin(314t + 90^\circ) \text{ (Ans)}$$

Assignment 15.3: A perfect capacitor is connected across 6V, 5KHz Supply, and the resulting current flow is 88.6 mA. Calculate the capacitance value.

Given: $V = 6V$ $f = 5KHz = 5 \times 10^3 \text{ Hz}$ $I = 88.6 \times 10^{-3} A$ Req: $C = ?$

Sol: we know that

$$X_c = 1 / 2\pi f C \text{ ----- } \textcircled{1}$$

Now we also know that

$$V = I X_c \Rightarrow X_c = V / I \text{ ----- } \textcircled{2} . \text{ Now put } \textcircled{2} \text{ in } \textcircled{1} \text{ we get as}$$

$$\Rightarrow V / I = 1 / 2\pi f C$$

$$\Rightarrow C = I / 2\pi f V$$

$$\Rightarrow C = 88.6 \times 10^{-3} A / 2 \times 3.14 \times 5 \times 10^3 \text{ Hz} \times 6V$$

$$\Rightarrow C = 0.47 \times 10^{-6} F \Rightarrow C = 0.47 \mu F \text{ Ans}$$

Example 15.5: A coil having a resistance of 75Ω and an inductance of 31.8mH is connected to $220V$, 50 Hz Supply. Calculate (a) the circuit current (b) phase angle (c) power factor (d) Power consumed.

Given: $L = 31.8 \times 10^{-3} H = 0.0318H$ $V = 220V$ $R = 75\Omega$ $f = 50\text{Hz}$

Req: (a) $I = ?$ (b) Phase angle $= \Phi = ?$ (c) Power factor $= ?$ (d) $P = ?$

Sol (a) The circuit is RL series circuit. For this we can write as

$$V = I Z \Rightarrow V = I \sqrt{(R^2 + X_L^2)}$$

$$\Rightarrow I = V / \sqrt{(R^2 + X_L^2)} \Rightarrow I = 220V / (\sqrt{R^2 + (2\pi f L)^2})$$

$$\Rightarrow I = 220V / \sqrt{(75)^2 + (2 \times 3.14 \times 50\text{Hz} \times 0.0318H)^2}$$

$$\Rightarrow I = 18.03 A \text{ Ans.}$$

(b): Now we find phase angle ϕ . For this we know that

$$\Phi = \tan^{-1}(X_L / R)$$

$$\Rightarrow \Phi = \tan^{-1}(2\pi f L / R) \Rightarrow \Phi = \tan^{-1}(2 \times 3.14 \times 50\text{Hz} \times 0.0318H)$$

$$\Rightarrow \Phi \approx 55^\circ \text{ Lag (Ans)}$$

Current will be in 75° lagging with phase angle $\Phi = 55^\circ$.

(c): Now we find power factor $= \cos\Phi$

$$\text{Power factor} = \cos(55^\circ) = (0.573 \text{ Lag}) \text{ Ans.}$$

(d) Now we find power loss. For this we know that

$$P = VI \cos\Phi$$

$$\Rightarrow P = 220V \times 18.03A \times 0.573$$

$$\Rightarrow P = 2272.8W \text{ Ans.}$$

Assignment 15.4: $220V$, 50 Hz , AC Source is connected to an inductance of $0.2H$ and a resistance of 20Ω in series. What is the current in the circuit?

Given: $V = 220V$. $f = 50\text{Hz}$ $L = 0.2H$. $R = 20\Omega$

Req: Current in inductive circuit $= I = ?$

Sol: we know that

$$V = I Z$$

$$\Rightarrow I = V/Z \quad \Rightarrow I = V / \sqrt{R^2 + X_L^2}$$

$$\Rightarrow I = V / \sqrt{R^2 + (2\pi fL)^2}$$

$$\Rightarrow I = 220V / \sqrt{((20)^2 + (2 \times 3.14 \times 50\text{Hz} \times 0.2\text{H})^2)} \Rightarrow I = 3.33\text{A Ans.}$$

Example 15.6: A 100V, 50 Hz AC Supply is applied to a capacitor of capacitance 79.5uF connected in series with a non-inductive resistance of 30Ω. Find ① impedance ② current ③ phase angle and ④ Equation for the instantaneous value of current.

Given: $V = 100\text{V}$ $R = 30\Omega$. $f = 50\text{Hz}$ $C = 79.5\mu\text{F} = 79.5 \times 10^{-6} \text{ F}$

Req: ① Impedance = $Z = ?$ ② Current = $I = ?$

③ Phase angle = Φ ④ Equation for current.

Sol: ① The given circuit is RC circuit. So in this case we can write as

$$Z = \sqrt{R^2 + X_C^2}$$

$$\Rightarrow Z = \sqrt{(30\Omega)^2 + (1/(2\pi fC))^2}$$

$$\Rightarrow Z = \sqrt{(30\Omega)^2 + (1/(2 \times 3.14 \times 50\text{Hz} \times 79.5 \times 10^{-6} \text{ F}))^2}$$

$$\Rightarrow Z = 50\Omega \text{ Ans.}$$

② We know that

$$V = IZ \quad \Rightarrow I = V/Z \quad \Rightarrow I = 100\text{V}/50\Omega \quad \Rightarrow I = 2\text{A Ans.}$$

③ We know that

$$\Phi = \tan^{-1}(X_C/R)$$

$$\Rightarrow \Phi = \tan^{-1}(1/(2\pi fCR))$$

$$\Rightarrow \Phi = \tan^{-1}(1/(2 \times 3.14 \times 50\text{Hz} \times 79.5 \times 10^{-6} \text{ F} \times 30\Omega))$$

$$\Rightarrow \Phi = 53^\circ \text{ (Lead) Ans. Current will be in leading.}$$

④ We know that $I = I_m \sin(\omega t + \Phi)$ —①

Now we find I_m . For this we know that

$$I_{\text{rms}} = I_m / \sqrt{2}$$

$$\Rightarrow I_m = \sqrt{2} I_{\text{rms}} \quad \Rightarrow I_m = \sqrt{2} \times I \quad \Rightarrow I_m = \sqrt{2} \times 2\text{A} \quad \Rightarrow I_m = 2.828 \text{ A} \text{ ----- ②}$$

$$\text{Now } \omega = 2\pi f \quad \Rightarrow \omega = 2 \times 3.14 \times 50 \text{ Hz} \quad \Rightarrow \omega = 314 \text{ rad/s} \text{ ----- ③}$$

Put ② and ③ in ①. We get as

$$I = 2.828 \sin(314t + 53^\circ) \text{ Ans.}$$

Example 15.7: A 220V, 50Hz AC supply is applied to a coil of 0.06 H inductance and 2.5Ω resistance connected in series with a 6.8uF capacitor. Calculate (a) impedance (b) current (c) phase angle b/w current and voltage (d) Power factor (e) Power consumed.

Given: $V = 220\text{V}$ $f = 50\text{Hz}$. $L = 0.06\text{H}$. $R = 2.5\Omega$ $C = 6.8 \times 10^{-4}$

Sol (a) The circuit is RLC series AC circuit. So for this we write as

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \text{ ----- ①}$$

$$\text{Now } X_L = 2\pi fL \quad \Rightarrow X_L = 2 \times 3.14 \times 50 \text{ Hz} \times 0.06\text{H} \quad \Rightarrow X_L = 18.85\Omega \text{ ---- ②}$$

$$\text{Now } X_C = 1/2\pi fC \quad \Rightarrow X_C = 1/(2 \times 3.14 \times 50\text{Hz} \times 6.8 \times 10^{-4} \text{ F}) \quad \Rightarrow X_C = 468\Omega \text{ ----- ③}$$

$$\text{Eq ① } \Rightarrow Z = \sqrt{(2.5\Omega)^2 + (18.85\Omega - 468\Omega)^2} \Rightarrow Z = 449.2\Omega. \text{ Ans}$$

(b) we know that

$$V = IZ$$

$$\Rightarrow I = V/Z \quad \Rightarrow I = 220V/449.2\Omega \quad \Rightarrow I = 0.489A. \text{ Ans}$$

(c) we know that $\Phi = \tan^{-1} (X_L - X_C)/R$

$$\Rightarrow \Phi = \tan^{-1}((18.85\Omega - 468\Omega)/2.5\Omega) \quad \Rightarrow \Phi = -89.7^\circ. \text{ Ans}$$

The negative sign show that the current is leading the voltage.

(d) we know that

$$\text{Power factor} = \cos\Phi$$

$$\Rightarrow \text{Power factor} = \cos(-89.7^\circ)$$

$$\Rightarrow \text{Power factor} = 0.0052$$

(e) Now we find power loss. For this we can write as

$$\text{Power} = VI \cos\Phi$$

$$\Rightarrow P = VI \cos\Phi$$

$$\Rightarrow P = 220V \times 0.489A \times 0.0052$$

$$\Rightarrow P = 0.60 \text{ W Ans}$$

Exercise MCQs

1) The rms value of a sine wave is 100A. Its peak value is

- a) 70.7A. b) 141.4A ✓ c) 150A d) 282.8A

Ans: As we know that $I_{rms} = I_m / \sqrt{2} \Rightarrow I_m = \sqrt{2} I_{rms} = \sqrt{2} \times 100A = 141.4A$

2) A sine wave has a frequency of 50 Hz. Its angular frequency is ---- rad/s.

- a) 100π ✓ b) 50π c) 25π d) 5π

Ans : $\omega = 2\pi f \Rightarrow \omega = 2 \times 3.14 \times 50 \text{ Hz} \Rightarrow \omega = 100\pi \text{ rad/s.}$

3) The power factor of partly resistive circuit is ----

- a) Zero b) unity ✓ c) lagging d) leading

Ans: phase angle $\Phi = 0^\circ$ b/w "V" and I in resistive circuit. So $\cos 0^\circ = 1$.

4) A capacitor is perfectly insulator for

- a) Direct current ✓ b) Alternating current
c) Direct as well as Alternating current d) Induced emf.

Ans: Because direct current can fully charge the capacitor which stops flow of current in the circuit.

5) The peak value of alternating current is $5\sqrt{2}$ A. The mean square value of current will be

- a) 5A ✓ b) 2.5A c) $5\sqrt{2}$ A d) 5^2

Ans: $I_{rms} = I_m / \sqrt{2} = 5\sqrt{2} / \sqrt{2} \Rightarrow I_{rms} = 5$

6) In choke coil the reactance X_L and resistance "R" are related as

- a) $X_L = R$. b) $X_L \ll R$ ✓ c) $X_L \gg R$. d) $X_L = \infty$

7) The phase difference between the current and voltage at resonance is

- a) 0 ✓ b) π c) $-\pi$ d) $\pi/2$

Ans: See explanation in topic.

8) In AC system we generate sine wave from because

- a) It can be easily draw

b) It produces least disturbance in electrical circuits ✓

c) It is nature standard

d) other waves can not be produced easily.

9) An alternating voltage is given by $20 \sin 157t$. The frequency of alternating voltage is

a) 50 Hz b) 25Hz ✓ c) 100Hz d) 75 Hz

Ans: We know that $\omega = 2\pi f \Rightarrow f = \omega/2\pi = 157 \text{ rad/s} / 2 \times 3.14 \Rightarrow f = 25\text{Hz}$

10) In LR current which one of the following statements is correct?

a) "L" and "R" opposes each other

b) 'P' value increases with frequency

c) The inductive reactance increases with frequency ✓

d) The inductive reactance decreases with frequency.

11) An alternating quantity (voltage or current) is completely known if we know its

a) Maximum value b) frequency and phase

c) effective value d) Both 'd' and 'b' ✓

12) For electromagnetic waves, Maxwell generalized

a) Gauss's law for magnetism b) Gauss's law for electricity

c) Faraday's law d) Ampere's law ✓

13) An electromagnetic waves goes from air to glass which of the following does not change?

a) Radio waves. (b) X-rays ✓ c) ultraviolet radiation d) Ultra sound waves.

a) wavelength (b) frequency c) Speed d) All

14) The circuit in which current and voltage are in phase, the power factor is?

a) zero (b) 1 ✓ c) -1 d) 2 Ans : $\cos\Phi = \cos 0^\circ = 1$.

15) With increase in frequency of an AC Supply, the impedance of an RLC series circuit

a) remain constant b) Increase ✓

c) decrease d) Capacitive reactance " X_c " decreases only.

Ans: $Z = \sqrt{R^2 + (X_L - X_c)^2}$

$X_L = 2\pi fL$ and $X_c = 1/2\pi fC$

$X_L \Rightarrow$ greater then $X_c \Rightarrow$ decrease . So $Z \Rightarrow$ increase

Conceptual Questions

Question 01: Sketch a graph of emf induced in an inductive coil against rate of change of current. What is the significance of the gradient?

Ans: As we know that the magnitude of self emf induced in an inductive coil due to a current change in it can be written as

$$\varepsilon = L \Delta I / \Delta t$$

Now from eq (1) we can write as $\varepsilon \propto \Delta I$ - (2).

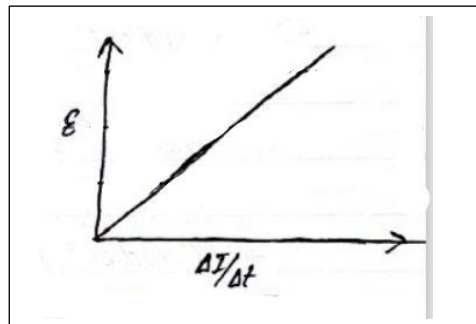
Now from relation (2) we can say that the graph between the emf induced in an inductive coil against rate of change of current will be straight line as shown in fig. Now we find gradient or slope of the graph. i.e

$$\text{Gradient} = \Delta Y / \Delta X$$

$$\Rightarrow \text{Gradient} = \varepsilon / (\Delta I / \Delta t)$$

$$\Rightarrow \text{Gradient} = L \text{ ----- (3)}$$

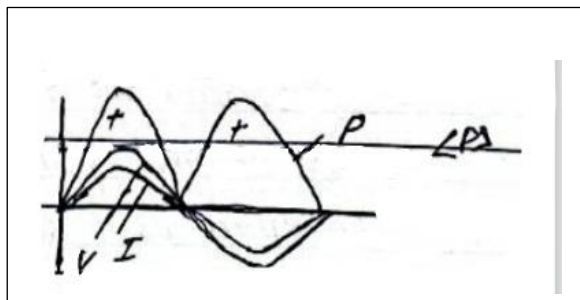
From eq (3) we can say that the significance of the gradient is that it will give us the value of the self inductance "L" of the coil.



Question 02: Current and voltage provided by an AC generator are sometimes negative and sometime positive. Why for, an AC generator connected to a resistor, power can never be negative?

Ans: Power is never negative in case of resistor connected to an AC generator because both current and voltage are in phase in a resistor. In case of resistor the time varying graph of current, voltage and power is shown in a figure. Now in case of resistor we know that

$$P = IV \text{ ----- (1)}$$



During first half cycle, both current and voltage are positive. So from eq (1) we can say that power is positive. During second half cycle, both current and voltage are negative. So from eq (1) we can say that the power is again positive.

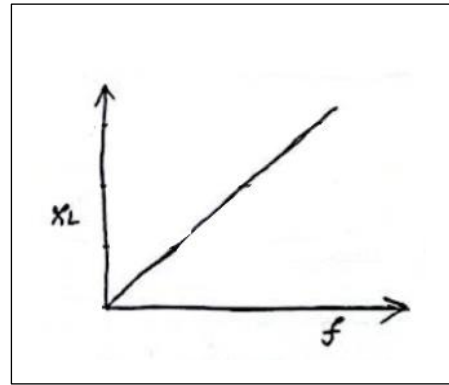
Question 03: What determines the gradient of a graph of inductive reactance against frequency?

Ans: As we know that the magnitude of inductive reactance can be written as

$$X_L = 2\pi fL \text{ ----- (1)}$$

Now from eq (1) we can write as $X_L \propto f$ - (2). Now from relation (2) we can say that the graph between the inductive reactance against frequency will be straight line as shown in figure. Now we find gradient or slope of the graph. i.e

$\text{Gradient} = \Delta Y / \Delta X$
 $\Rightarrow \text{Gradient} = X_L / f$
 $\Rightarrow \text{Gradient} = 2\pi f L / f$
 $\Rightarrow \text{Gradient} = 2\pi L$
 $\Rightarrow \text{Gradient} / 2\pi = L$
 $\Rightarrow \text{Gradient per cycle} = L$ --- (3) ($2\pi \text{ rad} = 1 \text{ cycle}$)
 From eq (3) we can say that gradient per cycle determines the value of self inductance of the inductor.



Question 04: How does doubling the frequency affect the reactance of (a) An inductor (b) a capacitor?

Ans: (a) As we know that the magnitude of inductive reactance can be written as $X_L = 2\pi f L$ ----- (1)

Now we find " X'_L " when we put $f = 2f$ in eq (1). i.e

$$X'_L = 2\pi (2f)L \Rightarrow X'_L = 2(2\pi f L) \Rightarrow X'_L = 2X_L \text{ ----- (2)}$$

So by doubling the frequency f , inductive reactance X_L is also doubled.

Now we know that the magnitude of capacitive reactance

$$X_C = 1 / 2\pi f C \text{ ----- (3)}$$

Now we find " X'_C " when we put $f = 2f$ in eq (3). i.e

$$X'_C = 1 / 2\pi (2f)C \Rightarrow X'_C = 1 / 2(2\pi f C) \Rightarrow X'_C = X_C / 2 \text{ ----- (4)}$$

From eq (4) we can say that by doubling the frequency, capacitive reactance " X_C " will become half.

Question 05: If the peak value of sin wave is 1000 volts, what is the effective value?

Ans: It is given that peak value of voltage is 1000 volts. i.e $V_m = 1000V$. Now we find its effective or rms value. For this we know that

$$V_{rms} = V_m / \sqrt{2}$$

$$\Rightarrow V_{rms} = 1000V / \sqrt{2}$$

$$\Rightarrow V_{rms} = 707V$$

Question 06: Show that reactance is measured in Ohms for both inductors and capacitors?

Ans: As we know that the inductive reactance can be written as

$$X_L = 2\pi f L$$

$$\Rightarrow X_L = \omega L \text{ ----- (1)}$$

We also know that

$$\omega L = V_m / I_m \text{ ----- (2). Put (2) in (1). We get as}$$

$$\Rightarrow X_L = V_m / I_m$$

$$\Rightarrow X_L = \text{volt} / \text{ampere} = \text{Ohm} \quad (V / I = R).$$

$$\Rightarrow X_L = \text{Ohm} \text{ ----- (3)}$$

From (3) we can say that inductive reactance is measured in Ohms.

Now we also know that the capacitive reactance can be written as

$$X_C = 1 / 2\pi f C$$

$$\Rightarrow X_C = 1 / \omega C \text{ ----- (4)}$$

We also know that

$$1/\omega C = V_m / I_m \text{ ----- (5). Put (5) in (4). We get as}$$

$$\Rightarrow X_C = V_m / I_m$$

$$\Rightarrow X_C = \text{volt} / \text{ampere}$$

$$\Rightarrow X_C = \text{Ohm} \text{ ----- (6)}$$

From (6) we can say that the capacitive reactance is measured in Ohm.

Question 07: What is the principle of ECG?

Ans: An electrocardiogram (ECG) is a medical test that measures the electrical activity which are produced due to the movement of ions (sodium, potassium, calcium and chloride) when the heart beats.

Then electrodes are placed on the skin to detect such electrical activity of the heart. Then such electrodes transmit the electrical activity to an ECG device, and then such device amplifies and processes this electrical activity. Then at last the processed electrical activity in the form of voltage is displayed on a paper in the form of wavy curves.

Note: In the numerical problems “V” will represent the “ V_{rms} ” and I will represent “ I_{rms} ”.

Numerical problems

Question 01: The peak voltage of an AC supply is 300V. What is the “ V_{rms} ” voltage?

Given: $V_m = 300V$ Req: V_{rms}

Sol: As we know that

$$V_{rms} = V_m / \sqrt{2} \quad \Rightarrow V_{rms} = 300V / \sqrt{2} \quad \Rightarrow V_{rms} = 212V \text{ Ans}$$

Question 02: The I_{rms} value of current in an AC circuit is 10A. What is the peak current?

Given : $I_{rms} = 10A$ Req : $I_m = ?$

Sol: As we know that

$$I_{rms} = I_m / \sqrt{2} \quad \Rightarrow I_m = \sqrt{2} I_{rms} \quad \Rightarrow I_m = \sqrt{2} \times 10A \quad \Rightarrow I_m = 14.1A \text{ Ans}$$

Question 03: The AC Voltage across a $0.5\mu F$ capacitor is

$16 \sin(2 \times 10^3 t)V$. Find the capacitive reactance (b) the peak value of current through capacitor.

Given : $C = 0.5 \times 10^{-6}F$ and $V = 16 \sin(2 \times 10^3 t)V$

Req: a) $X_C = ?$ and $I_m = ?$

Sol (a) It is given that

$$V = 16 \sin(2 \times 10^3 t) \text{ ----- (1). We also know that}$$

$$V = V_m \sin(\omega t) \text{ ----- (2)}$$

Comparing eq(1) and eq(2). We get as

$$V_m = 16V \text{ and } \omega = 2 \times 10^3 \text{ rad/s. Now we know that}$$

$$X_C = 1 / 2\pi fC$$

$$\Rightarrow X_C = 1 / \omega C$$

$$\Rightarrow X_C = 1 / (2 \times 10^3 \text{ rad/s} \times 0.5 \times 10^{-6} \text{ F})$$

$$\Rightarrow X_C = 1000 \Omega$$

Now we find I_m . For this we know that

$$V_m = I_m X_C$$

$$\Rightarrow I_m = V_m / X_C$$

$$\Rightarrow I_m = 16 \text{ V} / 1000 \Omega$$

$$\Rightarrow I_m = 16 \times 10^{-3} \text{ A} = 16 \text{ mA (Ans.)}$$

Question 04: The voltage across a $0.01 \mu\text{F}$ capacitor is $240 \sin(1.25 \times 10^4 t - 30^\circ) \text{ V}$. Write the mathematical expression for the current through it.

Given: $C = 0.01 \times 10^{-6} \text{ F}$ $V = 240 \sin(1.25 \times 10^4 t - 30^\circ)$

Req: Equation for current.

Sol: As we know that in capacitive circuit the current leads voltage by 90° . So equation for current can be written as

$$I = I_m \sin(\omega t - 30^\circ + 90^\circ) \Rightarrow I = I_m \sin(\omega t + 60^\circ) \text{ ----- (1)}$$

Now it is given that

$$V = 240 \sin(1.25 \times 10^4 t - 30^\circ) \text{ ----- (2)}$$

From eq(2) we can write as

$$V_m = 240 \text{ V and } \omega = 1.25 \times 10^4 \text{ rad/s.}$$

Now we know that

$$V_m = I_m X_C$$

$$\Rightarrow I_m = V_m / X_C$$

$$\Rightarrow I_m = V_m / (1 / 2\pi fC)$$

$$\Rightarrow I_m = 2\pi fC V_m$$

$$\Rightarrow I_m = \omega C V_m$$

$$\Rightarrow I_m = 1.25 \times 10^4 \text{ rad/s} \times 0.01 \times 10^{-6} \text{ F} \times 240 \text{ V}$$

$$\Rightarrow I_m = 0.03 \text{ A}$$

Now eq(1) we can write as

$$I = 0.03 \sin(1.25 \times 10^4 t + 60^\circ) \text{ Ans.}$$

Question 8.5: An inductor with an inductance of $100 \mu\text{H}$ passes a current of 10 mA . When its terminal voltage is 6.3 V , calculate the frequency of AC Supply.

Given: $L = 100 \times 10^{-6} \text{ H}$ $I = 10 \times 10^{-3} \text{ A}$ $V = 6.3 \text{ V}$ Req: $f = ?$

Sol: As we know that

$$V = I X_L$$

$$\Rightarrow V = I \times 2\pi f L$$

$$\Rightarrow f = V / 2\pi L I$$

$$\Rightarrow f = 6.3 \text{ V} / (2 \times 3.14 \times 100 \times 10^{-6} \times 10 \times 10^{-3} \text{ A}) \Rightarrow f = 10^6 \text{ Hz (Ans.)}$$

Question 8.6: (a) Calculate the inductive reactance of a 3mH inductor. When 60 Hz and 10KHz AC voltages are applied. (b) What is I_{rms} current at each frequency if the applied rms voltage is 120V?

Given: $L = 3 \times 10^{-3} \text{ H}$ $f_1 = 60 \text{ Hz}$ $f_2 = 10 \times 10^3 \text{ Hz}$ $V = 120\text{V}$.

Req: a) $(X_L)_1 = ?$ $(X_L)_2 = ?$ (b) $(I_{rms})_1 = ?$ $(I_{rms})_2 = ?$

Sol: we know that

$X_L = 2\pi fL$. Now we find as

$$(X_L)_1 = 2\pi f_1 L = 2 \times 3.14 \times 60 \text{ Hz} \times 3 \times 10^{-3} \text{ H} = 1.135 \Omega$$

$$(X_L)_2 = 2\pi f_2 L = 2 \times 3.14 \times 10 \times 10^3 \text{ Hz} \times 3 \times 10^{-3} \text{ H} = 188.4 \Omega$$

Now we also know that

$$V_{rms} = I_{rms} X_L$$

$$\Rightarrow I_{rms} = V_{rms} / X_L$$

$$\text{Now } (I_{rms})_1 = V_{rms} / (X_L)_1 = 120\text{V} / 1.135 \Omega = 106 \text{ A}$$

$$\text{Now } (I_{rms})_2 = V_{rms} / (X_L)_2 = 120\text{V} / 188.4 \Omega = 0.637 \text{ A}$$

Question 8.7: For the same RLC Series circuit having a 40Ω resistor, a 3mH inductor, and a $5 \mu\text{F}$ capacitor. (a) Find the resonant frequency. (b) Calculate I_{rms} at resonance if V_{rms} is 120V.

Given: $R = 40 \Omega$ $L = 3 \times 10^{-3} \text{ H}$ $C = 5 \times 10^{-6} \text{ F}$ $V_{rms} = 120\text{V}$

Req: (a) $f_r = ?$ (b) $I_{rms} = ?$

Sol: As we know that

$$f_r = 1 / 2\pi\sqrt{LC}$$

$$\Rightarrow f_r = 1 / (2 \times 3.14 \sqrt{(3 \times 10^{-3} \text{ H} \times 5 \times 10^{-6} \text{ F})}) \Rightarrow f_r = 1300 \text{ Hz}$$

Now we know that $V_{rms} = I_{rms} Z$ (in RLC Circuit).

$$\Rightarrow I_{rms} = V_{rms} / Z \text{ ----- (1)}$$

Now at Resonance $Z = R$ ----- (2)

Now put eq (2) in eq (1). We get as

$$I_{rms} = V_{rms} / R$$

$$\Rightarrow I_{rms} = 120 \text{ V} / 40 \Omega$$

$$\Rightarrow I_{rms} = 3 \text{ A Ans}$$

Question 8.8: A resistor of resistance 30Ω is connected in series with a capacitor of capacitance $79.5 \mu\text{F}$ across a power supply of 50Hz and 100V. Find (a) Impedance (b) current (c) phase angle and (d) Equation for the instantaneous value of current.

Given: $R = 30 \Omega$ $C = 79.5 \times 10^{-6} \text{ F}$ $f = 50 \text{ Hz}$ $V = 100\text{V}$

Req: (a) $Z = ?$ (b) $I = ?$ (c) $\Phi = ?$ (d) Equation for current.

Sol: (a) we know that in RC circuit as

$$Z = \sqrt{R^2 + X_C^2} \text{ ----- (1) Now}$$

$$X_C = 1 / 2\pi fC$$

$$\Rightarrow X_C = 1 / (2 \times 3.14 \times 50 \text{ Hz} \times 79.5 \times 10^{-6} \text{ F})$$

$$\Rightarrow X_C = 40 \Omega \text{ ----- (2)}$$

New eq (1) we can write as

$$Z = \sqrt{(30\Omega)^2 + (40\Omega)^2}$$

$$\Rightarrow Z = 50\Omega \text{ Ans}$$

b) Now we find current "I". For this we know that

$$V = I Z \quad \Rightarrow I = V/Z \quad \Rightarrow I = 100V/50 \Omega \quad \Rightarrow I = 2A \text{ Ans.}$$

c) Now we find phase angle. For this we know that

$$\Phi = \tan(R/Z) \quad \Rightarrow \Phi = \tan^{-1}(30\Omega/50\Omega) \quad \Rightarrow \Phi = 53.1^\circ \text{ Lead}$$

Now we find equation for current. For this we can write as

$$I = I_m \sin(\omega t + 53^\circ) \text{ ----- (3)}$$

Now we find "I_m". For this we can write as

$$I_{rms} = I_m/\sqrt{2}$$

$$\Rightarrow I_m = \sqrt{2} I_{rms}$$

$$\Rightarrow I_m = \sqrt{2} \times 2A$$

$$\Rightarrow I_m = 2.828A$$

Now we find " ω ". i.e

$$\omega = 2\pi f \quad \Rightarrow \omega = 2 \times 314 \times 50\text{Hz} \quad \Rightarrow \omega = 314 \text{ rad/s}$$

Now $I = 2.828 \sin(314t + 53.1^\circ)$ Ans.

Question 09: A coil having a resistance of 7Ω and an inductance of 31.8 mH is connected to $230V$, 50 Hz Supply. Calculate (a) the circuit current (b) Phase angle (c) power factor (d) Power consumed.

Given: $R = 7\Omega$ $L = 31.8 \times 10^{-3} \text{ H}$ $V = 230V$ $f = 50\text{Hz}$

Req: (a) $I = ?$ (b) $\Phi = ?$ (c) $\cos \Phi = ?$ (d) $P = ?$

Sol: As we know that

$$V = I Z \quad \Rightarrow I = V/Z \text{ ----- (1)}$$

Now in RL circuit we know that

$$Z = \sqrt{R^2 + X_L^2}$$

$$\Rightarrow Z = \sqrt{R^2 + (2\pi f L)^2}$$

$$\Rightarrow Z = \sqrt{(7\Omega)^2 + (2 \times 3.14 \times 50\text{Hz} \times 31.8 \times 10^{-3} \text{ H})^2}$$

$$\Rightarrow Z = \sqrt{(7\Omega)^2 + (10 \Omega)^2}$$

$$\Rightarrow Z = 12.2\Omega$$

Now equation (1) we can write as

$$I = 230V/12.2\Omega \quad \Rightarrow I = 18.8A$$

(b) we know that

$$\Phi = \tan^{-1}(R/Z) \quad \Rightarrow \Phi = \tan^{-1}(75/12.2\Omega) \quad \Rightarrow \Phi = 55^\circ \text{ Lag}$$

(c) power factor = $\cos \Phi = \cos 55^\circ = 0.574$.

(d) Now we find power consumed. For this we can write as

$$P = VI \cos \Phi$$

$$\Rightarrow P = 230V \times 18.8A \times 0.574$$

$$\Rightarrow P = 2481W$$

Question 10: A capacitor which has a resistance of 10Ω and a capacitance value of $100 \mu\text{F}$ is connected to a supply voltage given

as $V(t) = 100 \sin(314t)$. Calculate the capacitive reactance, circuit impedance and peak current flowing into the capacitor.

Given: $R = 10\Omega$ $C = 100 \times 10^{-6} \text{ F}$ $V(t) = 100 \sin(314t)$

Req: (a) X_C (b) Z (c) I_m

Sol. It is given that

$V(t) = 100 \sin(314t)$ and

Now $V_m = 100\text{V}$ and $\omega = 314 \text{ rad/s}$.

Now we find " X_C ". i.e

$$X_C = 1 / 2\pi fC$$

$$\Rightarrow X_C = 1 / \omega C$$

$$\Rightarrow X_C = 1 / 314 \text{ rad/s} \times 100 \times 10^{-6} \text{ F}$$

$$\Rightarrow X_C = 31.85 \Omega \text{ Ans.}$$

b) Now we find impedance " Z ". For RC circuit we can write as

$$Z = \sqrt{(R^2 + (X_C)^2)} \Rightarrow Z = \sqrt{(10\Omega)^2 + (31.85 \Omega)^2}$$

$$\Rightarrow Z = 33.4 \Omega \text{ Ans.}$$

c) Now we find peak current " I_m ". For this we can write as

$$V_m = I_m X_C$$

$$\Rightarrow I_m = V_m / X_C$$

$$\Rightarrow I_m = 100\text{V} / 31.8552$$

$$\Rightarrow I_m = 3.14\text{A} \text{ Ans.}$$

END OF CHAPTER 05